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Quantitative Methods,
Economics, and
Corporate Finance

Level I Book 1

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Regards,

A handwritten signature in dark ink, appearing to read "Derek Burkett".

Derek Burkett, CFA, FRM, CAIA
Vice President (Advanced Designations)

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Book 1: Quantitative Methods, Economics, and Corporate Finance

SchweserNotes™ 2027

Level I CFA®



SCHWESERNOTES™ 2027 LEVEL I CFA® BOOK 1: QUANTITATIVE METHODS, ECONOMICS,
AND CORPORATE FINANCE

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Learning Outcome Statements (LOS)

1. Returns of Financial Assets and Instruments

The candidate should be able to:

- a. describe, compare, and interpret returns.
- b. describe, compare, and interpret required rates of return, risk-free rates, risk premia, and inflation.

2. Types of Financial Returns

The candidate should be able to:

- a. calculate, compare, and interpret different types of returns for financial assets, instruments, and indicators.

3. Benchmarking Returns

The candidate should be able to:

- a. calculate and compare money-weighted and time-weighted rates of return.
- b. describe the choices and the implications of the different weighting methods used in index construction and management, and calculate, interpret, and explain the value and the returns of an index.

4. The Time Value of Money in Finance

The candidate should be able to:

- a. calculate and interpret the present value of fixed-income and equity instruments based on expected future cash flows.
- b. calculate and interpret the implied return of fixed-income instruments and required return and implied growth of equity instruments given their present value and cash flows.
- c. explain the cash flow additivity principle and its importance for the condition of no arbitrage, and explain its use in calculating implied forward interest rates, forward exchange rates, and option values.

5. Statistical Characteristics of Asset Returns

The candidate should be able to:

- a. calculate, interpret, and evaluate various measures of (1) central tendency and location and (2) dispersion.
- b. describe, interpret, and evaluate measures of skewness and kurtosis.
- c. calculate, interpret, and evaluate covariance and correlation.
- d. calculate, interpret, and evaluate semi-deviation and coefficient of variation.

6. Statistical Distributions for Financial Asset Prices and Returns

The candidate should be able to:

- a. calculate, interpret, and evaluate unconditional expected values for mean, variance, and covariance.
- b. calculate, interpret, and evaluate the principal moments of key statistical distributions used in finance.
- c. c
- d. formulate investment problems through Bayesian updating.

7. Estimation and Hypothesis Testing

The candidate should be able to:

- a. explain the central limit theorem and the application of confidence intervals and sampling methodologies.
- b. explain hypothesis testing and its components, including statistical significance, Type I and Type II errors, and the power of a test; construct appropriate hypothesis tests; and interpret the results.
- c. compare and contrast parametric and non-parametric tests, describe situations in which each is the more appropriate type of test, construct appropriate hypothesis tests, and interpret the results.

8. The Return and Risk of a Financial Portfolio

The candidate should be able to:

- a. calculate, interpret, and evaluate the expected return, variance, standard deviation, covariance, and correlation of portfolio returns.
- b. describe, calculate, and interpret the minimum-variance portfolio and portfolios that lie on the efficient frontier.
- c. explain the selection of an optimal portfolio, given an investor's risk aversion and the capital allocation line, and how this extends to the market portfolio and the capital market line.

9. Simulation of Financial Asset Prices and Returns

The candidate should be able to:

- a. describe historical simulation and explain how it can be used in investment applications.
- b. describe bootstrap resampling, and explain how it can be used in investment applications.
- c. describe Monte Carlo simulation and explain how it can be used in investment applications.

10. Applications of Simple Linear Regression in Finance

The candidate should be able to:

- a. describe, interpret, and explain simple linear regression, including coefficient estimation using the least squares criterion.

- b. describe and compare the assumptions of simple linear regression, identify violations through analyzing residuals, evaluate the estimated model's goodness-of-fit and regression coefficients, and results of ANOVA estimates.
- c. calculate and interpret predicted values, the standard error of the estimate, and prediction intervals for the dependent variable in a simple linear regression model and describe different functional forms.
- d. calculate and interpret the variable estimates of the capital asset pricing model (CAPM).

11. Introduction to Financial Data Science

The candidate should be able to:

- a. describe how big data, machine learning, and artificial intelligence are used in financial data science, fintech, and investment management.

12. The Firm and Market Structures

The candidate should be able to:

- a. determine and interpret breakeven and shutdown points of production, as well as how economies and diseconomies of scale affect costs under perfect and imperfect competition.
- b. describe characteristics of perfect competition, monopolistic competition, oligopoly, and pure monopoly.
- c. explain supply and demand relationships under monopolistic competition, including the optimal price and output for firms as well as pricing strategy.
- d. explain supply and demand relationships under oligopoly, including the optimal price and output for firms as well as pricing strategy.
- e. identify the type of market structure within which a firm operates and describe the use and limitations of concentration measures.

13. Understanding Business Cycles

The candidate should be able to:

- a. describe the business cycle and its phases.
- b. describe credit cycles.
- c. describe how resource use, consumer and business activity, housing sector activity, and external trade sector activity vary over the business cycle and describe their measurement using economic indicators.

14. Fiscal Policy

The candidate should be able to:

- a. compare monetary and fiscal policy.
- b. describe roles and objectives of fiscal policy as well as arguments as to whether the size of a national debt relative to GDP matters.

- c. describe tools of fiscal policy, including their advantages and disadvantages.
- d. explain the implementation of fiscal policy and difficulties of implementation as well as whether a fiscal policy is expansionary or contractionary.

15. Monetary Policy

The candidate should be able to:

- a. describe the roles and objectives of central banks.
- b. describe tools used to implement monetary policy tools and the monetary transmission mechanism, and explain the relationships between monetary policy and economic growth, inflation, interest, and exchange rates.
- c. describe qualities of effective central banks; contrast their use of inflation, interest rate, and exchange rate targeting in expansionary or contractionary monetary policy; and describe the limitations of monetary policy.
- d. explain the interaction of monetary and fiscal policy.

16. Introduction to Geopolitics

The candidate should be able to:

- a. describe geopolitics from a cooperation versus competition perspective.
- b. describe geopolitics and its relationship with globalization.
- c. describe functions and objectives of the international organizations that facilitate trade, including the World Bank, the International Monetary Fund, and the World Trade Organization.
- d. describe geopolitical risk.
- e. describe tools of geopolitics and their impact on regions and economies.
- f. describe the impact of geopolitical risk on investments.

17. International Trade

The candidate should be able to:

- a. describe the benefits and costs of international trade.
- b. compare types of trade restrictions, such as tariffs, quotas, and export subsidies, and their economic implications.
- c. explain motivations for and advantages of trading blocs, common markets, and economic unions.

18. Capital Flows and the FX Market

The candidate should be able to:

- a. describe the foreign exchange market, including its functions and participants, distinguish between nominal and real exchange rates, and calculate and interpret the percentage change in a currency relative to another currency.

- b. describe exchange rate regimes and explain the effects of exchange rates on countries' international trade and capital flows.
- c. describe common objectives of capital restrictions imposed by governments.

19. Exchange Rate Calculations

The candidate should be able to:

- a. calculate and interpret currency cross-rates.
- b. explain the arbitrage relationship between spot and forward exchange rates and interest rates, calculate a forward rate using points or in percentage terms, and interpret a forward discount or premium.

20. Organizational Forms, Corporate Issuer Features, and Ownership

The candidate should be able to:

- a. compare the organizational forms of businesses.
- b. describe key features of corporate issuers.
- c. compare publicly and privately owned corporate issuers.

21. Investors and Other Stakeholders

The candidate should be able to:

- a. compare the financial claims and motivations of lenders and shareholders.
- b. describe a company's stakeholder groups and compare their interests.
- c. describe environmental, social, and governance factors of corporate issuers considered by investors.

22. Corporate Governance: Conflicts, Mechanisms, Risks, and Benefits

The candidate should be able to:

- a. describe the principal-agent relationship and conflicts that may arise between stakeholder groups.
- b. describe corporate governance and mechanisms to manage stakeholder relationships and mitigate associated risks.
- c. describe potential risks of poor corporate governance and stakeholder management and benefits of effective corporate governance and stakeholder management.

23. Working Capital and Liquidity

The candidate should be able to:

- a. explain the cash conversion cycle and compare issuers' cash conversion cycles.
- b. explain liquidity and compare issuers' liquidity levels.
- c. describe issuers' objectives and compare methods for managing working capital and liquidity.

24. Capital Investments and Capital Allocation

The candidate should be able to:

- a. describe types of capital investments.
- b. describe the capital allocation process, calculate net present value (NPV), internal rate of return (IRR), and return on invested capital (ROIC), and contrast their use in capital allocation.
- c. describe principles of capital allocation and common capital allocation pitfalls.
- d. describe types of real options relevant to capital investments.

25. Capital Structure

The candidate should be able to:

- a. calculate and interpret the weighted-average cost of capital for a company.
- b. explain factors affecting capital structure and the weighted-average cost of capital.
- c. explain the Modigliani–Miller propositions regarding capital structure.
- d. describe optimal and target capital structures.

26. Business Models

The candidate should be able to:

- a. describe key features of business models.
- b. describe various types of business models.

READING 1

RETURNS OF FINANCIAL ASSETS AND INSTRUMENTS

MODULE 1.1: RATES OF RETURN

LOS 1.a: Describe, compare, and interpret returns.

We can evaluate returns on a financial asset, a financial instrument, or a financial indicator. These categories overlap somewhat, but we can define them as follows:

Financial assets are items investors can own that represent value, and primarily include:

- | cash
- | equity ownership
- | debt, which is an asset of the party to whom a debt is owed
- | securities that are hybrids of debt and equity

Financial instruments are standardized means of trading financial assets in markets, and primarily include:

- | equity and debt securities
- | tradable standardized contracts, such as some derivatives

Financial indicators are observable measures of value that do not generate cash flows, and include:

- | interest rates
- | currency exchange rates
- | market indexes

Returns in a Single Period

The two sources of asset returns are price changes and cash income. We speak of the **price return** (or **capital appreciation return**) as the percentage change in an asset's price between two points in time:

$$r_{\text{price}} = \frac{P_1 - P_0}{P_0}, \text{ where } P_0 \text{ is the beginning price and } P_1 \text{ is the ending price.}$$

The **capital distribution return** is the cash income an asset generates during a period, as a percentage of the asset's beginning price. Examples of capital distribution return we will see later in the Level I curriculum include the **dividend yield** of an equity security and the **current yield** of a fixed income security:

$$r_{\text{distribution}} = \frac{\text{income}}{P_0}$$

Total return on an asset during a period is the sum of its price return and capital distribution return for that period:

$$\begin{aligned} r_{\text{total}} &= r_{\text{price}} + r_{\text{distribution}} \\ &= \frac{P_1 - P_0 + \text{income}}{P_0} \end{aligned}$$



PROFESSOR'S NOTE

We often refer to this as an asset's **holding period return**. As we will see later in this reading, returns are typically stated on an annualized basis rather than on a holding period basis.

EXAMPLE: Price, capital distribution, and total return

A stock is valued at €20 at the beginning of the period, pays €1 in dividends over the period, and at the end of the period is valued at €22. Calculate its price return, capital distribution return, and total return for the period.

Answer:

$$\text{Price return} = \frac{22 - 20}{20} = 10\%$$

$$\text{Capital distribution return} = \frac{1}{20} = 5\%$$

$$\text{Total return} = 10\% + 5\% = 15\%, \text{ or } \frac{22 - 20 + 1}{20} = 15\%$$

A return calculated from known past prices and income is an **actual return**, or **ex post return**. We can also estimate an **expected return**, or **ex ante return**, for a future period. Investors make decisions based on expected returns, but actual returns most often differ from what was expected. The difference between expected returns and actual returns represents the concept of *risk*, which we will address in the next LOS.

Returns based on cash that an investor has received are **realized returns**. These include income received (capital distributions) and the proceeds from selling an investment. We may also speak of the **unrealized return** on an asset that an investor still holds, based on its current market price. The distinction between realized and unrealized returns is important for tax reasons. With some exceptions, tax authorities generally assess taxes on income in the period an investor receives it, and on price gains in the period when an investor realizes them by selling an asset.

Returns Across Multiple Periods

The **arithmetic mean return** is the simple average of a series of periodic returns:

$$\text{arithmetic mean return} = \frac{(R_1 + R_2 + R_3 + \dots + R_n)}{n}$$

The **geometric mean return**, or **compound average return**, is a compound rate:

$$\text{geometric mean return} = \sqrt[n]{(1 + R_1) \times (1 + R_2) \times (1 + R_3) \times \dots \times (1 + R_n)} - 1$$

EXAMPLE: Arithmetic and geometric mean return

For the last three years, the returns for Acme Corporation common stock have been -9.34%, 23.45%, and 8.92%. Calculate the arithmetic average and compound average rate of return over the three-year period.

Answer:

$$\text{arithmetic mean return} = \frac{-9.34 + 23.45 + 8.92}{3} = 7.68\%$$

geometric mean return

$$\begin{aligned} &= \sqrt[3]{(1 - 0.0934) \times (1 + 0.2345) \times (1 + 0.0892)} - 1 \\ &= \sqrt[3]{0.9066 \times 1.2345 \times 1.0892} - 1 \\ &= \sqrt[3]{1.21903} - 1 \\ &= 1.06825 - 1 = 6.825\% \end{aligned}$$



PROFESSOR'S NOTE

You can get a third root (or other higher-order root) with your calculator as follows:

- On the TI, enter 1.21903 [y^x] 3 [1/x] - 1 [=]
- On the HP, enter 1.21903 [ENTER] 3 [1/x] [y^x] 1 -

Notice in the example that the geometric mean return is less than the arithmetic mean return. The **arithmetic-geometric mean inequality principle** states that the geometric mean is always less than or equal to the arithmetic mean, and the difference increases as the variability of the observations increases. The only time the arithmetic and geometric means are equal is when there is no variability in the observations (i.e., all observations are equal).

An arithmetic mean return assumes the amount invested at the start does not change. However, that amount does change if an investment experiences any price appreciation or depreciation, or has any cash flows. For this reason, an arithmetic mean return can be misleading, as the following example shows.

EXAMPLE: Geometric and arithmetic mean returns

An investment of \$1,000 loses 25% in Period 1 and gains 25% in Period 2. Is the return over both periods more accurately represented by the arithmetic mean return or the geometric mean return?

Answer:

In Period 1, the investment's value decreases to $\$1,000(1 - 0.25) = \750 . In Period 2, its value increases to $\$750(1 + 0.25) = \937.50 .

The arithmetic mean return is $\frac{-25\% + 25\%}{2} = 0\%$. This is misleading because it suggests the investment's value did not change over the two periods.

The geometric mean return is $\sqrt{(1 - 0.25)(1 + 0.25)} - 1 = -3.175\%$. This accurately reflects the actual change in value over the two periods: $\$1,000(1 - 0.03175)(1 + 0.03175) = \937.51 (difference due to rounding).

Compounding Periods and Annualized Returns

Interest rates and market returns are typically stated as **annualized returns**, regardless of the actual length of the period over which they occur. To annualize a return that is realized over a specific number of days, use the following formula:

$$\text{annualized return} = (1 + \text{return during the period})^{365/\text{days}} - 1$$

EXAMPLE: Annualized return, shorter than one year

A saver deposits \$100 into a bank account. After 90 days, the account balance is \$100.75. What is the saver's annualized rate of return?

Answer:

$$\text{return over 90 days} = \frac{100.75}{100} - 1 = 0.0075 = 0.75\%$$

$$\text{annualized return} = (1 + 0.0075)^{365/90} - 1 = 0.0308 = 3.08\%$$

EXAMPLE: Annualized return, longer than one year

An investor buys a 500-day government bill for \$970 and redeems it at maturity for \$1,000. What is the investor's annualized return?

Answer:

$$\text{return over 500 days} = \frac{1000}{970} - 1 = 0.0309 = 3.09\%$$

$$\text{annualized return} = (1 + 0.0309)^{365/500} - 1 = 0.0225 = 2.25\%$$

In our earlier example of arithmetic and geometric means, the geometric mean resulted in an annual rate of return because the holding periods were years. If the holding periods are other than years, the geometric mean is *not* the same as the annual return. The root for the geometric mean is the number of *periods*, while the root for the annual return is the number of *years* from the start of the first holding period to the end of the last holding period.

EXAMPLE: Geometric mean vs. annual return

For the last four semiannual periods, the 6-month holding period returns on an investment were 2.0%, 0.5%, -1.0%, and 1.5%. Calculate the geometric mean and the annual rate of return.

Answer:

$$\begin{aligned}\text{geometric mean} &= \sqrt[4]{(1 + 0.02)(1 + 0.005)(1 - 0.01)(1 + 0.015)} - 1 \\ &= 0.007435 = 0.7435\%, \text{ which is the geometric mean} \\ &\quad \text{of the six-month holding period returns.}\end{aligned}$$

$$\begin{aligned}\text{annual return} &= \sqrt[2]{(1 + 0.02)(1 + 0.005)(1 - 0.01)(1 + 0.015)} - 1 \\ &= 0.0149 = 1.49\%\end{aligned}$$

The four semiannual periods equal two years, so to get an annual return, we use two as the root.

When we know the number of periods in a year (as we do with daily, weekly, monthly, or quarterly periods), we can easily annualize the returns for a period using the following formula:

$$\text{annualized return} = (1 + \text{return during the period})^c - 1$$

where:

c is the number of periods that make up a year.

EXAMPLE: Comparing returns on an annualized basis

An investor observes the following three securities:

- Security A produced a monthly return of 0.5%.
- Security B produced a quarterly (three-month) return of 1.5%.
- Security C produced a semiannual (six-month) return of 3.0%.

Which security produced the highest annualized return?

Answer:

- Security A annualized return $= (1 + 0.005)^{12} - 1 = 0.0617 = 6.17\%$
- Security B annualized return $= (1 + 0.015)^4 - 1 = 0.0614 = 6.14\%$
- Security C annualized return $= (1 + 0.03)^2 - 1 = 0.0609 = 6.09\%$

On an annualized basis, Security A produced the highest return.

While annualizing returns provides us with a consistent standard for comparing the performance of different investments, we must recognize that it assumes returns are repeatable. In the previous example, the annualized return on Security A is based on one month's return of 0.5%. To realize a return of 6.17% over a full year, the security would have to produce the same 0.5% return in each of another 11 months. Its actual performance is likely to differ—and even if its monthly returns average 0.5%, they are likely to exhibit some variability.

Continuous Compounding

In time value of money calculations (which we will address in more detail in our reading on The Time Value of Money in Finance), more frequent compounding has an impact on future value and present value computations. For example, we can calculate the present value of \$1,000 to be received one year from now using a stated annual interest rate of 6% with a range of compounding periods.

With semiannual compounding:

$$PV = 1,000 \left(1 + \frac{0.06}{2} \right)^{-2} = 942.60$$

With quarterly compounding:

$$PV = 1,000 \left(1 + \frac{0.06}{4} \right)^{-4} = 942.18$$

With monthly compounding:

$$PV = 1,000 \left(1 + \frac{0.06}{12} \right)^{-12} = 941.91$$

With daily compounding:

$$PV = 1,000 \left(1 + \frac{0.06}{365} \right)^{-365} = 941.77$$

The mathematical limit of shortening the compounding period is known as continuous compounding. Given the return for a period, we can use the natural logarithm (ln, or LN on your financial calculator) to state its associated **continuously compounded return** or **logarithmic return**:

$$R_{cc} = \ln(1 + \text{return during the period}) = \ln\left(\frac{\text{ending value}}{\text{beginning value}}\right)$$

EXAMPLE: Calculating continuously compounded returns

A stock was purchased for \$100 and sold one year later for \$120. Calculate the investor's annual rate of return on a continuously compounded basis.

Answer:

$$\ln\left(\frac{120}{100}\right) = 18.232\%$$

If we were given the return (20%) instead, the calculation is this:

$$\ln(1 + 0.20) = 18.232\%$$

A useful property of continuously compounded rates of return is that they are additive for multiple periods. That is, a continuously compounded return from $t = 0$ to $t = 2$ is the sum of the continuously compounded return from $t = 0$ to $t = 1$ and the continuously compounded return from $t = 1$ to $t = 2$.

EXAMPLE: Continuously compounded return over multiple periods

The value of an investment is 1,000 at $t = 0$, 1,150 at $t = 1$, and 1,250 at $t = 2$. Calculate its continuously compounded rates of return for each period and for the time from $t = 0$ to $t = 2$.

Answer:

$$\text{From } t = 0 \text{ to } t = 1: \ln\left(\frac{1,150}{1,000}\right) = 0.13976 = 13.976\%$$

$$\text{From } t = 1 \text{ to } t = 2: \ln\left(\frac{1,250}{1,150}\right) = 0.08338 = 8.338\%$$

$$\text{From } t = 0 \text{ to } t = 2: \ln\left(\frac{1,250}{1,000}\right) = 0.22314 = 22.314\%$$

$$\text{Or, } 13.976\% + 8.338\% = 22.314\%.$$



MODULE QUIZ 1.1

1. If an investment loses 3% of its value over 120 days, its annualized return is *closest* to:
 - A. -8.0%.
 - B. -8.5%.
 - C. -9.0%.
2. If a stock's initial price is \$20 and its price increases to \$23, its continuously compounded rate of return is *closest* to:
 - A. 13.64%.
 - B. 13.98%.
 - C. 15.00%.
3. An ex ante return is described *most accurately* as a return:

- A. expected over a future period.
- B. realized in a specified past period.
- C. that would be realized in a year if past performance continues.

MODULE 1.2: COMPONENTS AND MEASURES OF RETURN

LOS 1.b: Describe, compare, and interpret required rates of return, risk-free rates, risk premia, and inflation.

Interest rates measure the time value of money, although risk differences among financial securities lead to differences in their equilibrium interest rates.

Equilibrium interest rates are the **required rate of return** for a particular investment, in the sense that the market rate of return is the return that investors and savers require to get them to willingly lend their funds. Interest rates are also referred to as **discount rates** and, in fact, the terms are often used interchangeably. If an individual can borrow funds at an interest rate of 10%, then that individual should discount payments to be made in the future at that rate to get their equivalent value in current dollars or other currencies.

We can also view interest rates as the **opportunity cost** of current consumption. If the market rate of interest on 1-year securities is 5%, earning an additional 5% is the opportunity forgone when current consumption is chosen rather than saving (postponing consumption).

For any asset that has **risk**, or uncertainty about the amount or timing of its future cash flows, the interest rate (investors' required rate of return) will reflect the degree of risk. We can think of an interest rate as consisting of a **risk-free rate** and a **risk premium**.

Risk-Free Rates of Return

The risk-free rate (which we sometimes refer to as the **nominal risk-free rate**) has two components: a real risk-free rate and an expected inflation rate. When we speak of a "real" rate of return, we are referring to an investor's increase in purchasing power (i.e., after adjusting for inflation). The **real risk-free rate** is an interest rate that contains no expectation of inflation and no uncertainty about future cash flows. What the real risk-free rate represents in economic terms is

time preference, the degree to which current consumption is preferred to equal future consumption.

Because expected inflation in future periods is not zero, the rates we observe on instruments that are considered free of default risk (e.g., US Treasury bills) are essentially risk-free rates, but not real risk-free rates. Treasury bill rates are nominal risk-free rates because they contain an **inflation premium**.

The relation between nominal and real risk-free rates is as follows:

$$(1 + \text{nominal risk-free rate}) = (1 + \text{real risk-free rate})(1 + \text{expected inflation rate})$$

or,

$$\text{real risk-free rate} = \frac{1 + \text{nominal risk-free rate}}{1 + \text{inflation rate}} - 1$$

Often, including in many parts of the CFA curriculum, this relation is approximated as follows:

$$\text{real risk-free rate} \approx \text{nominal risk-free rate} - \text{expected inflation rate}$$

Risk Premia

Securities may have one or more types of risk. The risk premium includes each type of risk, so each added risk increases the required rate of return. A risk premium in the interest rate on a bond may include (but is not limited to) the following:

- | **Default risk premium.** This reflects the risk that a borrower will not make the promised payments in a timely manner.
- | **Liquidity risk premium.** This reflects the risk of receiving less than fair value for an investment if it must be sold quickly for cash.
- | **Maturity risk premium.** The prices of longer-term bonds are more sensitive to interest rate changes than those of shorter-term bonds. Longer-maturity bonds have more maturity risk than shorter-term bonds and require a maturity risk premium.

Examples of risk premia in the required rate of return on a stock include (but are not limited to) the following:

- | **Size risk premium.** This reflects the difference between the returns on large-capitalization stocks and small-capitalization stocks.

- | **Value risk premium.** This reflects the difference between the returns on value stocks (those with high ratios of book value to market value) and stocks with lower book-to-market value ratios.

Later in the Level I curriculum, we will discuss and use the concept of a **market risk premium**, which we define as the difference between the return on the “market portfolio” of all risky assets and the return on a risk-free asset.

Common Measures of Return

As we saw with risk-free returns, a **real return** is a nominal return adjusted for inflation. Consider an investor who earns a nominal return of 7% over a year when inflation is 2%. The investor’s approximate real return is simply $7 - 2 = 5\%$. The investor’s exact real return is slightly lower: $1.07 / 1.02 - 1 = 0.049 = 4.9\%$.

A real return measures the increase in an investor’s purchasing power—how much more goods she can purchase at the end of one year due to the increase in the value of her investments. If she invests \$1,000 and earns a nominal return of 7%, she will have \$1,070 at the end of the year. If the price of the goods she consumes has increased 2%, from \$1.00 to \$1.02 per unit, she will be able to consume $1,070 / 1.02 = 1,049$ units. She has given up consuming 1,000 units today, but instead, she is able to purchase 1,049 units at the end of one year. Her purchasing power has gone up 4.9%; this is her real return.

We often speak of an investment’s **excess return**, which is how much more (or less) it earns compared to a benchmark or a reference rate. For example, if the chosen benchmark (such as the nominal risk-free rate) is 3%, and an investment earns 8%, we can approximate its excess return as $8\% - 3\% = 5\%$. More exactly, its excess return is $1.08 / 1.03 - 1 = 4.85\%$.

Gross return refers to the total return on a security portfolio before deducting fees for the management and administration of the investment account. **Net return** refers to the return after these fees have been deducted. Both gross and net returns deduct commissions on trades and other costs that are necessary to generate the investment returns.

gross return = total return – transaction costs

net return = gross return – management fees and administrative costs

Pretax nominal return refers to the return before paying taxes. **After-tax nominal return** refers to the return after the tax liability is deducted. Tax rates on investments differ and may be affected by tax laws where the investment is made as well as those in the investor's home jurisdiction. Dividend income, interest income, short-term capital gains, and long-term capital gains may all be taxed at different rates. Many countries tax long-term capital gains at lower rates than short-term capital gains. Investors typically can use capital losses on some investments to offset capital gains on other investments for tax purposes. Capital gains are usually not taxed until the period when the investor realizes the gains.

Assuming a jurisdiction has one tax rate for capital gains and another single tax rate for capital distributions like interest and dividends, we can express the after-tax nominal rate of return as follows:

$$\text{after-tax return} = (\text{price return})(1 - \text{capital gains tax rate}) + (\text{capital distribution return})(1 - \text{capital distribution tax rate})$$

EXAMPLE: Pretax and after-tax nominal returns

An investor buys a stock for \$160 at the beginning of the year and sells it for \$200 at the end of the year. The stock paid \$8 in dividends during the year. The investor is subject to tax rates of 25% on capital gains and 40% on capital distributions. Calculate the investor's pretax and after-tax nominal rates of return.

Answer:

Price return = $(\$200 - \$160) / \$160 = 25\%$

Capital distribution return = $\$8 / \$160 = 5\%$

The pretax return is $25\% + 5\% = 30\%$.

The after-tax return is $25\%(1 - 0.25) + 5\%(1 - 0.40) = 21.75\%$.

In finance, *leverage* refers to using borrowed funds or derivatives to increase the amount of an asset an investor can control. A **leveraged return** refers to a return to an investor that is a multiple of the return on the underlying asset. The leveraged return is calculated as the gain or loss on the investment as a percentage of an investor's cash investment.

To illustrate the effect of leverage on returns, consider a fund that can invest the amount V_E in a portfolio without leverage, and earn the rate of return r_P on that

portfolio. The fund's unleveraged return (as a money amount) is simply $r_P \times V_E$. Now, let's say this fund can borrow the amount V_D at an interest rate of r_D , and can still earn r_P by investing the proceeds. The fund's leveraged return (again as a money amount), after subtracting the interest cost, then becomes $r_P \times (V_E + V_D) - (r_D \times V_D)$.

Thus, stated as a rate of return on the initial value of V_E , the leveraged rate of return is as follows:

$$\text{leveraged return} = \frac{r_P(V_D + V_E) - r_D V_D}{V_E}$$

EXAMPLE: Effect of leverage on a positive return

Mammoth Fund can earn an 8% return on an unleveraged portfolio of \$800 million. The fund can borrow up to \$200 million at a 5% interest rate. Calculate the fund's leveraged return if it borrows and invests this amount.

Answer:

$$\begin{aligned} \text{leveraged return} &= \frac{r_P(V_D + V_E) - r_D V_D}{V_E} \\ &= \frac{0.08(\$200 \text{ million} + \$800 \text{ million}) - 0.05(\$200 \text{ million})}{\$800 \text{ million}} = 8.75\% \end{aligned}$$

Using leverage increased the fund's rate of return from 8% to 8.75%.

EXAMPLE: Effect of leverage on a negative return

Continuing the previous example, calculate Mammoth Fund's leveraged return if it experienced a loss of 4% on its investment portfolio.

Answer:

$$\begin{aligned} \text{leveraged return} &= \frac{-0.04(\$200 \text{ million} + \$800 \text{ million}) - 0.05(\$200 \text{ million})}{\$800 \text{ million}} \\ &= -6.25\% \end{aligned}$$

Using leverage decreased the fund's rate of return from -4% to -6.25%.

These examples highlight the key point investment managers must understand about leverage: *It magnifies gains, but it also magnifies losses.*

An investment in a derivative security, such as a futures contract or an option, produces a leveraged return because the cash deposited is only a fraction of the value of the assets underlying the derivative.



MODULE QUIZ 1.2

1. An interest rate is *best* interpreted as a:
 - A. discount rate or a measure of risk.
 - B. measure of risk or a required rate of return.
 - C. required rate of return or the opportunity cost of consumption.
2. An interest rate from which the inflation premium has been subtracted is *most accurately* described as a:
 - A. real interest rate.
 - B. risk-free interest rate.
 - C. real risk-free interest rate.
3. The value of an investment increases 5% before commissions and fees. This 5% increase represents:
 - A. the investment's net return.
 - B. the investment's gross return.
 - C. neither the investment's net return nor its gross return.

KEY CONCEPTS

LOS 1.a

Total return = price return + capital distribution return.

Price return = $\frac{P_1 - P_0}{P_0}$, where P_0 is the beginning price and P_1 is the ending price.

Capital distribution return = $\frac{\text{income}}{P_0}$.

Actual return or ex-post return is calculated from actual past prices and income. Expected return or ex-ante return is an estimated future return.

Realized returns are based on cash an investor has received. Unrealized returns are based on the market prices of assets an investor still owns.

$$\text{arithmetic mean return} = \frac{(R_1 + R_2 + R_3 + \dots + R_n)}{n}$$

$$\text{geometric mean return} = \sqrt[n]{(1 + R_1) \times (1 + R_2) \times (1 + R_3) \times \dots \times (1 + R_n)} - 1$$

The geometric mean is less than the arithmetic mean, as long as there is any variability in the observed returns.

For returns over any number of days:

$$\text{annualized return} = (1 + \text{return during the period})^{365/\text{days}} - 1$$

For returns over a period, when c is the number of periods that make up a year:

$$\text{annualized return} = (1 + \text{return during the period})^c - 1$$

Continuously compounded return:

$$r_{cc} = \ln(1 + \text{return during the period}) = \ln\left(\frac{\text{ending value}}{\text{beginning value}}\right)$$

LOS 1.b

The required rate of return for a particular investment is the return that investors and savers require to get them to willingly lend their funds. We can also view interest rates as the opportunity cost of current consumption.

An interest rate or required rate of return for a risky asset consists of a risk-free rate and a risk premium.

The real risk-free rate contains no expectation of inflation and zero probability of default. A nominal risk-free rate includes an inflation premium, but no risk premium.

$$\text{real risk-free rate} = \frac{1 + \text{nominal risk-free rate}}{1 + \text{inflation rate}} - 1$$

Or, approximately, $\text{real risk-free rate} \approx \text{nominal risk-free rate} - \text{expected inflation rate}$.

Examples of risk premia for bonds include default risk, liquidity risk, and maturity risk. Examples of risk premia for stocks include size risk and value risk.

Real return is a nominal return adjusted for inflation.

Excess return is the difference between an investment's rate of return and that of a chosen benchmark.

Gross return is total return after deducting transaction costs, but before deducting management and administration fees. Net return is the return after these fees have been deducted.

Pretax nominal return is the return before paying taxes. After-tax nominal return is the return net of tax liability.

Leverage magnifies both gains and losses on investments.

Leveraged return = $\frac{r_P(V_D + V_E) - r_D V_D}{V_E}$, where r_P is the unleveraged return, V_E is the amount invested without leverage, r_D is the interest rate on borrowed funds, and V_D is the amount borrowed.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 1.1

- . C Annualized return = $(1 - 0.03)^{\frac{365}{120}} - 1 = -0.0885 = -8.85\%$. (LOS 1.a)
- . B $\ln\left(\frac{23}{20}\right) = 0.1398 = 13.98\%$. (LOS 1.a)
- . A An ex ante return is an expected return. A return realized in a past period is an ex post return. An annualized return is one that would be realized in a year if an investment's past performance continued over the year. (LOS 1.a)

Module Quiz 1.2

- . C Interest rates can be interpreted as required rates of return, discount rates, or opportunity costs of current consumption. A risk premium can be, but is not always, a component of an interest rate. (LOS 1.b)
- . A Real interest rates are those that have been adjusted for inflation. Inflation has been subtracted, but this does not mean the interest rate is risk-free. (LOS 1.b)
- . C Gross return is the total return after deducting commissions on trades and other costs necessary to generate the returns, but before deducting fees for the management and administration of the investment account. Net return is the return after management and administration fees have been deducted. (LOS 1.b)

READING 2

TYPES OF FINANCIAL RETURNS

MODULE 2.1: FINANCIAL ASSETS, INSTRUMENTS, AND INDICATORS



PROFESSOR'S NOTE

In this reading, we discuss the types of financial returns earned by the major types of financial instruments used in investment portfolios. These instruments, such as equities, bonds, and derivatives, are only informally introduced here; each will have its own topic area later where we will discuss them in much more detail. For now, focus on how different financial instruments have different components of returns, and on how this affects an investment manager's decisions.

LOS 2.a: Calculate, compare, and interpret different types of returns for financial assets, instruments, and indicators.

Types of Return

As discussed in the previous reading, investors earn returns from two major sources:

Price return (or **capital appreciation return**) comes from a change in the value of the investment.

Capital distribution return comes from the income that an investment periodically pays. This can be *fixed*, where the rate is stated at the outset, or *variable*, where the rate is uncertain (e.g., when it is linked to prevailing interest rates at the time a particular capital distribution is made).

The **total return** from an investment is the sum of its price return and capital distribution return.

Types of Financial Instruments

The main types of financial instruments are equities, bonds, and derivatives.

- | **Equities** (also referred to as *stocks* or *shares*) represent an ownership stake in the issuer of the securities, usually a corporation. The largest component of equity return is price return, driven by changes in the factors that affect the company's earnings. These factors can be company-specific, such as a new product launch, or broader market factors that affect all stocks, such as economic growth. Equity issuers may make **variable capital distributions** in the form of discretionary dividend payments.
- | **Bonds** (also referred to as *fixed income* or *debt securities*) are tradable debt instruments. A bondholder is a lender who is typically promised a series of regular fixed interest payments over the bond's life (referred to as *coupons*), and a repayment of the loan balance, (referred to as a bond's *principal*, *par value*, or *face value*), typically at the end of the bond's life (referred to as its *maturity*). The largest component of a bond's return is usually the coupon payment, which is a **fixed capital distribution**. However, a bond's market value can change during its life, leading to price return.
- | **Derivatives** are financial instruments that derive their value from the performance of a specified reference asset. For example, a derivative might lock in a price at which two parties will exchange a specific amount of gold on a specific date in the future. This contract will be profitable to the buyer if gold increases in price, while the seller of the contract would suffer a loss.

Cash (notes, coins, and non-interest-bearing deposits) is considered a financial asset that provides no return. **Hybrid financial assets** also exist that offer returns and risks that are a blend of equity and debt, such as a bond that the holder can convert into equity shares of the issuer.

Types of Financial Indicators

Financial indicators are metrics that can affect financial returns, but do not generate cash flows directly themselves. Examples include interest rates, foreign exchange rates, and market indexes.

Interest rates, paid to savers when they lend funds, are observed most explicitly when banks quote rates on interest-bearing deposits. In capital markets, interest rates are observed through the returns offered by securities issued by borrowers in the fixed income market.

An important interest rate is the rate of return offered on short-term government securities. For a developed economy, these rates are referred to as **risk-free rates** because it can be assumed that the government will make the promised payments on these securities, which can therefore be viewed as being free of **credit risk**.

A 3-month US Treasury bill (T-bill) is an example of such a security. T-bills are priced at a discount to their face value and pay their face value at maturity. The following example shows how we can derive a T-bill's rate of return from its price.

EXAMPLE: Rate of return for a 3-month US T-bill

A 3-month US T-bill with a principal value of \$1,000 will mature in 90 days. The T-bill's price is \$991.55. Calculate the annualized risk-free rate implied by this price, using a 365-day year.

Answer:

The total return of the T-bill over 90 days is calculated as:

$$r = \frac{P_1 - P_0 + \text{income}}{P_0} = \frac{1,000 - 991.55 + 0}{991.55} = 0.008522, \text{ or } 0.8522\%$$

The annualized return, r_{annual} , is calculated as follows:

$$r_{\text{annual}} = (1.008522)^{\frac{365}{90}} - 1 = 0.0350, \text{ or } 3.50\%.$$

If market interest rates increase, the value of fixed-income securities decreases, leading to negative price return.

EXAMPLE: Impact of increasing rates on price of a 3-month US T-bill

Calculate the change in price of the 90-day US T-bill from the previous example if the annualized 3-month risk-free rate (based on a 365-day year) increases to 4.0%.

Answer:

The new 90-day interest rate is calculated as:

$$r_{\text{annual}} = (1.04)^{\frac{90}{365}} - 1 = 0.009718, \text{ or } 0.9718\%$$

The new price of the T-bill is:

$$\text{price} = \frac{\text{principal}}{1 + r_{90\text{-day}}} = \frac{1,000}{1.009718} = 990.38$$

If the annualized 90-day risk-free rate increases from 3.5% to 4.0%, the price of a 90-day T-bill decreases by $\$991.55 - \$990.38 = \$1.17$.

As we will see in the Fixed Income topic area, price sensitivity to changes in market interest rates is more pronounced for longer-dated bonds, which have more fixed cash flows over a longer time frame.

Changes in the perceived creditworthiness of the issuer also affect the value of fixed-income securities. If the issuer's default risk increases, the value of the bond decreases. Debt issued by governments is usually considered to have lower default risk than corporate debt.

Foreign exchange rates represent the value of one country's currency expressed in the value of a different currency. For example, the amount of US dollars (USD) per British pound (GBP) could be quoted as 1 GBP = 1.35 USD. Changes in foreign exchange rates affect returns for international investors holding foreign assets. A weakening foreign currency decreases the returns in domestic currency terms of investments that are denominated in the weakening currency; a strengthening foreign currency increases returns for an investor who holds assets denominated in that currency.



PROFESSOR'S NOTE

Domestic currency return refers to returns in the investor's home currency and includes the impact of changes in exchange rates. The return of a foreign asset in foreign currency terms (i.e., ignoring the impact of foreign exchange rates) is called the *local currency return*. We will discuss foreign exchange rates in more detail, along with quotation conventions used on the Level I exam, in the Economics topic area. For now, simply pay attention to how changes in foreign exchange rates can affect returns.

EXAMPLE: Impact of foreign exchange rates on returns

A US investor invests in UK company shares at a price of GBP 4.00 per share, when the exchange rate is 1.400 USD per GBP. The investor holds the shares for a year, at the end of which the company pays a dividend of GBP 0.08 per

share, the share price is GBP 4.40, and the exchange rate is 1.3393 USD per GBP. Calculate the return to the investor in USD.

Answer:

Original cost of the shares in USD = GBP 4.00 $\times$ 1.4000 = USD 5.60

Final value of the shares plus the dividend income in USD = (GBP 4.40 + GBP 0.08) $\times$ 1.3393 = USD 6.00

Domestic return to the US investor is calculated as:

$$\begin{aligned} r_{\text{USD}} &= \frac{\text{ending USD value} - \text{starting USD value}}{\text{starting USD value}} \\ &= \frac{\text{USD } 6.00 - \text{USD } 5.60}{\text{USD } 5.60} = 0.0714, \text{ or } 7.14\% \end{aligned}$$

Another way to reach the same result is to combine the local return of the foreign asset, r_{GBP} , with the return of foreign exchange rate, r_{FX} :

$$\begin{aligned} \text{Local currency return of the foreign asset} = r_{\text{GBP}} &= \frac{P_1 - P_0 + \text{income}}{P_0} = \\ &= \frac{\text{GBP } 4.40 - \text{GBP } 4.00 + \text{GBP } 0.08}{\text{GBP } 4.00} = 0.12, \text{ or } 12\% \end{aligned}$$

$$\begin{aligned} \text{Return of the foreign exchange rate} = r_{\text{FX}} &= \frac{1.3393 - 1.4000}{1.4000} = -0.04336, \\ &\text{or } -4.336\% \end{aligned}$$

Then, the domestic return, r_{USD} , is calculated by compounding these two sources of return:

$$\begin{aligned} r_{\text{USD}} &= (1 + 0.12)(1 - 0.04336) - 1 \\ &= (1.12)(0.9566) - 1 = 0.0714, \text{ or } 7.14\% \end{aligned}$$

The domestic return to the US investor (7.14%) is lower than the local return in the foreign market (12%) due to a weakening of the foreign currency (-4.336%).

A **market index** represents the average performance of a group of investments that represent a market, sector, geography, or asset class. Examples include the S&P 500 Index in the United States, or the Bloomberg Global Aggregate Bond Index.

Indexes are more diversified than individual securities. Over time, this decreases the variability of index returns compared to individual securities, as we will show in upcoming readings. It also improves multiperiod returns on a compound geometric basis, as we demonstrate in the following example.

EXAMPLE: Index returns vs. individual stock returns

The returns of an individual stock, and an index for the market in which the stock trades, over three consecutive years are presented in the following table:

Year	Individual Stock Return	Index Return
1	50%	10%
2	−50%	−10%
3	15%	3%

Calculate the arithmetic average return and geometric average return for both the individual stock and the index.

Answer:

Arithmetic returns are calculated as simple averages:

$$\text{arithmetic average return of individual stock} = \frac{50\% - 50\% + 15\%}{3} = 5\%$$

$$\text{arithmetic average return of index} = \frac{10\% - 10\% + 3\%}{3} = 1\%$$

Geometric returns are calculated as follows:

$$\text{geometric average return of individual stock} = [(1.50)(0.50)(1.15)]^{1/3} - 1 = -0.0481, \text{ or } -4.81\%$$

$$\text{geometric average return of index} = [(1.10)(0.90)(1.03)]^{1/3} - 1 = 0.0065, \text{ or } 0.65\%$$

This example demonstrates clearly the negative impact of volatility in returns on multiperiod compound returns: greater volatility decreases multiperiod compound returns. The arithmetic mean return of the volatile individual stock (5%) is greater than the arithmetic mean return of the less volatile index (1%), but the compound geometric return of the index (0.65%) is greater than that of the individual stock (−4.81%). Arithmetic returns, which do not reflect compounding, do not capture this volatility drag over time, so they tend to overstate multiperiod investment performance for volatile assets.



MODULE QUIZ 2.1

1. Which of the following statements regarding different types of returns is *most accurate*?
 - A. Equities typically generate returns primarily through capital appreciation, with a smaller proportion of returns coming from fixed capital distribution.
 - B. Changes in foreign exchange rates affect both the domestic capital appreciation returns and domestic capital distribution returns of international investors.

- C. Fixed-income securities do not provide variable capital distribution returns.
- 2. For coupon-paying bonds, when interest rates increase:
 - A. capital appreciation returns are positive.
 - B. capital distribution returns are positive.
 - C. longer-dated bonds outperform shorter-dated bonds.
- 3. An analyst observes that for a particular historical period, geometric average returns are 12.26% for large-cap stocks and 11.23% for small-cap stocks. The relative outperformance of large-cap stocks compared to small-cap stocks is *closest* to:
 - A. 0.93%.
 - B. 1.03%.
 - C. 9.17%.

MODULE 2.2: LONG-TERM RETURNS ON FINANCIAL ASSETS

The relative historical performance of different asset classes can vary, depending on the period analyzed. However, the **risk-return tradeoff** in investing holds that rational investors should, over the long term, demand higher returns from riskier asset classes to compensate them for higher potential losses.

For example, this would suggest that, relative to large-cap stocks, investors should demand higher returns on small-cap stocks due to their higher volatility. It also suggests that, relative to government bonds, investors should demand a higher return on corporate bonds due to their higher default risk. (As we will see in Fixed Income, this extra return is referred to as a *credit spread*).

Analyzing the historical returns of asset classes between 1979 and 2023¹ provides the following observations:

- | Historical long-term asset returns show three distinct phases of risk and return. In Phase 1 (post-1995), stock performance and volatility increase, while bond total return indexes maintain steady growth. The global financial crisis of 2008 (Phase 2) saw all risky asset classes decline in value, with recoveries in the early 2010s. Post-2015 (Phase 3), stock returns and volatility once again increased relative to bonds.
- | Equities generate higher returns over time than bonds, as would be expected, given the higher volatility of equity prices compared to the relatively stable capital distribution returns of fixed income investments.
- | Longer-term fixed-income securities have outperformed shorter-term bonds. This is as expected, due to the higher inflation and interest rate risks associated

with locking in a fixed income over a longer time frame. The US 30-day T-bill return showed little excess return over inflation, which is to say it had nearly zero real return.

- | Dividend yields on equities have decreased over time as companies have increasingly chosen to return capital to shareholders through share buybacks rather than dividends. Investors may prefer buybacks because they support the value of shares and lead to capital gains, which are often subject to lower tax rates than dividends.
- | Capital distribution returns on bonds have generally declined over time, in line with a multidecade decrease in interest rates up to the early 2020s.



PROFESSOR'S NOTE

The curriculum learning module presents historical data showing large-cap stocks outperforming small-cap stocks and government bonds outperforming corporate bonds, contrary to expectations. This appears to be simply an artifact of the historical period used. It does not change the fact that investors should demand higher returns for riskier assets—and, therefore, should expect higher returns in the long term for small-cap stocks than large-cap stocks, and higher returns for corporate bonds than government bonds.

Risk

Risk relates to the chance of earning a return different to that which was expected. Risks can be broadly classified as **systematic risks** (or **macro risks**) that affect all investments to some degree, and **unsystematic risks** (or **micro risks**) that are specific to a particular security. As we will see in upcoming readings, diversifying investments in a portfolio can reduce unsystematic risk, but it does not reduce systematic risk.

Examples of systematic risk factors include:

- | **Inflation risk.** This is the risk that rising prices erode real returns.
- | **Interest rate risk.** This is the risk that increasing interest rates decrease the value of investments. As discussed earlier, interest rate risk is high for long-dated bonds. Increasing interest rates can also decrease the value of equity by increasing companies' financing costs and decreasing the present value of future cash flows to investors.

- | **Market risk.** This is the risk that a market downturn causes investment losses.

Examples of unsystematic risk include:

- | **Issuer risk.** These are risks that are specific to the issuer of a security. This risk may relate to the volatility in the issuer's earnings due to business conditions (**business risk**) or the issuer's use of financial leverage (**financial risk**). **Credit risk**, relating to a debt issuer's ability to meet debt payments when due, is also a key issuer risk.
- | **Sector risk or industry risk.** This is the risk that a downturn specific to an industry causes investment losses.

Analysts quantify risk by measuring how variable returns have been around their expected value. The simplest measure of variation is the **range** of returns (highest return minus lowest return). A wider range of returns around the expected return indicates more variability, so there is more risk associated with an investment. In upcoming readings, we will introduce risk measures more commonly used in financial analysis, such as standard deviation of returns.

The extra return investors demand for facing extra risk is referred to as a **risk premium**. Historical data show that, as expected, assets with higher variability in return—and, therefore, higher risk, such as equities—generate higher risk premiums over the long term than less risky asset classes, such as fixed income.

Asset Allocation

A portfolio manager's job is to construct an optimal portfolio of investments, given a client's objectives and constraints.

A client who requires high returns will need a larger allocation to riskier assets that provide price returns, such as equity. Clients with lower risk tolerance and the need for periodic income require a larger allocation to assets that provide fixed capital distributions, such as fixed-income securities. An investor who seeks both capital growth and income requires a more balanced portfolio with significant allocations to both equity and fixed income.

EXAMPLE: Asset allocation and portfolio expected return

A retired investor wishes to use his investment portfolio to generate income to meet his spending needs. Recommend an asset allocation from Allocations A,

B, and C in the following table, and calculate its total return, return from capital appreciation, and return from capital distributions.

Asset Class	Expected Returns		Asset Allocations		
	Capital Appreciation	Capital Distribution	A	B	C
Equities	8.0%	3.0%	85%	60%	40%
Bonds	0.5%	5.0%	10%	35%	50%
Cash (T-bills)	0.0%	2.5%	5%	5%	10%

Answer:

The most appropriate asset allocation choice is Allocation C. Because the investor is relying on his portfolio for retirement income, the portfolio should have a conservative allocation with a substantial allocation to fixed income securities.

The expected capital appreciation return of the portfolio is the weighted average of the capital appreciation returns of the individual asset classes:

$$(8.0\% \times 0.40) + (0.5\% \times 0.50) + (0.0\% \times 0.10) = 3.45\%$$

Similarly, the expected capital distribution of the portfolio is the weighted average of the capital appreciation returns of the individual asset classes:

$$(3.0\% \times 0.40) + (5.0\% \times 0.50) + (2.5\% \times 0.10) = 3.95\%$$

The total return of the portfolio is the sum of the capital appreciation and capital distribution returns:

$$\text{Total return of portfolio} = 3.45\% + 3.95\% = 7.40\%$$



MODULE QUIZ 2.2

- Which of the following asset classes in a developed-market country is expected to have the lowest long-term total return?
 - Equities.
 - Corporate bonds.
 - Government bonds.
- Which of the following bonds should be expected to have the highest return?
 - 2-year maturity US corporate bond with a coupon of 4%.
 - 5-year US government bond with a coupon of 4%.
 - 10-year maturity US corporate bond with a coupon of 5%.
- As a risk factor, inflation is considered to be:
 - a macro risk because it affects all types of investments.

- B. a micro risk because the impact of inflation on a particular issuer of securities is issuer specific.
- C. not a significant risk factor because rising prices increase the revenues of companies that issue securities.

KEY CONCEPTS

LOS 2.a

Equity investments primarily generate price return (capital appreciation), with a lower proportion of return coming from variable capital distribution in the form of dividend payments. Fixed-income investments, such as bonds, typically produce a higher component of fixed capital distribution return from the payment of a fixed periodic coupon, and a lower proportion of price return.

Financial indicators, such as interest rates, foreign exchange rates, and market indexes, affect returns but do not generate cash flows directly. An international investor earns higher domestic returns when foreign currencies strengthen. Most investments decrease in value when interest rates increase, particularly long-maturity bonds.

Corporate bond values are affected by changes in the perceived default risk of the issuer. Developed-market government bonds are deemed to have low default risk, with short-term bills being considered free of credit risk. To compensate investors for the higher default risk of corporate bonds, they offer a premium over government bond returns, known as a credit spread.

The risk-return tradeoff holds that rational investors should demand higher returns from investments with higher risk. Risk is measured by the variability of returns.

Risks can be divided into systematic (macro) risks that affect all investments, and unsystematic (micro) risks that are unique to an issuer. Systematic risks cannot be diversified away, but unsystematic risks are diversifiable.

The risk and return of different investments drive investors' asset allocation decisions. A risk-averse investor relying on a portfolio for income should allocate more to fixed-income securities. A less risk-averse investor looking for higher returns and capital appreciation should allocate more to equities.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 2.1

- . **B** Changes in foreign exchange rates affect both capital appreciation and capital distribution returns earned in foreign currencies. Equities offer variable capital distribution returns (not fixed capital distribution returns) in the form of dividends. Floating-rate notes pay a variable coupon linked to prevailing interest rates, so they offer variable capital distribution returns. (LOS 2.a)
- . **B** Capital distribution return is positive as long as a bond pays its periodic fixed coupon to investors. When interest rates increase, the value of fixed-income securities decreases, causing a negative capital appreciation return. Longer-dated bonds paying fixed coupons over more periods are more sensitive to changes in interest rates than shorter-dated bonds, so they fall by a larger amount when interest rates increase. (LOS 2.a)
- . **A** Outperformance, based on geometric compound returns, is calculated as follows:

$$\frac{1 + \text{large-cap returns}}{1 + \text{small-cap returns}} - 1 = \frac{1.1226}{1.1123} - 1 = 0.00926, \text{ or } 0.926\%.$$

(LOS 2.a)

Module Quiz 2.2

- . **C** Government bonds are expected to have the lowest returns because they have lower default risk than companies that issue bonds. Equities exhibit higher risk in terms of the variability in returns than either government or corporate bonds, so they would be expected to offer higher returns over the long term. (LOS 2.a)
- . **C** Corporate bonds are expected to have higher returns than government bonds due to higher default risk. Longer-maturity bonds have more sensitivity to interest rate changes (interest rate risk) and inflation risk. (LOS 2.a)
- . **A** Inflation risk, relating to rising prices leading to lower investment returns in real terms, affects the returns of all types of investments and is therefore considered a macro (systematic) risk. (LOS 2.a)

¹ Source: Bloomberg, as cited in the Level I CFA curriculum learning module, “Types of Financial

Returns.”

READING 3

BENCHMARKING RETURNS

MODULE 3.1: TIME-WEIGHTED AND MONEY-WEIGHTED RETURNS

LOS 3.a: Calculate and compare money-weighted and time-weighted rates of return.

The **money-weighted return** (or **dollar-weighted return**) applies the concept of the **internal rate of return (IRR)** to investment portfolios. An IRR is the interest rate at which a series of cash inflows and outflows sum to zero when discounted to their present value. That is, they have a **net present value (NPV)** of zero. The IRR and NPV are built-in functions on financial calculators that CFA Institute permits candidates to use for the exam.



PROFESSOR'S NOTE

We have provided an online video in our Resource Library on how to use the TI calculator. You can view it by logging in to your account at www.schweser.com.

The money-weighted rate of return is defined as the IRR on a portfolio, considering all cash inflows and outflows. The beginning value of the account is an inflow, as are all deposits into the account. All withdrawals from the account are outflows, as is the ending value.

EXAMPLE: Money-weighted rate of return

Assume an investor buys a share of stock for \$100 at $t = 0$. The share pays a \$2 cash dividend during the year, and the investor does not reinvest the dividend. At $t = 1$ (the end of the first year and the beginning of the second), she buys an additional share for \$120. The shares pay another cash dividend of \$2 per share during the year, and again, the investor does not reinvest the dividends. At $t = 2$

(the end of Year 2), the investor sells both shares for \$130 each. What is the money-weighted rate of return?

Answer:

Step 1: Determine the timing of each cash flow and whether the cash flow is an inflow (+), into the account, or an outflow (–), available from the account.

$t = 0$:	purchase of first share	=	+\$100.00	inflow to account
$t = 1$:	purchase of second share	=	+\$120.00	
	dividends from first share	=	<u>–\$2.00</u>	
	subtotal, $t = 1$		+\$118.00	inflow to account
$t = 2$:	dividend from two shares	=	–\$4.00	
	proceeds from selling shares	=	<u>–\$260.00</u>	
	subtotal, $t = 2$		–\$264.00	outflow from account

Step 2: Net the cash flows for each period and set the PV of cash inflows equal to the PV of cash outflows.

$$PV_{\text{inflows}} = PV_{\text{outflows}}$$

$$\$100 + \frac{\$118}{(1+r)} = \frac{\$264}{(1+r)^2}$$

Step 3: Solve for r to find the money-weighted rate of return. This can be done using trial and error, or by using the IRR function on a financial calculator or spreadsheet.

The intuition, here, is that we deposited \$100 into the account at $t = 0$, then added \$118 to the account at $t = 1$ (which, with the \$2 dividend, funded the purchase of one more share at \$120), and ended with a total value of \$264.

To compute this value with a financial calculator, use these net cash flows and follow the procedure(s) described to calculate the IRR:

$$\text{net cash flows: } CF_0 = +100; CF_1 = +120 - 2 = +118;$$

$$CF_2 = -260 + -4 = -264$$

The keystrokes to calculate the money-weighted return with the TI Business Analyst II Plus® are as follows. Note that the values for F01, F02, and so on are all equal to 1.

Keystrokes	Explanation	Display
[CF] [2 nd][CLR WORK]	Clear cash flow registers	CF0 = 0.00000
100 [ENTER]	Initial cash outlay	CF0 = +100.00000
[↓] 118 [ENTER]	Period 1 cash flow	C01 = +118.00000
[↓] [↓] 264 [+/-] [ENTER]	Period 2 cash flow	C02 = -264.00000
[IRR] [CPT]	Calculate IRR	IRR = 13.86122

The money-weighted rate of return for this problem is 13.86%.



PROFESSOR'S NOTE

In the preceding example, we entered the flows into the account as a positive and the ending value as a negative (the investor could withdraw this amount from the account). Note that there is no difference in the solution if we enter the cash flows into the account as negative values (out of the investor's pocket) and the ending value as a positive value (into the investor's pocket). If payments into the account and payments out of the account (including the ending value) are entered with opposite signs, the computed IRR will be correct.

We also assumed the investor did not reinvest the cash dividends. In the Equities topic area, we will examine the effect of reinvested dividends on an investor's return.

Time-weighted return measures compound growth and is the rate at which \$1 compounds over a specified performance horizon. Time-weighting is the process of averaging a set of values over time. The annual time-weighted return for an investment may be computed by performing the following steps:

- Step 1:* Value the portfolio immediately preceding significant additions or withdrawals. Form subperiods over the evaluation period that correspond to the dates of deposits and withdrawals.
- Step 2:* Compute the holding period return (HPR) of the portfolio for each subperiod.
- Step 3:* Compute the product of $(1 + \text{HPR})$ for each subperiod to obtain a total return for the entire measurement period [i.e., $(1 + \text{HPR}_1) \times (1 + \text{HPR}_2)$]

$\dots (1 + \text{HPR}_n)] - 1$. If the total investment period is greater than one year, you must take the geometric mean of the measurement period return, based on the number of years in the measurement period, to find the annual time-weighted rate of return.

EXAMPLE: Time-weighted rate of return

An investor purchases a share of stock at $t = 0$ for \$100. At the end of the year, $t = 1$, the investor buys another share of the same stock for \$120. At the end of Year 2, the investor sells both shares for \$130 each. At the end of both Years 1 and 2, the stock paid a \$2 per share dividend. What is the annual time-weighted rate of return for this investment? (This is the same investment as the preceding example.)

Answer:

Step 1: Break the evaluation period into two subperiods based on timing of cash flows.

Holding period 1: Beginning value = \$100

Dividends paid = \$2

Ending value = \$120

Holding period 2: Beginning value = \$240 (2 shares)

Dividends paid = \$4 (\$2 per share)

Ending value = \$260 (2 shares)

Step 2: Calculate the HPR for each holding period.

$$\text{HPR}_1 = [(\$120 + 2) / \$100] - 1 = 22\%$$

$$\text{HPR}_2 = [(\$260 + 4) / \$240] - 1 = 10\%$$

Step 3: Find the compound annual rate that would have produced a total return equal to the return on the account over the two-year period.

$$(1 + \text{total return})^2 = (1.22)(1.10) = 1.342$$

$$\text{The total return} = 1.342^{0.5} - 1 = 15.84\%.$$



PROFESSOR'S NOTE

The 0.5 exponent is the reciprocal of the number of years in the holding period ($1 / 2 = 0.5$). The number of years is not necessarily the same as the number of subperiods being considered during the overall holding period.

Unlike the money-weighted rate of return, the time-weighted rate of return is not affected by the timing of cash inflows and outflows. In the investment management industry, time-weighted return is the preferred method of performance measurement because portfolio managers typically do not control the timing of deposits to and withdrawals from the accounts they manage.

In the preceding examples, the time-weighted rate of return for the portfolio was 15.84%, while the money-weighted rate of return for the same portfolio was 13.86%. The results are different because the money-weighted rate of return gave a larger weight to the Year 2 HPR, which was 10%, versus the 22% HPR for Year 1. This is because there was more money in the account at the beginning of the second period.

If funds are contributed to an investment portfolio just before a period of relatively poor portfolio performance, the money-weighted rate of return will tend to be lower than the time-weighted rate of return. On the other hand, if funds are contributed to a portfolio at a favorable time (just before a period of relatively high returns), the money-weighted rate of return will be higher than the time-weighted rate of return. The use of the time-weighted return removes these distortions, so it provides a better measure of a manager's ability to select investments over the period. If the manager has complete control over money flows into and out of an account, the money-weighted rate of return would be the more appropriate performance measure.

CFA Institute has developed **Global Investment Performance Standards (GIPS)**, a voluntary set of performance presentation guidelines for investment management firms. Firms that claim compliance with GIPS must use time-weighted returns to present investment performance, except under a specific set of conditions that make money-weighted returns more appropriate. These include the investment manager controlling cash flows into the portfolio, in addition to the portfolio being closed end, having a fixed life, having committed capital from investors that the manager can call on demand, or consisting largely of illiquid investments.



PROFESSOR'S NOTE

GIPS are addressed in detail in the Level III CFA curriculum.



MODULE QUIZ 3.1

1. An investor buys a share of stock for \$40, buys another share of the same stock for \$50 at the beginning of the next period, and sells both shares for \$60 each at the end of that

period. The stock pays a dividend of \$1 per share at the end of each period, and the investor does not reinvest the dividends. The periodic money-weighted rate of return on the investment is *closest* to:

- A. 22.2%.
 - B. 23.0%.
 - C. 23.8%.
2. An investor buys a share of stock for \$40, buys another share of the same stock for \$50 at the beginning of the next period, and sells both shares for \$60 each at the end of that period. The stock pays a dividend of \$1 per share at the end of each period, and the investor does not reinvest the dividends. The time-weighted rate of return on the investment for the period is *closest* to:
- A. 24.7%.
 - B. 25.7%.
 - C. 26.8%.

MODULE 3.2: SECURITY MARKET INDEXES

LOS 3.b: Describe the choices and the implications of the different weighting methods used in index construction and management, and calculate, interpret, and explain the value and the returns of an index.

A **security market index** is used to represent the performance of an asset class, security market, or segment of a market. They are usually created as portfolios of individual securities, which are referred to as the **constituent securities** of the index. An index has a numerical value that is calculated from the market prices (actual when available, or estimated) of its constituent securities at a point in time. An index return is the percentage change in the index's value over a period.

An index return may be calculated using a **price index** or a **return index**. A price index uses only the prices of the constituent securities in the return calculation. A rate of return that is calculated based on a price index is referred to as a **price return**.

A return index includes both prices and income from the constituent securities. A rate of return that is calculated based on a return index is called a **total return**. If the assets in an index produce interim cash flows such as dividends or interest payments, the total return will be greater than the price return.

Once returns are calculated for each period, they then can be compounded together to arrive at the return for the measurement period:

$$R_P = (1 + R_{S1})(1 + R_{S2})(1 + R_{S3})(1 + R_{S4}) \dots (1 + R_{Sk}) - 1$$

where:

R_P = portfolio return during the measurement period

k = total number of subperiods

R_{Sk} = portfolio return during the subperiod k

For example, if the returns for the first two periods were 0.50% and 1.04%, they would be geometrically linked to produce 1.55%:

$$R_P = (1 + R_{S1})(1 + R_{S2}) - 1 = (1.005)(1.0104) - 1 = 0.0155, \text{ or } 1.55\%$$

If the starting index value is 100, its value after two periods would be $100 \times 1.0155 = 101.55$.

We can describe a process for constructing, calculating, and maintaining a security market index using the following five steps:

- Step 1: Objective of the index.* Specify the security market or sector the index will measure.
- Step 2: Constituent securities.* Define the criteria and select securities that meet them.
- Step 3: Weighting method.* Weighting schemes include price weighting, equal weighting, market capitalization weighting, float-adjusted market capitalization weighting, and fundamental weighting.
- Step 4: Calculations.* Determine the index value on an initial date and on subsequent dates.
- Step 5: Maintenance.* Over time, it may become necessary to rebalance or reconstitute an index (e.g., to reflect corporate actions such as mergers or bankruptcies), and to re-evaluate the security selection criteria and the index construction process.

Price Weighting

A **price-weighted index** is essentially an arithmetic average of the prices of the securities included in the index, with its denominator adjusted for changes in the composition of the index over time—such that the index value is unaffected by such changes. The denominator is adjusted when securities are added or deleted from the index, and when corporate actions change the proportion of a company that one share represents. These actions include **stock splits**, in which one share becomes a larger number of shares (e.g., a 2-for-1 stock split means one share

becomes two shares, but each of those shares represents half as much ownership of the company as one share represented before the split); **reverse stock splits**, in which some number of shares become one share; and **stock dividends**, which are similar to stock splits except that they are stated as a percentage (e.g., a 10% stock dividend means one share becomes 1.1 shares).



PROFESSOR'S NOTE

It is crucial to distinguish between stock dividends and cash dividends. A cash dividend pays part of a company's earnings to shareholders. A stock dividend does not pay cash or change a shareholder's ownership position; like a stock split, it merely changes what one share represents.

We will examine these transactions with shareholders in more detail in the Equities topic area, as well as in Financial Statement Analysis when we calculate earnings per share.

EXAMPLE: Price-weighted index

The three stocks presented here comprise a price-weighted index. On December 31, 20X6, the index divisor is set such that the index value equals 100. Calculate the price-weighted index on January 31, 20X7, and the index's return over the one-month period.

Index Firm Data

	Share Price December 31, 20X6	Share Price January 31, 20X7
Stock X	\$10	\$20
Stock Y	\$20	\$15
Stock Z	\$60	\$40

Answer:

We can determine the index divisor by solving for D in the following equation:

$$\frac{10 + 20 + 60}{D} = 100$$

$$D = \frac{10 + 20 + 60}{100} = 0.9$$

Because no index changes, stock splits, or stock dividends have occurred, we can use this divisor to calculate the index value on January 31:

$$\frac{20 + 15 + 40}{0.9} = 83.33$$

Hence, this is the 1-month percentage return of the price-weighted index:

$$\frac{83.33}{100.00} - 1 = -16.67\%$$

EXAMPLE: Adjusting a price-weighted index for stock splits

Stock A has a price of \$5, Stock B has a price of \$25, and Stock C has a price of \$90. The value of a price-weighted index of these three stocks is $(5 + 25 + 90) / 1.2 = 100$. If Stock C splits 2-for-1, effective on the next day, what is the new denominator for the index?

Answer:

The effect of the split on the price of Stock C, in the absence of any change from the price at the end of Day 1, would be to reduce it to $\$90 / 2 = \45 . The index denominator will be adjusted so that the index value would remain at 100 if there were no changes in the stock prices, other than to adjust for the split. The new denominator, D , must satisfy $(5 + 25 + 45) / D = 100$ and equals 0.75.

The returns on a price-weighted index could be matched by purchasing an equal number of shares of each stock represented in the index.

The advantage of a price-weighted index is that its computation is simple. One disadvantage is that a given percentage change in the price of a higher-priced stock has a greater impact on the index's value than does an equal percentage change in the price of a lower-priced stock. Put another way, higher-priced stocks have more weight in the calculation of a price-weighted index.

Additionally, the price of a single share does not indicate the market value of a company, because companies may have widely different numbers of shares outstanding. For this reason, a price-weighted index is typically a suitable benchmark only for a portfolio that is affected disproportionately by price changes of specific securities.

Two major price-weighted indexes are the Dow Jones Industrial Average (DJIA) and the Nikkei Stock Average (Nikkei 225). The DJIA is a price-weighted index based on 30 US stocks. The Nikkei is constructed from the prices of 225 stocks that trade in the first section of the Tokyo Stock Exchange.

Equal Weighting

An **equal-weighted index** places an equal weight on the returns of all index stocks, regardless of their prices or market values. A \$2 change in the price of a \$20 stock has the same effect on the index as a \$30 change in the price of a \$300 stock, regardless of the size of the company. Because price changes of an index's constituent securities cause their weights in the index to drift, maintaining an equal-weighted index requires frequent rebalancing.

The following example illustrates how an equal-weighted index is assigned a value at inception and how price changes affect the index.

EXAMPLE: Equal-weighted index

Calculate the equal-weighted index value for the three stocks described in the following table, assuming an initial index value of 100.

Equal-Weighted Index Data

Stock	Initial Price	Current Price	Price Change
A	\$12	\$15	+25.0%
B	\$52	\$48	-7.7%
C	\$38	\$45	+18.4%

Answer:

Initial Value

At inception, each stock in the index is assigned an equal reference value. The index includes the number of shares of each stock that is required to equal that reference value. Here, we will use 5,200 as the reference value (the value chosen is arbitrary).

Stock A: $5,200 / \$12 = 433.33$ shares

Stock B: $5,200 / \$52 = 100$ shares

Stock C: $5,200 / \$38 = 136.84$ shares

The aggregate reference value is $3 \times 5,200 = 15,600$. To set the initial index value to 100, we use a divisor of 156.

Stock	Price	Shares	Value	Weight %
A	\$12	433.33	\$5,200	33.33
B	\$52	100.00	\$5,200	33.33
C	\$38	136.84	\$5,200	33.33
Total			\$15,600	100
Divisor			156	
Index value			100	

Current Value

Stock A: 433.33 shares $\times$ \$15 = 6,500.0

Stock B: 100 shares $\times$ \$48 = 4,800.0

Stock C: 136.84 shares $\times$ \$45 = 6,158.0

Aggregate reference value = 6,500.0 + 4,800.0 + 6,158.0 = 17,458.0

Index value = 17,458.0 / 156 = 111.9

Stock	Price	Number	Value	Weight %
A	\$15	433.33	\$6,500	37.2
B	\$48	100.00	\$4,800	27.5
C	\$45	136.84	\$6,158	35.3
Total			\$17,458	100
Divisor			156	
Index value			111.9	

The index value increased by 11.9%. Notice that this equals the arithmetic average of the percentage changes in the stocks' prices:

$$\frac{25\% - 7.7\% + 18.4\%}{3} = 11.9\%$$

Thus, each stock's performance had an equal influence on the index's performance. For this to remain true going forward, however, the index would need to be rebalanced such that each stock's reference value was once again equal.

One complication with an equal-weighted index return is that a matching portfolio would have to be adjusted periodically (rebalanced) as prices change so that the values of all security positions are made equal each period. The portfolio

rebalancing required to match the performance of an equal-weighted index creates high transactions costs that decrease portfolio returns.

An equal-weighted index can be a suitable benchmark for an investment strategy that is designed to limit exposure to companies with the largest market capitalizations.

The Value Line Composite Average and the Financial Times Ordinary Share Index are well-known examples of equal-weighted indexes.

Market Capitalization Weighting

A **market-capitalization-weighted index** (or **value-weighted index**) has weights based on the market capitalization of each index stock (current stock price multiplied by the number of shares outstanding) as a proportion of the total market capitalization of all the stocks in the index. This weighting method more closely represents changes in aggregate investor wealth than price weighting. Because the weight of an index stock is based on a company's total market value, a market-capitalization-weighted index does not need to be adjusted for a stock split or a stock dividend, as those transactions do not change the value of a company. (The divisor does need to be adjusted when an index's components change.)

A market-capitalization-weighted index is calculated by summing the total value of all the stocks in the index. This sum is then divided by a divisor that gives the index its base period value (typically 100 or 1,000).

EXAMPLE: Market-capitalization-weighted index

In its base period on December 31, the total market value of an index portfolio is \$80 million. One month later, on January 31, the market value of the portfolio is \$95 million. If the base period index value is 100, calculate the index value on January 31.

Answer:

To set the base-year index to 100, the divisor is $80 \text{ million} / 100 = 800,000$.

$$\text{index on January 31} = \frac{95,000,000}{800,000} = 118.75$$

The following example of price weighting versus market value weighting shows how these two indexes are calculated and how they differ.

EXAMPLE: Price-weighted vs. market-capitalization-weighted indexes

Consider the three firms described in the following table. Compare the effects on a price-weighted index and a market-capitalization-weighted index if Stock A doubles in price, or if Stock C doubles in price. Assume the period shown in the table is the base period and the base value of each index is 100.

Index Firm Data

$$\text{index on January 31} = \frac{95,000,000}{800,000} = 118.75$$

Answer:

The price-weighted index divisor equals:

$$\frac{100 + 10 + 1}{100} = 1.11$$

If Stock A doubles in price to \$200, the price-weighted index value is:

$$\frac{200 + 10 + 1}{1.11} = 190.09$$

If Stock C doubles in price to \$2, the price-weighted index value is:

$$\frac{100 + 10 + 2}{1.11} = 100.90$$

If Stock A doubles in value, the index increases by 90 points; if Stock C doubles in value, the index only goes up by less than 1 point. Changes in the value of the firm with the highest stock price have a disproportionately large influence on a price-weighted index.

For a market-capitalization-weighted index, the base period market capitalization is $(100,000 \times \$100) + (1,000,000 \times \$10) + (20,000,000 \times \$1) = \$40,000,000$, so the index divisor equals $40,000,000 / 100 = 400,000$.

If Stock A doubles in price to \$200, the index goes to:

$$\frac{100,000 \times \$200 + 1,000,000 \times \$10 + 20,000,000 \times \$1}{400,000} = 125$$

If Stock C doubles in price to \$2, the index goes to:

$$\frac{100,000 \times \$100 + 1,000,000 \times \$10 + 20,000,000 \times \$2}{400,000} = 150$$

In the market-capitalization-weighted index, the returns on Stock C have the greatest influence on the index return because Stock C's market capitalization is larger than that of Stock A or Stock B.

The following example illustrates the difference between the price return and the total return of an index.

EXAMPLE: Market-capitalization-weighted index with capital distributions

The three stocks in the following table comprise a market-capitalization-weighted index with a value of 100 in its base period. Calculate the price return and total return of the index for the period shown in the table.

Company	Number of Shares Outstanding (000s)	Stock Price, Base Period	Stock Price, One Period Later	Capital Distributions (Per Share)
A	100	\$100.00	\$98.00	\$3
B	1,000	\$10.00	\$9.50	\$1
C	20,000	\$1.00	\$1.10	0

Answer:

As in the previous example, the base period market capitalization is $(100,000 \times \$100) + (1,000,000 \times \$10) + (20,000,000 \times \$1) = \$40,000,000$, so the index divisor equals $40,000,000 / 100 = 400,000$.

The index value at the end of the period is:

$$\frac{100,000 \times \$98 + 1,000,000 \times \$9.50 + 20,000,000 \times \$1.10}{400,000} = 103.25$$

The price return for the period is $\frac{103.25}{100} - 1 = 3.25\%$.

The index value, including capital distributions, is:

$$\frac{100,000 \times (\$98 + \$3) + 1,000,000 \times (\$9.50 + \$1) + 20,000,000 \times \$1.10}{400,000} = 106.50$$

The total return for the period is $\frac{106.50}{100} - 1 = 6.5\%$.

A market-capitalization-weighted index return can be matched with a portfolio in which the allocation to each security is the same as the proportion of that security's market capitalization to that of all the securities in the index.

An alternative to using a firm's market capitalization to calculate its weight is to use its **market float**. A firm's market float is the total value of the shares that are actually available to the investing public. This excludes the value of shares held by strategic stockholders because they are unlikely to sell their shares. Market float is often calculated also excluding shares held by corporations or governments. A market float calculation that excludes shares not available to foreign buyers is referred to as **free float**. The point of using a market float measure is to better match the index weights of stocks to their proportions of the total value of all the shares of index stocks that are available to investors.

A **float-adjusted market-capitalization-weighted index** is constructed like a market-capitalization-weighted index. The weights, however, are based on the proportionate value of each firm's shares available to investors to the total market value of the shares of index stocks available to investors. Firms with relatively large percentages of their shares held by controlling stockholders will have less weight than they have in an unadjusted market-capitalization index.

Advantages of market-capitalization-weighted indexes of either type are that index security weights represent proportions of total market value; that they tend to have infrequent turnover of their constituent securities; and that rebalancing happens automatically for a portfolio that tracks such an index. Their primary disadvantage is that the relative impact of a stock's return on the index increases as its price rises and decreases as its price falls. This means that stocks that are possibly overvalued are given disproportionately high weights in the index, and stocks that are possibly undervalued are given disproportionately low weights.

A market-capitalization-weighted index is typically a suitable benchmark for a passive investment strategy that is designed to earn the market rate of return. The Standard and Poor's 500 (S&P 500) Index Composite is an example of a market-capitalization-weighted index.

EXAMPLE: Effects of weighting methods on reported returns

Consider the three stocks in the following table:

	As of January 1		As of December 31	
	Share Price	Number of Shares Outstanding (Thousands)	Share Price	Number of Shares Outstanding (Thousands)
Stock A	\$22	1,500	\$28	1,500
Stock B	\$40	10,000	\$50	10,000
Stock C	\$34	3,000	\$30	3,000

Calculate the 1-year return on a price-weighted index, an equal-weighted index, and a market-capitalization-weighted index of these three stocks.

Answer:

For each weighting method, set the beginning index value to 100 and then calculate the ending value.

Price-weighted index:

$$\text{Divisor: } \frac{22 + 40 + 34}{100} = 0.96$$

$$\text{Index value on December 31: } \frac{28 + 50 + 30}{0.96} = 112.5$$

$$\text{Rate of return} = \frac{112.5}{100} - 1 = 12.5\%$$

Equal-weighted index:

To avoid starting with fractional shares, we will choose an initial reference value of $22 \times 40 \times 34 = 29,920$ for each stock.

$$\text{Stock A: } 29,920 / 22 = 1,360 \text{ shares}$$

$$\text{Stock B: } 29,920 / 40 = 748 \text{ shares}$$

$$\text{Stock C: } 29,920 / 34 = 880 \text{ shares}$$

$$\text{Aggregate reference value} = 3 \times 29,920 = 89,760$$

$$\text{Divisor} = 89,760 / 100 = 897.6$$

Index value on December 31:

$$\text{Stock A: } 1,360 \times 28 = 38,080$$

$$\text{Stock B: } 748 \times 50 = 37,400$$

$$\text{Stock C: } 880 \times 30 = 26,400$$

$$\text{Index value on December 31: } \frac{38,080 + 37,400 + 26,400}{897.6} = 113.5$$

$$\text{Rate of return} = \frac{113.5}{100} - 1 = 13.5\%$$

Market-capitalization-weighted index:

Total portfolio value January 1:

$$22(1,500) + 40(10,000) + 34(3,000) = \$535,000$$

$$\text{Divisor} = 535,000 / 100 = 5,350$$

Total portfolio value December 31:

$$28(1,500) + 50(10,000) + 30(3,000) = \$632,000$$

$$\text{Index value on December 31: } \frac{632,000}{5,350} = 118.13$$

$$\text{Rate of return} = \frac{118.13}{100} - 1 = 18.13\%$$

Fundamental Weighting

An index that uses **fundamental weighting** uses weights based on firm fundamentals, such as earnings, dividends, or cash flow. In contrast to market capitalization index weights, these weights are unaffected by the share prices of the index stocks (although related to them over the long term). Fundamental weights can be based on a single measure or some combination of fundamental measures. A fundamental-weighted index can be used as a performance benchmark for an active fund manager who believes changes in fundamental factors drive future stock performance and cause individual firms' securities to outperform or underperform the market.

An advantage of a fundamental-weighted index is that it focuses on factors beyond share prices and market capitalization. A disadvantage is that a fundamental-weighted index may exclude non-dividend-paying stocks or firms that do not currently have positive earnings, which may bias the index toward or away from some market sectors. A fundamental-weighted index typically has a value tilt, overweighting firms with high value-based metrics such as book-to-market ratios or earnings yields. Thus, it can be a suitable benchmark for investors who are employing value-driven strategies.

Rebalancing and Reconstitution of an Index

Rebalancing refers to adjusting the weights of securities in a portfolio to their target weights after price changes have affected the weights. For index calculations, rebalancing to target weights on the index securities is done on a periodic basis, usually quarterly. Because the weights in price- and value-weighted indexes (portfolios) are adjusted to their correct values by changes in prices, rebalancing is an issue primarily for equal-weighted and fundamental-weighted indexes.

As noted previously, the weights on security returns in an (initially) equal-weighted portfolio do not remain equal as securities prices change over time. Therefore, rebalancing the portfolio at the end of each period used to calculate index returns is necessary for the portfolio return to match the index return. For a portfolio that tracks an equal-weighted index, this tends to result in selling stocks that have increased in value and buying stocks that have decreased in value—both of which are contrary to a trend-following strategy, which would instead “cut losses short and let profits run.”

EXAMPLE: Rebalancing an equal-weighted index

Using the stocks from our earlier example of an equal-weighted index, determine what adjustments would be needed to rebalance the index, if the stock prices change as indicated in the following table.

Equal-Weighted Index Data

Stock	Base Period Price	Shares in Index	Value	Weight	Current Price
A	\$12	433.33	\$5,200	33.3%	\$15
B	\$52	100	\$5,200	33.3%	\$48
C	\$38	136.84	\$5,200	33.3%	\$45

Answer:

Before rebalancing, the stocks' weights are as follows:

Stock	Current Price	Shares	Value	Weight (Before Rebalancing)
A	\$15	433.33	6,500.0	37.2%
B	\$48	100	4,800.0	27.5%
C	\$45	136.84	<u>6,157.8</u>	35.3%
			17,457.8	

To rebalance the index, each stock needs to have a value of $17,457.8 / 3 = 5,819.27$. Therefore, to rebalance the index, the portfolio needs the following numbers of shares:

$$\text{Stock A: } \frac{\$5,819.27}{\$15} = 387.95 \text{ shares}$$

$$387.95 - 433.33 = -45.38 \text{ shares}$$

$$\text{Stock B: } \frac{\$5,819.27}{\$48} = 121.23 \text{ shares}$$

$$121.23 - 100 = +21.23 \text{ shares}$$

$$\text{Stock C: } \frac{\$5,819.27}{\$45} = 129.32 \text{ shares}$$

$$129.32 - 136.84 = -7.52 \text{ shares}$$

The rebalanced portfolio has fewer shares of the two stocks that increased in price (Stocks A and C) and more shares of the stock that decreased in price (Stock B).

Index **reconstitution** refers to periodically adding and deleting securities that make up an index. Securities are deleted if they no longer meet the index criteria (e.g., a company in the S&P 500 Index may fall out of the top 500 companies by market capitalization) and are replaced by other securities that do. Indexes are also reconstituted to reflect corporate events, such as bankruptcy or delisting of index firms. The index's divisor is adjusted at the date of the reconstitution to keep the index constant.

When a security is added to an index, its price tends to rise as portfolio managers who are seeking to track that index in a portfolio buy the security. The prices of deleted securities tend to fall as portfolio managers sell them. Additions and deletions also require that the weights on the returns of other index stocks be adjusted to conform to the desired weighting scheme.



MODULE QUIZ 3.2

1. Market float of a stock is *best* described as its:
 - A. total outstanding shares.
 - B. shares that are available to domestic investors.
 - C. outstanding shares, excluding those held by controlling shareholders.
2. For which of the following indexes will rebalancing occur *most* frequently?
 - A. A price-weighted index.
 - B. An equal-weighted index.
 - C. A market-capitalization-weighted index.

3. The publisher of an index that includes 50 corporate bonds removes three bonds that are nearing maturity and one whose issuer has defaulted, and selects four actively traded bonds to replace them in the index. This bond index is said to have been:
- A. redefined.
 - B. rebalanced.
 - C. reconstituted.
4. Consider a market-capitalization-weighted index comprising the following three stocks:

Stock	Shares Outstanding	Price Per Share, Beginning of Period	Price Per Share, End of Period
X	10,000,000	40	42
Y	20,000,000	25	24
Z	30,000,000	12	13

The return on the index for this period is *closest* to:

- A. 2.4%.
 - B. 2.6%.
 - C. 3.1%.
5. Consider an equal-weighted index comprising the following three stocks:

Stock	Shares Outstanding	Price Per Share, Beginning of Period	Price Per Share, End of Period
X	10,000,000	40	42
Y	20,000,000	25	24
Z	30,000,000	12	13

The return on the index for this period is *closest* to:

- A. 2.4%.
- B. 2.6%.
- C. 3.1%.

KEY CONCEPTS

LOS 3.a

The money-weighted rate of return is the IRR calculated using periodic cash flows into and out of an account and is the discount rate that makes the present value of cash inflows equal to the present value of cash outflows.

The time-weighted rate of return measures compound growth and is the rate at which money compounds over a specified performance horizon.

If funds are added to a portfolio just before a period of poor performance, the money-weighted return will be lower than the time-weighted return. If funds are added just before a period of high returns, the money-weighted return will be higher than the time-weighted return.

The time-weighted return is the preferred measure of a manager's ability to select investments. If the manager controls the money flows into and out of an account, the money-weighted return is the more appropriate performance measure.

LOS 3.b

A security market index represents the performance of an asset class, security market, or segment of a market. The performance of the market or segment over a period is represented by the percentage change in (i.e., the return on) the value of the index.

A price index uses only the prices of the constituent securities in the return calculation. The rate of return is called a price return. A total return index uses both the price of and the income from the index securities in the return calculation. The rate of return is called a total return.

Constructing, calculating, and managing an index involves five steps:

Step 1: State the objective and specify the market or sector.

Step 2: Select the constituent securities.

Step 3: Determine the weighting method.

Step 4: Calculate the index value on the initial date and subsequent dates.

Step 5: Rebalance and reconstitute the index when needed and evaluate the construction and selection process.

A price-weighted index is an arithmetic mean of the prices of the index securities. Its divisor must be adjusted for stock splits and changes in the composition of the index over time. Higher-priced stocks have more weight in the returns of a price-weighted index than lower-priced stocks.

An equal-weighted index places an equal weight on the returns of all index stocks, regardless of their prices or market values. Because price changes of an index's constituent securities cause their weights in the index to drift, maintaining an equal-weighted index requires frequent rebalancing.

A market-capitalization-weighted index gives each constituent security a weight equal to its proportion of the total market value of all securities in the index. Market capitalization can be adjusted for a security's market float or free float to reflect the fact that not all outstanding shares are available for purchase.

A fundamental-weighted index uses weights that are independent of security prices, such as company earnings, revenue, assets, or cash flow.

Index providers periodically rebalance the weights of the constituent securities. This is most important for equal-weighted and fundamental-weighted indexes. Reconstitution refers to changing the securities that are included in an index. This is necessary when securities mature, or when they no longer have the required characteristics to be included.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 3.1

- . **C** Using the cash flow functions on your financial calculator, enter:

$$CF_0 = -40; CF_1 = -50 + 1 = -49; CF_2 = 60 \times 2 + 2 = 122; \text{ and}$$

$$\text{CPT IRR} = 23.82\%. \text{ (LOS 3.a)}$$

$$\text{HPR}_1 = \frac{50 + 1}{40} - 1 = 27.5\%$$

- . **A** $\text{HPR}_2 = \frac{120 + 2}{100} - 1 = 22.0\%$

$$\text{TWR} = \sqrt{(1 + 0.275)(1 + 0.22)} - 1 = 24.72\%$$

$$\text{(LOS 3.a)}$$

Module Quiz 3.2

- . **C** Market float represents shares available to the investing public and excludes shares held by controlling shareholders. Free float is a narrower measure that also excludes shares that are not available to foreign investors. (LOS 3.b)
- . **B** An equal-weighted index will be rebalanced most frequently because as stock prices change, their representation in the index needs to be adjusted. Price-weighted and market-capitalization-weighted indexes do not usually need rebalancing. (LOS 3.b)

- . C Changing the constituent securities of an index is referred to as reconstituting the index. Rebalancing refers to adjusting the index weights to their target levels. (LOS 3.b)
- . A The total market capitalization of the index at the beginning of the period is $10,000,000 \times 40 + 20,000,000 \times 25 + 30,000,000 \times 12 = 1,260,000,000$. To set the initial value of the index to 100, use a divisor of $1,260,000,000 / 100 = 12,600,000$.

The index value at the end of the period is:

$$\frac{10,000,000 \times 42 + 20,000,000 \times 24 + 30,000,000 \times 13}{12,600,000} = 102.38$$

$$\text{Rate of return} = \frac{102.38}{100} - 1 = 2.38\%.$$

(LOS 3.b)

- . C To construct an equal-weighted index, give each stock the same initial reference value. While the reference value chosen is arbitrary, we will use a value of $40 \times 25 \times 12 = 12,000$.

Stock X: $12,000 / 40 = 300$ shares

Stock Y: $12,000 / 25 = 480$ shares

Stock Z: $12,000 / 12 = 1,000$ shares

The aggregate reference value is $3 \times 12,000 = 36,000$, so to set the initial value of the index to 100, we will use a divisor of $36,000 / 100 = 360$.

The index value at the end of the period is:

$$\frac{300 \times 42 + 480 \times 24 + 1,000 \times 13}{360} = 103.11$$

$$\text{Rate of return} = \frac{103.11}{100} - 1 = 3.11\%.$$

(LOS 3.b)

READING 4

THE TIME VALUE OF MONEY IN FINANCE



PROFESSOR'S NOTE

The examples we use in this reading are meant to show how the time value of money appears throughout finance. Don't worry if you are not yet familiar with the securities we describe in this reading. We will see these examples again when we cover bonds and forward interest rates in Fixed Income, stocks in Equities, foreign exchange in Economics, and options in Derivatives.

WARM-UP: USING A FINANCIAL CALCULATOR

For the exam, you must be able to use a financial calculator when working time value of money problems. You simply do not have the time to solve these problems any other way.

CFA Institute allows only two types of calculators to be used for the exam: (1) the Texas Instruments® TI BA II Plus™ (including the BA II Plus Professional™) and (2) the HP® 12C (including the HP 12C Platinum). This reading is written primarily with the TI BA II Plus in mind. If you do not already own a calculator, purchase a TI BA II Plus—but if you already own the HP 12C and are comfortable with it, by all means, continue to use it.

Before we begin working with financial calculators, you should familiarize yourself with your TI BA II Plus by locating the keys noted next. Located on the third row, these are the only keys you need to know to calculate virtually all the time value of money problems:

- | N = number of compounding periods
- | I/Y = interest rate per compounding period
- | PV = present value
- | FV = future value

- | PMT = annuity payments, or constant periodic cash flow
- | CPT = compute



PROFESSOR'S NOTE

We have provided an online video in our Resource Library on how to use the TI calculator. You can view it by logging in to your account at

MODULE 4.1: DISCOUNTED CASH FLOW VALUATION

LOS 4.a: Calculate and interpret the present value of fixed-income and equity instruments based on expected future cash flows.

The **time value of money** refers to the principle that individuals value a given amount of money in the present more highly than they value the same amount of money in the future. To be willing to give up money today in exchange for the promise of money in the future, individuals must expect to earn a **rate of return** on that money. We use the rate of return, or **discount rate**, to calculate the **present value** of a future sum or the **future value** of a present sum.

We can state the relationship between present values and future values as follows:

$$FV = PV (1 + r)^t$$

$$PV = \frac{FV}{(1 + r)^t} = FV (1 + r)^{-t}$$

where:

r = interest rate per compounding period

t = number of compounding periods

where:

r = interest rate per compounding period

t = number of compounding periods

If we are using continuous compounding, this is the relationship:

$$FV = PV \times e^{rt}$$

$$PV = FV \times e^{-rt}$$

Fixed-Income Securities

One of the simplest examples of the time value of money concept is a **pure discount** debt instrument, such as a **zero-coupon bond**. With a pure discount instrument, the investor pays less than the face value to buy the instrument and receives the face value at maturity. The price the investor pays depends on the instrument's **yield to maturity** (the discount rate applied to the face value) and the time until maturity. The amount of interest the investor earns is the difference between the face value and the purchase price.

EXAMPLE: Zero-coupon bond

A zero-coupon bond with a face value of \$1,000 will mature 15 years from today. The bond has a yield to maturity of 4%. Assuming annual compounding, what is the bond's price?

Answer:

$$PV = \frac{\$1,000}{(1 + 0.04)^{15}} = \$555.26$$

We can infer a bond's yield from its price using the same relationship. Rather than solving for r with algebra, we typically use our financial calculators. For this example, if we were given the price of \$555.26, the face value of \$1,000, and annual compounding over 15 years, we would enter the following:

$$PV = -555.26$$

$$FV = 1,000$$

$$PMT = 0$$

$$N = 15$$

Then, to get the yield, CPT I/Y = 4.00.



PROFESSOR'S NOTE

Remember to enter cash outflows as negative values and cash inflows as positive values. From the investor's point of view, the purchase price (PV) is an outflow, and the return of the face value at maturity (FV) is an inflow.

In some circumstances, interest rates can be negative. A zero-coupon bond with a negative yield would be priced at a **premium**, which means its price is greater than its face value.

EXAMPLE: Zero-coupon bond with a negative yield

If the bond in the previous example has a yield to maturity of -0.5% , what is its price, assuming annual compounding?

Answer:

$$PV = \frac{\$1,000}{(1 - 0.005)^{15}} = \$1,078.09$$

A **fixed-coupon bond** is only slightly more complex. With a coupon bond, the investor receives a cash interest payment each period in addition to the face value at maturity. The bond's **coupon rate** is a percentage of the face value and determines the amount of the interest payments. For example, a 3% annual coupon, $\$1,000$ bond pays 3% of $\$1,000$, or $\$30$, each year.

The coupon rate and the yield to maturity are two different things. We only use the coupon rate to determine the coupon payment (PMT). The yield to maturity (I/Y) is the discount rate implied by the bond's price.

EXAMPLE: Price of an annual coupon bond

Consider a 10-year, $\$1,000$ par value, 10% coupon, annual-pay bond. What is the value of this bond if its yield to maturity is 8% ?

Answer:

The coupon payments will be $10\% \times \$1,000 = \100 at the end of each year. The $\$1,000$ par value will be paid at the end of Year 10, along with the last coupon payment.

The value of this bond with a discount rate (yield to maturity) of 8% is:

$$\frac{100}{1.08} + \frac{100}{1.08^2} + \frac{100}{1.08^3} + \dots + \frac{100}{1.08^9} + \frac{1,100}{1.08^{10}} = 1,134.20$$

The calculator solution is:

$$N = 10; PMT = 100; FV = 1,000; I/Y = 8; CPT PV = -1,134.20$$

The bond's value is $\$1,134.20$.

**PROFESSOR'S NOTE**

For this reading where we want to illustrate time value of money concepts, we are only using annual coupon payments and compounding

periods. In the Fixed Income topic area, we will also perform these calculations for semiannual-pay bonds.

In contrast to fixed-coupon bonds, a **floating-rate bond** has coupon payments that change during the bond's life, based on changes in a market reference rate. For example, a floating-rate bond might pay a margin of 120 basis points over its market reference rate. A **basis point** is one hundredth of a percentage point, so 120 basis points represent 1.2%. If the market reference rate is 3.0%, the coupon rate for the next payment will be $3.0\% + 1.2\% = 4.2\%$.

Some bonds exist that have no maturity date. We refer to these as **perpetual bonds** or **perpetuities**. We cannot speak meaningfully of the future value of a perpetuity, but its present value simplifies mathematically to the following:

$$\text{PV of a perpetuity} = \frac{\text{payment}}{r}$$

An **amortizing bond** is one that pays a constant amount each period, including its maturity period. The difference between an amortizing bond and a fixed-coupon bond is that for an amortizing bond, each payment includes some portion of the principal. With a fixed-coupon bond, the entire principal is paid to the investor on the maturity date.

While the payment is fixed, the proportions of interest and principal within the payment change over the life of the bond. In early periods, when the outstanding principal is high, the fixed payment contains a greater proportion of interest. As principal is repaid over the life of the bond, the outstanding principal decreases, resulting in the proportion of interest declining and the proportion of principal increasing with each fixed payment.

Amortizing bonds and standard fixed-rate mortgages are examples of **annuity** instruments. An annuity consists of a number of constant payments, with constant time intervals between each payment. For an annuity, the payment each period is calculated as follows:

$$\text{annuity payment} = \frac{r \times \text{PV}}{1 - (1 + r)^{-t}}$$

where:

r = interest rate per period

t = number of periods

PV = present value (principal)

We can also determine an annuity payment using a financial calculator.

EXAMPLE: Computing a loan payment

Suppose you are considering applying for a \$2,000 loan that will be repaid with equal end-of-year payments over the next 13 years. If the annual interest rate for the loan is 6%, how much are your payments?

Answer:

The size of the end-of-year loan payment can be determined by inputting values for the three known variables and computing PMT. Note that $FV = 0$ because the loan will be fully paid off after the last payment:

$$N = 13; I/Y = 6; PV = 2,000; FV = 0; CPT \rightarrow PMT = -\$225.92$$

Equity Securities

As with fixed-income securities, we value **equity securities** such as common and preferred stock as the present value of their future cash flows. The key differences are that equity securities typically do not mature, and their cash flows may change over time.

Preferred stock pays a fixed dividend that is stated as a percentage of its **par value** (similar to the face value of a bond). As with bonds, we must distinguish between the stated percentage that determines the cash flows and the discount rate we apply to the cash flows. We say that equity investors have a **required return** that will induce them to own an equity share. This required return is the discount rate we use to value equity securities.

Because we can consider a preferred stock's fixed stream of dividends to be infinite, we can use the perpetuity formula to determine its value:

$$\text{preferred stock value} = \frac{D_p}{r_p}$$

where:

D_p = dividend per period

r_p = the market's required return on the preferred stock

EXAMPLE: Preferred stock valuation

A company's \$100 par preferred stock will pay a \$5 annual dividend and has a required return of 8%. Calculate the value of the preferred stock.

Answer:

$$\text{Value of the preferred stock: } \frac{D_p}{r_p} = \frac{5.00}{0.08} = \$62.50$$

Common stock is a residual claim to a company's assets after it satisfies all other claims. Common stock typically does not promise a fixed dividend payment. Instead, the company's management decides whether and when to pay common dividends.

Because the future cash flows are uncertain, we must use models to estimate the value of common stock. Here, we will look at three approaches analysts use frequently, which we call **dividend discount models (DDMs)**. We will return to these examples in the Equities topic area and explain when each model is appropriate.

Assume a constant future dividend. Under this assumption, we can value a common stock the same way we value a preferred stock, using the perpetuity formula.

Assume a constant growth rate of dividends. With this assumption, we can apply the **constant growth DDM**, also known as the **Gordon growth model**. In this model, which is a perpetuity model adjusted for constant growth in the cash flows, we state the value of a common share as follows:

$$V_0 = \frac{D_1}{r_e - g_c}$$

where:

V_0 = value of a share *this* period

D_1 = dividend expected to be paid *next* period

r_e = required return on common equity

g_c = constant growth rate of dividends

In this model, V_0 represents the PV of all the dividends in future periods, beginning with D_1 . Note that r_e must be greater than g_c , or the math will not work.

EXAMPLE: Gordon growth model valuation

Calculate the value of a stock that is expected to pay a \$1.62 dividend next year, if dividends are expected to grow at 8% forever and the required return on equity is 12%.

Answer:

$$\begin{aligned}\text{Calculate the stock's value} &= \frac{D_1}{(r_e - g_c)} \\ &= \frac{\$1.62}{(0.12 - 0.08)} \\ &= \$40.50\end{aligned}$$

Assume a changing growth rate of dividends. This can be done in many ways. The example we will use here (and the one that is required for the Level I CFA exam) is known as a **multistage DDM**. Essentially, we assume a pattern of dividends in the short term, such as a period of high growth, followed by a constant growth rate of dividends in the long term.

To use a multistage DDM, we discount the expected dividends in the short term as individual cash flows, then apply the constant growth DDM to the long term. As we saw in the previous example, the constant growth DDM gives us a value for an equity share *one period before* the dividend we use in the numerator.

EXAMPLE: Multistage growth

Consider a stock with dividends that are expected to grow at 15% per year for two years, after which they are expected to grow at 5% per year, indefinitely. The last dividend paid was \$1.00, and the required return on equity is 11%. Calculate the value of this stock using the multistage growth model.

Answer:

Calculate the dividends over the high growth period:

$$D_1 = D_0(1 + g^*) = 1.00(1.15) = \$1.15$$

$$D_2 = D_1(1 + g^*) = 1.15(1.15) = 1.15^2 = \$1.32$$

Calculate the first dividend of the constant growth period:

$$D_3 = D_2(1 + g) = 1.32 \times 1.05 = \$1.386$$

Use the constant growth model to get P_2 , a value for all the (infinite) dividends expected from time = 3 onward:

$$P_2 = \frac{D_3}{r_e - g_c} = \frac{1.386}{0.11 - 0.05} = \$23.10$$

Finally, we can sum the present values of Dividends 1 and 2 and of P_2 to get the present value of all the expected future dividends during both the high

growth and constant growth periods:

$$\frac{1.15}{1.11} + \frac{1.32 + 23.10}{(1.11)^2} = \$20.86$$



MODULE QUIZ 4.1

1. Terry Corporation preferred stock is expected to pay a \$9 annual dividend in perpetuity. If the required rate of return on an equivalent investment is 11%, one share of Terry preferred should be worth:
A. \$81.82.
B. \$99.00.
C. \$122.22.
2. Dover Company wants to issue a \$10 million face value of 10-year bonds with an annual coupon rate of 5%. If the investors' required yield on Dover's bonds is 6%, the amount the company will receive when it issues these bonds (ignoring transactions costs) will be:
A. less than \$10 million.
B. equal to \$10 million.
C. greater than \$10 million.
3. Wilson Company paid a dividend of \$2.00 in the most recent year. The dividend is expected to grow at a 10% rate for the next three years, after which it is expected to grow at a constant rate of 4%. If the required rate of return on Wilson stock is 9%, its value today is *closest* to:
A. \$48.87.
B. \$55.37.
C. \$62.65.

MODULE 4.2: IMPLIED RETURNS AND CASH FLOW ADDITIVITY

LOS 4.b: Calculate and interpret the implied return of fixed-income instruments and required return and implied growth of equity instruments given their present value and cash flows.

The examples we have seen so far illustrate the relationships among present value, future cash flows, and the required rate of return. We can easily rearrange these relationships and solve for the required rate of return, given a security's price and its future cash flows.

EXAMPLE: Rate of return for a pure discount bond

A zero-coupon bond with a face value of \$1,000 will mature 15 years from today. The bond's price is \$650. Assuming annual compounding, what is the

investor's annualized return?

Answer:

$$\frac{\$1,000}{(1 + r)^{15}} = \$650$$

$$(1 + r)^{15} = \frac{\$1,000}{\$650} = 1.5385$$

$$r = 1.5385^{1/15} - 1 = 0.0291 = 2.91\%$$

EXAMPLE: Yield of an annual coupon bond

Consider the 10-year, \$1,000 par value, 10% coupon, annual-pay bond we examined in an earlier example, when its price was \$1,134.20 at a yield to maturity of 8%. What is its yield to maturity if its price immediately decreases to \$1,085.00?

Answer:

$$N = 10; PMT = 100; FV = 1,000; PV = -1,085; CPT I/Y = 8.6934$$

The bond's yield to maturity increased to 8.69%.

Notice that the relationship between prices and yields is inverse. *When the price decreases, the yield to maturity increases; when the price increases, the yield to maturity decreases.* Or, equivalently, *when the yield increases, the price decreases; when the yield decreases, the price increases.* We will use this concept repeatedly when we study bonds in the Fixed Income topic area.

In our examples for equity share values, we assumed a value for the investor's required rate of return. In practice, the required rate of return on equity is not directly observable. Instead, we use share prices that we can observe in the market to derive implied required rates of return on equity, given our assumptions about their future cash flows.

For example, if we assume a constant rate of dividend growth, we can rearrange the constant growth DDM to solve for the required rate of return:

$$V_0 = \frac{D_1}{r_e - g_c}$$

$$r_e - g_c = \frac{D_1}{V_0}$$

$$r_e = \frac{D_1}{V_0} + g_c$$

That is, the required rate of return on equity is the ratio of the expected dividend to the current price (which we refer to as a share's **dividend yield**) plus the assumed constant growth rate.

We can also rearrange the model to solve for a stock's **implied growth rate**, given the required rate of return:

$$r_e = \frac{D_1}{V_0} + g_c$$

$$g_c = r_e - \frac{D_1}{V_0}$$

That is, the implied growth rate is the required rate of return minus the dividend yield.



PROFESSOR'S NOTE

In the Equities topic area, we will revisit this method of estimating the required return on equity and illustrate how it can also be estimated using price multiples such as the price-to-earnings ratio.

LOS 4.c: Explain the cash flow additivity principle and its importance for the condition of no arbitrage, and explain its use in calculating implied forward interest rates, forward exchange rates, and option values.

The **cash flow additivity principle** refers to the fact that the PV of any stream of cash flows equals the sum of the PVs of the individual cash flows. If we have two series of cash flows, the sum of the PVs of the two series is the same as the PVs of the two series taken together, adding cash flows that will be paid at the same point in time. We can also divide up a series of cash flows any way we like, and the PV of the “pieces” will equal the PV of the original series.

EXAMPLE: Cash flow additivity principle

A security will make the following payments at the end of the next four years: \$100, \$100, \$400, and \$100. Calculate the PV of these cash flows using the concept of the PV of an annuity when the appropriate discount rate is 10%.

Answer:

We can divide the cash flows in a way that lets us calculate the PV of an annuity:

$t = 1$	$t = 2$	$t = 3$	$t = 4$	
100	100	100	100	Cash Flow Series 1
0	0	300	0	Cash Flow Series 2
<u>\$100</u>	<u>\$100</u>	<u>\$400</u>	<u>\$100</u>	

The additivity principle tells us that to get the PV of the original series, we can just add the PVs of Cash Flow Series 1 (a 4-period annuity) and Cash Flow Series 2 (a single payment three periods from now).

For the annuity: $N = 4$; $PMT = 100$; $FV = 0$; $I/Y = 10$; $CPT \rightarrow PV = -\$316.99$

For the single payment: $N = 3$; $PMT = 0$; $FV = 300$; $I/Y = 10$; $CPT \rightarrow PV = -\$225.39$

The sum of these two values is $316.99 + 225.39 = \$542.38$.

The sum of these two (present) values is identical (except for rounding) to the sum of the present values of the payments of the original series:

$$\frac{100}{1.1} + \frac{100}{1.1^2} + \frac{400}{1.1^3} + \frac{100}{1.1^4} = \$542.38$$

This is a simple example of **replication**. In effect, we created the equivalent of the given series of uneven cash flows by combining a 4-year annuity of 100 with a 3-year zero-coupon bond of 300.

We rely on the cash flow additivity principle in many of the pricing models we see in the Level I CFA curriculum. It is the basis for the **no-arbitrage principle**, or “law of one price,” which says that if two sets of future cash flows are identical under all conditions, they will have the same price today (or if they don’t, investors will quickly buy the lower-priced one and sell the higher-priced one, which will drive their prices together).

Three examples of valuation based on the no-arbitrage condition are forward interest rates, forward exchange rates, and option pricing using a binomial model. We will explain each of these examples in greater detail when we address the related concepts in the Fixed Income, Economics, and Derivatives topic

areas. For now, just focus on how they apply the principle that equivalent future cash flows must have the same present value.

Forward Interest Rates

A **forward interest rate** is the interest rate for a loan to be made at some future date. The notation used must identify both the length of the loan and when in the future the money will be borrowed. Thus, $1y1y$ is the rate for a 1-year loan to be made one year from now; $2y1y$ is the rate for a 1-year loan to be made two years from now; $3y2y$ is the 2-year forward rate three years from now; and so on.

By contrast, a **spot interest rate** is an annualized interest rate for a loan to be made today. We will use the notation S_1 for a 1-year rate today, S_2 for a 2-year rate today, and so on.

The way the cash flow additivity principle applies here is that, for example, borrowing for three years at the 3-year spot rate, or borrowing for one-year periods in three successive years, should have the same cost today. This relation is illustrated as follows: $(1 + S_3)^3 = (1 + S_1)(1 + 1y1y)(1 + 2y1y)$.

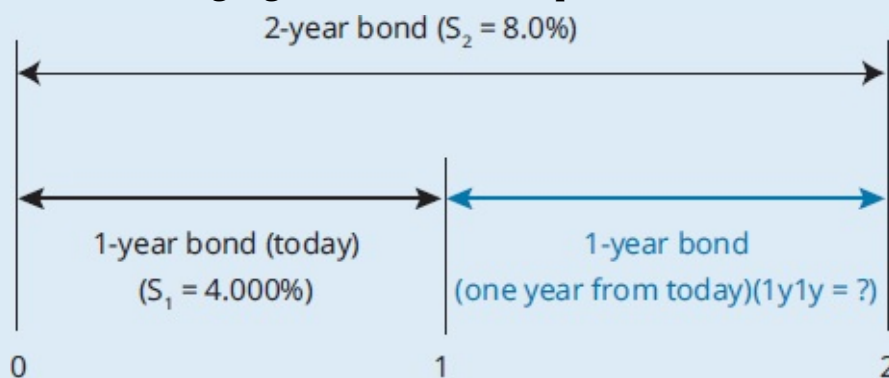
In fact, any combination of spot and forward interest rates that cover the same period should have the same cost. Using this idea, we can derive **implied forward rates** from spot rates that are observable in the fixed-income markets.

EXAMPLE: Computing a forward rate from spot rates

The 2-year spot rate, S_2 , is 8%, and the 1-year spot rate, S_1 , is 4%. Calculate the one-year forward rate one year from now, $1y1y$.

Answer:

The following figure illustrates the problem.



Because we know that both ways of borrowing for two years have the same cost today:

$$(1.08)^2 = (1.04)(1 + 1y1y)$$

$$(1 + 1y1y) = \frac{(1.08)^2}{(1.04)}$$

$$1y1y = \frac{(1.08)^2}{(1.04)} - 1 = \frac{1.1664}{1.04} - 1 = 12.154\%$$

In other words, investors are willing to accept 4.0% on the 1-year bond today (when they could get 8.0% on the 2-year bond today) only because they can get 12.154% on a 1-year bond one year from today. This future rate that can be locked in today is a forward rate.

Forward Currency Exchange Rates

An **exchange rate** is the price of one country's currency in terms of another country's currency. For example, an exchange rate of 1.416 USD/EUR means that one euro (EUR) is worth 1.416 US dollars (USD). The Level I CFA curriculum refers to the currency in the numerator (USD, in this example) as the *price currency*, and the one in the denominator (EUR in this example) as the *base currency*.

Like interest rates, exchange rates can be quoted as spot rates for currency exchanges to be made today, or as forward rates for currency exchanges to be made at a future date.

The percentage difference between forward and spot exchange rates is approximately the difference between the two countries' interest rates. This is because there is an arbitrage trade with a riskless profit to be made when this relation does not hold.

The possible arbitrage is as follows: borrow Currency A at Interest Rate A, convert it to Currency B at the spot rate and invest it to earn Interest Rate B, and sell the proceeds from this investment forward at the forward rate to turn it back into Currency A. If the forward rate does not correctly reflect the difference between interest rates, such an arbitrage could generate a profit to the extent that the return from investing Currency B and converting it back to Currency A with a forward contract is greater than the cost of borrowing Currency A for the period.

For spot and forward rates expressed as price currency per unit of base currency, the no-arbitrage relation is as follows:

$$\frac{\text{forward}}{\text{spot}} = \frac{(1 + \text{interest rate}_{\text{price currency}})}{(1 + \text{interest rate}_{\text{base currency}})}$$

This formula can be rearranged as necessary to solve for specific values of the relevant terms.

EXAMPLE: Calculating the arbitrage-free forward exchange rate

Consider two currencies, the ABE and the DUB. The spot ABE/DUB exchange rate is 4.5671, the 1-year riskless ABE rate is 5%, and the 1-year riskless DUB rate is 3%. What is the 1-year forward exchange rate that will prevent arbitrage profits?

Answer:

Rearranging our formula, we have:

$$\text{forward} = \text{spot} \left(\frac{1 + I_{\text{ABE}}}{1 + I_{\text{DUB}}} \right)$$

And we can calculate the forward rate as:

$$\text{forward} = 4.5671 \left(\frac{1.05}{1.03} \right) = 4.6558 \text{ ABE / DUB}$$

As you can see, the forward rate is greater than the spot rate by $4.6558 / 4.5671 - 1 = 1.94\%$. This is approximately equal to the interest rate differential of $5\% - 3\% = 2\%$.

Option Pricing Model

An **option** is the right, but not the obligation, to buy or sell an asset on a future date for a specified price. The right to buy an asset is a **call option**, and the right to sell an asset is a **put option**.

Valuing options is different from valuing other securities because the owner can let an option expire unexercised. A call option owner will let the option expire if, at expiration, the underlying asset can be bought in the market for less than the price specified in the option. A put option owner will let the option expire if the underlying asset can be sold in the market for more than the price specified in the option. In these cases, we say an option is *out of the money*. If a call option is *in the money* on its expiration date, the owner has the right to buy the asset for

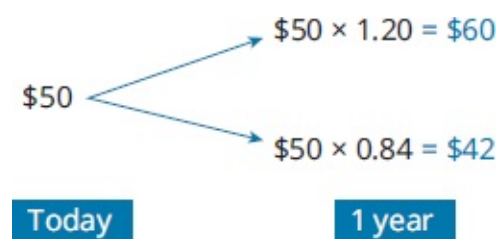
less than its market price, and therefore will exercise the option. If a put option is in the money on its expiration date, the owner has the right to sell the asset for more than its market price, so the owner will exercise.

An approach to valuing options that we will use in the Derivatives topic area is a **binomial model**. A binomial model is based on the idea that, over the next period, an asset's price will change to one of two possible values. To construct a one-period binomial model for pricing an option, we need the following:

- ▮ A value for the underlying asset at the beginning of the period
- ▮ An exercise price for the option; the exercise price can be different from the value of the underlying, and we assume the option expires one period from now
- ▮ Returns that will result from an up move and a down move in the value of the underlying over one period
- ▮ The risk-free rate over the period

As an example, we can model a one-period call option value at initiation by constructing a risk-free portfolio. Let us assume that the current price of the stock is \$50 and in one period, when the call option expires, it will either increase by 20% to \$60 or decrease by 16% to \$42.

Figure 4.1: One-Period Binomial Tree



Let us also assume that the exercise price of the option is \$55 and the risk-free interest rate is 3%.

Now that we have the exercise price and the values the stock can take at expiration, we can compute the value (payoff) of the option at its expiration. For call options, the value at expiration is the greater of the stock price minus the exercise price, or zero. (For put options, it is the greater of the exercise price minus the stock price, or zero.) Here is the breakdown:

$$\text{call option value at expiration with an up move in stock price} = \$60 - \$55 = \$5$$

call option value at expiration with a down move in stock price: $\$42 - \$55 = -\$13$, so the option value is zero (the greater of $-\$13$ or zero); this is because the holder is not required to exercise the option (he can simply let it expire)

We can now construct a risk-free portfolio by *writing* one call option (that is, we grant someone else the right to buy the stock from us) and buying a fraction (h) of the underlying stock. To be risk free, the portfolio must have the same value at the option's expiration, regardless of whether the underlying stock rises or falls in value. We need to solve for the h that will give the portfolio the same value in either case.

The portfolio value at expiration = h units of stock $\times$ stock price at option expiration – value of the call option. We subtract the value of the call option because we are short the option.

value of the portfolio if the stock price rises = $h \times \$60 - \5

value of the portfolio if the stock price falls = $h \times \$42 - \0

Remember that the value of the portfolio needs to be identical, regardless of whether the stock rises or falls, which gives us the following:

$$h(\$60) - \$5 = h(\$42)$$

$$h(\$60) - h(\$42) = \$5$$

$$h = \frac{\$5}{(\$60 - \$42)} = 0.278$$

This result is known as the *hedge ratio* for this option. It represents the fraction of stock that must be purchased for each option written to create the risk-free portfolio. So, for every one option shorted, 0.278 shares of stock must be purchased.

We can now compute the value of the portfolio at option expiration as shown in the following table.

Stock price at expiration	\$60	\$42
Long stock	$0.278 \times \$60 = \16.68	$0.278 \times \$42 = \11.68
Short option value	$-\$5$	$\$0$
Portfolio value	$\$11.68$	$\$11.68$

As shown in the table, the portfolio has a value of \$11.68, regardless of whether the stock price rises or falls. A portfolio with a certain future value is risk free, and as a result must generate a risk-free rate of return (3%, in this example).

Using this logic, we can compute the value of the portfolio at initiation:

$$V_0 = \frac{\$11.68}{1.03} = \$11.34$$

From this, we can calculate the value of the option at initiation. The value of 0.278 shares of the underlying stock is $0.278 \times \$50 = \13.90 . We know that the value of the portfolio at initiation is \$11.34, so the difference must be the premium we received for writing the option:

$$\text{option value} = \$13.90 - \$11.34 = \$2.56$$



PROFESSOR'S NOTE

The concepts for binomial option pricing are quite complex. We will revisit this example in the Derivatives topic area, where we address options in more detail.



MODULE QUIZ 4.2

1. For an equity share with a constant growth rate of dividends, we can estimate its:
 - A. value as the next dividend discounted at the required rate of return.
 - B. growth rate as the sum of its required rate of return and its dividend yield.
 - C. required return as the sum of its constant growth rate and its dividend yield.
2. An investment of €5 million today is expected to produce a one-time payoff of €7 million three years from today. The annual return on this investment, assuming annual compounding, is *closest* to:
 - A. 12%.
 - B. 13%.
 - C. 14%.

KEY CONCEPTS

LOS 4.a

The value of a fixed-income instrument or an equity security is the present value of its future cash flows, discounted at the investor's required rate of return:

$$PV = \frac{FV}{(1 + r)^t} = FV(1 + r)^{-t}$$

where:

r = interest rate per compounding period

t = number of compounding periods

$$\text{annuity payment} = \frac{r \times PV}{1 - (1 + r)^{-t}}$$

where:

r = interest rate per period

t = number of periods

PV = present value (principal)

The PV of a perpetual bond or a preferred stock = $\frac{\text{payment}}{r}$, where r = required rate of return.

The PV of a common stock with a constant growth rate of dividends is:

$$V_0 = \frac{D_1}{r_e - g_c}$$

LOS 4.b

By rearranging the present value relationship, we can calculate a security's required rate of return based on its price and its future cash flows. The relationship between prices and required rates of return is inverse.

For an equity share with a constant rate of dividend growth, we can estimate the required rate of return as the dividend yield plus the assumed constant growth rate, or we can estimate the implied growth rate as the required rate of return minus the dividend yield.

LOS 4.c

Using the cash flow additivity principle, we can divide up a series of cash flows any way we like, and the present value of the pieces will equal the present value of the original series. This principle is the basis for the no-arbitrage condition, under which two sets of future cash flows that are identical must have the same present value.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 4.1

- . A $\frac{9}{0.11} = \$81.82$. (LOS 4.a)
- . A Because the required yield is greater than the coupon rate, the present value of the bonds is less than their face value: $N = 10$; $I/Y = 6$; $PMT = 0.05 \times \$10,000,00 = \$500,000$; $FV = \$10,000,000$; and $CPT PV = -\$9,263,991$. (LOS 4.a)
- . A Dividends during the high growth period:

$$D_1 = \$2.00 \times 1.10 = \$2.20$$

$$D_2 = \$2.20 \times 1.10 = \$2.42$$

$$D_3 = \$2.42 \times 1.10 = \$2.662$$
 First dividend of the constant growth period:

$$D_4 = \$2.662 \times 1.04 = \$2.7685$$

$$\text{Stock value one period earlier (t = 3)} = \frac{\$2.7685}{0.09 - 0.04} = \$55.3696$$

$$\text{Stock value today (t = 0)} = \frac{\$2.20}{1.09} + \frac{\$2.42}{(1.09)^2} + \frac{\$2.662 + \$55.3696}{(1.09)^3}$$

$$= \$48.8663$$
 (LOS 4.a)

Module Quiz 4.2

- . C Using the constant growth dividend discount model, we can estimate the required rate of return as $k_e = \frac{D_1}{V_0} + g_c$. The estimated value of a share is *all* its future dividends discounted at the required rate of return, which simplifies to $V_0 = \frac{D_1}{k_e - g_c}$ if we assume a constant growth rate. We can estimate the constant growth rate as the required rate of return *minus* the dividend yield. (LOS 4.b)
- . A $\left(\frac{7}{5}\right)^{\frac{1}{3}} - 1 = 0.1187$.
(LOS 4.b)

READING 5

STATISTICAL CHARACTERISTICS OF ASSET RETURNS

MODULE 5.1: MEASURES OF CENTRAL TENDENCY

LOS 5.a: Calculate, interpret, and evaluate various measures of (1) central tendency and location and (2) dispersion.

Measures of central tendency identify the center, or average, of a dataset. This central point can then be used to represent the typical, or expected, value.

Two concepts are important here: the population and the sample. The **population** represents the entire set of data points or elements that can be observed, such as the entire history of percentage changes in the price of oil. A **sample** is a subset of a population, such as percentage changes in the price of oil over the last five years.

The **arithmetic mean** is the sum of the observed values divided by the number of observations. It is the most widely used measure of central tendency. An example of an arithmetic mean is a **sample mean**, which is the sum of all the values in a sample, divided by the number of observations in the sample, n . It is used to make inferences about the population mean.

The sample mean is:

$$\bar{X} = \sum_{i=1}^n \frac{x_i}{n}$$

The population mean is:

$$\mu = \sum_{i=1}^N \frac{x_i}{N}$$

The **median** is the midpoint of a dataset when the data is arranged in ascending or descending order. Half of the observations lie above the median, and half are below. To determine the median, arrange the data from the highest to lowest value, or lowest to highest value, and find the middle observation (if the dataset has an odd number of observations) or the average of the middle two observations (if the dataset has an even number of observations).

EXAMPLE: Median using an odd number of observations

What is the median return for five portfolio managers with a 10-year annualized total returns record of 30%, 15%, 25%, 21%, and 23%?

Answer:

First, arrange the returns in descending (or ascending) order:

30%, 25%, 23%, 21%, 15%

Then, select the observation that has an equal number of observations above and below it—the one in the middle. For this dataset, the third observation, 23%, is the median.

EXAMPLE: Median using an even number of observations

Suppose we add a sixth manager to the previous example with a return of 28%. What is the median return?

Answer:

Arranging the returns in descending order gives us this:

30%, 28%, 25%, 23%, 21%, 15%

With an even number of observations, there is no single middle value. The median value, in this case, is the arithmetic mean of the two middle observations, 25% and 23%. Thus, the median return for the six managers

is $\frac{25\% + 23\%}{2} = 24\%$.

The median is important because the arithmetic mean can be affected by **outliers**, which are extremely large or small values. When this occurs, the median is a better measure of central tendency than the mean because it is not affected by extreme values that might result from measurement error, data entry error, sampling errors, or simply from the actual variability of the data.

Consider the sample data: 12%, 14%, 16%, 17%, and 19%, for which the mean is 15.6% and the median is 16%. If we replace 19% with an outlier value of 30%, the median is unchanged at 16%, while the mean is increased to 17.8%.

When a statistical estimate produces reliable results in the presence of outliers, or when the data are from a different distribution than the researcher has assumed, we refer to it as a **robust estimator**. The median is a more robust estimator of central tendency than the mean.

The **mode** is the value that occurs most frequently in a dataset. A dataset may have more than one mode, or no mode. The dataset 1, 1, 1, 3, 5, 6, 6, 6 has two modes, 1 and 6, and is referred to as bimodal.

EXAMPLE: Mode

What is the mode of the following dataset?

Dataset: [30%, 28%, 25%, 23%, 28%, 15%, 5%]

Answer:

The mode is 28% because it is the value appearing most frequently.

For continuous data, such as investment returns, we typically do not identify a single outcome as the mode. Instead, we divide the relevant range of outcomes into intervals, and we identify the **modal interval** as the one into which the largest number of observations fall.

Methods for Dealing With Outliers

In some cases, a researcher may decide that outliers should be excluded from a measure of central tendency. One technique for doing so is to use a **trimmed mean**. A trimmed mean excludes a stated percentage of the most extreme observations. A 1% trimmed mean, for example, would discard the lowest 0.5% and the highest 0.5% of the observations.

Another technique is to use a **winsorized mean**. With a winsorized mean, instead of discarding the highest and lowest observations, we substitute values for them. To calculate a 90% winsorized mean, for example, we would determine the 5th and 95th percentile of the observations, substitute the 5th percentile value for any values lower than that, substitute the 95th percentile

value for any values higher than that, and then calculate the mean of the revised dataset. Percentiles are measures of location, and we address them next.

Quantiles

Quantile is the general term for a value at or below which a stated proportion of the data in a distribution lies.

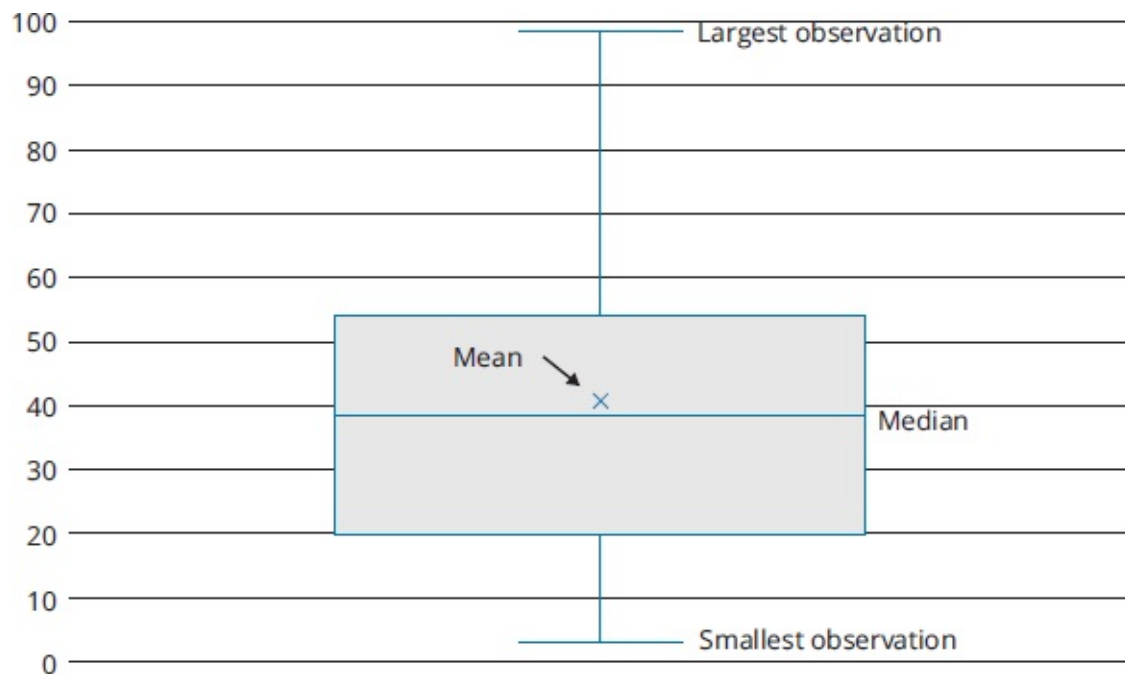
Examples of quantiles include the following:

- | **Quartile.** The distribution is divided into quarters.
- | **Quintile.** The distribution is divided into fifths.
- | **Decile.** The distribution is divided into tenths.
- | **Percentile.** The distribution is divided into hundredths (percentages).

Any quantile may be expressed as a percentile. For example, the third quartile partitions a distribution such that three-fourths, or 75%, of the observations fall below that value. Thus, the third quartile is the 75th percentile. The difference between the third quartile and the first quartile (25th percentile) is known as the **interquartile range**.

To visualize a dataset, we can create a **box and whisker plot**, as shown in Figure 5.1. In a box and whisker plot, the box represents the central portion of the data, such as the interquartile range. The vertical line represents the entire range. In Figure 5.1, we can see that the largest observation is farther away from the center than the smallest observation. This suggests that the data might include one or more outliers on the upper end.

Figure 5.1: Box and Whisker Plot



If the data are measured correctly (no data errors), it may be best to include all data values, so as not to discard any valid data points and reflect the characteristics of the distribution accurately. Trimming the data by discarding upper and lower values offers an objective identification of which observations are considered outliers, but it may remove valuable information about the distribution. Winsorizing the data, replacing high and low outliers with percentile values, preserves more information than a trimmed mean. With a small number of outliers, the winsorized mean will be a better measure of a distribution's central tendency than either an arithmetic mean or a trimmed mean.

Statistical analysis works best when data are **stationary**, with a mean and variance of the underlying distribution that do not change over time. When the characteristics of a distribution vary over time, the distribution is termed **nonstationary**, making statistical analysis and tests less reliable.

As an example, a price series that increases over time, such as stock prices or index values, is nonstationary. If we instead examine weekly or monthly percentage changes in prices, we may find that the distribution of the data is stationary over longer time periods.



MODULE QUIZ 5.1

1. A dataset has 100 observations. Which of the following measures of central tendency will be calculated using a denominator of 100?

- A. The winsorized mean, but not the trimmed mean.
 - B. Both the trimmed mean and the winsorized mean.
 - C. Neither the trimmed mean nor the winsorized mean.
2. Observe the following dataset:
- 5, 3, –1, 6, 4, –1, 3, 6, 2, 3
- The value of 2 is this dataset's:
- A. mode.
 - B. median.
 - C. arithmetic mean.

MODULE 5.2: DISPERSION, SKEWNESS, AND KURTOSIS

Dispersion is defined as variability around the central tendency. The common theme in finance and investments is the tradeoff between reward and risk, where central tendency of returns is the measure of reward and dispersion of returns is the measure of risk.

The **range** is a relatively simple measure of variability, but when used with other measures, it provides useful information. The range is the distance between the largest and the smallest value in a dataset.

$$\text{range} = \text{maximum value} - \text{minimum value}$$

EXAMPLE: Range

What is the range for the 5-year annualized total returns for five investment managers if the managers' individual returns were 30%, 12%, 25%, 20%, and 23%?

Answer:

$$\text{range} = 30 - 12 = 18\%$$

The **mean absolute deviation (MAD)** is the average of the absolute values of the deviations of individual observations from the arithmetic mean.

$$\text{MAD} = \frac{\sum_{i=1}^n |X_i - \bar{X}|}{n}$$

The reason MAD uses absolute values of each deviation is because the sum of actual deviations from the arithmetic mean is always zero.

EXAMPLE: Mean absolute deviation

What is the MAD of the investment returns for the five managers discussed in the preceding example? How is it interpreted?

Answer:

Annualized returns: [30%, 12%, 25%, 20%, 23%]

$$\bar{X} = \frac{(30 + 12 + 25 + 20 + 23)}{5} = 22\%$$

$$\text{MAD} = \frac{[|30 - 22| + |12 - 22| + |25 - 22| + |20 - 22| + |23 - 22|]}{5}$$

$$\text{MAD} = \frac{[8 + 10 + 3 + 2 + 1]}{5} = 4.8\%$$

This result can be interpreted to mean that, on average, an individual return will deviate $\pm 4.8\%$ from the mean return of 22%.

Another measure of dispersion is the **median absolute deviation**, which is the median of the absolute deviations from the data's median:

$$\text{median absolute deviation} = \text{Med}(|X_i - \text{Med}(X)|)$$

Compared to the MAD, the median absolute deviation is a more robust measure of dispersion.

Variance and Standard Deviation

The two most widely used measures of dispersion, variance and standard deviation, are based on squared deviations from the mean.

Population variance, σ^2 , is defined as the average of the squared deviations around the population mean, μ :

$$\sigma^2 = \sum_{i=1}^n \frac{(x_i - \mu)^2}{n}$$

The variance calculation squares the deviations around the mean to resolve the problem that the sum of negative and positive deviations is always zero. The variance of financial asset returns measures the variability of the asset returns, with a higher variance indicating greater risk. Variance is sensitive to outliers because squaring the deviations magnifies the importance of outliers.

Population standard deviation, σ , is the positive square root of the population variance:

$$\sigma = \sqrt{\sum_{i=1}^n \frac{(x_i - \mu)^2}{n}}$$

Unlike variance, standard deviation is in the same units as the original data (e.g., percentage returns, dollars, euros), rather than those units squared.

The **sample variance**, s^2 , is the measure of dispersion that applies when we are evaluating a sample of n observations from a population. Sample variance is calculated using the following formula:

$$s^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}$$

The denominator for s^2 is $n - 1$, one less than the sample size n . Based on statistical theory, using the entire number of sample observations, n , instead of $n - 1$, as the divisor in the computation of s^2 , would systematically *underestimate* the population variance—particularly for smaller sample sizes. This systematic underestimation causes the sample variance to be a **biased estimator** of the population variance. Using $n - 1$ instead of n in the denominator (an adjustment known as **Bessel's correction**) improves the statistical properties of s^2 as an estimator of the population variance.

EXAMPLE: Sample variance

Assume that the 5-year annualized total returns for the five investment managers used in the preceding examples represent only a sample of the managers at a large investment firm. What is the sample variance of these returns?

Answer:

$$\begin{aligned}\bar{X} &= \frac{[30 + 12 + 25 + 20 + 23]}{5} = 22\% \\ s^2 &= \frac{[(30 - 22)^2 + (12 - 22)^2 + (25 - 22)^2 + (20 - 22)^2 + (23 - 22)^2]}{5 - 1} \\ &= 44.5(\%^2)\end{aligned}$$

Thus, the sample variance of $44.5(\%^2)$ can be interpreted to be an unbiased estimator of the population variance. Note that 44.5 “percent squared” is

0.00445, and you will get this value if you put the percentage returns in decimal form—for example, $(0.30 - 0.22)^2$.

The **sample standard deviation** is the square root of the sample variance. The sample standard deviation, s , is calculated as follows:

$$s = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}}$$

We say the sample standard deviation is an **unbiased estimator** of the true population standard deviation, because the expected value of the statistic is equal to the population standard deviation.

EXAMPLE: Sample standard deviation

Compute the sample standard deviation based on the result of the preceding example.

Answer:

Because the sample variance for the preceding example was computed to be 44.5(%²), this is the sample standard deviation:

$$s = [44.5(\%^2)]^{1/2} = 6.67\%, \sqrt{0.00445} = 0.0667$$

This means that on average, an individual return from the sample will deviate $\pm 6.67\%$ from the mean return of 22%.

In statistical theory, the **root of time law** holds that variance increases linearly with time. For example, if the variance of daily returns is σ^2 , the variance of weekly returns is $5 \times \sigma^2$. Because we would calculate standard deviation of daily returns as the square root of $\sigma^2 = \sigma$, the standard deviation of weekly returns is the square root of $(5 \times \sigma^2)$, which equals $\sqrt{5} \times \sigma$. However, this relation does not always hold in practice. The variance of returns for longer time periods tends to be less than predicted by statistical theory, because of differences between the characteristics of the observed data and the assumptions underlying the theory.

LOS 5.b: Describe, interpret, and evaluate measures of skewness and kurtosis.

A distribution is symmetrical if it is shaped identically on both sides of its mean. Distributional symmetry implies that intervals of losses and gains will exhibit the same frequency. For example, a symmetrical distribution with a mean return of zero will have losses in the -6% to -4% interval as frequently as it will have gains in the $+4\%$ to $+6\%$ interval. The extent to which a returns distribution is symmetrical is important because the degree of symmetry tells analysts whether deviations from the mean are more likely to be above it or below it.

Skewness, or skew, refers to the extent to which a distribution is not symmetrical. Nonsymmetrical distributions may be either positively or negatively skewed and result from the occurrence of outliers in the dataset.

- | A *positively skewed* distribution is characterized by outliers greater than the mean (in the upper region, or right tail). A positively skewed distribution is said to be “skewed right” because of its relatively long upper (right) tail.
- | A *negatively skewed* distribution has a disproportionately large number of outliers less than the mean that fall within its lower (left) tail. A negatively skewed distribution is said to be “skewed left” because of its long lower tail.

Skewness affects the location of the mean, median, and mode of a distribution:

- | For a symmetrical distribution, the mean, median, and mode are equal.
- | For a positively skewed, unimodal distribution, the mode is less than the median, which is less than the mean. The mean is affected by outliers; a positively skewed distribution has large positive outliers, which will tend to pull the mean upward, or more positive.
- | For a negatively skewed, unimodal distribution, the mean is less than the median, which is less than the mode. In this case, large negative outliers tend to pull the mean downward (to the left).

An example of a positively skewed distribution is that of housing prices. Suppose you live in a neighborhood with 100 homes: 99 of them sell for \$100,000, and one sells for \$1,000,000. The median and the mode will be \$100,000, but the mean will be \$109,000. Hence, the mean has been pulled upward (to the right) by the existence of one expensive home (outlier) in the neighborhood.

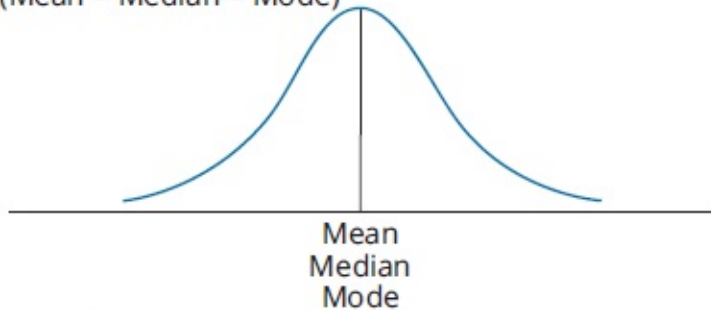
The key to remembering how measures of central tendency are affected by skewed data is to recognize that skew affects the mean more than the median and

mode, and the mean is pulled in the direction of the skew. The relative location of the mean, median, and mode for different distribution shapes is shown in Figure 5.2. Note that the median is between the other two measures for both positively and negatively skewed distributions.

Figure 5.2: Effect of Skewness on Mean, Median, and Mode

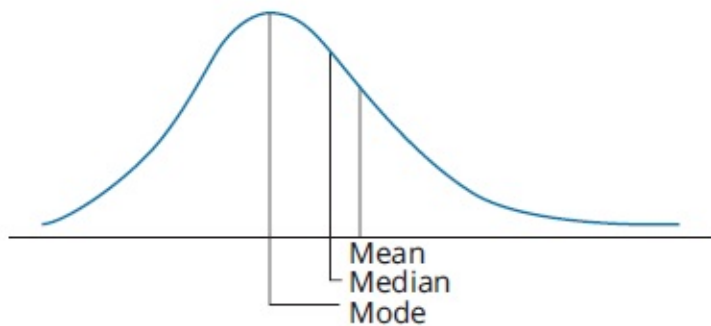
Symmetrical

(Mean = Median = Mode)



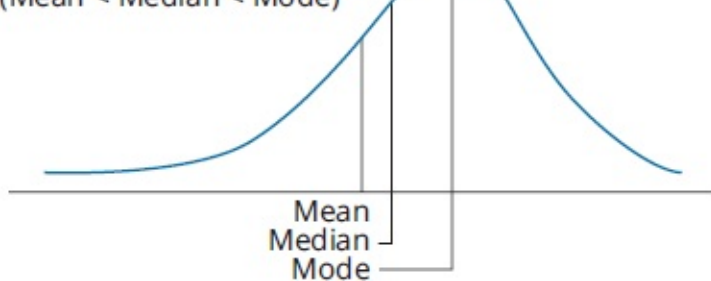
Positive (right) skew

(Mean > Median > Mode)



Negative (left) skew

(Mean < Median < Mode)



Sample skewness is calculated as follows:

$$\text{skewness} = \frac{1}{n} \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{s} \right)^3$$

where:

s = sample standard deviation



PROFESSOR'S NOTE

The LOS requires us to “describe, interpret, and evaluate” measures of skewness and kurtosis, but not to calculate them.

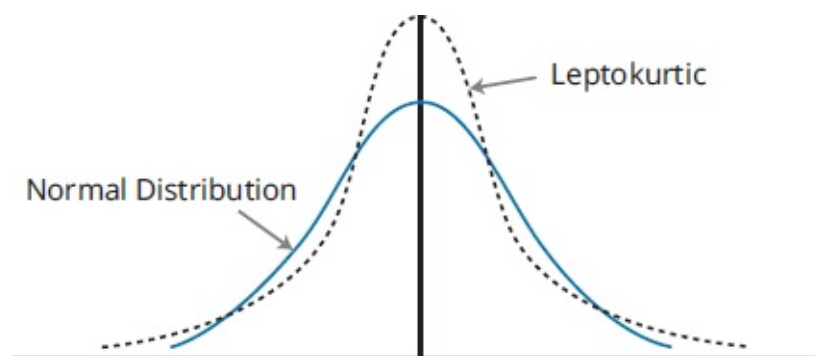
When a distribution is right skewed, sample skewness is positive because the deviations above the mean are larger, on average. A left-skewed distribution has a negative sample skewness. If skewness is equal to zero, the distribution is symmetrical.

Kurtosis is a measure of the degree to which a distribution is more or less peaked than a *normal distribution* (which we will describe in our next reading).

Leptokurtic describes a distribution that is more peaked than a normal distribution, whereas **platykurtic** refers to a distribution that is less peaked, or flatter, than a normal one. A distribution is **mesokurtic** if it has the same kurtosis as a normal distribution.

As indicated in Figure 5.3, a leptokurtic return distribution will have more returns clustered around the mean (more peaked) and more returns with large deviations from the mean (fatter tails). Relative to a normal distribution, a leptokurtic distribution will have a greater percentage of small deviations from the mean and a greater percentage of extremely large deviations from the mean. This means that there is a relatively greater probability of an observed value being either close to the mean or far from the mean. Regarding an investment returns distribution, a greater likelihood of a large deviation from the mean return is often perceived as an increase in risk.

Figure 5.3: Kurtosis



Kurtosis is calculated as follows:

$$\text{kurtosis} = \frac{1}{n} \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{s} \right)^4$$

A distribution is said to exhibit **excess kurtosis** if it has *more* or *less* kurtosis than a normal distribution. The kurtosis of a normal distribution equals three. Statisticians, however, sometimes report excess kurtosis, which is defined as kurtosis minus three. Thus, a normal distribution has excess kurtosis equal to zero, a leptokurtic distribution has excess kurtosis greater than zero, and a platykurtic distribution has excess kurtosis less than zero.

$$\text{excess kurtosis} = \frac{1}{n} \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{s} \right)^4 - 3$$

Kurtosis is critical in a risk management setting. Most research about the distribution of securities returns has shown that returns are not normally distributed. Actual securities returns tend to exhibit both skewness and kurtosis. When securities returns are modeled using an assumed normal distribution, predictions from the models will not consider the potential for extremely large negative outcomes. In fact, most risk managers put little emphasis on the mean and standard deviation of a distribution and focus more on the distribution of returns in the tails of the distribution, because that is where the risk is. In general, greater excess kurtosis and more negative skew in returns distributions indicate increased risk.



MODULE QUIZ 5.2

1. XYZ Corp. Annual Stock Returns

20X1	20X2	20X3	20X4	20X5	20X6
22%	5%	-7%	11%	2%	11%

The sample standard deviation is *closest* to:

- A. 9.8%.
 - B. 72.4%.
 - C. 96.3%.
2. For a statistical distribution with skew of -0.753 and excess kurtosis of 2.87, an analyst would *most likely* conclude that the underlying distribution, compared to a normal distribution, is characterized by:
- A. more frequent and more extreme positive returns.
 - B. more frequent and extreme negative returns.
 - C. fewer negative returns but more extreme returns, both positive and negative.

MODULE 5.3: COVARIANCE, CORRELATION, AND ALTERNATIVE MEASURES OF DISPERSION

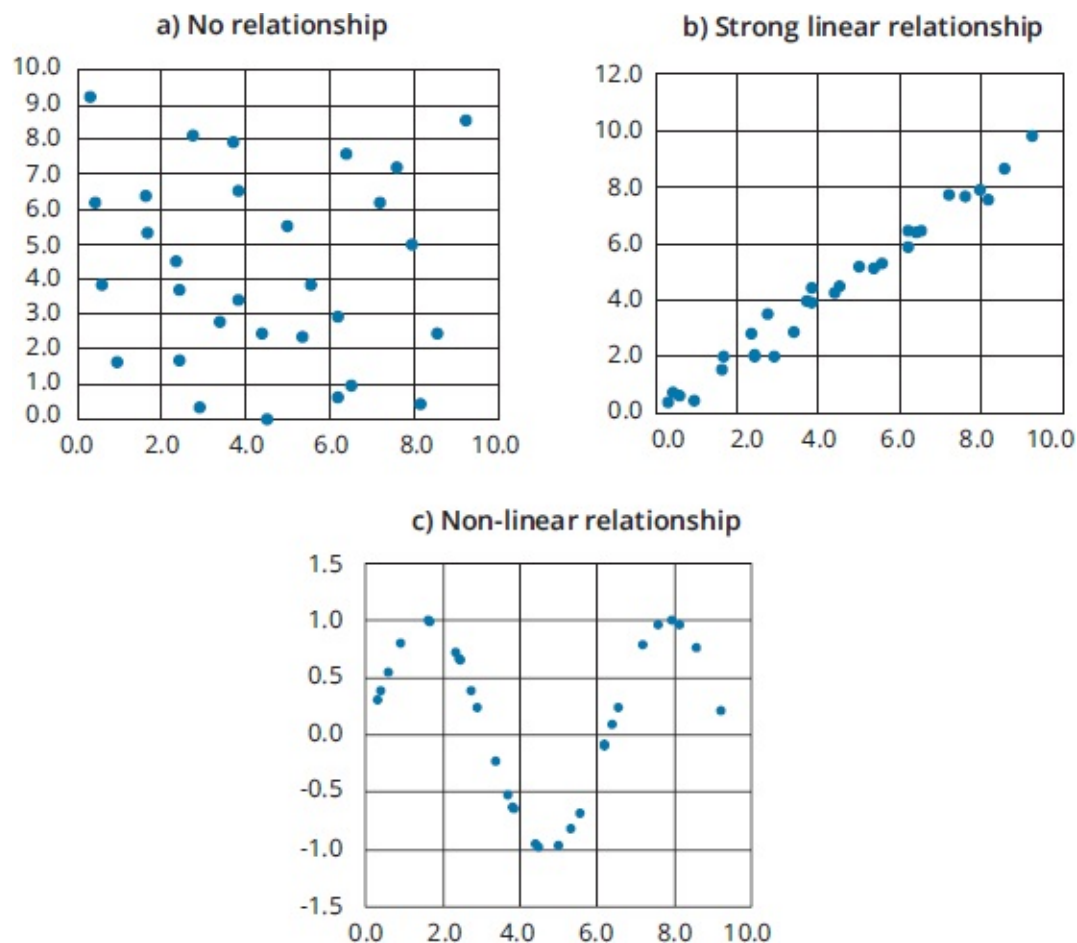
LOS 5.c: Calculate, interpret, and evaluate covariance and correlation.

Scatter plots are a method for displaying the relationship between two variables. With one variable on the vertical axis and the other on the horizontal axis, their paired observations can each be plotted as a single point. For example, in Panel A of Figure 5.4, the point farthest to the upper right shows that when one of the variables (on the horizontal axis) equaled 9.2, the other variable (on the vertical axis) equaled 8.5.

The scatter plot in Panel A is typical of two variables that have no clear relationship. Panel B shows two variables that have a strong linear relationship. As we will see shortly, this implies that these variables exhibit a high correlation coefficient.

A key advantage of creating scatter plots is that they can reveal *nonlinear* relationships, which are not described by the correlation coefficient. Panel C illustrates such a relationship. Although the correlation coefficient for these two variables is close to zero, their scatter plot shows clearly that they are related in a predictable way.

Figure 5.4: Scatter Plots



Covariance is a measure of how two variables move together. Covariance is calculated using the following formula:

$$\sigma_{XY} = \frac{\sum_{i=1}^n (X_i - \mu_X)(Y_i - \mu_Y)}{n}$$

where:

X_i = an observation of variable X

Y_i = an observation of variable Y

μ_X = mean of variable X

μ_Y = mean of variable Y

n = number of periods



PROFESSOR'S NOTE

This formula is population covariance. A sample covariance, s_{XY} , would use deviations from the sample means ($\bar{X}$ and $\bar{Y}$) and a denominator of $n - 1$.

In practice, covariance is difficult to interpret. The value of covariance depends on the units of the variables. For example, the covariance of daily price changes of two securities priced in yen will be much greater than their covariance if the securities are priced in dollars. Like the variance, the units of covariance are the square of the units used for the data.

Additionally, we cannot interpret the relative strength of the relationship between two variables using their covariance. Knowing that the covariance of X and Y is 0.8756 tells us only that they tend to move together, because the covariance is positive.

A standardized measure of the linear relationship between two variables is called the **correlation coefficient**, or simply *correlation*. The correlation between two variables, X and Y , is calculated as follows:

$$\rho_{XY} = \frac{s_{XY}}{s_X s_Y}$$

This implies:

$$s_{XY} = \rho_{XY} s_X s_Y$$

The properties of the correlation of two random variables, X and Y , are summarized here:

- | Correlation measures the strength of the *linear* relationship between two random variables.
- | Correlation has no units; it is a scale-free measure.
- | Possible values of the correlation coefficient range from -1 to $+1$. That is, $-1 \leq \rho_{XY} \leq +1$.
- | If $\rho_{XY} = 1.0$, the random variables have perfect positive correlation. This indicates that a movement in one random variable relative to its mean results in a proportional positive movement in the other relative to its mean.
- | If $\rho_{XY} = -1.0$, the random variables have perfect negative correlation. This indicates that a movement in one random variable relative to its mean results in an exact opposite proportional movement in the other relative to its mean.
- | If $\rho_{XY} = 0$, there is no linear relationship between the variables, indicating that a prediction of Y cannot be made on the basis of X using linear methods.

EXAMPLE: Correlation

The variance of returns on Stock A is 0.0028, the variance of returns on Stock B is 0.0124, and their covariance of returns is 0.0058. Calculate and interpret the correlation of the returns for Stocks A and B.

Answer:

First, it is necessary to convert the variances to standard deviations:

$$s_A = (0.0028)^{1/2} = 0.0529$$

$$s_B = (0.0124)^{1/2} = 0.1114$$

Now, the correlation between the returns of Stock A and Stock B can be computed as follows:

$$\rho_{AB} = \frac{0.0058}{(0.0529)(0.1114)} = 0.9842$$

The fact that this value is close to +1 indicates that the linear relationship is not only positive, but also very strong.

Analysts should take care when drawing conclusions based on correlation. Causation is not implied simply from significant correlation. Even if it were, correlation does not reveal which variable is driving change in the other. It is more prudent to say that two variables exhibit positive (or negative) association, suggesting that the nature of any causal relationship is to be investigated separately or based on theory that can be subject to additional tests.

One question that can be investigated is the role of outliers in the correlation of two variables. If removing the outliers reduces the calculated correlation significantly, further inquiry is necessary into whether the outliers provide information or are caused by noise (randomness) in the data used.

Spurious correlation refers to correlation that is either the result of chance or present due to changes in both variables over time that is caused by their association with a third variable. For example, we can find instances where two variables that are both related to the inflation rate exhibit significant correlation, but for which causation in either direction is not present.

In his book *Spurious Correlation*,¹ Tyler Vigen presents the following examples. The correlation between the age of each year's Miss America and the number of films Nicolas Cage appeared in that year is 87%. This seems a bit random. The correlation between the United States spending on science, space, and

technology and suicides by hanging, strangulation, and suffocation over the 1999–2009 period is 99.87%. It is an impressive correlation, but both variables increased in an approximately linear fashion over the period.

LOS 5.d: Calculate, interpret, and evaluate semi-deviation and coefficient of variation.

When we use variance or standard deviation as risk measures, we calculate risk based on outcomes both above and below the mean. In some situations, it may be more appropriate to consider only outcomes less than the mean (or some other specific value) when calculating a risk measure. In this case, we are measuring **downside risk**.

The **semivariance** is the variance of only those outcomes below the mean, calculated as follows:

$$\sigma_{\text{semi}}^2 = \sum_{i: X_i < \bar{X}} \frac{(X_i - \bar{X})^2}{n}$$

A related measure of downside risk is **target downside deviation**, which is also known as **target semideviation**. Calculating target downside deviation is similar to calculating standard deviation—but in this case, we choose a target value against which to measure each outcome and only include deviations if the outcomes are below that target.

The formula for target downside deviation is stated as follows:

$$s_{\text{target}} = \sqrt{\frac{\sum_{\text{all } X_i < B} (X_i - B)^2}{n - 1}}$$

where B = the target

The denominator remains the sample size n minus one, even though we are not using all the observations in the numerator.

EXAMPLE: Target downside deviation

Calculate the target downside deviation based on the data in our earlier examples for measures of dispersion, for a target return equal to the mean (22%), and for a target return of 24%.

Answer:

Return	Deviation From Mean	Deviation From Target Return
30%	30% – 22% = 8%	30% – 24% = 6%
12%	12% – 22% = –10%	12% – 24% = –12%
25%	25% – 22% = 3%	25% – 24% = 1%
20%	20% – 22% = –2%	20% – 24% = –4%
23%	23% – 22% = 1%	23% – 24% = –1%

$$s_{22\%} = \sqrt{\frac{(-10)^2 + (-2)^2}{5 - 1}} = 5.10\%$$

$$s_{24\%} = \sqrt{\frac{(-12)^2 + (-4)^2 + (-1)^2}{5 - 1}} = 6.34\%$$

A direct comparison between two or more measures of dispersion may be difficult. For instance, suppose you are comparing the annual returns distribution for retail stocks with a mean of 8% and an annual returns distribution for a real estate portfolio with a mean of 16%. Comparing the dispersions of the two distributions directly is not meaningful because of the relatively large difference in their means. To make a meaningful comparison, we must use a relative measure of dispersion. **Relative dispersion** is the amount of variability in a distribution around a reference point or benchmark.

Relative dispersion is commonly measured with the **coefficient of variation (CV)**, which is computed as follows:

$$CV = \frac{s_{\bar{X}}}{\bar{X}} = \frac{\text{standard deviation of } \bar{x}}{\text{average value of } \bar{x}}$$

CV measures the amount of dispersion in a distribution relative to its mean. This is useful because it enables us to compare dispersion across different sets of data. In an investment setting, the CV is used to measure the risk (variability) per unit of expected return (mean). A lower CV is better.

EXAMPLE: Coefficient of variation

During an observed period of time, the mean monthly return on T-bills is 0.25% with a standard deviation of 0.36%, and the mean monthly return for the S&P 500 is 1.09% with a standard deviation of 7.30%. Calculate and interpret the CV for these two investments.

Answer:

$$CV_{\text{T-bills}} = \frac{0.36}{0.25} = 1.44$$

$$CV_{\text{S\&P 500}} = \frac{7.30}{1.09} = 6.70$$

These results indicate that there is less dispersion (risk) per unit of monthly return for T-bills than for the S&P 500 (1.44 vs. 6.70).

MODULE QUIZ 5.3

1. The correlation between two variables is +0.25. The *most appropriate* way to interpret this value is that:
 - A. a scatter plot of the two variables is likely to show a strong linear relationship.
 - B. when one variable is above its mean, the other variable tends to also be above its mean.
 - C. a change in one of the variables usually causes the other variable to change in the same direction.

2. XYZ Corp. Annual Stock Returns

20X1	20X2	20X3	20X4	20X5	20X6
22%	5%	-7%	11%	2%	11%

Assume an investor has a target return of 11% for XYZ stock. What is the stock's target downside deviation?

- A. 9.39%.
 - B. 12.10%.
 - C. 14.80%.
3. An analyst is comparing three investment portfolios.

Portfolio	Average Return (%)	Standard Deviation (%)
A	8	3
B	12	4
C	14	2

Which portfolio has the *best* risk-return tradeoff?

- A. Portfolio A.
- B. Portfolio B.
- C. Portfolio C.

KEY CONCEPTS**LOS 5.a**

The arithmetic mean is the average of observations. The sample mean is the arithmetic mean of sample data.

The median is the midpoint of a dataset when the data is arranged in order.

The mode of a dataset is the value (or values) that occur most frequently. The modal interval is a measure of mode for continuous data.

A trimmed mean omits outliers. A winsorized mean replaces outliers with given values. Both reduce the effect of outliers on the mean.

A quartile divides a distribution into quarters. A percentile divides a distribution into hundredths (percentages). The interquartile range is the difference between the 75th and 25th percentiles; it represents the central 50% of the data.

The range is the difference between the largest and smallest values in a dataset.

Mean absolute deviation is the average of the absolute values of the deviations from the arithmetic mean.

Variance is the mean of the squared deviations from the arithmetic mean.

Standard deviation is the positive square root of the variance, and it is frequently used as a measure of risk.

LOS 5.b

Skewness describes the degree to which a distribution deviates from symmetry. Zero skew indicates symmetry around a central value. Right or positive skew indicates more outcomes in the upper part of the distribution. Left or negative skew indicates the opposite. For a positively skewed, unimodal distribution, the mean is greater than the median, which is greater than the mode. For a negatively skewed, unimodal distribution, the mean is less than the median, which is less than the mode.

Kurtosis measures the peakedness of a distribution and the probability of extreme outcomes (thickness of tails). Excess kurtosis is measured relative to the shape of a normal distribution, which has a kurtosis of 3. Positive excess kurtosis indicates a distribution that has fatter tails than a normal distribution. Negative excess kurtosis indicates thinner tails than a normal distribution.

LOS 5.c

Correlation is a standardized measure of association between two random variables. It is independent of scale and ranges in value from -1 to $+1$.

Scatter plots are useful for revealing nonlinear relationships that are not captured by correlation.

Correlation does not imply that changes in one variable cause changes in the other. Spurious correlation may result from chance, or from two variables being related to a third variable.

LOS 5.d

The coefficient of variation is the ratio of the standard deviation of the sample to its mean and is used to compare the risk-return tradeoffs among distributions of returns.

Semideviation and target downside deviation are measures of downside risk that do not include deviations above a given value.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 5.1

- . A The winsorized mean substitutes a value for some of the largest and smallest observations, so there are 100 observations. The trimmed mean removes some of the largest and smallest observations, so there are less than 100 datapoints. (LOS 5.a)

- . C Arranged in order, this is the dataset:

−5, −1, −1, 2, 3, 3, 3, 4, 6, 6

The mode is the most frequently occurring value, which is 3. With an even number of observations, the median is the average of the two middle observations, which is also 3.

The arithmetic mean is $\frac{-5 + -1 + -1 + 2 + 3 + 3 + 3 + 4 + 6 + 6}{10} = 2.8$.

(LOS 5.a)

Module Quiz 5.2

- . A The sample mean is $(22\% + 5\% + -7\% + 11\% + 2\% + 11\%) / 6 = 7.3\%$.
The sample standard deviation is the square root of the sample variance:

$$s = \sqrt{\frac{(22 - 7.3)^2 + (5 - 7.3)^2 + (-7 - 7.3)^2 + (11 - 7.3)^2 + (2 - 7.3)^2 + (11 - 7.3)^2}{6 - 1}} \\ = \sqrt{96.3} = 9.8\%.$$

(LOS 5.b)

- . **B** Relative to a normal distribution, the negative skew indicates more frequent negative returns. The positive excess kurtosis indicates fat tails (i.e., more extreme observations at both ends of the distribution). (LOS 5.b)

Module Quiz 5.3

- . **B** A correlation of +0.25 indicates a positive linear relationship between the variables; one tends to be above its mean when the other is above its mean. The value 0.25 indicates that the linear relationship is not particularly strong. Correlation does not imply causation. (LOS 5.c)

- . **A** Deviations from the target return:

$$22\% - 11\% = 11\%$$

$$5\% - 11\% = -6\%$$

$$-7\% - 11\% = -18\%$$

$$11\% - 11\% = 0\%$$

$$2\% - 11\% = -9\%$$

$$11\% - 11\% = 0\%$$

Target downside deviation =

$$\sqrt{\frac{(-6)^2 + (-18)^2 + (-9)^2}{6 - 1}} = \sqrt{88.2} = 9.39\%.$$

(LOS 5.d)

- . **C** Portfolio C has the lowest coefficient of variation—that is, the least risk per unit of return, indicating the preferred risk-return tradeoff.

$$\text{Portfolio A: } CV = \frac{3}{8} = 0.375.$$

$$\text{Portfolio B: } CV = \frac{4}{12} = 0.333.$$

$$\text{Portfolio C: } CV = \frac{2}{14} = 0.143.$$

(LOS 5.d)

¹ “Spurious Correlations,” Tyler Vigen, www.tylervigen.com

READING 6

STATISTICAL DISTRIBUTIONS FOR FINANCIAL ASSET PRICES AND RETURNS

MODULE 6.1: PROBABILITY DISTRIBUTIONS AND EXPECTED VALUES

LOS 6.a: Calculate, interpret, and evaluate unconditional expected values for mean, variance, and covariance.

A **random variable** is a variable with an uncertain outcome, also referred to as a **stochastic variable**. If the number of possible outcomes is countable, the variable is referred to as a **discrete random variable**. An example of a discrete random variable is the number of days it will rain in a month, because the possible outcomes are countable, ranging from zero to the number of days in the month. A random variable that has an infinite number of possible outcomes, such as the monthly percentage change in a stock index, is referred to as a **continuous random variable**.

A **probability distribution** describes the probabilities of all the possible outcomes for a random variable. The probabilities of all possible outcomes must sum to 1. A simple discrete probability distribution is the outcome of the roll of one fair die. This has six possible outcomes, and each has a probability of one-sixth, so their probabilities sum to 1.

A **probability function**, denoted $p(x)$, specifies the probability that a random variable is equal to a specific value. More formally, $p(x)$ is the probability that random variable X takes on the value x , or $p(x) = P(X = x)$. We refer to a probability function for a discrete random variable as a **probability mass function (PMF)**, and to a probability function for a continuous random variable as a **probability density function (PDF)**.

A probability function has two key properties:

Each outcome has a probability between zero and one inclusive, or $0 \leq p(x) \leq 1$.

The sum of the probabilities for all possible outcomes equals 1, or $\sum p(x) = 1$.

EXAMPLE: Evaluating a probability function

Consider the following function: $X = \{1, 2, 3, 4\}$, $p(x) = \frac{x}{10}$, else $p(x) = 0$

Determine whether this function satisfies the conditions for a probability function.

Answer:

All the probabilities ($1/10$, $2/10$, $3/10$, and $4/10$) are between 0 and 1, and the sum of all the probabilities ($1/10 + 2/10 + 3/10 + 4/10$) equals 1, so both conditions for a probability function are satisfied.

For a *discrete* distribution, $p(x) = 0$ when x cannot occur, or $p(x) > 0$ if it can. Recall that $p(x)$ is read as “the probability that random variable $X = x$.” For example, the probability of it raining on 33 days in June is zero because June has only 30 days, but the probability of it raining on 25 days in June has some positive value.

For a *continuous* distribution, $p(x) = 0$, even though x can be the outcome, because the number of possible outcomes is infinite. We can only meaningfully consider the probability that an outcome will fall within a range, or $P(x_1 \leq X \leq x_2)$, where x_1 and x_2 are actual numbers. For example, the probability of receiving exactly two inches of rain in June is effectively zero, because two inches is a single point in an infinite range of possible values. However, the probability of the amount of rain being between 1.99 and 2.01 inches has some positive value. In the case of continuous distributions, $P(x_1 \leq X \leq x_2)$ is equal to $P(x_1 < X < x_2)$, because $p(x_1) = p(x_2) = 0$.

In finance, we treat some discrete distributions as though they are continuous because the number of possible outcomes is very large. For example, the increase or decrease in the price of a stock traded on an American exchange is recorded in dollars and cents. Analysts generally do not believe their pricing models are precise enough to assign a probability to an outcome such as a change of exactly \$1.47. It is customary, therefore, to speak in terms of the probability of a range of possible price changes—say, between \$1.00 and \$2.00.

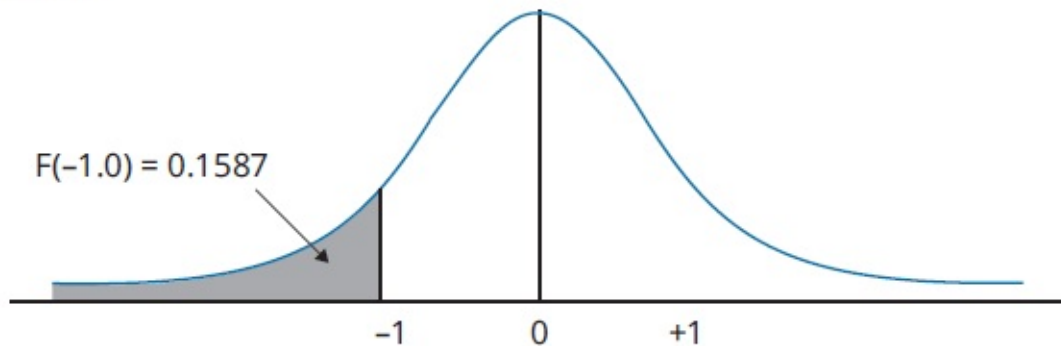
A **cumulative distribution function (CDF)**, or simply **distribution function**, defines the probability that a random variable X takes on a value less than or equal to a specific value, x . It represents the sum, or cumulative value, of the probabilities for the outcomes up to and including a specified outcome. The CDF for a random variable, X , may be expressed as $F(x) = P(X \leq x)$.

Consider the probability function we defined in our earlier example, $X = \{1, 2, 3, 4\}$, $p(x) = x / 10$. For this distribution, $F(3) = 0.1 + 0.2 + 0.3 = 0.6$, and $F(4) = 0.1 + 0.2 + 0.3 + 0.4 = 1$. This means that $F(3)$ is the cumulative probability that outcomes 1, 2, or 3 occur, and $F(4)$ is the cumulative probability that one of the four possible outcomes occurs.

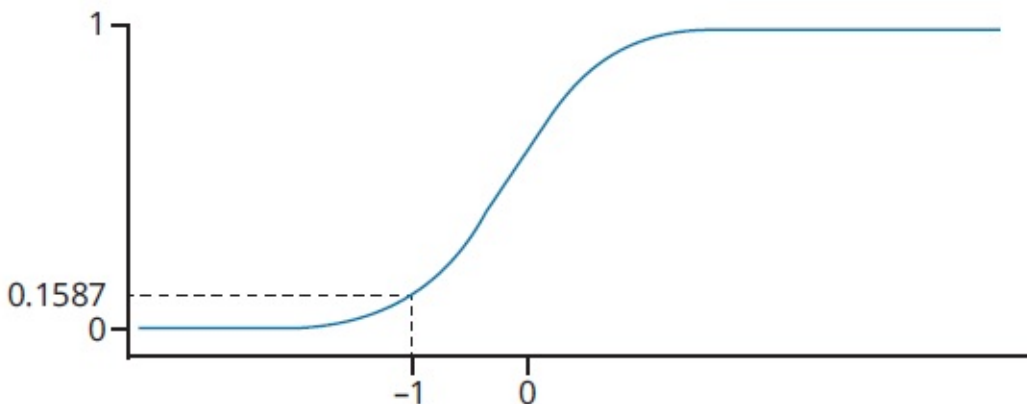
Figure 6.1 shows an example of a CDF (for a *standard normal distribution*, which we will describe later in this reading). It has a 15.87% probability of a value less than -1 . This is the total area to the left of -1 in the PDF in Panel (a), and the y-axis value of the CDF for a value of -1 in Panel (b).

Figure 6.1: Standard Normal Probability Density and Cumulative Distribution Functions

(a) PDF



(b) CDF



EXAMPLE: Cumulative distribution function

Cermak Company's return on equity is a continuous distribution over the range from -20% to +30% and has a CDF of $F(x) = (x + 20) / 50$. Calculate the probability that its return on equity will be between 0% and 15%.

Answer:

To determine the probability of an outcome between 0% and 15%, we can first calculate the probability that return on equity will be less than or equal to 15%, or $F(15)$, and then from that, subtract the probability that ROE will be less than zero, or $F(0)$:

$$P(0 \leq x \leq 15) = F(15) - F(0)$$

$$F(15) = (15 + 20) / 50 = 0.70$$

$$F(0) = (0 + 20) / 50 = 0.40$$

$$F(15) - F(0) = 0.70 - 0.40 = 0.30 = 30\%$$

An **unconditional mean**, **unconditional expectation**, or **expected value** is the probability-weighted average of a random variable's possible outcomes. For a discrete random variable X , where x_i is a possible value of X and $P(x_i)$ is its probability, we can express the expected value as follows:

$$\mu_X = E(X) = \sum_i [x_i \times P(x_i)]$$

For continuous variables, the expected value is expressed mathematically as follows:

$$\mu_X = E[X] = \int_{-\infty}^{\infty} x \times f_X(x) dx$$

where $f_X(x)$ is the PDF of X .



PROFESSOR'S NOTE

This notation comes from calculus, which candidates are not required to perform on the CFA exams.

An **unconditional variance** is the squared spread of a random variable around its expected value, and an **unconditional standard deviation** is the square root of unconditional variance:

$$\begin{aligned}\sigma_X^2 &= \text{Var}(X) = E[X - E(X)]^2 \\ &= E(X^2) - [E(X)]^2\end{aligned}$$

**PROFESSOR'S NOTE**

Unconditional mean and variance from a probability model can be computed on the BA II+ calculator. We illustrate the steps in our Resource Library videos on calculator functions.

For two random variables X and Y , their **unconditional covariance** measures the strength and direction of the linear relationship between them. It is expressed as follows:

$$\begin{aligned}\sigma_{XY} &= \text{Cov}(X, Y) = E\{[X - E(X)] \times [Y - E(Y)]\} \\ &= E(XY) - E(X)E(Y)\end{aligned}$$

EXAMPLE: Unconditional mean, variance, and covariance

An analyst estimates the following probabilities for the returns of two securities:

Probability	0.3	0.5	0.2
Security P return	−4%	2%	5%
Security Q return	−2%	1%	3%

Calculate the unconditional means, variances, and standard deviations of each security, and their unconditional covariance.

Answer:

Unconditional mean return of Security P = $E(P) = 0.3(-4\%) + 0.5(2\%) + 0.2(5\%) = 0.8\%$

Unconditional mean return of Security Q = $E(Q) = 0.3(-2\%) + 0.5(1\%) + 0.2(3\%) = 0.5\%$

Unconditional variance of Security P = $E(P^2) - [E(P)]^2$
 $= 0.3(-4^2) + 0.5(2^2) + 0.2(5^2) - 0.8^2 = 11.16$

Alternatively: $\text{Var}(P) = E[P - E(P)]^2$
 $= 0.3(-4.0 - 0.8)^2 + 0.5(2.0 - 0.8)^2 + 0.2(5.0 - 0.8)^2 = 11.16$

Unconditional variance of Security Q = $E(Q^2) - [E(Q)]^2$
 $= 0.3(-2^2) + 0.5(1^2) + 0.2(3^2) - 0.5^2 = 3.25$

Alternatively: $\text{Var}(Q) = E[Q - E(Q)]^2$
 $= 0.3(-2.0 - 0.5)^2 + 0.5(1.0 - 0.5)^2 + 0.2(3.0 - 0.5)^2 = 3.25$

The units for variance are *not* percentages because we squared the percentages we used as inputs.

Unconditional standard deviation of Security P = $\sqrt{11.16} = 3.34\%$

Unconditional standard deviation of Security Q = $\sqrt{3.25} = 1.80\%$

Converting the variances to standard deviations changes the units back to percentages.

Unconditional covariance of Securities P and Q = $E(PQ) - E(P)E(Q)$

$$E(PQ) = 0.3(-4)(-2) + 0.5(2)(1) + 0.2(5)(3) = 6.4$$

$$E(PQ) - E(P)E(Q) = 6.4 - (0.8)(0.5) = 6.0$$

Alternatively: $\text{Cov}(P, Q) = E\{[P - E(P)] \times [Q - E(Q)]\}$
 $= 0.3(-4.0 - 0.8)(-2.0 - 0.5) + 0.5(2.0 - 0.8)$
 $(1.0 - 0.5) + 0.2(5.0 - 0.8)(3.0 - 0.5)$
 $= 6.0$

The positive covariance implies that the returns of these two securities tend to change in the same direction. As with the variance, the units of covariance are not percentages because we multiplied two percentages together.



MODULE QUIZ 6.1

1. The expected value of a single throw using a fair six-sided die is:
A. 3.0.

- B. 3.5.
C. 4.0.
2. A brokerage firm rates the stocks in an index of 500 stocks using a rating system from 1 (best) to 5 (worst). The number of stocks with each rating is presented as follows:

Rating	Number of Stocks
1(best)	47
2	102
3	220
4	107
5 (worst)	24

The probability that a randomly selected stock chosen at random has a rating of 3 or better is *closest* to:

- A. 26%.
B. 30%.
C. 74%.
3. The following are probabilities for the returns of an investment:

Rate of return	-5%	-2%	3%	6%
Probability	0.10	0.25	0.50	0.15

The unconditional standard deviation of the investment's returns is *closest* to:

- A. 3.4%.
B. 7.4%.
C. 11.4%.

MODULE 6.2: DISCRETE AND CONTINUOUS PROBABILITY DISTRIBUTIONS

LOS 6.b: Calculate, interpret, and evaluate the principal moments of key statistical distributions used in finance.



PROFESSOR'S NOTE

We do not believe candidates are (or should be) required to memorize or dive deeply into the formulas for the principal moments of all these distributions. While we reproduce these formulas for reference, our text describing the individual distributions will indicate those features and moments we think are important for understanding their characteristics.

Discrete Distributions

Figure 6.2 lists the mathematical expressions for the four key moments of three kinds of discrete distributions: a discrete uniform distribution, a binomial distribution, and a Poisson distribution.

Figure 6.2: Moments for Discrete Statistical Distributions

	Mean	Variance	Skewness	Excess Kurtosis	Where:
Discrete uniform	$\frac{a+b}{2}$	$\frac{(b-a+1)^2-1}{12}$	Zero	$-\frac{6(n^2+1)}{5(n^2-1)}$	n = number of outcomes a = minimum outcome b = maximum outcome
Binomial	np	np(1-p)	$\frac{1-2p}{\sqrt{np(1-p)}}$	$\frac{1-6p(1-p)}{np(1-p)}$	n = number of trials p = probability of success (1-p) = probability of failure
Poisson	λ	λ	$\frac{1}{\sqrt{\lambda}}$	$\frac{1}{\lambda}$	λ = constant rate of occurrence

Discrete Uniform Distribution

A **discrete uniform distribution** is one for which the probabilities for all possible outcomes are equal. It has n possible outcomes ranging from a to b , and the probability of each outcome is $1/n$.

For example, consider a discrete uniform probability distribution defined as $X = \{1, 2, 3, 4, 5\}$, $p(x) = 0.2$. The expected value of X is $(a+b)/2 = 3$, which is also the mean of the equally likely possible outcomes.

The variance of a discrete uniform distribution is given by:

$$\sigma^2 = \frac{(b-a+1)^2-1}{12}$$

For the discrete uniform distribution $X = \{1, 2, 3, 4, 5\}$, the variance equals:

$$\sigma^2 = \frac{(5-1+1)^2-1}{12} = \frac{24}{12} = 2$$

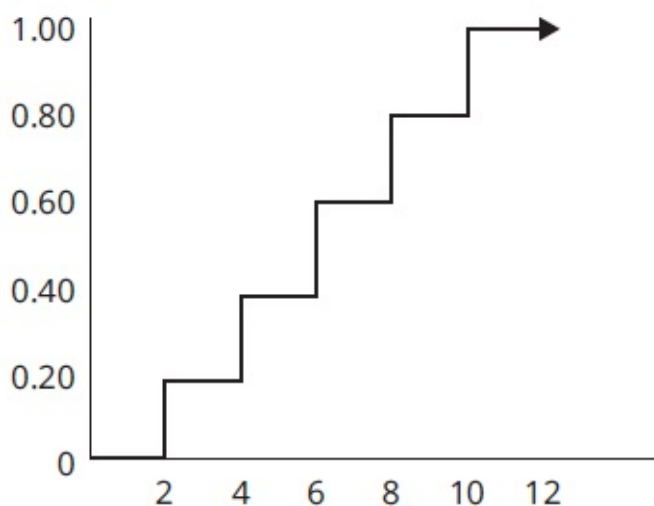
The CDF for the n th outcome, $F(x_n)$, is $np(x)$. The probability for a range of outcomes is $kp(x)$, where k is the number of possible outcomes in the range. Figure 6.3 illustrates the CDF for this discrete uniform variable.

Figure 6.3: Probability and CDFs

$X = x$	Probability of x Prob ($X = x$)	CDF Prob ($X < x$)
1	0.20	0.20
2	0.20	0.40
3	0.20	0.60
4	0.20	0.80
5	0.20	1.00

Cumulative Distribution Function for $X \sim \text{Uniform } \{2, 4, 6, 8, 10\}$

Prob($X \leq x$)



Binomial Distribution

A **binomial distribution** is also a discrete distribution. The random variable is the number of “successes” from a given number of trials, for which the outcome of each trial can be either “success” or “failure.” The probability of success, p , is constant for each trial, and the trials are independent (i.e., success or failure in the next trial is not affected by the outcomes of previous trials). A binomial random variable for which the number of trials is 1 is called a **Bernoulli distribution**.

For a given series of n trials, the expected number of successes, or $E(X)$, is given by this formula:

$$E(X) = np$$

The intuition is straightforward. If we perform n trials and the probability of success on each trial is p , we expect np successes.

The variance of a binomial random variable is given by this formula:

variance of $X = np(1 - p)$

EXAMPLE: Expected value of a binomial random variable

Based on empirical data, an analyst estimates that the probability of the Dow Jones Industrial Average (DJIA) increasing on any given day is 0.67.

Assuming the only other outcome is that it decreases, we can state $p(\text{Up}) = 0.67$ and $p(\text{Down}) = 0.33$. Further, assume that movements in the DJIA are independent (i.e., an increase in one day is independent of what happened on another day).

Calculate the expected value and the variance of the number of up days in a five-day period.

Answer:

Using binomial terminology, we define success as Up, so $p = 0.67$:

$$E(X \mid n = 5, p = 0.67) = (5)(0.67) = 3.35$$

Because the binomial distribution is a discrete distribution, the result $X = 3.35$ is not a possible outcome. However, if we were to record the results of many five-day periods, the average number of up days (successes) would converge to 3.35 as the number of 5-day periods gets large.

The variance of the random variable is $5(0.67)(1 - 0.67) = 1.1055$

Binomial random variables can be used to model possible stock prices over multiple periods. In the binomial model, the price of the stock can move up or down at each time step. The stock price at each node is calculated as follows:

$$S_{i,j} = S_0(u^i)(d^j)$$

where:

S_0 = initial stock price

i = number of up movements

j = number of down movements

n = number of movements, or $i + j$

u = the model's up factor (e.g., $u = 1.05$ is a 5% increase)

d = the model's down factor (e.g., $d = 0.95$ is a 5% decrease)

EXAMPLE: Binomial model for a stock price

A binomial model for the price of Orland Company stock has a monthly up factor of 1.02 and a monthly down factor of $1 / 1.02 = 0.9804$. The stock's price today is \$10.00. According to this model, what will the stock price be after two up months and three down months?

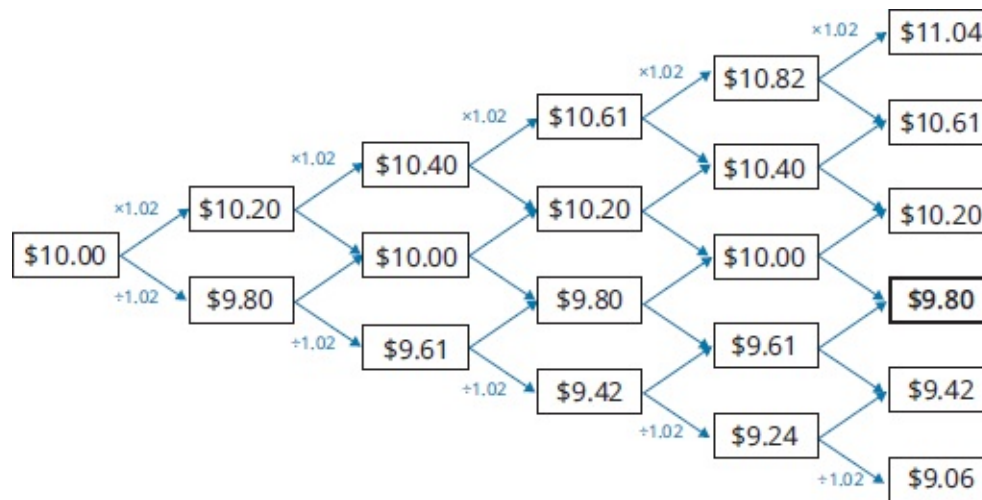
Answer:

$$S_{2,3} = \$10.00(1.02^2)(0.9804^3) = \$9.80$$

We can illustrate a binomial model with a **probability tree** or **decision tree**. Figure 6.4 shows a probability tree for the stock price model in our previous example. Because this model defined the up factor and the down factor as reciprocals, an up move followed by a down move or a down move followed by an up move results in no change in the price—and two up moves and three down moves result in a price of \$9.80, regardless of the order in which the moves occur.

In our example, we did not assign probabilities to the price paths. An analyst can design a binomial model to exhibit various probability functions by choosing different probabilities of success.

Figure 6.4: Probability Tree



Poisson Distribution

The **Poisson distribution** is often used to model the number of times a specific event is expected to occur within a given interval of time. It is assumed that the events happen independently at a known constant rate, which we denote as λ . The probability of observing k events in a time interval t is stated as follows:

$$P(k \text{ events in interval } t) = \frac{(\lambda t)^k}{k!} e^{-\lambda t}$$

where:

e = the base of the natural logarithm, or 2.71828 . . .

$k!$ = k factorial = $k \times (k - 1) \times (k - 2) \times \dots \times 1$

For a single unit of time ($t = 1$), the mean and variance of a Poisson distribution are both equal to λ . It has skewness of $\frac{1}{\sqrt{\lambda}}$ and excess kurtosis of

1. The average rate of occurrence λ = number of bonds $\times$ probability

$$\lambda = 800 \times 0.005 = 4$$

$$P(k \text{ events in interval } t) = \frac{(\lambda t)^k}{k!} e^{-\lambda t}$$

The probability of *no* default ($k = 0$) is $\frac{(\lambda t)^0}{0!} e^{-\lambda t}$. Because any value to the power of zero equals one, and zero factorial also equals one, this reduces to simply $e^{-\lambda t}$. Therefore, the probability of default, which is one minus the probability of no default, must equal $1 - e^{-\lambda t}$.

The probability of default in one year ($t = 1$) = $1 - e^{-4} = 0.982$, or 98.2%.

2. The probability of exactly three defaults in a year ($k = 3$, $t = 1$)

$$= P(X = k = 3) = \frac{\lambda^k}{k!} e^{-\lambda} = \frac{4^3}{3!} e^{-4} = 0.195, \text{ or } 19.5\%.$$

Analysts use the Poisson distribution to model how frequently a rare event can be expected to occur. The Poisson distribution can be thought of as the mathematical limit of the binomial distribution, as the number of trials n approaches infinity and the probability p of an event approaches zero.

EXAMPLE: Using a Poisson distribution to predict bond default

A portfolio contains 800 bonds. The probability of any single bond defaulting within one year is 0.5%. Calculate (1) the probability of a default occurring within one year, and (2) the probability of exactly three bonds defaulting within a year.

Answer:

1. The average rate of occurrence $\lambda = \text{number of bonds} \times \text{probability}$

$$\lambda = 800 \times 0.005 = 4$$

$$P(k \text{ events in interval } t) = \frac{(\lambda t)^k}{k!} e^{-\lambda t}$$

The probability of *no* default ($k = 0$) is $\frac{(\lambda t)^0}{0!} e^{-\lambda t}$. Because any value to the power of zero equals one, and zero factorial also equals one, this reduces to simply $e^{-\lambda t}$. Therefore, the probability of default, which is one minus the probability of no default, must equal $1 - e^{-\lambda t}$.

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$$= P(X = k = 3) = \frac{\lambda^k}{k!} e^{-\lambda} = \frac{4^3}{3!} e^{-4} = 0.195, \text{ or } 19.5\%.$$

Continuous Distributions

Figure 6.5 lists the mathematical expressions for the key moments of four kinds of continuous distributions: a continuous uniform distribution, a normal distribution, a log-normal distribution, and a logistic distribution.

Figure 6.5: Moments for Continuous Statistical Distributions

	Mean	Variance	Skewness	Excess Kurtosis	Where:
Continuous uniform	$\frac{a + b}{2}$	$\frac{(b - a)^2}{12}$	Zero	-1.2	$a = \text{minimum outcome}$ $b = \text{maximum outcome}$
Normal	μ	σ^2	Zero	Zero	
Log-normal	$e^{\mu + \frac{\sigma^2}{2}}$	$(e^{\sigma^2} - 1)e^{2\mu + \sigma^2}$	$\sqrt{-1 + e^{\sigma^2}}(2 + e^{\sigma^2})$	$-3 + 3e^{2\sigma^2} + 2e^{3\sigma^2} + e^{4\sigma^2}$	
Logistic	μ	$\frac{s^2 \pi^2}{3}$	Zero	1.2	$s = \text{scale parameter}$

Continuous Uniform Distribution

A **continuous uniform distribution** is defined over a range that spans between some lower limit, a , and some upper limit, b . Outcomes can only occur between a and b , and because the distribution is continuous, even within the range from a to b , the probability of any single outcome is zero. We can only speak meaningfully of the probability that an outcome will fall within a range.

Formally, the probability of outcomes between x_1 and x_2 (assuming x_1 and x_2 fall between the boundaries a and b) equals $\frac{x_2 - x_1}{b - a}$.

Do not miss how simple this is just because the notation is mathematical. For a continuous uniform distribution, the probability of outcomes in a range that is one-half the whole range is 50%. The probability of outcomes in a range that is one-quarter of the possible range is 25%.

EXAMPLE: Continuous uniform distribution

X is uniformly distributed between 2 and 12. Calculate the probability that X will be between 4 and 8.

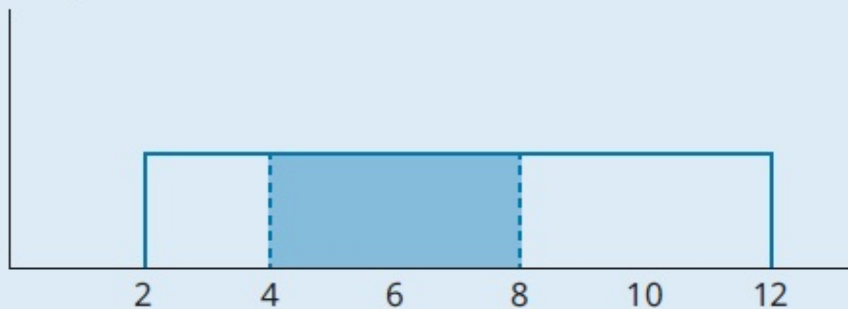
Answer:

$$\frac{8 - 4}{12 - 2} = \frac{4}{10} = 40\%$$

Figure 6.6 illustrates this continuous uniform distribution. Note that the area bounded by 4 and 8 is 40% of the total probability between 2 and 12 (which is 100%).

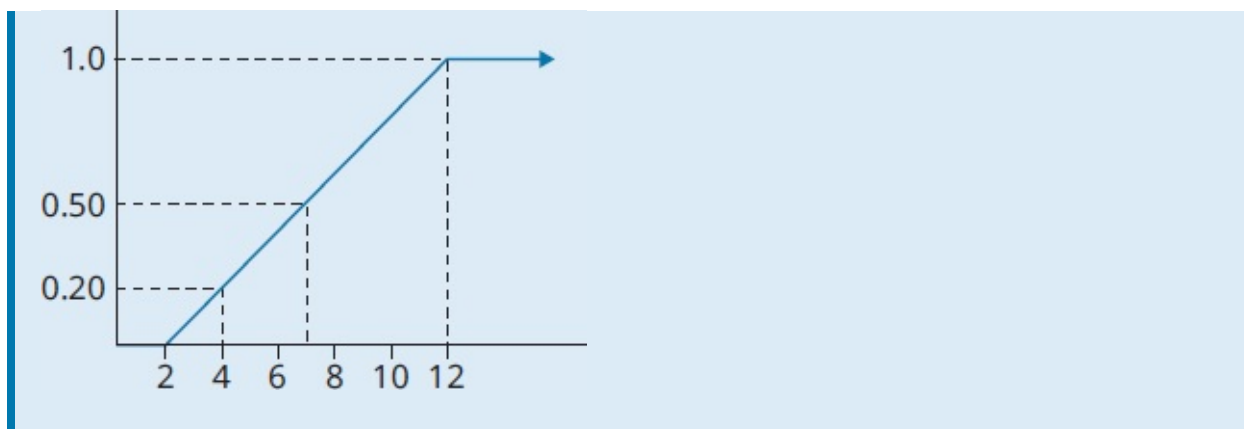
Figure 6.6: Continuous Uniform Distribution

Probability



Because outcomes are equal over equal-size possible intervals, the CDF is linear over the variable's range. The CDF for the distribution in the example, $\text{Prob}(X < x)$, is shown in Figure 6.7.

Figure 6.7: CDF for a Continuous Uniform Variable



The mean of a continuous uniform distribution is its midpoint, $\frac{b-a}{2}$. Its variance, $\frac{(b-a)^2}{12}$, is proportional to the squared width of the interval from a to b . Its skew is zero (the distribution is symmetric), and its kurtosis is 1.8 (therefore, excess kurtosis is $1.8 - 3 = -1.2$).

Uniform distributions are used for large-scale **Monte Carlo simulation**, a computational method that uses repeated random sampling to produce a distribution of the outcomes from a complex process, such as the value of an option based on multiple stochastic inputs. We can then use the distribution of simulated outcomes to calculate the expected value of the option as the mean of the many outcomes, and the expected variance of the option value as the variance of the many outcomes.



PROFESSOR'S NOTE

We will discuss Monte Carlo simulation in our reading on the simulation of financial asset prices and returns.

Normal Distribution

The well-known **normal distribution** (also known as a **Gaussian distribution**) is a symmetrical, bell-shaped curve. The bell shape indicates that outcomes are more likely to occur closer to the mean.

A normal distribution is defined completely by its mean and standard deviation, expressed as $N(\mu, \sigma)$, because its skewness is zero and its excess kurtosis is zero (i.e., it exhibits kurtosis of 3).

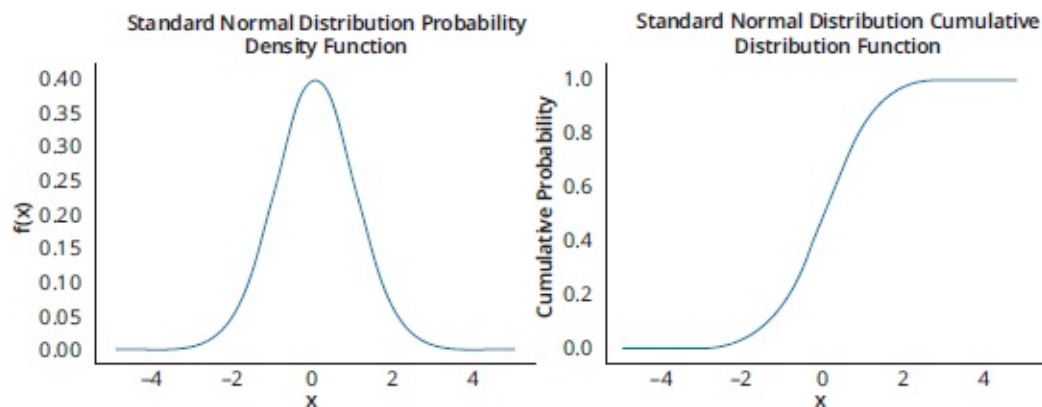
The normal distribution is important for many reasons. Many of the random variables that are relevant to finance and other professional disciplines follow a

normal distribution. In the area of investment and portfolio management, the normal distribution plays a central role in portfolio theory.

Many independent random variables, or sums of them, tend toward a normal distribution, even if the original variables themselves are not normally distributed. Binomial distributions converge to normal distributions as the number of trials, n , gets large.

A normal distribution with a mean of zero and a variance (and standard deviation) of 1, or $N(0,1)$, is termed a **standard normal distribution**. Figure 6.8 illustrates the PDF and CDF of a standard normal distribution, where the horizontal axes represent numbers of standard deviations from the mean.

Figure 6.8: Standard Normal Distribution¹



In subsequent readings in statistical analysis, we will use two facts about a normal distribution to draw statistical inferences:

Approximately 68% of the outcomes are within one standard deviation of its mean.

Approximately 95% of the outcomes are within two standard deviations of its mean.

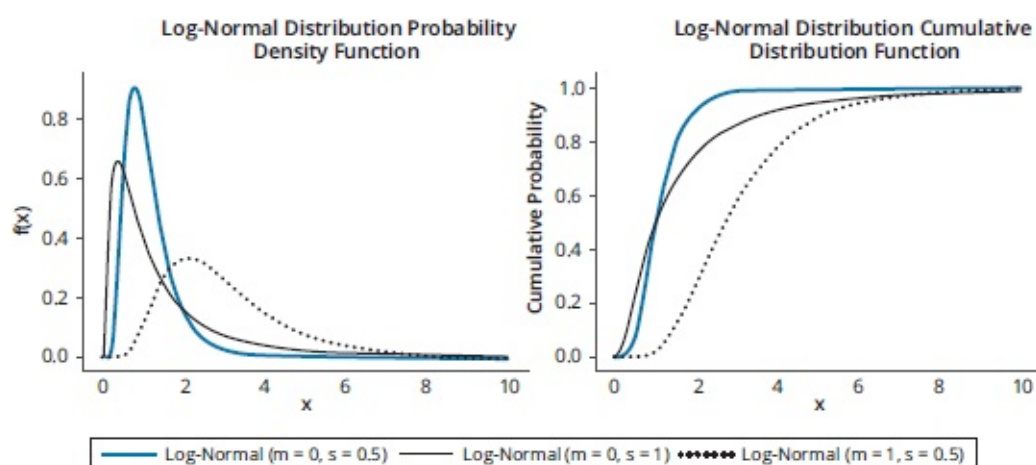
Log-Normal Distribution

A **log-normal distribution** is used to model data where only positive values can occur, and observations may differ by several degrees of magnitude. It is the probability distribution of a random variable where the natural logarithm ($\ln$) of the variable is normally distributed.

A log-normal distribution has outcomes equal to e^x , where x follows a normal distribution and e is the base of the natural logarithm—a constant approximately equal to 2.7183.

Because the value e^x cannot be negative (e to the power negative infinity is zero), a log-normal variable cannot produce outcomes less than zero. This makes a log-normal distribution well suited to model asset prices, which do not take on negative values. The positive values are not bounded, so the right-hand tail is long, and the distribution is right-skewed. Figure 6.9 illustrates these characteristics for three log-normal distributions.

Figure 6.9: Log-Normal Distributions²



PROFESSOR'S NOTE

A log-normal distribution is described by two parameters, mean and standard deviation, referred to as the location parameter and scale parameter. This is not the standard deviation of the x variable, but the standard deviation of the variable once it has been transformed into a normal distribution—in other words, $\ln(x)$. In Figure 6.9, the location parameter and scale parameter are given by m and s , respectively.

The mean and variance of a log-normal random variable Y , where $Y = e^X$, are expressed mathematically as follows:

$$\mu_Y = e^{\mu_X + \frac{\sigma_X^2}{2}}$$

$$\sigma_Y^2 = (e^{\sigma_X^2} - 1) e^{2\mu_X + \sigma_X^2}$$

We can use logarithmic annual returns, $\ln(P_1 / P_0)$, to predict future stock prices with the advantage that they will not create negative stock prices. The “drift

term” in the following formula represents the expected growth rate adjusted for volatility in a geometric Brownian motion model:

$$\ln(S_T) = \ln(S_0) + T\left(\mu - \frac{\sigma^2}{2}\right),$$

where:

$$T\left(\mu - \frac{\sigma^2}{2}\right) = \text{drift term}$$

EXAMPLE: Modeling an asset price with a log-normal distribution

Assume a stock’s price follows a log-normal distribution, has a price today of £10, and has a mean logarithmic annual return of 6% with a standard deviation of 12%. Calculate the expected stock price in three years.

Answer:

$$\ln(S_T) = \ln(S_0) + T\left(\mu - \frac{\sigma^2}{2}\right)$$

$$\ln(S_3) = \ln(10) + 3\left(0.06 - \frac{0.12^2}{2}\right)$$

$$\ln(S_3) = 2.3026 + 3(0.0528) = 2.4610$$

$$S_3 = e^{2.4610} = \text{£}11.72$$

Logistic Distribution

Another distribution that is useful for financial analysis is the **logistic distribution**. A logistic distribution resembles a normal distribution in shape, but it has fatter tails. Like the normal distribution, the logistic distribution is centered on its mean. A logistic distribution has a scale parameter s , that affects only its variance, which equals $\frac{s^2\pi^2}{3}$.

A logistic distribution is bell-shaped and symmetric like a normal distribution, so its skew is zero. Because it has fatter tails than a normal distribution, it has positive excess kurtosis, which is equal to 1.2.

In financial analysis, the logistic distribution is employed in logistic regression, in which the predicted (dependent) variable is binary (often 0 and 1, for “will not happen” and “will happen”) and is determined by one or more independent

variables. The logistic distribution is also used in machine learning and AI applications.



MODULE QUIZ 6.2

1. A trader makes 10 trades a day. The probability that each trade will not be profitable is 48%, and the outcomes of the trades follow a binomial distribution. The average number of profitable trades in a five-day work week is:
 - A. 20.
 - B. 24.
 - C. 26.
2. A trader makes 20 trades a day. The probability that each trade will be profitable is 52%, and the outcomes of the trades follow a binomial distribution. The variance of the number of profitable trades each day is *closest* to:
 - A. 2.24.
 - B. 5.00.
 - C. 5.42.
3. When the number of impactful news items arriving each day affecting a stock's value is modeled using a Poisson distribution with $\lambda = 3$, the average number of news items that arrive in a day and the variance of the number of news arrivals are:
 - A. 0.33 and 3.
 - B. 3 and 3.
 - C. 3 and 9.
4. For a normal distribution, it is *most likely* that:
 - A. excess kurtosis equals 3.
 - B. exactly half the outcomes are less than the mean.
 - C. we must know its skewness to fully define the distribution.
5. A logistic distribution has excess kurtosis:
 - A. less than zero.
 - B. equal to zero.
 - C. greater than zero.

MODULE 6.3: CONDITIONAL EXPECTATIONS AND BAYESIAN UPDATING

LOS 6.c: Calculate, interpret, and evaluate conditional expectations, variances, and covariances.

A **joint probability distribution** describes the probabilities for multiple variables and, importantly, how they vary over time relative to each other. A **bivariate distribution** is for two variables; it requires information on their

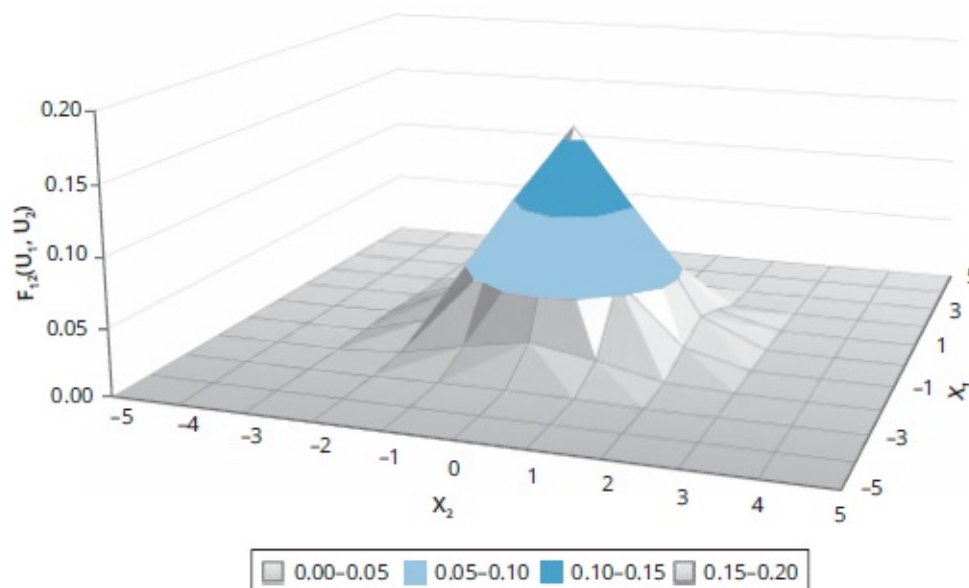
means and variances and how they are distributed relative to each other, by either their covariance or their correlation.

Specifically, a joint bivariate CDF, $F_{12}(x_1, x_2)$, quantifies the joint probability of outcomes in which the random variable X_1 does not exceed a value x_1 , and the random variable X_2 does not exceed a value x_2 . Formally, we can describe a joint CDF for the two variables as:

$$F_{12}(x_1, x_2) = P(X_1 \leq x_1, X_2 \leq x_2)$$

Figure 6.10 illustrates such a joint probability function where the height of the plot is the probability of joint outcomes of X_1 and X_2 . For continuous random variables, the height is proportional to the probability in a small range of outcomes around the specific values of X_1 and X_2 .

Figure 6.10: Joint PDF³



For discrete random variables, instead of using a three-dimensional graph, we can represent a joint probability distribution with a **contingency table** that summarizes their relationship with the frequencies at which certain combinations of the random variables occur. Figure 6.11 illustrates such a table for the stocks in an index, with ranges of returns outcomes for two categories of firms.

Figure 6.11: Contingency Table for Discrete Variables

Rate of Return	Firm Type		Total
	Financial	Nonfinancial	
Greater than 8%	120	140	260
Less than or equal to 8%	180	60	240
Total	300	200	500

The sums in the bottom row are the marginal distributions by firm type, and those in the right-most column are marginal distributions by rate of return. The probability that a randomly selected firm in the index is a financial firm is $300 / 500 = 60\%$. The probability that a firm in the index will have returns less than or equal to 8% is $240 / 500 = 48\%$. The probability that a randomly selected firm is nonfinancial and has returns greater than 8% is $140 / 500 = 28\%$.

A **conditional mean** is the expected value of a random variable given that a certain condition is true. In general, we can express this for random variable X as $E(X|S)$, which we read as the expected value of X given that the condition S is met.

For a discrete random variable X , the conditional mean is the sum of the possible values of X times their conditional probabilities.

$$E(X|S) = \sum_{i=1}^n X_i \times P(X_i|S)$$

The **conditional variance** measures the variation of a random variable around its conditional mean:

$$\text{Var}(X|S) = E\{[X - E(X|S)]^2\}$$

The **conditional covariance** measures how two variables move together if a condition S is met:

$$\text{Cov}(X|S, Y|S) = E\{[X - E(X|S)] \times [Y - E(Y|S)]\}$$



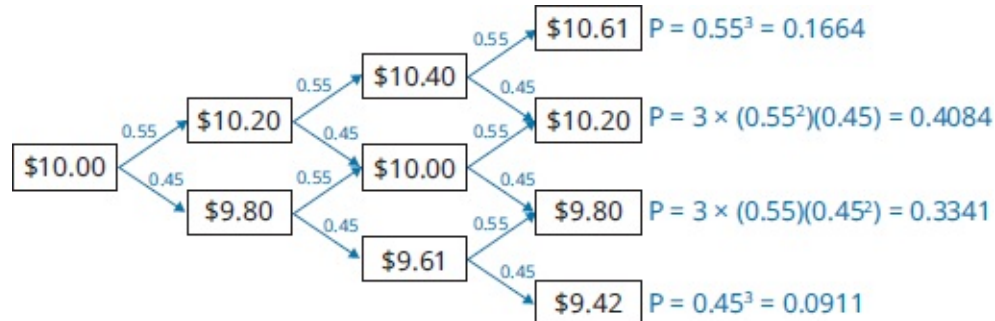
PROFESSOR'S NOTE

No example or question that requires calculating conditional covariance appears in the Level I CFA curriculum.

Recall the probability tree we used earlier to display a model of possible paths for a stock price. Now, let us assume the probability of an up move is 0.55, and the probability of a down move is 0.45. Over the first three periods, this gives us the probabilities shown in Figure 6.12 for the four possible price paths. Notice

that the middle two outcomes can each be reached by three possible paths, while the highest and lowest outcomes can only be reached by one possible path.

Figure 6.12: Stock Price Model With Probabilities



At the beginning period, the *unconditional* expectation and variance for the stock price are as follows:

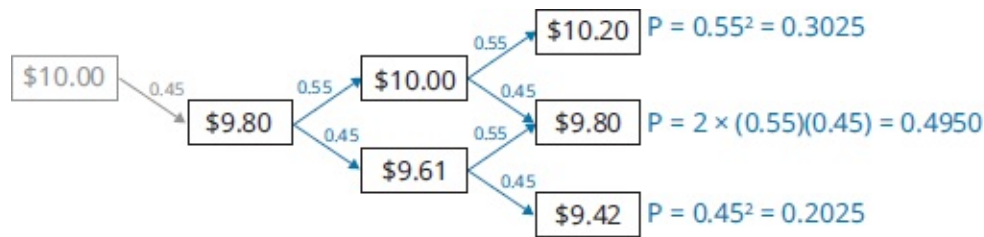
$$E(P_3) = 0.1664(\$10.61) + 0.4084(\$10.20) + 0.3341(\$9.80) + 0.0911(\$9.42) \\ = \$10.06$$

$$\text{Var}(P_3) = E(P_3^2) - [E(P_3)]^2 \\ = 0.1664(10.61^2) + 0.4084(10.20^2) + 0.3341(9.80^2) \\ + 0.0911(9.42^2) - 10.06^2 \\ = 0.1892$$

We can also calculate *conditional* expectations and variances from the model. For example, given the condition that the price after one period decreased to \$9.80, we can determine the conditional expectation $E(P_3 | P_1 = \$9.80)$ and the conditional variance $\text{Var}(P_3 | P_1 = \$9.80)$.

In effect, we are now only considering those branches of the probability tree that follow from $P_1 = \$9.80$. Figure 6.13 shows the updated probabilities for the three remaining possible outcomes after three periods. Now, we are left with two paths to the middle outcome and one path each to the highest and lowest outcomes.

Figure 6.13: Probabilities Given $P_1 = \$9.80$



$$E(P_3 | P_1 = \$9.80) = 0.3025(\$10.20) + 0.4950(\$9.80) + 0.2025(\$9.42) = \$9.84$$

$$\begin{aligned} \text{Var}(P_3 | P_1 = \$9.80) &= E(P_3^2 | P_1 = \$9.80) - [E(P_3 | P_1 = \$9.80)]^2 \\ &= 0.3025(10.20^2) + 0.4950(9.80^2) + 0.2025(9.42^2) - 9.84^2 \\ &= 0.1554 \end{aligned}$$

LOS 6.d: Formulate investment problems through Bayesian updating.

Bayes' formula is used to update a **prior probability**, or a **priori probability**, of a given event for new information. The updated probability is referred to as a **posterior probability**. We can derive Bayes' formula using the multiplication rule for conditional probability.

The multiplication rule is that $P(B|A) \times P(A) = P(BA)$. That is, the conditional probability of B occurring given A, times the unconditional probability that A will occur, equals the unconditional probability that both B and A will occur. By the same logic, we can also state that $P(A|B) \times P(B) = P(AB)$.

We also know that the joint probability of A and B occurring is the same as the joint probability of B and A occurring: $P(BA) = P(AB)$.

From these facts, we can write $P(A|B) P(B) = P(B|A) P(A)$, as both sides of the equation are equal to the joint probability that A and B will both occur.

Bayes' formula is derived by dividing both sides of this equality by the unconditional probability $P(B)$:

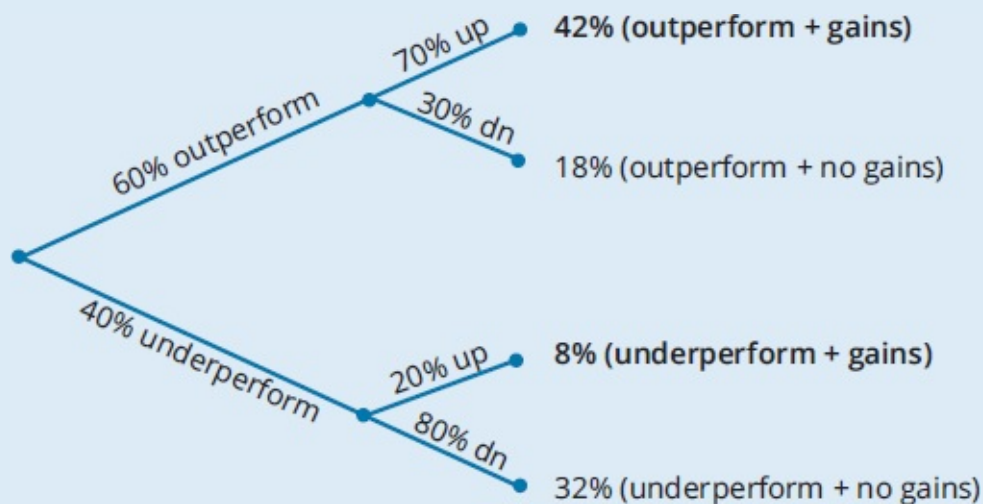
$$\frac{P(A|B)P(B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$$

Therefore: $P(A|B) = P(B|A) \frac{P(A)}{P(B)}$

EXAMPLE: Bayes' formula

There is a 60% probability that the economy will outperform—and if it does, there is a 70% probability a stock will have gains and a 30% probability the stock will have no gains. There is a 40% probability the economy will underperform, and if it does, there is a 20% probability the stock in question will have gains and an 80% probability it will not. Given that the stock has gained, calculate the updated probability that the economy outperformed.

Answer:



In the figure just presented, we have multiplied the probabilities to calculate the probabilities of each of the four outcome pairs. Note that these sum to 1. Given that the stock has gains, what is our updated probability of an outperforming economy? We sum the probability of stock gains in both states (outperform and underperform) to get $42\% + 8\% = 50\%$, which is the unconditional probability of gains. Given that the stock has gains and using Bayes' formula, the probability that the economy has outperformed is $\frac{42\%}{50\%} = 84\%$. Given that the stock has gained in value, the market will have outperformed 84% of the time—which is our updated (posterior) probability that the economy will have outperformed after learning that the stock has gains.



MODULE QUIZ 6.3

1. An analyst collects data on companies' earnings growth and dividend yields, and it categorizes the data in the following contingency table:

Earnings Growth	Dividend Yield			Total
	Low	Medium	High	
Low	28	53	42	123
Medium	42	32	39	113
High	49	25	14	88
Total	119	110	95	324

The probability that a randomly selected company will have high earnings growth and less than a high dividend yield is *closest* to:

- A. 4.3%.
 - B. 22.8%.
 - C. 84.1%.
2. An analyst believes that Davies Company has a 40% probability of earning more than \$2 per share. She estimates that the probability that Davies Company's credit rating will be upgraded is 70% if its earnings per share (EPS) are greater than \$2, and 20% if its EPS are \$2 or less. Given the information that Davies Company's credit rating has been upgraded, what is the updated probability that its EPS are greater than \$2?
- A. 50%.
 - B. 60%.
 - C. 70%.

KEY CONCEPTS

LOS 6.a

A discrete random variable has a countable number of possible outcomes. A continuous random variable has an infinite number of possible outcomes within a defined range.

The probability density function for continuous variables is proportional to the probability of outcomes in an interval around a given value.

A cumulative distribution function, or simply distribution function, defines the probability that a random variable takes on a value equal to or less than a specific value.

The unconditional expected value of a random variable is the weighted average of its possible outcomes:

$$E(X) = \sum P(x_i)x_i = P(x_1)x_1 + P(x_2)x_2 + \dots + P(x_n)x_n$$

Unconditional variance is the probability-weighted sum of the squared deviations from the mean or expected value. The unconditional standard

deviation is the positive square root of the unconditional variance.

LOS 6.b

The four principal moments of a distribution are its mean, variance, skew, and kurtosis.

Distributions commonly used in finance are the discrete uniform, binomial, continuous uniform, Poisson, normal, log-normal, and logistic:

- | A discrete uniform distribution is one for which the probabilities for all possible outcomes are equal.
- | A binomial distribution has two possible outcomes and can be used to model asset prices over multiple periods.
- | A continuous uniform distribution is used to generate random outcomes, such as those used in Monte Carlo simulations.
- | A Poisson distribution models the number of times an event occurs over a specified time period.
- | A normal distribution is completely described by its mean and variance and is often used for hypothesis testing and statistical inference.
- | A log-normal distribution is bounded from below by zero, which makes it suitable for modeling asset prices over time.
- | A logistic distribution is employed in regression models with a binary dependent variable and is used in machine learning and AI.

LOS 6.c

The joint density of two random variables describes the probability distribution of all possible pairs of outcomes.

A conditional distribution is the distribution of a random variable, given a restriction on the values of other variables in a joint probability distribution.

Conditional expectations represent the expected value of a random variable under certain conditions using the appropriate conditional distribution.

Probability trees illustrate the probabilities of possible outcomes over multiple periods.

LOS 6.d

Bayes' formula provides a method to update probabilities for the effect of new information. The conditional probability $P(A|B)$ is the posterior (updated)

probability of A, given that B has occurred, and can be calculated as follows:

$$P(A|B) = P(B|A) \frac{P(A)}{P(B)}$$

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 6.1

- . **B** The mean, or expected value, of a single throw of a fair six-sided die is calculated as the probability-weighted sum of the possible outcomes:

$$E[X] = \frac{1}{6} \times (1 + 2 + 3 + 4 + 5 + 6) = \frac{21}{6} = 3.5$$

(LOS 6.a)

- . **C** The probability of getting a rating of 3 or is the sum of the stocks rated 1, 2, or 3 divided by the total number of stocks:

$$\frac{47 + 102 + 220}{500} = 0.738 = 73.8\%$$

(LOS 6.a)

- . **A** The unconditional expectation of returns is $0.10(-5\%) + 0.25(-2\%) + 0.50(3\%) + 0.15(6\%) = 1.4\%$.

$$\begin{aligned}\text{Unconditional variance} &= E(X^2) - [E(X)]^2 \\ &= 0.10(-5^2) + 0.25(-2^2) + 0.50(3^2) + 0.15(6^2) - 1.4^2 \\ &= 11.44\end{aligned}$$

$$\text{Unconditional standard deviation} = \sqrt{11.44} = 3.38\%. \text{ (LOS 6.a)}$$

Module Quiz 6.2

- . **C** The probability of a profitable trade is $1 - 48\% = 52\%$, so the average number of profitable trades when making $5 \times 10 = 50$ trades is the expected number of successes for a binomial random variable:

$$\mu = np = 50 \times 0.52 = 26.$$

(LOS 6.b)

- . **B** The variance is calculated using the formula for a binomial distribution:

$$\sigma^2 = np(1 - p) = 20 \times 0.52 \times 0.48 = 5.0.$$

(LOS 6.b)

- . **B** For a Poisson distribution with a rate $\lambda = 3$, the mean and variance both equal 3. (LOS 6.b)
- . **B** A normal distribution is symmetrical about the mean. It is defined completely by its mean and variance because it has zero skewness and zero excess kurtosis (its kurtosis equals 3). (LOS 6.b)
- . **C** The logistic distribution has fatter tails; that is, positive excess kurtosis. (LOS 6.b)

Module Quiz 6.3

- . **B** Probability of high earnings yield and either low or medium dividend yield:

$$\frac{49 + 25}{324} = 0.2284 = 22.84\%.$$

(LOS 6.c)

- . **C** This is an application of Bayes' formula. The updated probability that EPS are greater than \$2 is:

$$P(\text{EPS} > \$2 \mid \text{credit upgrade}) =$$

$$P(\text{credit upgrade} \mid \text{EPS} > \$2) \frac{P(\text{EPS} > \$2)}{P(\text{credit upgrade})}$$

$$P(\text{credit upgrade} \mid \text{EPS} > \$2) = 70\% \text{ (given)}$$

$$P(\text{EPS} > \$2) = 40\% \text{ (given)}$$

$$P(\text{credit upgrade}) = 70\%(40\%) + 20\%(1 - 40\%) = 28\% + 12\% = 40\%$$

$$P(\text{EPS} > \$2 \mid \text{credit upgrade}) = 70\% \times \frac{40\%}{40\%} = 70\%$$

(LOS 6.d)

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² Reproduced from the Level I CFA curriculum learning module, "Statistical Distributions for Financial Asset Prices and Returns," with permission from CFA Institute.

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READING 7

ESTIMATION AND HYPOTHESIS TESTING

MODULE 7.1: THE CENTRAL LIMIT THEOREM, CONFIDENCE INTERVALS, AND SAMPLING

LOS 7.a: Explain the central limit theorem and the application of confidence intervals and sampling methodologies.

The **central limit theorem (CLT)** states that, for simple random samples of size n from any population with a mean μ and a finite variance σ^2 , the observed sample mean, $\bar{x}$, will be approximately normally distributed with a mean equal to the population mean μ and a variance equal to $\frac{\sigma^2}{n}$ as the sample size becomes large.

The central limit theorem implies the following:

- | If the sample size n is sufficiently large ($n \geq 30$), the sampling distribution of the sample means will be approximately normal. Random samples of size n are repeatedly being taken from an overall larger population. Each of these random samples has its own mean $\bar{x}$, which is itself a random variable, and *this set of sample means has a distribution that is approximately normal*.
- | The mean of the population, μ , and the mean of the distribution of all possible sample means, $E[\bar{x}]$, are equal.
- | The variance of the distribution of sample means, $\sigma_{\bar{x}}^2$, is equal to $\frac{\sigma^2}{n}$, the population variance divided by the sample size.



PROFESSOR'S NOTE

Appreciate what the CLT is saying, because it is the foundation to the estimation and hypothesis testing material that follows.

For example, assume the true population mean of a distribution of annual investment returns, μ , is 5% and the true variance of returns, σ^2 , is 196 (expressed in whole percentage squared terms). If we observe a sample of 30 annual returns, the sample mean, $\bar{x}$, while not necessarily equal to 5%, is expected to be 5% (because the average sample mean is the population mean) and has a variance of $196 / 30 = 6.53$ around the true population mean of 5%. We know that the sample mean $\bar{x}$ is approximately normally distributed around the true population mean, μ .

Usually we do not know μ and σ^2 , so we use this insight about how the observed sample mean behaves relative to the unknown true population parameters μ and σ^2 to estimate confidence intervals and perform hypothesis tests on their likely value.

The observed sample mean, $\bar{x}$, is being used here as an **estimator** for the true population parameter, μ . The difference between the observed sample mean $\bar{x}$ and the true population parameter, μ , is called **sampling error**. The sampling error we expect to observe is equal to the standard deviation of $\bar{x}$ around the true population parameter μ . The standard deviation of the estimator is referred to as the **standard error of the estimate**, $\sigma_{\bar{x}}$. We know from the CLT that the variance of the sample mean, $\sigma_{\bar{x}}^2$, is equal to $\frac{\sigma^2}{n}$, and because standard deviation is the square root of variance:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

where:

$\sigma_{\bar{x}}$ = standard error of the sample mean

σ = standard deviation of the population

n = size of the sample

In practice, the *population's standard deviation σ is almost never known*. Instead, the standard error of the sample mean must be estimated by dividing the standard deviation of *the sample*, s , by $\sqrt{n}$:

$$s_{\bar{x}} = \frac{s}{\sqrt{n}}$$

EXAMPLE: Standard error of sample mean (unknown population variance)

A sample contains the past 30 monthly returns for McCreary, Inc. The sample mean return is 2%, and the sample standard deviation is 20%. Calculate and

interpret the standard error of the sample mean.

Answer:

Because σ is unknown, the standard error of the sample mean is calculated using the sample standard deviation:

$$s_x = \frac{s}{\sqrt{n}} = \frac{20\%}{\sqrt{30}} = 3.6\%$$

This implies that if we took all possible samples of the size of 30 from McCreary's monthly returns and prepared a distribution of the sample means, the mean of this distribution would be 2% with a standard error of 3.6%.

EXAMPLE: Effect of sample size on standard error of sample mean

Continuing with our example, suppose that instead of a sample size of 30, we take a sample of the past 200 monthly returns for McCreary, Inc. To highlight the effect of sample size on the sample standard error, let's assume that the mean return and standard deviation of this larger sample remain at 2% and 20%, respectively. Now, calculate the standard error of the sample mean for the 200-return sample.

Answer:

The standard error of the sample mean is:

$$s_x = \frac{s}{\sqrt{n}} = \frac{20\%}{\sqrt{200}} = 1.4\%$$

Notice that the standard error of the sample mean decreased from 3.6% to 1.4% as the sample size increased from 30 to 200. As the sample size increases, the sample mean, $\bar{x}$, gets closer, on average, to the true mean of the population, μ (sampling error decreases). The distribution of the sample means about the population mean gets narrower as the standard error of the sample mean decreases, and our estimates of μ get more accurate.

When using sample statistics as estimators for population parameters in this way, it is desirable for our estimators to be unbiased, efficient, consistent, and robust.

- | An **unbiased estimator** is one for which the expected value of the estimator is equal to the parameter being estimated. We know from the CLT that the expected value of the sample mean is equal to the population mean.

- | An **efficient estimator** has a variance of its sampling distribution that is smaller than any other estimator for the parameter being estimated. If an estimator is *both* unbiased and efficient, it is referred to as the **minimum variance unbiased estimator (MVUE)**. The **best linear unbiased estimator (BLUE)** is the unbiased estimator with lowest variance that is calculated through a linear combination of sample data.
- | A **consistent estimator** is one whose accuracy increases as the sample size increases (i.e., sampling error decreases). The sample mean is a consistent estimator because it is unbiased, and the standard error of the sample mean decreases as the sample size increases.
- | A **robust estimator** is one that is not sensitive to outliers or non-normality in the underlying data. The sample mean, being an arithmetic average, is relatively sensitive to outliers in the data and is, therefore, not considered a robust estimator.

Estimation: Confidence Intervals

Individual values derived from observed sample data, such as a single sample mean, are referred to as **point estimates** for the population parameters being estimated (e.g., μ). The CLT can be used to construct ranges of values around a point estimate that have a specified level of confidence that they contain the true (unknown) population parameter. These are known as **confidence intervals**.

To construct a confidence interval, a **confidence level** must be specified (usually 95% or 99%). The confidence level is often expressed as $(1 - \alpha)$, where α is called the **significance level**. For example, we might estimate that the population mean of a random variable, μ , lies in the range of 15–25 with a 95% degree of confidence, or at the 5% level of significance.

To understand how we construct confidence intervals for the true population mean, we rely on the CLT, and the key well-known features of the normal distribution:

- | The CLT states that the sample mean is approximately normally distributed about the true population mean with a standard error of the sample mean, calculated as $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.
- | 95% of the time, a normally distributed random variable lies within 1.96 standard deviations of the mean. These distances for the standard normal distribution are called **z-values**.

The two points just listed combine to imply that when we observe a sample mean, 95% of the time we are observing a sample mean that is located within 1.96 standard errors of the true population mean. *Therefore, if we construct a range that covers 1.96 standard errors either side of this observed sample mean, we will be 95% confident that this interval includes the true population mean, even though we don't know what that actually is.*

Confidence intervals take on the following form:

point estimate $\pm$ (reliability factor $\times$ standard error)

where:

point estimate = value of the estimator calculated from sample data

reliability factor = number that depends on the distribution of the estimator and the confidence level ($1 - \alpha$) (in our example, for 95% confidence, we use a z-value of 1.96)

standard error = standard deviation of the point estimate

The distance added to and subtracted from the point estimate (reliability factor $\times$ standard error) to create the confidence interval is referred to as the **margin of error**.

When we use the sample mean as an estimator for the true population mean, a confidence interval for the true population mean is calculated as:

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

where:

$\bar{x}$ = point estimate of the population mean (sample mean)

$z_{\alpha/2}$ = reliability factor, a standard normal random variable for which the probability in the right-hand tail of the distribution is $\alpha/2$; this is the z-score that leaves $\alpha/2$ of probability in the upper tail

$\frac{\sigma}{\sqrt{n}}$ = standard error of the sample mean

These are the most commonly used standard normal distribution reliability factors:

- | $z_{\alpha/2} = 1.65$ for 90% confidence intervals (10% significance level; 5% in each tail)
- | $z_{\alpha/2} = 1.96$ for 95% confidence intervals (5% significance level; 2.5% in each tail)
- | $z_{\alpha/2} = 2.58$ for 99% confidence intervals (1% significance level; 0.5% in each tail)

PROFESSOR'S NOTE



You should learn these frequently used critical distances for the exam. These will also come in handy during the hypothesis testing material later in this reading.

EXAMPLE: Confidence interval for a population mean (population variance known)

A practice exam was administered to 36 Level I candidates. The mean score on this practice exam was 70. Assuming a population standard deviation equal to 20, construct and interpret a 99% confidence interval for the true mean practice exam score.

Answer:

At a confidence level of 99%, $z_{\alpha/2} = z_{0.005} = 2.58$. So, the 99% confidence interval is calculated as:

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 70 \pm 2.58 \frac{20}{\sqrt{36}} = 70 \pm 8.6$$

Thus, the 99% confidence interval for the true population mean exam score, μ , is 61.4 to 78.6.

It is usually the case that the population standard deviation is not known when constructing confidence intervals, in which case we use the sample standard deviation, s , when calculating standard error. This means that the sample mean no longer follows a normal distribution. Instead, it follows a **Student's t -distribution**, so we need to use reliability factors based on this distribution. The confidence interval becomes:

$$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$

where:

$\bar{x}$ = the point estimate of the population mean

$t_{\alpha/2}$ = the t -reliability factor (a.k.a. t -statistic or critical t -value) corresponding to a t -distributed random variable with $n - 1$ degrees of freedom, where n is the sample size. The area under the tail of the t -distribution to the right of $t_{\alpha/2}$ is $\alpha/2$.

$\frac{s}{\sqrt{n}}$ = standard error of the sample mean

s = sample standard deviation

The t -distribution is a symmetrical bell-shaped curve with fatter tails than the normal distribution, but it tends toward a standard normal distribution as the sample size (and, therefore, degrees of freedom) increase. The critical t -value

distances are then higher than those for critical z -values, giving wider confidence intervals, but they decrease and tend toward the relevant z -value as the sample size increases. For example, with a 95% confidence interval, the critical t -values for 10, 20, and 30 degrees of freedom are 2.228, 2.086, and 2.042, respectively (recall the relevant z -value here is 1.96). It is common to assume that standard normal z -value distances can be used in constructing confidence intervals when the sample size is large, which is defined as $n \geq 30$.



PROFESSOR'S NOTE

The curriculum liberally uses the standardized normal z -value distances as approximations to t -values when the sample standard deviation is used to calculate the standard error and the sample size is large ($n \geq 30$). While it's useful to know that t -values technically should be used, they only absolutely must be used when sample sizes are small and we are using s in the calculation of standard error. When the sample size is large ($n \geq 30$), we can use the well-known z -values listed earlier. We provide a table of critical t -values in Appendix B of this book.

If an exam question requires you to use a t -value, it will provide the relevant segment of the t -table. Make sure that you are able to select the correct t -value from a table, as defined by the area in the tail ($\alpha/2$) and the degrees of freedom ($n - 1$).

EXAMPLE: Confidence intervals

A sample contains the past 31 monthly returns for McCreary, Inc. The mean return is 2%, and the sample standard deviation is 20%. Because the population variance is unknown, the standard error of the sample must be estimated by dividing the standard deviation of the sample by $\sqrt{n}$:

$$s_x = \frac{s}{\sqrt{n}} = \frac{20\%}{\sqrt{31}} = 3.6\%$$

Calculate a 95% confidence interval for the mean monthly return.

Answer:

We will use the t -reliability factor because the population variance is unknown. Because there are 31 observations, the degrees of freedom are $31 - 1 = 30$. This is a two-tailed test at the 95% confidence level—the probability under each tail must be $\alpha/2 = 2.5\%$ for a total of 5%. Using the one-tailed probability tables for a Student's t -distribution, we find the critical t -value (reliability

factor) for $\alpha/2 = 0.025$ and $df = 30$ to be $t_{30, 2.5} = 2.04$. The 95% confidence interval for the population mean is:

$$2\% \pm 2.04 \left(\frac{20\%}{\sqrt{31}} \right) = 2\% \pm 2.04(3.592\%) = 2\% \pm 7.33\%$$

The 95% confidence has a lower limit of -5.33% and an upper limit of $+9.33\%$.

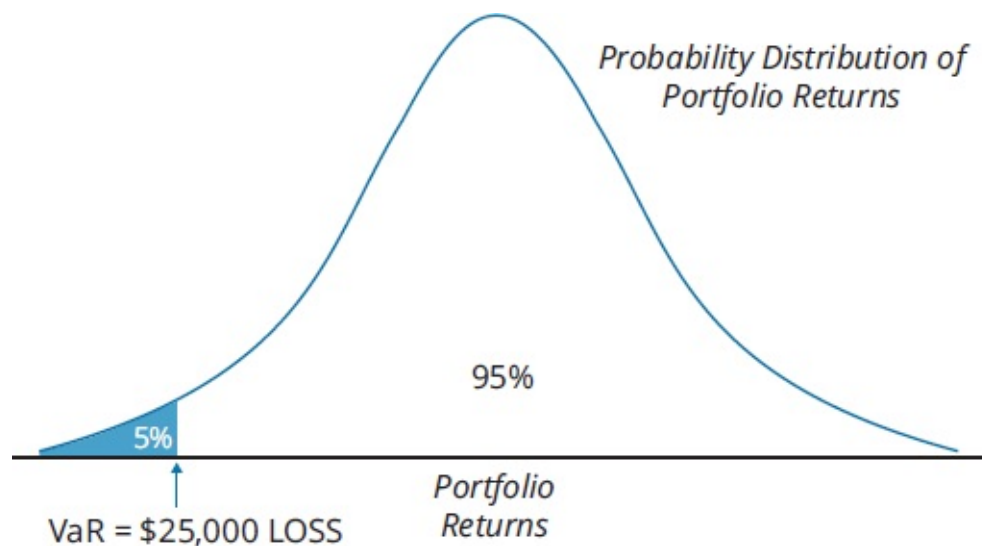
We interpret this confidence interval by saying that we are 95% confident that the true population mean, μ , for monthly returns for McCreary stock is between -5.33% and $+9.33\%$.

Value at Risk

Value at risk (VaR) is a tool used in risk management that applies the idea of one-tailed confidence intervals to investment returns.

The VaR of a portfolio measures a threshold for losses in a portfolio over a specified time frame with a specified probability. Consider the statement, “There is a 5% probability that the portfolio will experience a loss of \$25,000 or more in any given month.” This is the same as stating that the monthly 5% VaR is \$25,000. The statement is equivalent to stating that we can be 95% confident that monthly losses will *not* be \$25,000 or worse. This situation is pictured in Figure 7.1.

Figure 7.1: Monthly 5% VaR



Several methods exist for estimating the distribution of portfolio returns to calculate VaR. One of the simplest methods, the **parametric method**, assumes portfolio returns are normally distributed, so we can calculate the VaR of a portfolio using our familiar critical z-values.

If we assume that the short-term mean return is zero (a prudent assumption, often used in practice by risk management departments), then the VaR for a significance level of α for a portfolio simplifies to:

$$\text{VaR} = z_{\alpha} \times \sigma \times P$$

where:

z_{α} = the critical z-value corresponding to a significance of α in a single tail

σ = the standard deviation of the portfolio

P = current portfolio value

Because the full significance level α is in a single tail here, these are the appropriate z-values:

- | $z_{\alpha} = 1.28$ for 90% confidence intervals (significance level of 10% in a single tail)
- | $z_{\alpha} = 1.65$ for 95% confidence intervals (significance level of 5% in a single tail)
- | $z_{\alpha} = 2.33$ for 99% confidence intervals (significance level of 1% in a single tail)

The formula just listed assumes that portfolio volatility is expressed in percentage terms, and the result will give a value for the VaR in dollar loss terms. VaR can also be stated in percentage terms by leaving out the portfolio value.



PROFESSOR'S NOTE

Notice the difference between these single-tail z-values and the two-tailed z-values presented earlier. Focus on the required area in the tail(s) rather than the confidence level of the test when selecting a z-value. For example, with 95% confidence, the appropriate z-value would be 1.96 for a two-tailed confidence interval (with 2.5% in each tail) or 1.65 for a single tail of 5%. The confidence level is not enough to identify the correct z-value; you must also specify the number of tails.

EXAMPLE: Value at risk

A portfolio with a value of \$100 million has daily standard deviation of 0.95%. Calculate and interpret the daily 1% VaR for this portfolio using the parametric

method.

Answer:

The relevant z-value for 1% in a single tail is 2.33. Under the parametric method, the daily 1% VaR is calculated as:

$$\text{VaR} = 2.33 \times 0.0095 \times \$100,000,000 = \$2,213,500$$

We can interpret this figure as follows: On 1% of days, we expect to see a loss equal to \$2,213,500 or worse. Equivalently, we are 99% confident that on any given day, the portfolio will *not* lose \$2,213,500 or more.

While the parametric method for calculating VaR is quick and simple, it does have drawbacks:

- | The return distributions of many types of investment are not normal, exhibiting skewness and kurtosis.
- | Distribution parameters may change over time.
- | Reliance on historical data to estimate parameters is not appropriate if new or extreme market conditions evolve.
- | The method does not work well for assets with nonlinear returns, such as options.

Sampling Methods

To generate representative samples from a population, two general sampling methods are available: probability sampling and nonprobability sampling.

- | **Probability sampling** techniques ensure each population member has a known, nonzero chance of being selected in a sample. Methods include the following:
 - **Simple random sampling.** Each population member has an equal probability of being selected.
 - **Systematic sampling.** Every *k*th member from a population is selected.
 - **Stratified random sampling.** The population is divided into key groups or *strata*, and random samples are taken in each stratum, which are combined in the same proportion as the overall population. This ensures that the key strata (characteristics) are reflected in the sample. For example, imagine that we believe gender is an important factor in CFA results, and that we know 60%

of the population of CFA candidates is male and 40% female. To ensure this characteristic is reflected in a sample of candidates, 60% of sample observations would be selected from the male strata and 40% from the female strata.

- **Cluster sampling.** A population is divided into smaller groups (called *clusters*), each of which is assumed to be representative of the overall population. Clusters are randomly selected, and a random sample is taken from selected clusters. The CLT will apply to cluster sampling if the size of the clusters is large enough.
- **Nonprobability sampling** techniques do not have known nonzero probabilities of selection for all population members. Methods include the following:
 - **Convenience sampling.** Selection of sample elements is based on ease of access, using data that are readily available. Because such a sample is typically not random, sampling error will likely be greater.
 - **Judgmental sampling.** Sample elements are selected from the population based on experience and judgment. This process may introduce biases of the researcher, but the researcher's intuition could be critical when dealing with complex research issues.

Probability sampling is typically more accurate than nonprobability sampling, which tends not to produce representative samples but benefits from lower cost. The CLT does not apply to nonrepresentative samples.



MODULE QUIZ 7.1

1. Which of the following is *least likely* one of the four desirable properties of an estimator?
 - A. Reliability.
 - B. Efficiency.
 - C. Consistency.
2. A random sample of 100 computer store customers spent an average of \$75 at the store. Assuming the distribution is normal and the population standard deviation is \$20, the 95% confidence interval for the population mean is *closest* to:
 - A. \$71.08 to \$78.92.
 - B. \$73.89 to \$80.11.
 - C. \$74.56 to \$79.44.
3. What is the *most appropriate* test statistic for constructing confidence intervals for the population mean when the population is normally distributed, but the variance is unknown?
 - A. The z-statistic at α with n degrees of freedom.
 - B. The t-statistic at $\alpha/2$ with n degrees of freedom.

- C. The t -statistic at $\alpha/2$ with $n - 1$ degrees of freedom.
- 4. Weekly 5% VaR of £1 million indicates:
 - A. a maximum allowable loss of £1 million in 5% of weeks.
 - B. that the largest weekly loss is £1 million, or 5% of portfolio value.
 - C. a 5% probability of a loss equal to or greater than £1 million in any given week.
- 5. A portfolio has annual standard deviation of 9% and is currently valued at \$6.4 million. What is the 5% annual VaR of the portfolio using the parametric method, assuming mean returns are zero?
 - A. \$320,000.
 - B. \$576,000.
 - C. \$950,400.

MODULE 7.2: HYPOTHESIS TESTS OF THE POPULATION MEAN

LOS 7.b: Explain hypothesis testing and its components, including statistical significance, Type I and Type II errors, and the power of a test; construct appropriate hypothesis tests; and interpret the results.



PROFESSOR'S NOTE

This reading has various different hypothesis tests—some of which can look quite daunting at first. Focus on understanding hypothesis testing for a single mean before moving on to the more exotic kinds of hypothesis tests. Remember that the process is the same for all of the tests introduced in this reading, including the nonparametric tests toward the end.

Hypothesis Testing

A **hypothesis** is a statement about the value of a population parameter made to test a theory or belief. Hypotheses are usually stated in terms of the population parameter to be tested, such as the population mean. For example, a researcher may be interested in the mean daily return on a trading strategy. The researcher may state a hypothesis that the mean daily risk-adjusted return of the strategy is less than or equal to zero, then take a sample of data to look for evidence to reject this hypothesis, which would suggest that the strategy produces positive risk-adjusted returns.

Hypothesis tests take two primary forms:

Parametric tests rely on assumptions regarding the distribution of the population and are specific to population parameters, such as mean and variance. Hypotheses are made about potential values these parameters could take. Then, the data are sampled to test the likely validity of these hypotheses.

Nonparametric tests either do not consider a particular population parameter or have few assumptions about the population that is sampled. Nonparametric tests are used for quantities other than parameters of a statistical distribution, like mean and variance. Later in this reading, we describe two types of nonparametric tests that test the ranking of data (rather than values) and analyze categorized data to look for dependency between categories.

The basic steps in all hypothesis tests are as follows:

Step 1: State the null and alternative hypotheses.

Step 2: Identify the test statistic and its distribution.

Step 3: Specify the level of significance.

Step 4: Establish the decision rule.

Step 1: State the Null and Alternative Hypotheses

A hypothesis test requires two hypotheses: a null and an alternative hypothesis.

The **null hypothesis**, H_0 , is the hypothesis that the researcher is typically looking to *reject* in favor of the alternative hypothesis. It usually represents a lack of evidence that anything unusual is being observed in the data.

The **alternative hypothesis**, H_A , is the opposite of the null hypothesis. It implies that evidence in the data is sufficient to reject the null hypothesis. The alternative hypothesis typically represents the claim for which the researcher has set out to find evidence.

The null and alternative hypotheses must be *mutually exclusive and exhaustive*. They include all possible outcomes (exhaustive), and they cannot both be true at the same time (mutually exclusive). For example, a null hypothesis might state that a true mean is equal to zero, and the alternative hypothesis would be that the true mean is different than zero.

Step 2: Identify the Test Statistic and Its Distribution

An analyst takes a sample and calculates a **test statistic** from the data. For a test of the population mean, the test statistic measures how many standard errors the observed sample mean is from the analyst's hypothesized population mean.

$$\text{test statistic} = \frac{\text{sample mean} - \text{hypothesized value}}{\text{standard error of sample mean}}$$

In our example, the hypothesized value for the true mean is 0. Assume that a sample of size 60 is taken and the sample mean is 1.12%, with a sample standard deviation of 2.88%.

Recall from our earlier discussion of the CLT that the standard error of the sample mean is calculated as $s/\sqrt{n}$ when the true population standard deviation is unknown. In our example this is $2.88\% / \sqrt{60} = 0.372\%$.

We can now calculate the test statistic as follows:

$$\text{test statistic} = \frac{1.12\% - 0}{0.372\%} = 3.01$$

This test statistic shows we have observed a sample mean that is 3.01 standard errors away from the hypothesized value for the true population mean.

Step 3: Specify the Level of Significance

The analyst must specify a significance level, α , when designing the test. This is a small probability level (values commonly used are 1% or 5%) which we can informally think of as a probability threshold: if we observe a sample statistic that is less likely than the significance level under the null hypothesis, then we reject the null hypothesis.

Just as for confidence intervals earlier in the reading, a significance level of α implies a confidence level of $1 - \alpha$.

For our example, let us assume a significance level of 5%.

Step 4: Establish the Decision Rule

We know from the CLT that the sample mean is approximately normally distributed around the true population mean with a standard error of $\frac{\sigma}{\sqrt{n}}$ (approximated in our test with $\frac{s}{\sqrt{n}}$).

We also know from the features of the normal distribution that 95% of the distribution is observed within 1.96 standard deviations of the mean. For our test, this implies that if the null were true, 95% of the time we would observe a

sample mean that is within 1.96 standard errors of the hypothesized mean. Because the true population standard deviation is not known, theoretically we should be using a t -distribution, but with a large sample size a z -distribution is acceptable.

Our decision rule is as follows:

Reject H_0 if test statistic < -1.96 or > 1.96

This decision rule can be visualized as shown in Figure 7.2.

Figure 7.2: Two-Tailed Hypothesis Test Using the Standard Normal (z) Distribution

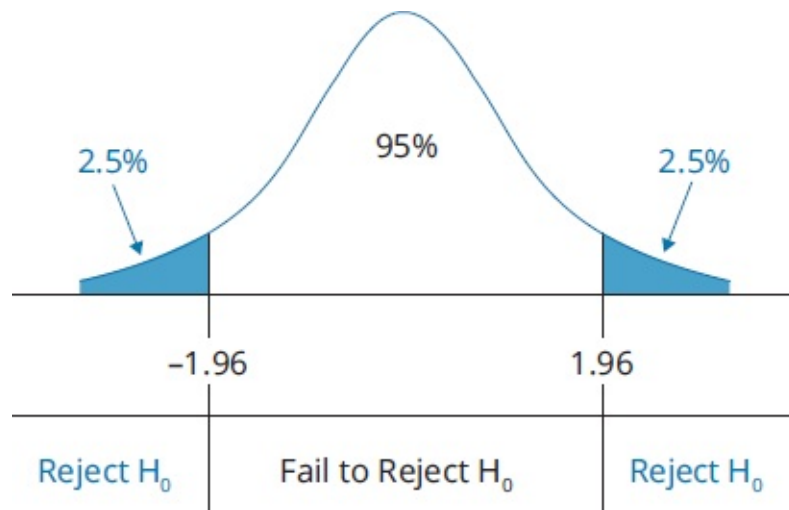
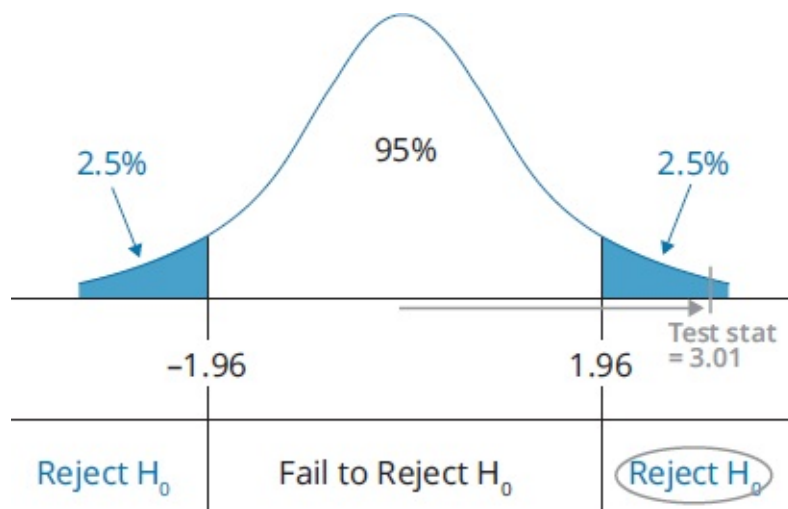


Figure 7.2 shows the standard normal distribution for a two-tailed hypothesis test using the z -distribution. The significance level of 0.05 means each tail has $0.05 / 2 = 0.025$ probability (area) beyond ± 1.96 .

In our example, we calculated a test statistic of 3.01. This is well inside the right-hand rejection zone.

Figure 7.3: Decision for a Two-Tailed Hypothesis Test Using the Standard Normal (z) Distribution



Our decision is to reject the null hypothesis (with 5% significance). The test statistic (3.01) is greater than the critical value for the test (1.96) because we have observed a sample mean that is less than 5% likely if the null hypothesis were true.

The test just described is a two-tailed test. When the null hypothesis states that the true mean is equal to a specified value, there are two rejection zones because we would reject the null if we saw a sample mean that was either too low or too high.

One-tailed hypothesis tests also exist, which look for evidence that a population parameter is less than, or greater than, a hypothesized value. The following example illustrates a one-tailed hypothesis test.

EXAMPLE: One-tailed test of a population mean

An analyst wishes to test whether the mean weekly return of a risk factor is greater than zero. State the hypotheses for this test.

Answer:

The analyst is looking for evidence that the mean weekly return of the risk factor is greater than zero. Because this is what we wish to find evidence for, we should make this the alternative hypothesis and look to reject a null relating to the opposite of this claim. The null and alternative hypotheses are as follows:

H_0 : mean weekly return from risk factor is equal to zero or less: $\mu \leq 0$

H_A : mean weekly return from risk factor is greater than zero: $\mu > 0$



PROFESSOR'S NOTE

The null hypothesis always contains some form of “equal to” condition. A two-tailed test has “=”, and one-tailed tests will have either “ $\leq$ ” or “ $\geq$ ” depending on which tail the rejection zone is in.

EXAMPLE: One-tailed test of a population mean

An analyst wishes to test whether the mean weekly return of a risk factor is greater than zero. She collects a sample of 150 weeks of data and calculates the sample mean to be 0.15% with sample standard deviation of 1%. Calculate the test statistic for this test and determine the decision to be made. Assume a significance level of 1%.

Answer:

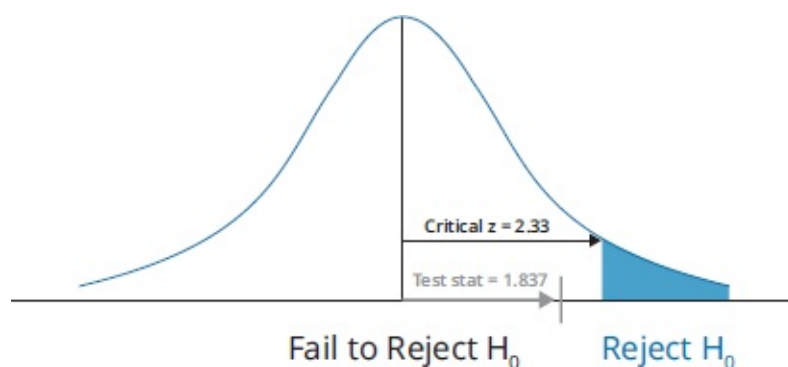
The test statistic is calculated as follows:

$$\begin{aligned}\text{test statistic} &= \frac{\text{sample mean} - \text{hypothesized value}}{\text{standard error of sample mean}} = \frac{0.15\% - 0}{1\% / \sqrt{150}} = \frac{0.15\%}{0.0816\%} \\ &= 1.837\end{aligned}$$

The critical z-value for a one-tailed test with a significance of 1% is 2.33. Because we used sample standard deviation to calculate standard error, a t -distribution should be used. However, with a sample size of 150, a z -distribution is acceptable.

Because the test statistic is not greater than the critical z -statistic, we fail to reject the null hypothesis. Figure 7.4 shows the distribution for this test.

Figure 7.4: Decision for a One-Tailed Hypothesis Test Using the Standard Normal (z) Distribution



The example has the rejection zone in the right tail because we are looking for evidence that the true mean is greater than a hypothesized value. One-tailed tests can also look for evidence that the true mean is less than a specified value, in which case the single rejection zone would be in the left tail of the distribution.



PROFESSOR'S NOTE

Ensure you are fully secure on the hypothesis test of a single mean before moving on to other, more complex hypothesis tests that we describe in our next module. The other tests all use the same basic four-step process. The only major differences are in the calculation of the test statistic and the distributions that the test statistics follow.

Type I and Type II Errors

Hypothesis tests can lead to incorrect conclusions. Two types of errors can occur:

Type I error. This is rejecting the null hypothesis when it is actually true.

Type II error. This is failing to reject the null hypothesis when it is actually false.

As an example, consider the “spam filter” on your email inbox. We can think of the filter software as testing the null hypothesis, “this email = spam.”

- | Allowing a spam email to reach your inbox is a Type I error, rejecting the null when it is actually true.
- | Diverting a legitimate email to your spam folder is a Type II error, failing to reject the null when it is actually false.

While these errors make the hypothesis testing process imperfect, a researcher can measure the probability of making these errors, and managing these probabilities will play a large part in the design of the test.

The probability of making a Type I error is equal to the significance level. This is because even when the null is actually true, we still expect to observe extreme samples that cause test statistics to be located in rejection zones.

Statisticians tend to dislike Type II errors, where a chance to reject an incorrect null was missed, more than they dislike Type I errors. The **power of a test** is the probability of correctly rejecting the null hypothesis when it is false. The higher the power of a test, the less likely a Type II error because $\text{power} = 1 - P(\text{Type II})$.

error). When more than one test statistic may be used, the power of the test for the competing test statistics may be useful in deciding which test statistic to use. Ordinarily, we wish to use the test statistic that provides the most powerful among all possible tests.

Figure 7.5 shows the relationship between the level of significance, the power of a test, and the two types of errors.

Figure 7.5: Type I and Type II Errors in Hypothesis Testing

Decision	True Condition	
	H_0 is true	H_0 is false
Do not reject H_0	Correct decision	Incorrect decision Type II error
Reject H_0	Incorrect decision Type I error Significance level, α , $= P(\text{Type I error})$	Correct decision Power of the test $= 1 - P(\text{Type II error})$

Sample size and the choice of significance level (Type I error probability) together determine the probability of a Type II error. The relation is not simple, however, and calculating the probability of a Type II error in practice is difficult. Decreasing the significance level (probability of a Type I error) from 5% to 1%, for example, will increase the probability of failing to reject a false null (Type II error), and therefore reduce the power of the test. Conversely, for a given sample size, we can increase the power of a test only at the cost of increasing the probability of rejecting a true null (Type I error). For a given significance level, we can decrease the probability of a Type II error and increase the power of a test only by increasing the sample size.



MODULE QUIZ 7.2

- Which of the following statements about hypothesis testing is *most accurate*?
 - A Type II error is rejecting the null when it is actually true.
 - The significance level equals one minus the probability of a Type I error.
 - A two-tailed test with a significance level of 5% has z-critical values of ± 1.96 .
- For a hypothesis test with a probability of a Type II error of 60% and a probability of a Type I error of 5%, which of the following statements is *most accurate*?
 - The power of the test is 40%, and there is a 5% probability that the test statistic will exceed the critical value(s).

- B. There is a 95% probability that the test statistic will be between the critical values if this is a two-tailed test.
 - C. There is a 5% probability that the null hypothesis will be rejected when actually true, and the probability of rejecting the null when it is false is 40%.
3. If the significance level of a test is 0.05 and the probability of a Type II error is 0.15, what is the power of the test?
- A. 0.850.
 - B. 0.950.
 - C. 0.975.
4. A researcher has 28 quarterly excess returns to an investment strategy and believes these returns are approximately normally distributed. The mean return on this sample is 1.645%, and the standard deviation is 5.29%. For a test with a 5% significance level of the hypothesis that excess returns are less than or equal to zero, the researcher should:
- A. reject the null hypothesis because the critical value for the test is 1.645.
 - B. not draw any conclusion because the sample size is less than 30.
 - C. fail to reject the null because the critical value is greater than 1.645.
5. An analyst wants to test a hypothesis concerning the population mean of monthly returns for a composite that has existed for 24 months. The analyst may appropriately use:
- A. a *t*-test but not a *z*-test, if returns for the composite are normally distributed.
 - B. either a *t*-test or a *z*-test, if returns for the composite are normally distributed.
 - C. a *t*-test but not a *z*-test, regardless of the distribution of returns for the composite.

MODULE 7.3: OTHER PARAMETRIC HYPOTHESIS TESTS

We can extend the hypothesis test process to conduct hypothesis testing of the following:

- | The difference between two means for independent samples
- | The mean difference for paired observations
- | A variance of a single distribution
- | The difference between the variances of two different distributions
- | Correlation of two distributions
- | Statistical independence of two distributions

Keep in mind with each of the following tests that we are simply applying the same four-step hypothesis test process we have outlined.

Testing the Difference Between Means (Independent Samples)

We frequently want to know if there is evidence of a difference between the means of two populations. The test for **differences between means** requires that we are reasonably certain that our samples are independent and that they are taken from two populations that are normally distributed.

The test statistic for the difference between the means of two different populations is as follows:

test statistic =

$$\frac{\text{difference between sample means} - \text{hypothesized difference between means}}{\text{standard error of the difference in sample means}}$$

The hypothesized difference between means is often zero, which simplifies the numerator. If we take samples from Population 1 and Population 2, then the test statistic, t , is calculated as follows:

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where:

$\bar{X}_1 - \bar{X}_2$ = observed difference in sample means

$\mu_1 - \mu_2$ = the hypothesized difference between means (often equal to zero)

s_1^2 = variance of the first sample

s_2^2 = variance of the second sample

n_1 = number of observations in the first sample

n_2 = number of observations in the second sample

The denominator is summing the squared standard errors of the two sample means, then taking the square root. Recall that for a single mean, the degrees of freedom for the sample statistic were $n - 1$. With two samples, our degrees of freedom total $(n_1 - 1) + (n_2 - 1)$, which equals $n_1 + n_2 - 2$.

EXAMPLE: Hypothesis test of the difference between means

An analyst is investigating whether differences in abnormal returns exist between acquiring firms in horizontal mergers versus vertical mergers. State the hypotheses for this test.

Answer:

The analyst is looking for evidence that the true population means are different. The hypotheses are as follows:

H_0 : the two population means are equal: $\mu_1 - \mu_2 = 0$

H_A : the two population means are not equal: $\mu_1 - \mu_2 \neq 0$

where:

μ_1 = mean abnormal return for horizontal mergers

μ_2 = mean abnormal return for vertical mergers

EXAMPLE: Hypothesis test of the difference between means

An analyst is investigating whether the abnormal returns differ for acquiring firms during merger announcement periods for horizontal mergers versus vertical mergers. The analyst collects two samples of returns of firms during merger announcement periods, one from each population, with the following results:

	Sample Size	Sample Mean	Sample Standard Deviation
Horizontal mergers	31	2.0%	20%
Vertical mergers	32	1.0%	10%

Calculate the test statistic and determine the decision to be made. Assume a significance level of 1%.

Answer:

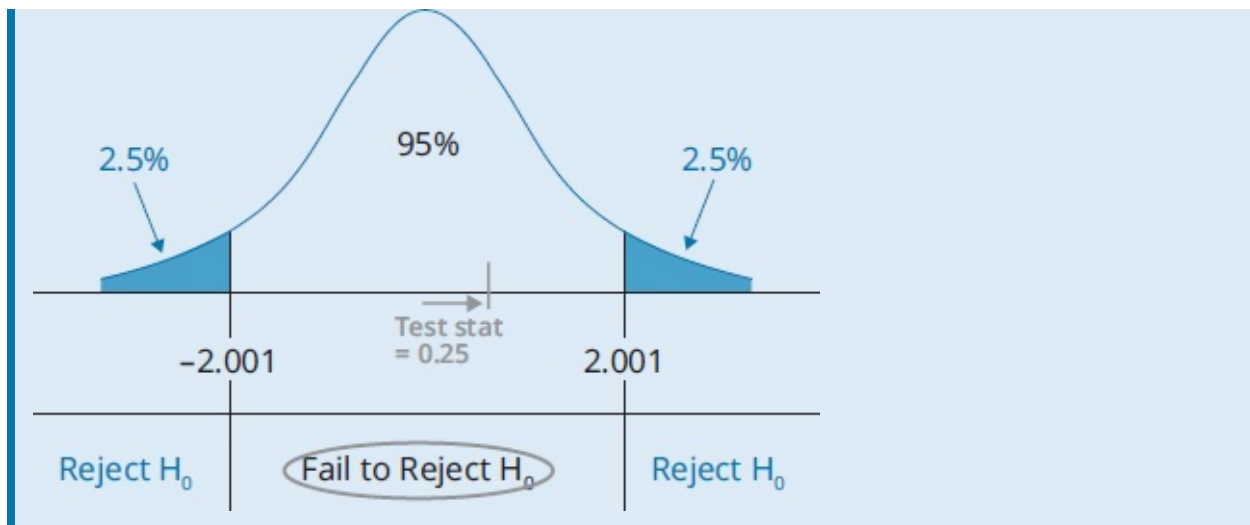
The test statistic is calculated as:

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{(2.0 - 1.0) - 0}{\sqrt{\frac{20^2}{31} + \frac{10^2}{32}}} = \frac{1.0}{4.004} = 0.249$$

Degrees of freedom for the t -stat are $n_1 + n_2 - 2 = 31 + 30 - 2 = 59$. The critical t -value for 95% confidence with two tails and 59 degrees of freedom is 2.001.

We fail to reject the null because the test statistic (0.249) is less than the critical value, and conclude that we have not found evidence that the means of the two distributions are different.

Critical Values and Rejection Regions



PROFESSOR'S NOTE

When the variance of the two distributions is assumed to be the same, a pooled variance estimator, s_p^2 , can be calculated by combining the two samples when calculating standard error. This calculation is mentioned in the curriculum learning module but is not actually carried out, so do not concern yourself with its mechanics.

Paired Comparisons (Means of Dependent Samples)

The previous test was of the difference between the means of two independent samples. Sometimes, our two samples may be dependent. If the observations in the two samples both depend on some other factor, we can construct a **paired comparisons test** of whether the *mean difference* between observations is different from zero.

For an example of a paired comparisons test, consider a test of the scores earned by CFA exam candidates before and after sitting an educational course. To test this, we measure the scores of a sample of candidates before the course, then measure the scores of the same candidates after taking the course. We then calculate each candidate's difference in scores. This becomes our sample. The obvious connection between the two samples is that the same population members are being observed twice. We are looking for evidence that the mean difference in scores (before and after) is nonzero to conclude that the course has some value.



PROFESSOR'S NOTE

Appreciate how this test of mean difference is distinct from testing the difference between means. They sound similar, but they are different ideas. Previously, we took two samples from independent populations and tested to see if their means were different. Now, we are taking two sample readings (usually from the same population members under different conditions) and calculating a difference in value for each member of the sample. We then take the sample mean of these differences and test its value, like we did for a single population mean.

For a two-tailed test, the hypotheses are as follows:

- | H_0 : the mean difference is equal to a hypothesized value (usually zero)
- | H_A : the mean difference is not equal to the hypothesized value

For the paired comparisons test, the t -statistic with $n - 1$ degrees of freedom is computed as follows:

$$t = \frac{\text{sample mean of differences} - \text{hypothesized mean of differences}}{\text{standard error of the sample mean difference}} = \frac{\bar{d} - \mu_0}{\frac{s_d}{\sqrt{n}}}$$

EXAMPLE: Paired comparisons test

An analyst is interested in whether a drug increases the amount of sleep in insomnia patients. The analyst collects data from a sample of 50 patients and measures the difference between the amount of sleep after the drug is taken versus the amount of sleep before taking the drug for each patient. The analysis shows the following:

- Sample mean of differences (hours) = 0.75
- Sample standard deviation of differences (hours) = 1.5

Perform a hypothesis test of whether the true mean difference in sleep times is zero, with a significance level of 1%.

Answer:

Because the mean difference may be positive or negative, a two-tailed test is in order. Thus, the hypotheses are structured as:

$$H_0: \mu_d = 0 \text{ versus } H_a: \mu_d \neq 0$$

There are $50 - 1 = 49$ degrees of freedom. The t -value for 49 degrees of freedom with 1% significance and two tails is 2.680.

Calculating the test statistic:

$$t = \frac{\text{sample mean of differences} - \text{hypothesized mean of differences}}{\text{standard error of the sample mean difference}}$$
$$= \frac{\bar{d} - \mu_0}{\frac{s_d}{\sqrt{n}}} = \frac{0.75 - 0}{1.5 / \sqrt{50}} = \frac{0.75}{0.21} = 3.53$$

The test statistic (3.53) is greater than the critical value of the test (2.680). We can reject the null hypothesis that the mean difference is zero and conclude that the mean difference is nonzero (hence, that the drug has an effect).

Testing the Variance of a Distribution

We can also perform hypothesis tests on the variance of a distribution. The process remains the same, but the calculation of the test statistic and the distribution it follows are different.

Letting σ^2 represent the true population variance and σ_0^2 represent the hypothesized variance, the hypotheses for a two-tailed test of a single population variance are structured as follows:

- | H_0 : true population variance is equal to a specified value: $\sigma^2 = \sigma_0^2$
- | H_A : true population variance is not equal to the specified value $\sigma^2 \neq \sigma_0^2$

The hypotheses for one-tailed tests are structured as follows:

- | $H_0: \sigma^2 \leq \sigma_0^2$ versus $H_A: \sigma^2 > \sigma_0^2$

or:

- | $H_0: \sigma^2 \geq \sigma_0^2$ versus $H_A: \sigma^2 < \sigma_0^2$

Hypothesis testing of the population variance requires the use of a chi-square distributed test statistic, denoted χ^2 . The **chi-square distribution** is asymmetrical and bounded below by zero, and it is often used to model statistics related to squared measures like variance.

To illustrate the chi-square distribution, consider a two-tailed test with a 5% level of significance and 30 degrees of freedom. As displayed in Figure 7.6, the critical chi-square values are 16.791 and 46.979 for the lower and upper bounds, respectively. These values are obtained from a chi-square table, which is used in

the same manner as a *t*-table. (We present a chi-square table in Appendix E of this book.)

The chi-square values in Figure 7.7 correspond to the probabilities in the right tail of the distribution. The 16.791 is from the column headed 0.975 because 95% + 2.5% of the probability is to the right of it. The 46.979 is from the column headed 0.025 because only 2.5% probability is to the right of it. Similarly, at a 5% level of significance with 10 degrees of freedom, Figure 7.7 shows that the critical chi-square values for a two-tailed test would be 3.247 and 20.483.

Figure 7.6: Decision Rule for a Two-Tailed Chi-Square Test

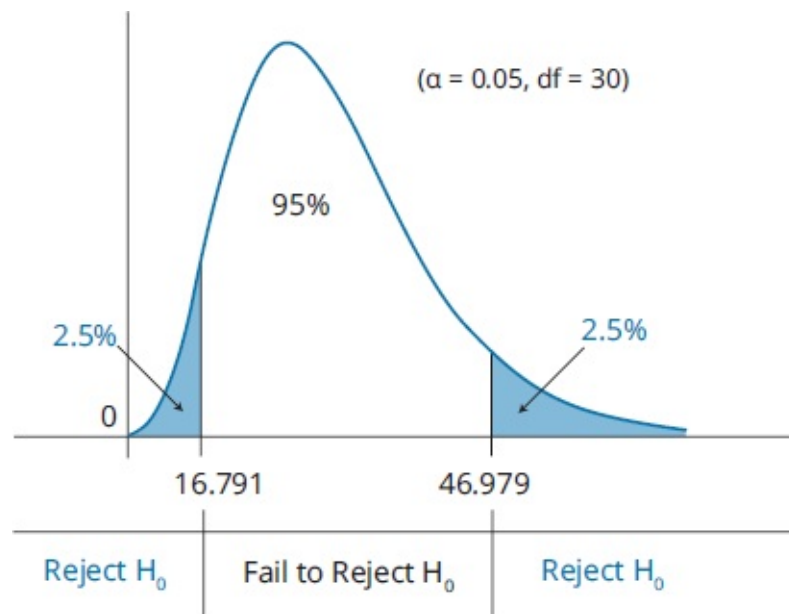


Figure 7.7: Chi-Square Table

Degrees of Freedom	Probability in Right Tail					
	0.975	0.95	0.90	0.10	0.05	0.025
9	2.700	3.325	4.168	14.684	16.919	19.023
10	3.247	3.940	4.865	15.987	18.30	20.483
11	3.816	4.575	5.578	17.275	19.675	21.920
30	16.791	18.493	20.599	40.256	43.773	46.979

The chi-square test statistic, χ^2 , with $n - 1$ degrees of freedom, is:

$$\chi^2_{n-1} = \frac{(n-1)s^2}{\sigma_0^2}$$

where:

n = sample size

s^2 = sample variance

σ_0^2 = hypothesized value for the population variance

Like other hypothesis tests, the chi-square test compares the test statistic, χ^2_{n-1} , to a critical chi-square value at a given level of significance and $n - 1$ degrees of freedom. Because the chi-square distribution is bounded below by zero, chi-square values cannot be negative.



PROFESSOR'S NOTE

You should be starting to see the pattern in these tests now: We specify hypotheses, then calculate sample test statistics from the given formulae and compare them to critical values, which come from statistical tables. While the types of calculation and distributions for test statistics change, the process is always the same.

EXAMPLE: Chi-square test for a single population variance

The High-Return Equity Fund advertises that its monthly returns have a standard deviation equal to 4%. An analyst collects monthly returns for the 24-month period between 20X5 and 20X7 and calculates a standard deviation of monthly returns of 3.8%.

State the hypotheses for a test of whether the true monthly standard deviation of the fund is 4%. Calculate the test statistic, and using a 5% significance level, determine whether the data challenge the fund's claim.

Answer:

The null hypothesis is that the population standard deviation is equal to 4%, or that the variance of monthly returns for the population is $(0.04)^2 = 0.0016$. Because the analyst simply wants to test whether the standard deviation differs from 4%, he will use a two-sided test. The hypothesis test structure takes this form:

$$H_0: \sigma^2 = 0.0016$$

$$H_A: \sigma^2 \neq 0.0016$$

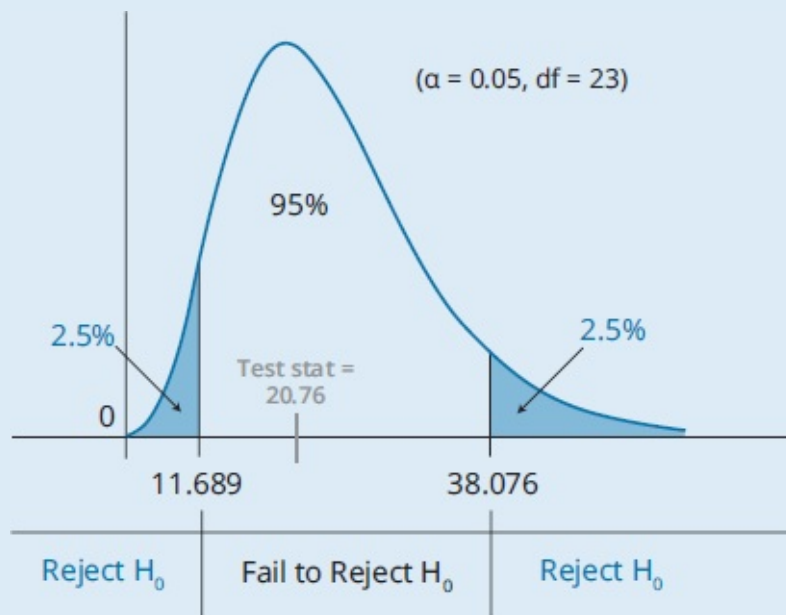
The chi-square statistic is calculated as follows:

$$\chi^2_{n-1} = \frac{(n-1)s^2}{\sigma_0^2} = \frac{(24-1) \times 0.038^2}{0.04^2} = \frac{0.0332}{0.0016} = 20.76$$

With a 24-month sample, degrees of freedom equal 23. For 23 degrees of freedom and probabilities of 0.975 and 0.025, we find two critical values: 11.689 and 38.076. The decision rule:

Reject H_0 if $\chi^2 < 11.689$, or $\chi^2 > 38.076$

Decision Rule for a Two-Tailed Chi-Square Test of a Single Population Variance



Because the computed test statistic falls between the two critical values, we cannot reject the null hypothesis that the variance is equal to 0.0016. The sample standard deviation is close enough to the claimed true standard deviation that we cannot say that it is different, at a 5% level of significance.

Comparing Two Population Variances

Hypotheses concerned with the equality of the variances of two populations are tested with an F -distributed test statistic (referred to as an **F -test**). The F -test assumes the populations from which samples are drawn are normally distributed, and that the samples are independent.

If we let σ_1^2 and σ_2^2 represent the variances of Population 1 and Population 2, respectively, the hypotheses for a two-tailed F -test of differences between their variances can be structured as follows:

$$\begin{aligned} H_0: \sigma_1^2 &= \sigma_2^2 \\ H_A: \sigma_1^2 &\neq \sigma_2^2 \end{aligned}$$

One-sided tests can be specified as follows:

$$H_0: \sigma_1^2 \leq \sigma_2^2 \text{ versus } H_A: \sigma_1^2 > \sigma_2^2$$

or:

$$H_0: \sigma_1^2 \geq \sigma_2^2 \text{ versus } H_A: \sigma_1^2 < \sigma_2^2$$

The test statistic for the F -test is the ratio of the sample variances. The F -statistic is computed as follows:

$$F = \frac{s_1^2}{s_2^2}$$

where:

S_1^2 = variance of the sample of n_1 observations drawn from Population 1

S_2^2 = variance of the sample of n_2 observations drawn from Population 2

Degrees of freedom used to identify the appropriate critical value from the F -table are $n_1 - 1$ and $n_2 - 1$.

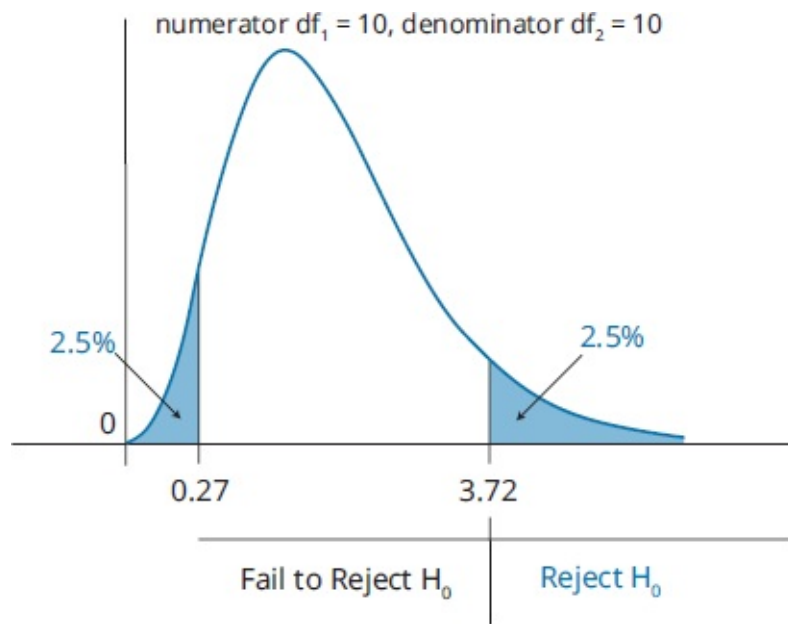


PROFESSOR'S NOTE

If we always put the larger variance in the numerator (s_1^2), the test statistic will always be greater than 1, and we only need to consider the critical value for the right-hand tail. Note, however, that a two-tailed test will still only have an area of $\alpha/2$ in the rejection zone; we are just constructing the test such that we only need to consider rejection in the right-hand tail. If conducting a one-tailed test, by putting the higher sample variance on the top, again, we only need to consider the right-hand tail—but this time, with the full significance of α in the tail.

An F -distribution is presented in Figure 7.8. As indicated, the F -distribution is right skewed and is bounded by zero on the left-hand side. The shape of the F -distribution is determined by two separate degrees of freedom: the numerator degrees of freedom, $df_1 = n_1 - 1$, and denominator degrees of freedom, $df_2 = n_2 - 1$.

Figure 7.8: F -Distribution



When the sample variances are equal, the value of the test statistic is 1. The upper critical value is always greater than one (the numerator is significantly greater than the denominator), and the lower critical value is always less than one (the numerator is significantly smaller than the denominator). In fact, the lower critical value is the reciprocal of the upper critical value. For this reason, in practice, we put the larger sample variance in the numerator and consider only the upper critical value.

EXAMPLE: *F*-test for equal variances

An analyst is examining company earnings for two different industries. The analyst believes the variance of earnings in the semiconductor industry is different from the variance in the biotechnology industry. To test this, the analyst collects a sample of 31 semiconductor companies and a sample of 41 biotech companies. The sample standard deviation of earnings across the semiconductor industry is \$4.30, and the sample standard deviation for the biotech companies is \$3.80. Using a 5% significance level, determine whether the earnings of the two industries have different variances.

Answer:

The test hypotheses are structured as follows:

- H_0 : variances of the two distributions are the same: $\sigma_1^2 = \sigma_2^2$
- H_A : variances of the two distributions are not the same: $\sigma_1^2 \neq \sigma_2^2$

The appropriate test statistic is:

$$F = \frac{s_1^2}{s_2^2}$$

where s_1^2 is the larger sample variance. This is calculated to as $\$4.30^2 / \pounds 3.80^2 = 1.28$.

The critical F -value for our test is found in an F -table:

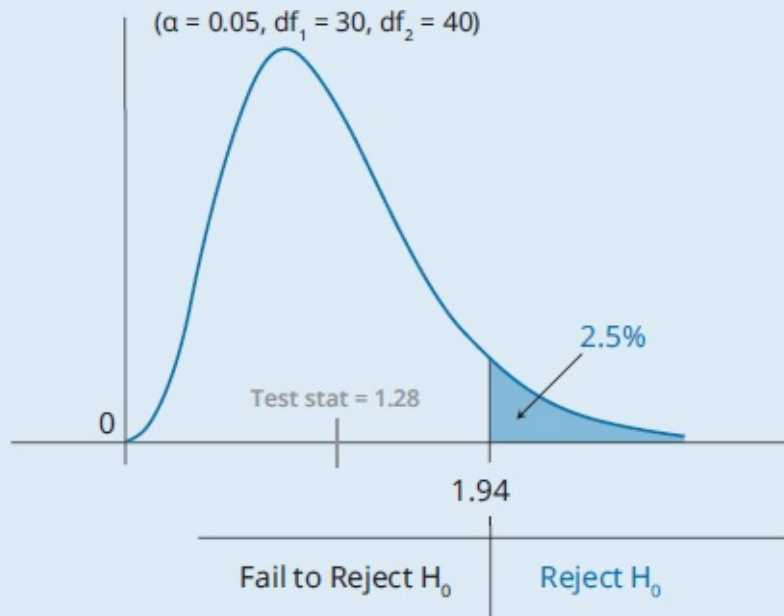
F -Table, Critical Values, 2.5% in Upper Tail

Degrees of Freedom for the Denominator	Degrees of Freedom for the Numerator		
		30	40
30		2.07	2.01
40		1.94	1.88
60		1.82	1.74

Given 30 degrees of freedom in the numerator and 40 degrees of freedom in the denominator, the critical F -value is 1.94. The decision rule, illustrated in the following diagram, can be stated as follows:

Reject H_0 if F -stat > 1.94

Decision Rule for F -Test



Because the calculated F -statistic of 1.28 is less than the critical F -statistic of 1.94, the analyst cannot reject the null hypothesis and should conclude that the earnings variances of the industries are not significantly different from one another at a 5% level of significance.

Parametric Test of Correlation

The correlation coefficient, ρ , sometimes referred to as the Pearson correlation, measures the strength of a linear relationship between two variables. The coefficient takes a value between -1 , which represents a perfect negative linear relationship, and $+1$, representing a perfect positive linear relationship. If the correlation between two variables is zero, they have no linear relationship.

We often wish to test whether a nonzero sample correlation coefficient, r , implies that the true correlation coefficient is nonzero. These are the appropriate hypotheses for this test:

- | H_0 : true population correlation coefficient is zero: $\rho_{X,Y} = 0$
- | H_A : true population coefficient is not zero: $\rho_{X,Y} \neq 0$

The test statistic for the correlation coefficient is calculated as:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

where:

r = sample correlation

n = sample size

This test statistic follows a t -distribution with $n - 2$ degrees of freedom. The test statistic increases, not only with the sample correlation coefficient, but also with sample size.



PROFESSOR'S NOTE

It makes sense that the test statistic increases with n , because the larger the sample size, the more confidence we can have in our observed sample correlation not just being random noise. As the observed sample correlation approaches $+1$ or -1 , the denominator gets smaller, once again increasing the test statistic and making it more likely we will reject the null hypothesis that the true correlation coefficient is zero.

EXAMPLE: Test of the hypothesis that the population correlation coefficient equals zero

A researcher computes the sample correlation coefficient for two normally distributed random variables as 0.35, based on a sample size of 42. Determine

whether to reject the hypothesis that the population correlation coefficient is equal to zero at a 5% significance level.

Answer:

$$\text{Our test statistic is } \frac{0.35 \sqrt{42 - 2}}{\sqrt{1 - 0.35^2}} = 2.363$$

Using the t -table with $42 - 2 = 40$ degrees of freedom for a two-tailed test and a significance level of 5%, the critical value is 2.021. Because our computed test statistic of 2.363 is greater than 2.021, we reject the hypothesis that the population correlation coefficient is zero. That is, the two populations are correlated—in this case, positively.

p -Hacking

A **p -value** is the probability of obtaining a test statistic that would lead to a rejection of the null hypothesis, assuming the null hypothesis is true. It is the smallest level of significance for which the null hypothesis can be rejected.

The expression **p -hacking** refers to illegitimate manipulation of the statistical testing process, typically carried out to generate a false positive result—in other words, to fabricate evidence that a claim is true when in fact it is not true. Examples where this has been known to occur in investment management include the following:

- | Selective disclosure by managers to present performance metrics more favorably than they really are, such as by testing only their best years of performance data or removing the worst performance days as outliers.
- | “Data mining” by performing many tests until the analyst finds evidence to reject the null hypothesis. This is likely to lead to false positives (Type I errors). Recall that with a significance level of 5%, we expect to make a Type I error once every 20 tests on average (i.e., 5% of the time). Therefore, doing many tests will lead to more Type I errors, rather than increase accuracy.
- | Manipulating performance data to inflate the apparent risk-adjusted return of strategies. An example is by using appraisal-based data, which have artificially low volatility, when estimating Sharpe ratios (excess return per unit of risk).



MODULE QUIZ 7.3

1. Which of the following assumptions is *least likely* required for the difference-in-means test based on two samples?

- A. The two samples are independent.
 - B. The two populations are normally distributed.
 - C. The two populations have known variances.
2. The appropriate test statistic for a test of the equality of variances for two normally distributed random variables, based on two independent random samples, is the:
- A. t -test.
 - B. F -test.
 - C. χ^2 test.
3. For a parametric test of whether a correlation coefficient is equal to zero, it is *least likely* that:
- A. degrees of freedom are $n - 1$.
 - B. the test statistic follows a t -distribution.
 - C. the test statistic increases with a greater sample size.

MODULE 7.4: NONPARAMETRIC HYPOTHESIS TESTS

LOS 7.c: Compare and contrast parametric and non-parametric tests, describe situations in which each is the more appropriate type of test, construct appropriate hypothesis tests, and interpret the results.

All the tests discussed so far in this reading have been parametric tests because they have made assumptions about the distribution of the data and directly tested assumptions about parameters of these distributions (e.g., mean and variance). We now turn our attention to **nonparametric hypothesis tests**, which are used for quantities other than the parameters of a statistical distribution, or when distributional assumptions behind parametric tests do not hold.

An example of a nonparametric test would be a hypothesis test of the mean for a variable that comes from a non-normal distribution and based on a small sample, so that neither the t -test nor the z -test is appropriate because the CLT does not apply. While nonparametric tests are more flexible, they tend to be less accurate than parametric tests when the assumptions of parametric tests are met by the data.

Spearman Rank Correlation Test

The **Spearman rank correlation test** can be used to test whether two sets of ranks are related, where ranks are simply ordered values. The rank correlation coefficient takes a value between -1 and $+1$ and measures the strength of the

monotonic relationship between two variables. If two variables have a perfect monotonic relationship, when one of them increases, the other one either always increases or always decreases.

The Spearman rank correlation, r_s (when all ranks are integer values), is calculated as follows:

$$r_s = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$$

where:

r_s = rank correlation

n = sample size

d_i = difference between two ranks of paired observations from two populations



PROFESSOR'S NOTE

While it might not be obvious, the formula just listed results from calculating the Pearson correlation of the ranks of our sample of paired observations in the usual way. This formula is not used directly for any calculations in the curriculum learning module, so it should not be prioritized. We only restate it here for completeness.

We can test the significance of the Spearman rank correlation with the same test statistic we used for estimating the significance of a parametric (Pearson) correlation coefficient:

$$\frac{r_s \sqrt{n-2}}{\sqrt{1-r_s^2}}$$

For large samples ($n > 30$), the test statistic follows a t -distribution with $n - 2$ degrees of freedom.

Tests of Independence

When faced with data that group observations into categories, we can use a nonparametric test for independence between categories.

The data needed for a test of independence are presented in a **contingency table**, which classifies observations for two potentially related categories. Figure 7.9 is a contingency table where the categories are earnings growth (low, medium, or high) and dividend yield (low, medium, or high). We can use the data in the table

to test the hypothesis that the two categories, earnings growth and dividend yield, are independent.

- | H_0 : two categories are independent
- | H_A : two categories are not independent

Figure 7.9: Contingency Table for Categorical Data

Earnings Growth	Dividend Yield			Total
	Low	Medium	High	
Low	28	53	42	123
Medium	42	32	39	113
High	49	25	14	88
Total	119	110	95	324

We index our three categories of earnings growth from low to high with $i = 1, 2$, or 3 , and our three categories of dividend yield from low to high with $j = 1, 2$, or 3 . From the table, we see in Cell 1,1 that 28 firms have both low earnings growth and low dividend yield. We see in Cell 3,2 that 25 firms have high earnings growth and medium dividend yields.

To carry out our test for independence, we must calculate the expected number of observations in each cell *if the categories were independent*, $E_{i,j}$:

$$E_{i,j} = \frac{\text{total for Row } i \times \text{total for Column } j}{\text{total number of observations}}$$

For example, the expected number of observations for Cell 1,1 if earnings growth and dividend yield are independent is:

$$E_{1,1} = \frac{\text{total for Row 1} \times \text{total for Column 1}}{\text{total number of observations}} = \frac{123 \times 119}{324} = 45.2$$



PROFESSOR'S NOTE

The intuition here is that if earnings growth and dividend yield are not related, then we would expect the 123 companies with low earnings growth to be spread across dividend yield categories in the same proportions as the general split across dividend yield categories when we ignore earnings growth (i.e., split proportionally as per column totals). If actual observations for Cell 1,1 differ significantly from these expectations, then it implies the two categories may be related and

therefore are not independent.

Figure 7.10 shows the expected frequencies for each cell in our earnings growth and dividend yield contingency table.

Figure 7.10: Contingency Table for Expected Frequencies

Earnings Growth	Dividend Yield		
	Low	Medium	High
Low	45.2	41.8	36.1
Medium	41.5	38.4	33.1
High	32.3	29.9	25.8

For our test, we compare the actual observations, $O_{i,j}$ to expected values $E_{i,j}$ if the two characteristics were independent. The test statistic is a chi-square statistic calculated as follows:

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

where:

O_{ij} = number of observations in Cell i,j : Row i and Column j (i.e., observed frequency)

E_{ij} = expected number of observations for Cell i,j (calculated as per the equation just listed)

r = number of row categories

c = number of column categories

The degrees of freedom are $(r - 1) \times (c - 1)$, which is $(3 - 1) \times (3 - 1) = 4$ in our example.



PROFESSOR'S NOTE

All this formula is doing is looking at the difference between the number of actual observations and expected observations under the assumption of independence, squaring this difference, then expressing this as a proportion of the amount expected. This is done for each cell and then summed across all cells. The more our actual observations deviate from what is expected under independence, the higher this statistic will be, and the more likely we are to reject the null hypothesis of independence.

For example, calculating our test statistic for Cell 1,1 gives:

$$\frac{(28 - 45.2)^2}{45.2} = 6.5451$$

Figure 7.11 shows the results of applying this calculation to each cell of the contingency table. The test statistic is calculated as the sum of all nine cells, equal to 27.47.

Figure 7.11: Squared Differences from Contingency Table

Earnings Growth	Dividend Yield		
	Low	Medium	High
Low	6.5451	3.0010	0.9643
Medium	0.0060	1.0667	1.0517
High	8.6344	0.8030	5.3969
Sum = $\chi^2 = 27.4691$			

The critical value for a significance level of 5% (from a chi-square table) with 4 degrees of freedom is 9.488. Based on our sample data, we can reject the hypothesis that the earnings growth and dividend yield categories are independent.

To identify which cells are contributing to the lack of independence, the standardized residual (Pearson residual) for each cell can be calculated as follows:

$$\text{standardized residual} = \frac{O_{ij} - E_{ij}}{\sqrt{E_{ij}}}$$

This standardized residual for each cell follows approximately a standard normal distribution, so we can conclude that a cell is significantly contributing to independence with 95% confidence when it has a value of less than -2 or greater than 2.

For example, the standardized residual for Cell 1,1 is calculated as:

$$\frac{28 - 45.2}{\sqrt{45.2}} = -2.56$$

Because this standardized residual is less than -2, we can conclude that there were significantly fewer observations of low earning growth stocks with low dividend yield.

Figure 7.12 shows the standardized residuals for the example involving earnings growth and dividend yield across all cells of the contingency table.

Figure 7.12: Standardized Residuals for Contingency Table

Earnings Growth	Dividend Yield		
	Low	Medium	High
Low	−2.56	1.74	0.99
Medium	0.08	−1.03	1.02
High	2.93	−0.89	−2.32

We can see from the table just listed that the significant deviations from expectations which caused the rejection of the null hypothesis of independence occurred in Cell 1,1 (−2.56), Cell 3,1 (2.93) and Cell 3, 3 (−2.32).



PROFESSOR'S NOTE

The absolute size of the standardized residual is just the square root of the numbers used to calculate the chi-square statistic in Figure 7.11. Hence, a shortcut is to simply square root the numbers in Figure 7.11, and if they are greater than 2, then they are contributing significantly to a lack of independence.



MODULE QUIZ 7.4

1. The test statistic for a Spearman rank correlation test for a sample size greater than 30 follows a:
 - A. *t*-distribution.
 - B. normal distribution.
 - C. chi-square distribution.
2. A contingency table can be used to test:
 - A. a null hypothesis that rank correlations are equal to zero.
 - B. whether multiple characteristics of a population are independent.
 - C. the number of *p*-values from multiple tests that are less than adjusted critical values.
3. The nonparametric test for independence of categorical data is a:
 - A. *z*-test.
 - B. chi-squared test.
 - C. *F*-test.
4. The test statistic for a nonparametric test of categorical data is calculated as the sum of the:
 - A. squared deviations of actual versus expected observations as a proportion of expected observation.

- B. squared deviations of actual versus expected observations as a proportion of expected observation squared.
 - C. deviations of actual versus expected observations.
5. An analyst is testing whether high accruals in corporate earnings lead to poor subsequent stock price performance. They take a sample of 300 companies and divide them into categories of low accruals, high accruals, good subsequent performance, and poor subsequent performance. The total number of companies observed with low accruals is 120, and the total number of companies with good performance is 160. What is the expected number of companies with both low accruals and good performance under the assumption of independence?
- A. 64.
 - B. 140.
 - C. 280.

KEY CONCEPTS

LOS 7.a

The central limit theorem states that for any population with a mean μ and a finite variance σ^2 , the distribution of the sample mean of all possible samples of size n (for $n \geq 30$) will be approximately normally distributed with a mean equal to μ and a variance equal to σ^2/n .

The standard error of the sample mean is calculated as $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$ (where the population standard deviation is known) or as $s_{\bar{X}} = \frac{s}{\sqrt{n}}$ (where the population standard deviation is unknown).

Desirable statistical properties of an estimator include unbiasedness (expected value of estimator is the true population parameter), efficiency (lower sampling error than any other estimator), consistency (variance of sampling error decreases with larger sample size), and robustness (estimator is not sensitive to outliers in the data).

Confidence intervals are ranges of values within which the actual value of the parameter is expected to lie with a given probability:

$$\text{confidence interval} = \text{point estimate} \pm (\text{reliability factor} \times \text{standard error})$$

For a normally distributed population, a confidence interval for its mean can be constructed using a z -statistic when variance is known, and a t -statistic when the variance is unknown.

$$\bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \text{ or}$$

$$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$

The z-statistic is acceptable if the sample size is large (30+).

Value at risk (VaR) measures a threshold for losses in a portfolio over a given time frame with a specified probability. Under the parametric method that assumes normality, and when the mean return is assumed to be zero, the parametric $\alpha\%$ VaR for a portfolio of size P is calculated as:

$$\text{VaR} = z_{\alpha} \times \sigma \times P$$

Probability sampling methods ensure that each population member has a known nonzero chance of being selected in a sample. Methods include simple random sampling, systematic sampling, stratified random sampling, and cluster sampling. Nonprobability sampling techniques include convenience sampling and judgmental sampling.

LOS 7.b

Hypothesis tests require a statement of a null and an alternative hypothesis, the selection of the appropriate test statistic, specification of the significance level, and a decision rule.

A two-tailed test results from a two-sided alternative hypothesis (e.g., $H_A: \mu \neq \mu_0$). A one-tailed test results from a one-sided alternative hypothesis (e.g., $H_A: \mu > \mu_0$, or $H_A: \mu < \mu_0$).

The test statistic is the calculated value upon which a decision about a null hypothesis will be based. For a test of the mean of a distribution:

$$\text{test statistic} = \frac{\text{sample statistic} - \text{hypothesized value}}{\text{standard error of the sample statistic}}$$

A hypothesis about a population parameter is rejected when the sample statistic lies outside a confidence interval around the hypothesized value for the chosen level of significance.

A Type I error is rejecting the null hypothesis when it is actually true. A Type II error is failing to reject the null hypothesis when it is actually false.

The significance level can be interpreted as the probability that a test statistic will reject the null hypothesis by chance when it is true (i.e., the probability of a

Type I error). A significance level must be specified to select the critical values for the test.

The power of a test is the probability of rejecting the null when it is false. The power of a test = $1 - P(\text{Type II error})$.

For two independent samples from two normally distributed populations, the difference in means can be tested with a t -statistic with degrees of freedom $n_1 + n_2 - 2$.

A paired comparisons test is concerned with the mean of the differences between the paired observations of two dependent, normally distributed samples. This test can be conducted on the sample of differences in the same way as a t -test for a single mean.

A hypothesis test of the population variance for a normally distributed population uses a chi-square test statistic. Degrees of freedom are $n - 1$.

A hypothesis test comparing two variances based on independent samples from two normally distributed populations uses an F -distributed test statistic.

To test a hypothesis that a population correlation coefficient equals zero, the appropriate test statistic is a t -statistic with $n - 2$ degrees of freedom.

The term p -hacking refers to describe deliberate manipulation of hypothesis testing to generate false positives (i.e., Type I errors).

LOS 7.c

A nonparametric test of correlation can be performed on ranked data. The Spearman rank correlation test examines whether the ranks for multiple periods are correlated. The test statistic follows a t -distribution for samples sizes greater than 30.

A contingency table is used to test the hypothesis that two characteristics of a sample of items are independent. A contingency table shows categorical data, counting the number of sample items with different levels of the characteristics. The test statistic follows a chi-square distribution with degrees of freedom equal to $(r - 1) \times (c - 1)$. If the test statistic is greater than the critical chi-square value for a given level of significance, we reject the hypothesis that the two characteristics are independent.

The standardized residual for each cell of a contingency table follows a standard normal distribution. We can conclude that cells with a standardized residual of less than -2 or greater than $+2$ are contributing to a finding of dependence between categories.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 7.1

- . **A** Efficiency, consistency, unbiasedness, and robustness are the four desirable properties of an estimator. (LOS 7.a)
- . **A** Because the population variance is known and $n \geq 30$, the confidence interval is determined as $\bar{x} \pm z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$. $z_{\frac{\alpha}{2}} = z_{0.025} = 1.96$. So, the confidence interval is $75 \pm 1.96(20/10) = 75 \pm 3.92 = 71.08$ to 78.92 . (LOS 7.a)
- . **C** Use the t -statistic at $\alpha/2$ and $n - 1$ degrees of freedom when the population variance is unknown. While the z -statistic is acceptable when the sample size is large, sample size is not given here, and the t -statistic is always appropriate under these conditions. (LOS 7.a)
- . **C** VaR states a minimum value that we could lose with a specified probability. If the weekly 5% VaR of a portfolio is £1 million, this means there is a 5% chance in any given week that we could lose £1 million or more. (LOS 7.a)
- . **C** VaR is the loss level for the portfolio where there is a 5% chance of experiencing this loss (or worse) in any given year. Under the parametric method, we assume portfolio returns are normal. Therefore, to identify the 5% VaR, we need to move $z_{5\%} = 1.65$ standard deviations from the mean returns to arrive at the 5% VaR. With a mean of zero, the 5% VaR is then calculated as:

$$5\% \text{ VaR} = z_{5\%} \times \sigma \times P = 1.65 \times 0.09 \times \$6.4 \text{ million} = \$950,400. \text{ (LOS 7.a)}$$

Module Quiz 7.2

- . **C** Rejecting the null when it is actually true is a Type I error. A Type II error is failing to reject the null hypothesis when it is false. The significance level equals the probability of a Type I error. (LOS 7.b)

- . **C** A Type I error is rejecting the null hypothesis when it is true. The probability of rejecting a false null is $[1 - \text{Prob Type II}] = [1 - 0.60] = 40\%$, which is called the power of the test. The other answer choices are not necessarily true because the null may be false, and the probability of rejection may be unknown. (LOS 7.b)
- . **A** The power of a test is $1 - P(\text{Type II error}) = 1 - 0.15 = 0.85$. (LOS 7.b)
- . **C** The standard error is $\frac{5.29}{\sqrt{28}} = 1.00$. Test statistic = $1.645 / 1.0 = 1.645$. The critical value for a t -test is greater than the critical value for a z -test at a 5% significance level (1.645 for a one-tailed test), so the calculated test statistic of 1.645 must be less than the critical value for a t -test (1.703 for a one-tailed test with 27 degrees of freedom), and we cannot reject the null hypothesis that mean excess return is less than or equal to zero. (LOS 7.b)
- . **A** With a small sample size, a t -test may be used if the population is approximately normally distributed. If the population has a nonnormal distribution, no test statistic is available unless the sample size is large. (LOS 7.b)

Module Quiz 7.3

- . **C** The difference-in-means test does not require the two population variances to be known. (LOS 7.b)
- . **B** The F -test is the appropriate test. (LOS 7.b)
- . **A** Degrees of freedom are $n - 2$ for a test of the hypothesis that correlation is equal to zero. The test statistic increases with sample size (degrees of freedom increase) and follows a t -distribution. (LOS 7.b)

Module Quiz 7.4

- . **A** The test statistic for a Spearman rank correlation test follows a t -distribution. (LOS 7.c)
- . **B** A contingency table is used to determine whether two characteristics of a group are independent. (LOS 7.c)
- . **B** The nonparametric test for independence of categorized data is a chi-squared test with $(r - 1) \times (c - 1)$ degrees of freedom, where r and c are the

number of rows and columns respectively in the contingency table. (LOS 7.c)

- . **A** The test statistic sums the squared deviations of actual versus expected observations as a proportion of expected observations for each cell in the contingency table. (LOS 7.c)
- . **A** For this test, we can assume rows in the contingency table related to low/high accruals, and columns relate to good/bad subsequent performance. Then:

$$\begin{aligned}\text{expected observations} &= \frac{\text{total for row} \times \text{total for column}}{\text{total number of observations}} \\ &= \frac{120 \times 160}{300} = 64\end{aligned}$$

This calculation is simply saying that if there were no relationship between accruals and performance, then we would expect the 120 companies with low accruals to be split proportionally to the overall number of companies that have good performance versus bad performance, regardless of the accruals level. (LOS 7.c)

READING 8

THE RETURN AND RISK OF A FINANCIAL PORTFOLIO



PROFESSOR'S NOTE

This reading introduces modern portfolio theory. This material is the basis for the Portfolio Construction topic area, and we will repeat much of it there.

While the calculations are necessary (and testable on the exam), they are there to support concepts that are quite straightforward. We will state the basics of those concepts here, and develop them throughout this reading and in Portfolio Construction. Keeping these key points in mind as you go along will make it easier to grasp what the calculations are telling us.

- *Risk*, in the context we use it here, refers to variability of an asset's or portfolio's rate of return. We measure it as the standard deviation of returns. Defined this way, risk includes both upside and downside.
- If an investment has no risk, we expect it to earn the risk-free rate of return.
- A single asset's risk has two components: *unsystematic risk* that is specific to that asset, and *systematic risk* that comes from the market in which it exists.
- Risk and expected return are a tradeoff. The more risk an investment has, the more return an investor must expect the investment to provide, for the investor to be willing to accept that risk.
- If two investments have the same expected return, a rational investor will prefer the one that has less risk. If two investments have the same risk, a rational investor will prefer the one with a greater expected return.

- If two risky assets have rates of return that are not perfectly correlated, it is possible to combine them in a portfolio that will have less risk than the weighted average of the securities' risks. This is the *diversification effect*.
- The diversification effect diminishes as we add more assets to a portfolio. After about 30 assets, a portfolio's risk and expected return approach those of the overall market. At that point, we say a portfolio has diversified away the unsystematic risk of the assets in it.
- Because it is possible to diversify away unsystematic risk, the market does not reward investors for taking it. Therefore, an asset's expected return only reflects its degree of exposure to the market (systematic risk).
- The best risk-return tradeoff an investor can achieve is some combination of a fully diversified portfolio and a risk-free asset. What combination is optimal depends on how risk averse an investor is.

MODULE 8.1: PORTFOLIO EXPECTED RETURN

LOS 8.a: Calculate, interpret, and evaluate the expected return, variance, standard deviation, covariance, and correlation of portfolio returns.

Effective portfolio management depends on various types of portfolio statistics, including expected returns, standard deviations, and correlations. These statistics can either be historical (which factors in past risk and return information) or forward-looking (which normally requires model-based forecasts).

Analysts can use historical returns to guide realistic expectations, but the data may be biased if the history is not long. Forward-looking returns depend on economic or financial forecasts, but these may incorporate historical information. With either methodology, analysts must re-assess their expectations regularly to see whether they need to make tactical or strategic adjustments.

The **historical return of a portfolio** for a period is simply the weighted average of the realized returns for each asset in the portfolio:

$$r_{P,t} = \sum_{i=1}^N w_{i,t-1} \times r_{i,t}$$

where:

N = number of assets in portfolio

$w_{i,t-1}$ = weight of asset i at the beginning of the period

$r_{i,t}$ = realized return of the asset i in period t

If we are evaluating multiple periods, we must incorporate the cumulative return of the assets over all periods. If we do not rebalance the portfolio during the entire time period (T), the cumulative portfolio return remains simply the weighted average of the assets' returns over the period, using the initial weighting of each asset:

$$r_{P,T} = \sum_{i=1}^N w_{i,0} \times r_{i,T}$$

This cumulative return differs from the arithmetic average of returns across all of these periods because the arithmetic average return does not incorporate compounding.

To make the effect of compounding explicit, we can express a portfolio's compound return over T periods as follows:

$$r_{P,T} = \sum_{i=1}^N w_{i,0} \left[\prod_{t=1}^T (1 + r_{i,t}) - 1 \right]$$

where:

$\prod_{t=1}^T (1 + r_{i,t}) - 1$ is the cumulative (or aggregate) return of each asset over all periods

This cumulative approach assumes no reallocation of capital, and that all distributions are reinvested into the same asset.

The geometric average of the portfolio returns can be calculated as follows:

$$\overline{r}_{gP} = (1 + r_{P,T})^{\frac{1}{T}} - 1$$

As the markets fluctuate over time, the weights of the individual assets in the portfolio will change as some assets decline in value, while others increase or stay the same. This is known as **portfolio drift**. Without rebalancing, over time this can lead to **style drift**, where the portfolio moves out of alignment with its original strategy. For example, in a portfolio that is meant to be balanced

between stocks and bonds, a strong equity market can result in an overweight stock position.

Style drift can also occur if the individual holdings have changes to their investment characteristics. One example is when a stock moves into a different capitalization category due to the company's growth, such as a small-cap stock moving to the mid-cap category.

While portfolio drift might sometimes offset style drift, it is appropriate for a portfolio manager to regularly monitor and rebalance holdings to ensure that a portfolio is maintaining its original investment goals.

Because portfolio drift changes the weights for the various assets, return calculations must be adjusted each period to reflect the new weightings. This is the cumulative return for a portfolio with changing weights:

$$r_{P,T} = \sum_{t=1}^T \sum_{i=1}^N w_{i,t-1} \times r_{i,t}, \text{ which equals } \sum_{t=1}^T r_{P,t}$$

The arithmetic average with changing weights is calculated as follows:

$$\bar{r}_P = \frac{1}{T} \sum_{t=1}^T r_{P,t}$$

This is the geometric or cumulative return over T periods:

$$r_{P,T} = \prod_{t=1}^T \left(1 + \sum_{i=1}^N w_{i,t-1} \times r_{i,t} \right) - 1$$

Finally, this is the geometric average return with changing weights:

$$\bar{r}_{g,P} = \left[\prod_{t=1}^T \left(1 + \sum_{i=1}^N w_{i,t-1} \times r_{i,t} \right) \right]^{\frac{1}{T}} - 1$$

EXAMPLE: Calculating returns with changing weights

Maya Rivera has been managing a two-asset portfolio for the past three quarters. Rivera needs to show her client both their quarterly returns and the cumulative return over this time frame. The initial allocation was \$100 for each asset. The individual returns for the assets are as follows:

Asset	Quarter 1	Quarter 2	Quarter 3
A	5%	-2%	2%
B	16%	-8%	9%

Calculate the arithmetic and geometric returns over the three quarters.

Answer:

Step 1: Calculate the return for Quarter 1 using the original weightings.

$$\text{Portfolio return} = (0.50 \times 5\%) + (0.50 \times 16\%) = 10.5\%$$

$$\text{Value of Asset A at the end of Quarter 1} = \$100 \times 1.05 = \$105.00$$

$$\text{Value of Asset B at the end of Quarter 1} = \$100 \times 1.16 = \$116.00$$

$$\text{New weight of Asset A at the beginning of Quarter 2} = 105.00 / 221.00 = 47.51\%$$

$$\text{New weight of Asset B at the beginning of Quarter 2} = 116.00 / 221.00 = 52.49\%$$

Step 2: Calculate the return for Quarter 2.

$$\text{Portfolio return} = (0.4751 \times -2\%) + (0.5249 \times -8\%) = -5.15\%$$

$$\text{Value of Asset A at the end of Quarter 2} = \$105.00 \times (1 - 0.02) = \$102.90$$

$$\text{Value of Asset B at the end of Quarter 2} = \$116.00 \times (1 - 0.08) = \$106.72$$

$$\text{New weight of Asset A at the beginning of Quarter 3} = 102.90 / 209.62 = 49.09\%$$

$$\text{New weight of Asset B at the beginning of Quarter 3} = 106.72 / 209.62 = 50.91\%$$

Step 3: Calculate the return for Quarter 3.

$$\text{Portfolio return} = (0.4909 \times 2\%) + (0.5091 \times 9\%) = 5.56\%$$

$$\text{The cumulative arithmetic return over the three periods is } 10.5\% - 5.15\% + 5.56\% = 10.91\%.$$

$$\text{The cumulative geometric return over the three periods is } (1 + 0.105) \times (1 - 0.0514) \times (1 + 0.056) - 1 = 10.69\%.$$

As mentioned, without rebalancing, a portfolio composition may deviate from its original risk-return profile; this, in turn, will affect the portfolio's performance. The degree of impact will depend on the weighting approaches that were used in the original allocation. With an equal-weighted portfolio, portfolio drift causes securities with high returns to increase in weight and causes the portfolio's performance to become biased toward higher-growth or higher-volatility assets. With a portfolio that is not equally weighted, portfolio drift causes assets with large allocations to increase in weight. This causes the portfolio's performance to

resemble that of a momentum-driven strategy, and it may also increase the portfolio's concentration risk.



MODULE QUIZ 8.1

1. A portfolio has the following returns:

Period 1: 11.0%

Period 2: 6.0%

Period 3: -3.0%

The geometric average return over this time frame is *closest* to:

A. 4.5%.

B. 4.7%.

C. 6.6%.

2. Due to significant growth in their consumer products division, Able Corporation has recently been reclassified from a small-cap company to a mid-cap company. For investors in portfolios that include Able, this *most likely* represents:

A. style drift.

B. portfolio drift.

C. a rebalancing effect.

MODULE 8.2: PORTFOLIO RISK MEASURES

The risk of assets and portfolios is measured in terms of how much their returns differ from the mean. Variance and standard deviation are the two metrics that quantify this risk. The **variance of portfolio returns**, identified as σ_P^2 , is calculated using the weights of the assets in the portfolio, the variances of the individual assets, and the covariances between pairs of assets. The formula is:

$$\sigma_P^2 = \sum_{i=1}^N \sum_{j=1}^N w_i w_j \sigma_{i,j}$$

where:

$\sigma_{i,j}$ = covariance between the returns for assets i and j

For a two-asset portfolio, this can be written more simply as:

$$\sigma_P^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \text{Cov}_{A,B}$$

The standard deviation is the square root of the variance. Because the standard deviation is measured in the same units as the portfolio returns, it is a preferred method to show the overall risk:

$$\sigma_P = \sqrt{\sigma_P^2} = \sqrt{\sum_{i=1}^N \sum_{j=1}^N w_i w_j \sigma_{i,j}}$$

Covariances between the assets' returns may not be known. An analyst can use expected values for the assets' covariances of returns to model an expected value for the portfolio variance.

We can express covariance in terms of the correlation between assets' returns:

$$\sigma_{ij} = \rho_{ij} \times \sigma_i \times \sigma_j$$

where:

ρ_{ij} = the correlation of returns between assets i and j

This calculation shows that covariance is dependent on the correlation between the assets and the standard deviation. However, this cannot be applied to an expected value for covariance because joint variabilities and dependencies must be taken into account. All three of the variables in this equation have interdependencies and nonlinear interactions that affect the calculation.

EXAMPLE: Calculate risk and return when weights of assets are varied

Portfolio manager Marcus White wants to build a presentation that provides examples of different risk-return relationships when allocations are altered. Asset 1 has an expected return of 10.25% and a standard deviation of 16.1%. Asset 2 has an expected return of 7.4% and a standard deviation of 14.5%. The two assets have a covariance of returns of 0.014. White is considering the following portfolio mixes:

Portfolio	Asset 1	Asset 2
1	100%	0%
2	75%	25%
3	25%	75%
4	0%	100%

Calculate the expected return and expected variance of these four portfolio combinations.

Answer:

Portfolio 1: Expected return = $(1.0 \times 10.25\%) + (0.0 \times 7.4\%) = 10.25\%$

Expected variance = $(1.0^2 \times 0.161^2) + (0.0^2 \times 0.145^2) + (2 \times 1.0 \times 0.0 \times 0.014) = 0.0259$

Expected standard deviation = $\sqrt{0.0259} = 16.09\%$

Portfolio 2: Expected return = $(0.75 \times 10.25\%) + (0.25 \times 7.4\%) = 9.54\%$

Expected variance = $(0.75^2 \times 0.161^2) + (0.25^2 \times 0.145^2) + (2 \times 0.75 \times 0.25 \times 0.014) = 0.0211$

Expected standard deviation = $\sqrt{0.0211} = 14.54\%$

Portfolio 3: Expected return = $(0.25 \times 10.25\%) + (0.75 \times 7.4\%) = 8.11\%$

Expected variance = $(0.25^2 \times 0.161^2) + (0.75^2 \times 0.145^2) + (2 \times 0.25 \times 0.75 \times 0.014) = 0.0187$

Expected standard deviation = $\sqrt{0.0187} = 13.66\%$

Portfolio 4: Expected return = $(0.0 \times 10.25\%) + (1.0 \times 7.4\%) = 7.4\%$

Expected variance = $(0.0^2 \times 0.161^2) + (1.0^2 \times 0.145^2) + (2 \times 0 \times 1 \times 0.014) = 0.0210$

Expected standard deviation = $\sqrt{0.0210} = 14.5\%$



MODULE QUIZ 8.2

1. In a portfolio of small-cap and mid-cap stocks, the small-cap stocks have an expected return of 14% with a standard deviation of 16%, while the mid-cap stocks have an expected return of 12% with a standard deviation of 8%. Mid-cap and small-cap stocks have a returns correlation of 0.40. The covariance of returns between mid-cap and small-cap stocks is *closest* to:
A. 0.0051.
B. 0.0067.
C. 0.0096.
2. The Smith family portfolio contains two assets. The initial weighting of each asset was \$100, and the performance over three periods was as follows:

Period	Return on Asset 1	Return on Asset 2
1	6.3%	2.1%
2	8.5%	2.7%

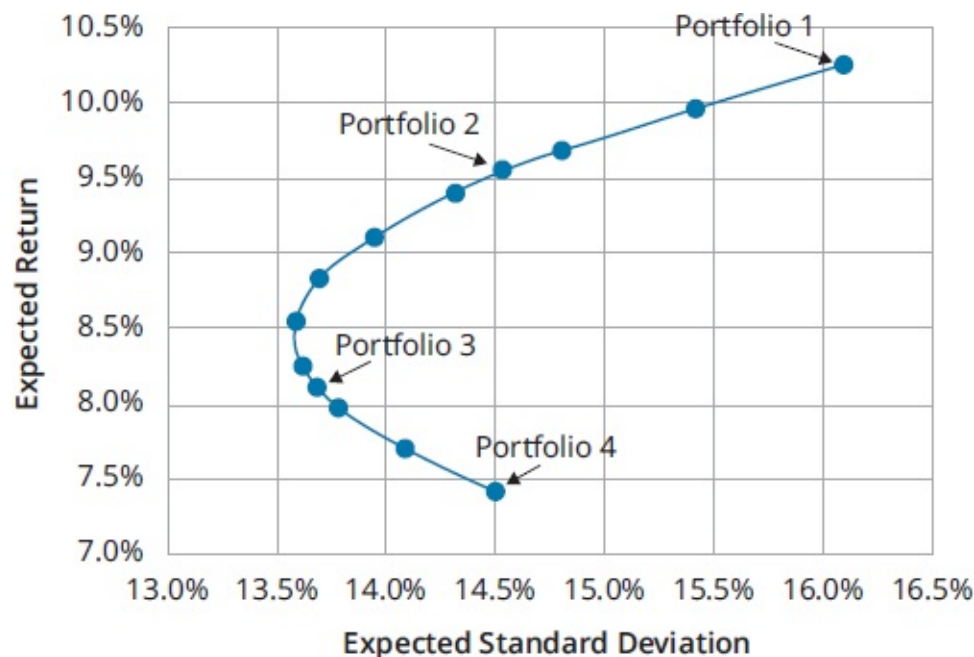
What is the approximate weight of Asset 1 at the end of Period 2?

- A. 47.62%.
- B. 51.01%.
- C. 52.38%.

MODULE 8.3: CORRELATION AND DIVERSIFICATION BENEFITS

Changing the weightings of assets in a portfolio affects both its risk profile and its expected returns. If the assets do not have a high correlation to each other, combining the assets creates a **diversification benefit** that can help to reduce the overall risk of the portfolio. This benefit is reflected in the concave shape of the curve shown in Figure 8.1, which is a mix of various weights of the two assets in the portfolio we used in our earlier example. Moving along the curve, note that portfolio risk decreases more than the decrease in portfolio return.

Figure 8.1: Diversification Benefit Using Different Portfolio Weights



The curve shown in Figure 8.1 shows the risk and return characteristics of the **feasible portfolios** an investor can create using these two assets. Of these portfolios, the **efficient set** includes those that have the highest return for any given level of risk.



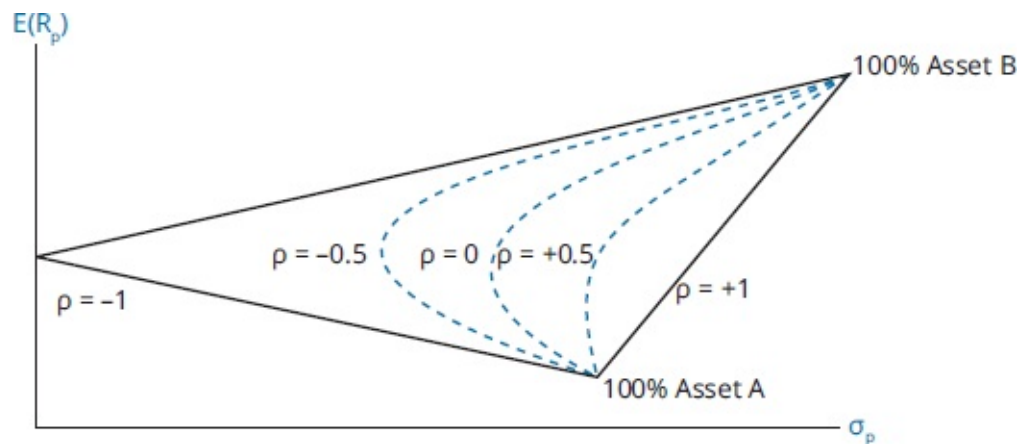
PROFESSOR'S NOTE

Later in this reading, we will use this concept to define the *efficient frontier* from among portfolios that theoretically include all investable assets.

Correlations between assets' returns drive the risk-return relationship in portfolios. Portfolio risk decreases as the correlation between the assets' returns decreases. That is, the lower the correlation of asset returns, the greater the risk reduction (diversification) benefit of combining assets in a portfolio. If asset returns were perfectly negatively correlated, portfolio risk could be eliminated altogether for a specific set of asset weights.

We show these relations graphically in Figure 8.2 by plotting the portfolio risk and return for all portfolios of two risky assets, for specific values of the assets' returns correlation.

Figure 8.2: Risk and Return for Different Values of ρ



Increasing the number of assets in the portfolio creates more complexity in assessing its risk and return. For example, if we add a third asset, we need to account for all the elements in the **variance-covariance matrix** shown in the first panel of Figure 8.3. Adding a fourth asset increases the number of covariances, as shown in the second panel.

Figure 8.3: Variance-Covariance Matrix for Three-Asset and Four-Asset Portfolios

σ_1^2	$\sigma_{1,2}$	$\sigma_{1,3}$
$\sigma_{2,1}$	σ_2^2	$\sigma_{2,3}$
$\sigma_{3,1}$	$\sigma_{3,2}$	σ_3^2

σ_1^2	$\sigma_{1,2}$	$\sigma_{1,3}$	$\sigma_{1,4}$
$\sigma_{2,1}$	σ_2^2	$\sigma_{2,3}$	$\sigma_{2,4}$
$\sigma_{3,1}$	$\sigma_{3,2}$	σ_3^2	$\sigma_{3,4}$
$\sigma_{4,1}$	$\sigma_{4,2}$	$\sigma_{4,3}$	σ_4^2

The number of covariances increases exponentially as the portfolio size increases. For a portfolio of N assets, the number of covariances is $(N^2 - N) / 2$. A portfolio of 30 assets has $(900 - 30) / 2 = 435$ unique covariances.

We can normalize a covariance matrix to create a **correlation matrix**. This matrix can help investors identify which assets have the lowest returns correlations and, thus, contribute to reducing overall portfolio risk.

EXAMPLE: Portfolio variance

The Carter family portfolio contains three equally weighted assets. The expected returns and standard deviations are presented here:

Asset	Expected Return (%)	Standard Deviation (%)
1	14%	20%
2	6%	4%
3	10%	10%

The three assets have the following correlations:

Correlation	Asset 1	Asset 2	Asset 3
Asset 1	1	0.4	0.7
Asset 2	0.4	1	0.8
Asset 3	0.7	0.8	1

Calculate the expected portfolio return and expected portfolio variance.

Answer:

The expected portfolio return with equal weighting is:

$$\frac{1}{3} \times 14\% + \frac{1}{3} \times 6\% + \frac{1}{3} \times 10\% = 10\%$$

Using returns correlations, we can calculate portfolio variance with the formula:

$$\sigma_P^2 = \sum_{i=1}^3 \sum_{j=1}^3 w_i w_j \sigma_i \sigma_j \rho_{i,j}$$

Expanding this formula for three assets gives us:

$$\begin{aligned} &w_1 w_1 \sigma_1 \sigma_1 \rho_{1,1} + w_1 w_2 \sigma_1 \sigma_2 \rho_{1,2} + w_1 w_3 \sigma_1 \sigma_3 \rho_{1,3} + \\ &w_2 w_1 \sigma_2 \sigma_1 \rho_{2,1} + w_2 w_2 \sigma_2 \sigma_2 \rho_{2,2} + w_2 w_3 \sigma_2 \sigma_3 \rho_{2,3} + \\ &w_3 w_1 \sigma_3 \sigma_1 \rho_{3,1} + w_3 w_2 \sigma_3 \sigma_2 \rho_{3,2} + w_3 w_3 \sigma_3 \sigma_3 \rho_{3,3} \end{aligned}$$

Because the correlation of an asset's returns with themselves is 1, and the covariance of an asset's returns with themselves is the asset's variance of returns, this reduces to:

$$\begin{aligned} &w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + w_3^2 \sigma_3^2 + 2(w_1 w_2 \sigma_1 \sigma_2 \rho_{1,2}) + 2(w_1 w_3 \sigma_1 \sigma_3 \rho_{1,3}) + \\ &2(w_2 w_3 \sigma_2 \sigma_3 \rho_{2,3}) \\ \sigma_P^2 &= (0.333^2 \times 0.20^2) + (0.333^2 \times 0.04^2) + (0.333^2 \times 0.10^2) + (2 \times \\ &0.333 \times 0.333 \times 0.20 \times 0.04 \times 0.4) + (2 \times 0.333 \times 0.333 \times \\ &0.20 \times 0.10 \times 0.7) + (2 \times 0.333 \times 0.333 \times 0.04 \times 0.10 \times 0.8) \\ &= 0.00444 + 0.00018 + 0.00111 + 0.00071 + 0.00311 + 0.00071 \\ \sigma_P^2 &= 0.01026 \end{aligned}$$

$$\text{Portfolio standard deviation} = \sqrt{0.01026} = 0.1012 = 10.13\%.$$



PROFESSOR'S NOTE

Some of the related material in the curriculum learning module expresses these calculations using matrix notation. Our understanding is that matrix notation will not be tested on the Level I CFA exam.

Expected Portfolio Variance

We can express portfolio variance on a forward-looking basis, using assets' expected returns and expected variances of returns. Portfolio returns can incorporate different economic scenarios and their associated probabilities, which sum to 1.

The first step is to calculate the portfolio's expected return under each scenario, which is the weighted average of the assets' expected returns under that scenario. Then, to arrive at the expected portfolio return across all scenarios, we take a probability-weighted average of all the scenarios' expected returns.

To calculate the expected portfolio variance, for each scenario we square the deviation between the scenario's expected return and the portfolio expected return. We then multiply the result by the scenario's probability. The sum of these probability-weighted squared deviations gives us the expected variance.

$$\sigma_p^2 = \sum_{m=1}^M P(m) \times (r_{p,m} - E[r_p])^2$$

where m represents a scenario in the model

EXAMPLE: Expected portfolio return

To prepare their investment forecast for the upcoming year, Trident Investors has built a probability distribution for potential market environments:

Outcome	Probability	Expected Returns		
		Asset A	Asset B	Asset C
Growth	45%	12%	5%	16%
Flat	35%	5%	7%	4%
Recession	20%	-8%	12%	-10%

The portfolio is allocated 50% to Asset A and 25% each to Assets B and C. Calculate the portfolio's expected return and expected standard deviation of returns.

Answer:

First, calculate the portfolio's expected return under each scenario:

Expected return, growth = $(0.50 \times 12\%) + (0.25 \times 5\%) + (0.25 \times 16\%) = 11.25\%$

Expected return, flat = $(0.50 \times 5\%) + (0.25 \times 7\%) + (0.25 \times 4\%) = 5.25\%$

Expected return, recession = $(0.50 \times -8\%) + (0.25 \times 12\%) + (0.25 \times -10\%) = -3.50\%$

Next, weight the scenarios' expected returns by their probabilities to get the portfolio's expected return:

$E(r_p) = 0.45 \times 11.25\% + 0.35 \times 5.25\% + 0.20 \times -3.50\% = 6.20\%$

To calculate the expected portfolio variance, square each scenario's deviation from the portfolio expected return, weight the results by the scenario probabilities, then sum the results:

$$\text{Growth: } (11.25\% - 6.2\%)^2 \times 0.45 = 11.4761$$

$$\text{Flat: } (5.25\% - 6.2\%)^2 \times 0.35 = 0.3159$$

$$\text{Recession: } (-3.50\% - 6.2\%)^2 \times 0.20 = 18.8180$$

$$11.4761 + 0.3159 + 18.8180 = 30.61$$

The expected portfolio standard deviation = $\sqrt{30.61} = 5.53\%$

Diversification

As the number of assets in a portfolio increases, each asset's contribution to the total portfolio variance decreases. For an equally weighted portfolio with a large number of assets, the variance of returns is approximately the product of the average variance of its assets' returns and their average correlation:

$$\sigma_P^2 \approx \bar{\rho} \sigma^2$$



PROFESSOR'S NOTE

A more exact expression for portfolio variance, assuming an equally weighted portfolio of $\frac{\sigma^2}{N} + \left(1 - \frac{1}{N}\right)\bar{\rho}\sigma^2$. The approximation should be good enough for a multiple-choice question.

EXAMPLE: Estimating portfolio variance with a large number of assets

Sanford Asset Management is reviewing the annual statistics for their domestic equity portfolio.

- Number of assets = 75
- Average correlation between the assets = 0.45
- Average variance of individual assets = 0.15

Estimate the portfolio variance.

Answer:

The portfolio variance can be estimated as: $\sigma_P^2 \approx \bar{\rho} \sigma^2$

$$0.45 \times 0.15 = 0.0675 = 6.75\%$$

Note: The more exact formula results in 6.86%.

A common rule in the investment management industry is that a portfolio of 20 to 30 well-chosen assets should capture most of the benefits of diversification. A portfolio of uncorrelated assets reduces the risk that is specific to individual assets, known as their **unsystematic risk**—but this does not eliminate the risk of the broader market, or **systematic risk**. Beyond about 30 assets, the marginal diversification benefit of each additional asset diminishes to near zero.



MODULE QUIZ 8.3

1. A portfolio of 30 assets is *least likely* to be characterized by:
 - A. style drift.
 - B. systematic risk.
 - C. unsystematic risk.
2. For a portfolio of 25 assets, the average variance of the individual assets is calculated to be 0.07. The average correlation between the assets is estimated to be 0.20. The standard deviation of the portfolio is *closest* to:
 - A. 1%.
 - B. 6%.
 - C. 12%.
3. Analyst Ravi Gupta estimates the following returns for his portfolios in various economic scenarios with the expected probability of each scenario:

Scenario	Probability	Asset 1	Asset 2	Asset 3
Growth	50%	12%	8%	14%
Flat	35%	4%	2%	6%
Recession	15%	−3%	0%	−6%

Gupta analyzes a portfolio with asset weights of $w_1 = 30\%$, $w_2 = 60\%$, and $w_3 = 10\%$. The expected portfolio standard deviation is *closest* to:

- A. 4.32%.
- B. 4.66%.
- C. 4.83%.

MODULE 8.4: THE MINIMUM-VARIANCE PORTFOLIO AND THE EFFICIENT FRONTIER

LOS 8.b: Describe, calculate, and interpret the minimum-variance portfolio and portfolios that lie on the efficient frontier.

Investors have constraints that limit which investments are appropriate to meet their objectives. Chief among these constraints are their return requirements and their preferred level of risk. **Portfolio optimization** works to find the most advantageous combination of assets in a portfolio, given an investor's return and risk constraints.

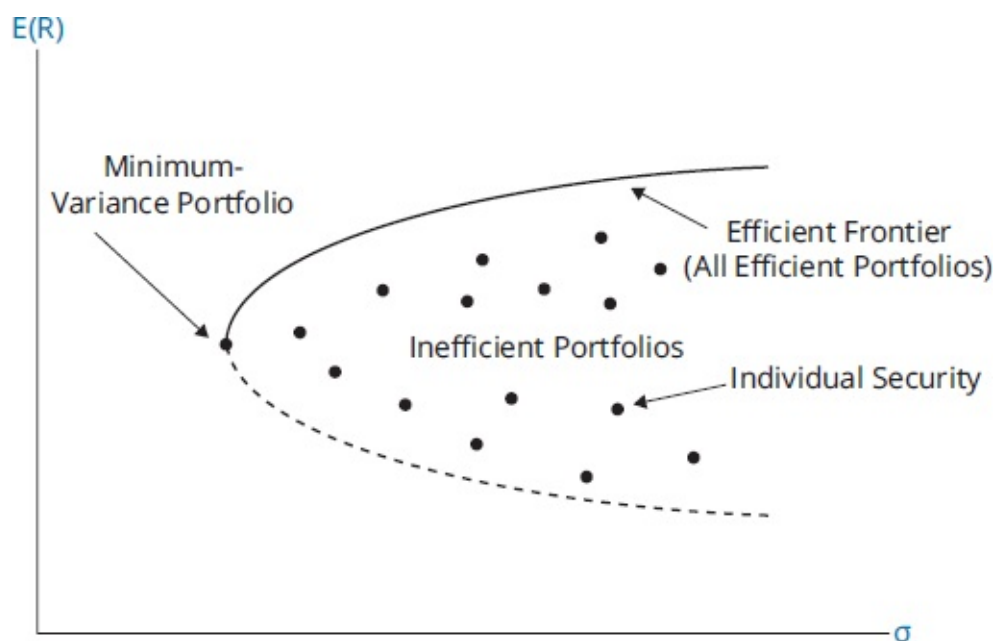
This can involve many possible asset mixes that offer a wide range of risk and return characteristics. Each of these individual feasible portfolios is considered a **risky portfolio**, as each one has uncertain return and exposes the investor to potential gains or losses.

For any given level of risk, one portfolio must offer a higher expected return than any others. These optimal portfolios form the **efficient frontier**, which is the upper bound of the feasible portfolios. Portfolios on the efficient frontier represent the highest expected return for a given level of risk.

By contrast, portfolios that have the lowest risk for a given level of expected return form the **minimum-variance frontier**. The efficient frontier is the segment of the minimum-variance frontier that only includes efficient portfolios. In Figure 8.4, the curved line to the left of the attainable portfolios is the minimum-variance frontier, and the solid segment of the curve is the efficient frontier.

The area to the left of the minimum-variance frontier (and above the efficient frontier) represents risk-return combinations that cannot be created with the available set of assets.

Figure 8.4: Feasible Portfolios and Efficient Frontier



In a simple case of a portfolio with only two risky assets, we can calculate the weights of each asset that will form the minimum-variance portfolio. If we know the returns variance of each asset (σ_A^2 , σ_B^2) and the assets' covariance of returns ($\sigma_{A,B}$), we select one of the assets and calculate its weight as follows:

$$w_A = \frac{\sigma_B^2 - \sigma_{A,B}}{\sigma_A^2 + \sigma_B^2 - 2\sigma_{A,B}}$$

Then, because the weights must sum to 100%, $W_B = 1 - W_A$.



PROFESSOR'S NOTE

The LOS says “calculate . . . the minimum-variance portfolio,” so this formula is testable on the exam. The curriculum learning module includes a derivation of this formula that uses calculus, which we believe is beyond the scope of the exam, but it's available for candidates who are interested.

EXAMPLE: Minimum-variance portfolio of two risky assets

A US investor wants to build a simple global portfolio with an allocation to both the domestic S&P 500 Index, and to the FTSE 100. The investor estimates that the S&P 500 has an expected return of 10.1% and standard deviation of 13.7%, while the FTSE 100 has an expected return of 7.4% and a standard deviation of 14.5%. The correlation between the two indexes is estimated to be 0.09. Determine the asset weights of the minimum-variance portfolio.

Answer:

To calculate the minimum-variance portfolio, we will estimate the allocation to the S&P 500 (Asset A):

$$\begin{aligned}w_A &= \frac{\sigma_B^2 - \sigma_{A,B}}{\sigma_A^2 + \sigma_B^2 - 2\sigma_{A,B}}, \text{ where: } \sigma_{A,B} = \sigma_A \sigma_B \rho_{A,B} \\&= \frac{0.145^2 - (0.137)(0.145)(0.09)}{0.137^2 + 0.145^2 - 2(0.137)(0.145)(0.09)} = 0.5311 \\w_B &= 1 - 0.5311 = 0.4689\end{aligned}$$

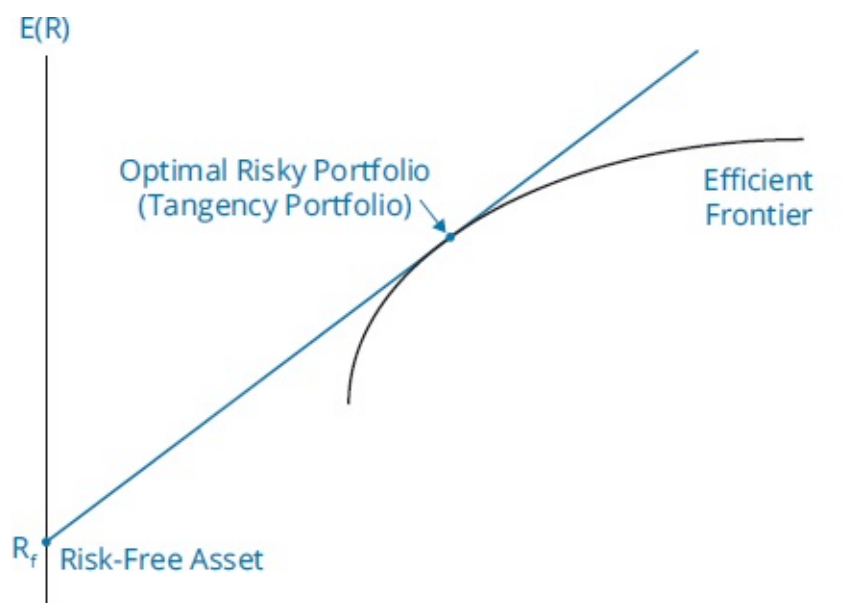
In a theoretical portfolio that includes all investable assets, the minimum-variance portfolio is known as the **global minimum-variance portfolio**, and the portfolios that have the highest expected return for each level of risk make up the **Markowitz efficient frontier**.

Adding a Risk-Free Asset

So far, we have been discussing portfolios that only consist of risky assets. Next, we consider how an investor can select an optimal portfolio if a **risk-free asset**, one that has zero variance of returns and earns the risk-free rate of return, can be combined with a portfolio of risky assets.

As we show in Figure 8.5, the risk-free asset lies on the vertical axis (zero risk). If we draw a straight line from the risk-free asset that intersects the efficient frontier at exactly one point (i.e., is tangent to the efficient frontier), that point denotes the **optimal risky portfolio** (also known as the **tangency portfolio**).

Figure 8.5: Optimal Risky Portfolio With a Risk-Free Asset



The tangency portfolio is “optimal” in the sense that it is the efficient portfolio with the greatest **Sharpe ratio**, or excess expected return (above the risk-free rate) per unit of risk:

$$\text{Sharpe ratio} = \frac{E(r_P) - r_f}{\sigma_P}$$

By combining a risky portfolio with the risk-free asset, investors can create a range of portfolios, shown by the straight line in Figure 8.5 (which we will define later in this reading as the *capital allocation line* associated with this risky asset), and select the one that best reflects their desired risk-return profile. The expected return on the combined portfolio is a weighted average of the risky portfolio’s expected return and the risk-free rate of return.

- ▮ Portfolios to the left of the tangency portfolio have weights between zero and 100% for the risk-free asset and the risky asset, and represent combinations of risk-free lending and investing in the tangency portfolio.
- ▮ Portfolios to the right of the tangency portfolio have negative weights of the risk-free asset and weights greater than 100% in the risky asset, which represent borrowing at the risk-free rate and investing the borrowed funds in the tangency portfolio.

Because the risk-free asset has zero standard deviation of returns—and, therefore, zero correlation with a risky portfolio’s returns—the standard deviation of the combined portfolio is simply the weight of the risky portfolio times its standard deviation of returns.

Given a target expected rate of return r_p for the combined portfolio, we can calculate the weight of the risky portfolio that is consistent with that expected rate of return as follows:

$$w_{\text{risky}} = \frac{r_p - r_f}{r_{\text{risky}} - r_f}, \text{ and } w_{\text{risk-free}} = 1 - w_{\text{risky}}$$

EXAMPLE: Combining a risky portfolio with a risk-free asset

An investment advisor has been mandated to combine the S&P 500 with a risk-free asset to achieve a target return of 9%. The risk-free asset has a rate of return of 3%. The advisor estimates the return on the S&P 500 to be 10.1% with a standard deviation of 12.2%. Determine the appropriate weights of the S&P 500 and the risk-free asset and calculate the standard deviation of returns on the resulting portfolio.

Answer:

Given our target return, we calculate the weights of the S&P 500 as:

$$\begin{aligned} w_{\text{S\&P}} &= \frac{r_p - r_f}{r_{\text{S\&P}} - r_f} \\ w_{\text{S\&P}} &= \frac{9\% - 3\%}{10.1\% - 3\%} = 0.845 = 84.5\% \\ w_{\text{rf}} &= 1 - 0.845 = 0.155 = 15.5\% \end{aligned}$$

Because the risk-free asset has a 0% standard deviation and zero correlation of returns with the S&P 500, the portfolio standard deviation becomes:

$$\begin{aligned} \sigma_p &= w_{\text{S\&P500}} \times \sigma_{\text{S\&P 500}} \\ &= 0.845 \times 0.122 = 0.1031 = 10.31\% \end{aligned}$$

Portfolio Optimization

Approaches to selecting the most efficient allocation of assets for a portfolio include the following:

- ▮ **Return-constrained optimization.** This minimizes portfolio risk for a given level of return.
- ▮ **Risk-constrained optimization.** This finds the highest expected return for a given level of risk and will identify combinations of portfolios that lie on the efficient frontier.
- ▮ **Risk-adjusted return optimization.** This maximizes the Sharpe ratio.

The key inputs into the portfolio optimization process are as follows:

- | Expected returns for each individual asset, which can be forecast based on their short-term or long-term historical performance, incorporating economic and financial indicators, industry trends, and company-specific factors
- | Volatility for each asset, which is commonly measured as the standard deviation of returns and can be determined based on historical data or forward-looking expectations
- | Asset correlations, which quantify the co-movements between asset returns

Investor-specific factors also come into play with portfolio optimization. An investor's **risk tolerance**, which is their ability and willingness to absorb losses, plays a key role in asset selection, as does an investor's time horizon and cash return requirements. Other examples of portfolio constraints include asset types, limits on exposure to individual assets and sectors, and nonnegative asset weights (i.e., no **short selling**, or borrowing a security to sell and then repurchase later). Such constraints may move the asset weights away from their otherwise optimal combination.



PROFESSOR'S NOTE

We will discuss investors' requirements and constraints in more detail in the Portfolio Construction topic area.



MODULE QUIZ 8.4

1. A portfolio on the efficient frontier has an expected return of 12% for a standard deviation of 16%. Which of the following portfolios is *most likely* to be unattainable with the assets available in the market? A portfolio with an expected return of:
 - A. 10%, with a standard deviation of 16%.
 - B. 14%, with a standard deviation of 16%.
 - C. 12%, with a standard deviation of 17%.
2. The risky portfolio on the efficient frontier that maximizes the Sharpe ratio is also known as the:
 - A. tangency portfolio.
 - B. minimum-variance portfolio.
 - C. global minimum-variance portfolio.
3. One of the key inputs for the portfolio optimization process is:
 - A. the efficient frontier.
 - B. the risk-free asset.
 - C. asset return correlations.

MODULE 8.5: RISK AVERSION

LOS 8.c: Explain the selection of an optimal portfolio, given an investor's risk aversion and the capital allocation line, and how this extends to the market portfolio and the capital market line.

The theory behind selecting an optimal investment portfolio rests on several key assumptions:

- | *Investors are rational and risk averse.* All investors make decisions aimed at maximizing their satisfaction (utility). They prefer higher returns and lower risk. They will choose investments that offer the best risk-return tradeoff according to their preferences.
- | *Investors have a single-period time horizon.* They will make an investment decision at the beginning of a period and evaluate their results at the end of that period.
- | *There are no taxes or transaction costs.* This simplifies the analysis to focus solely on investment performance and market behavior.
- | *A risk-free asset can be purchased or sold.* Investors can borrow and lend at a constant risk-free rate.

Investor Risk Preferences

An individual's attitude toward risk when making investment decisions is known as his **risk preference**. This preference determines how much risk (variability in returns) an individual is willing to assume in exchange for potentially larger gains. We can divide possible risk preferences into three categories:

Risk seeking. These are investors who prefer uncertain outcomes over guaranteed outcomes. They derive extra utility from the uncertainty of their investment and may even be willing to accept a lower return. A common example of risk-seeking is gambling in casinos, where the expected return is usually negative.

Risk neutral. These are investors who are indifferent between an uncertain outcome and a guaranteed outcome with the same expected value. A risk-neutral investor would focus on returns only, without regard to the level of risk. This is a more common attitude when an investment represents a small fraction of an individual's overall wealth.

Risk averse. These investors prefer guaranteed outcomes over uncertainty, even if a gamble has a high expected value. They aim to minimize risk for a given level of return.

Most real-world investors fall into the risk-averse category. However, risk-neutral investors can offer insights in theoretical settings. This kind of research focuses on large institutional investors that execute large numbers of small transactions. Over time, the variance of their trades averages out and renders their expected return nearly certain.

Utility Functions

We can characterize an investor's preferences as a **utility function** that factors in both expected return and risk. While several forms of investor utility functions have been posited, a typical utility function might be expressed as follows:

$$U = E(r) - \frac{1}{2}A\sigma^2$$

where:

U = utility

$E(r)$ = expected return

σ^2 = variance of returns

A = investor's coefficient of risk aversion

The more risk averse an investor is, the greater the A in the investor's utility function. While a risk-neutral investor would have an A equal to zero and a risk-seeking investor would have a negative A , portfolio theory assumes all investors are risk averse (A greater than zero). Research suggests investors' coefficients of risk aversion tend to be between 2 and 3 and are rarely below 1.

Another way of expressing a utility function is **mean-variance utility**, which represents an investor's preferences regarding only two factors, expected return and risk:

$$U_p = U [E(R_p), \sigma_p]$$

This function ignores other factors, such as an investor's total wealth, that are thought to affect an investor's behavior in economic theory.

Risk preference relates closely to the concept of **risk tolerance**, which is the extent to which an investor is willing to accept risk to achieve greater expected return. Risk tolerance is the inverse of risk aversion: the lower the risk aversion, the higher the risk tolerance.

An individual's risk tolerance tends to change over time due to factors such as age, wealth level, and the state of the economy. Risk tolerance also varies across countries and cultures.

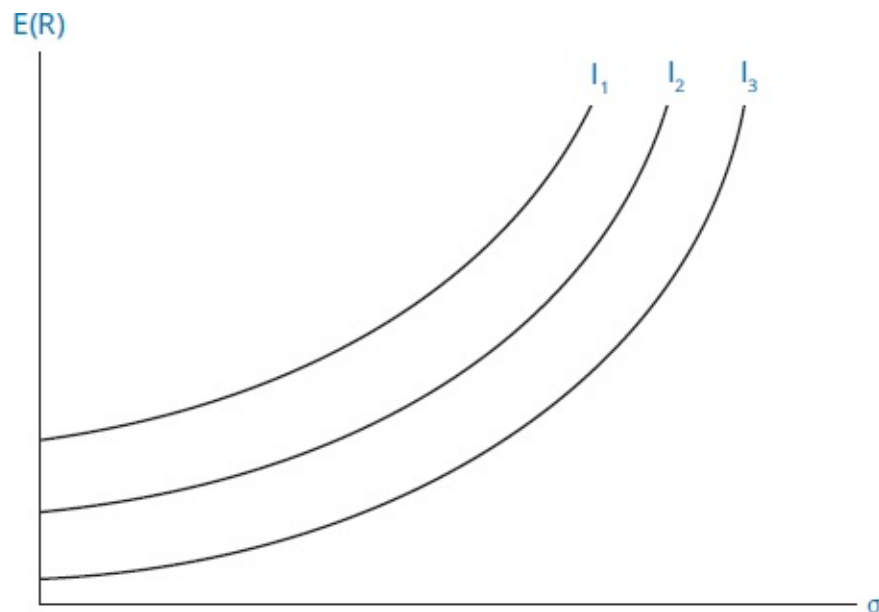
Approaches to observing investors' risk aversion include **absolute risk aversion**, which refers to how an investor's utility changes with each increment of additional risk, assuming the investor's wealth remains constant; or **relative risk aversion**, which assumes an investor's wealth changes and observes how this affects the proportion of their wealth an investor is willing to risk.

Indifference Curves

An **indifference curve** is a tool from economics that, in this application, plots combinations of risk (standard deviation) and expected returns among which an investor is indifferent. In constructing indifference curves for portfolios based on only their expected return and standard deviation of returns, we are assuming that these are the only portfolio characteristics that investors care about.

In Figure 8.6, we show three indifference curves for a risk-averse investor. The investor's expected utility is the same for all points (portfolios) along any single indifference curve. Portfolios along indifference curve I_1 in Figure 8.6 are preferred to all portfolios along I_2 , which are preferred to all portfolios along I_3 .

Figure 8.6: Risk-Averse Investor's Indifference Curves



The slope of the indifference curves represents an investor's degree of risk aversion, or how much additional risk an investor is willing to take on for additional return while maintaining the same level of utility. A steeper slope implies greater risk aversion. A theoretically risk-neutral investor would have flat indifference curves, while a risk-seeking investor's indifference curves would slope downward.



MODULE QUIZ 8.5

1. An investment advisor defines a client's utility using the function $U = E(r) - \frac{1}{2} A \sigma^2$ and determines that one client has a risk aversion coefficient of 2. Investment P has an expected return of 8% with an estimated variance of 4%. Investment Q has an expected return of 6% with an estimated variance of 1%. Investment R has an expected return of 5% with an estimated variance of 2%. Which investment will provide this client with the highest utility?
 - A. Investment P.
 - B. Investment Q.
 - C. Investment R.
2. An investor is found to have indifference curves such that if risk increases, higher returns are needed to maintain the same utility level. This investor would be considered:
 - A. risk averse.
 - B. risk seeking.
 - C. risk neutral.
3. The theory of selecting an optimal investment portfolio requires the assumption that investors are rational and risk averse. This implies that any investment decision focuses on:
 - A. minimizing transaction costs.
 - B. minimizing the investor's utility.
 - C. maximizing the investor's satisfaction.

MODULE 8.6: CAPITAL ALLOCATION LINE AND CAPITAL MARKET LINE

As we saw earlier in this reading, we can combine a risk-free asset with a risky portfolio to expand an investor's opportunity set beyond the efficient frontier of risky assets. This combined portfolio has an expected return that is the weighted average of the expected returns on the risky portfolio and the risk-free asset, and a standard deviation that is the weight of the risky portfolio times its standard deviation of returns.

When we do this, we assume the following:

- ▮ The risk-free asset has zero standard deviation of returns.
- ▮ The standard deviation and expected return of the risky portfolio are both greater than the risk-free rate.
- ▮ The risk-free asset and the risky tangency portfolio are the only two assets available to the investor.
- ▮ The investor can borrow or lend at the risk-free rate.

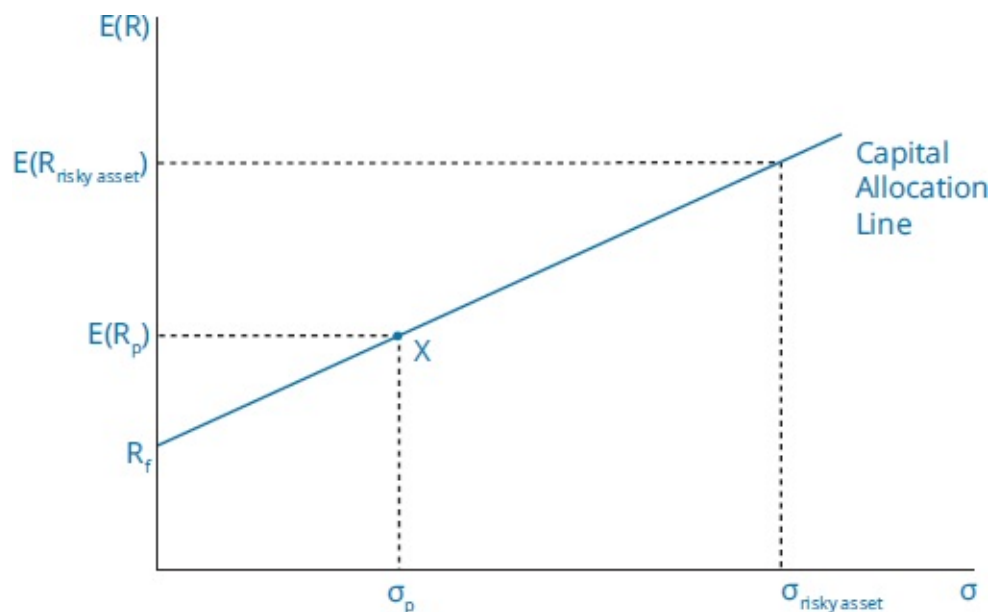
We can state the expected return on a portfolio of a risky asset and a risk-free asset as a straight-line relationship with the portfolio's risk:

$$E(r_P) = r_f + \frac{E(R_{\text{risky asset}}) - r_f}{\sigma_{\text{risky asset}}} \sigma_P$$

The line representing these possible combinations of risk-free assets and a risky asset is referred to as the risky asset's **capital allocation line (CAL)**. A CAL intercepts the vertical axis at the risk-free rate of return, and its slope is the risky asset's Sharpe ratio.

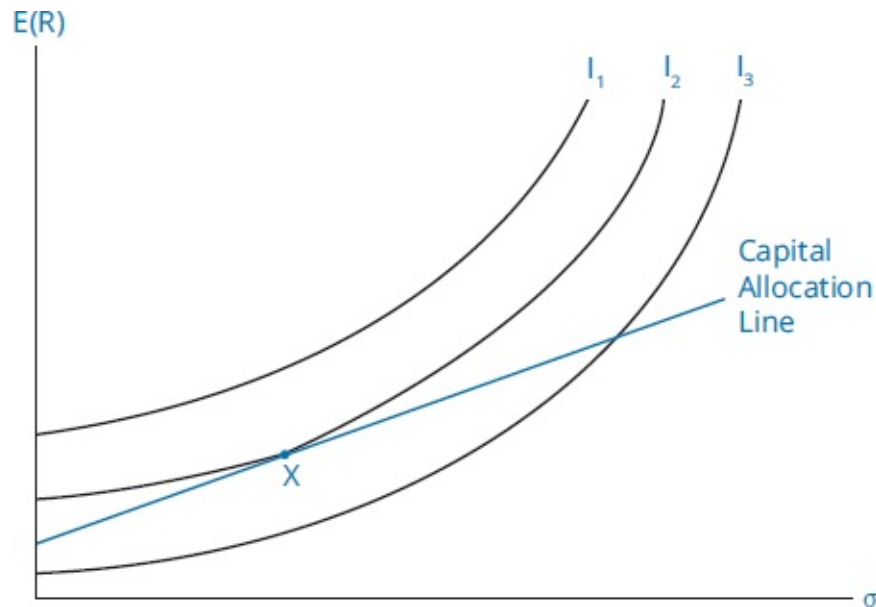
Point X on the CAL in Figure 8.7 represents a portfolio that is 40% invested in the risky asset (tangency portfolio) and 60% invested in the risk-free asset. Its expected return will be $0.40[E(R_{\text{risky asset}})] + 0.60(R_f)$, and its standard deviation will be $0.40(\sigma_{\text{risky asset}})$.

Figure 8.7: CAL and Risky Asset Weights



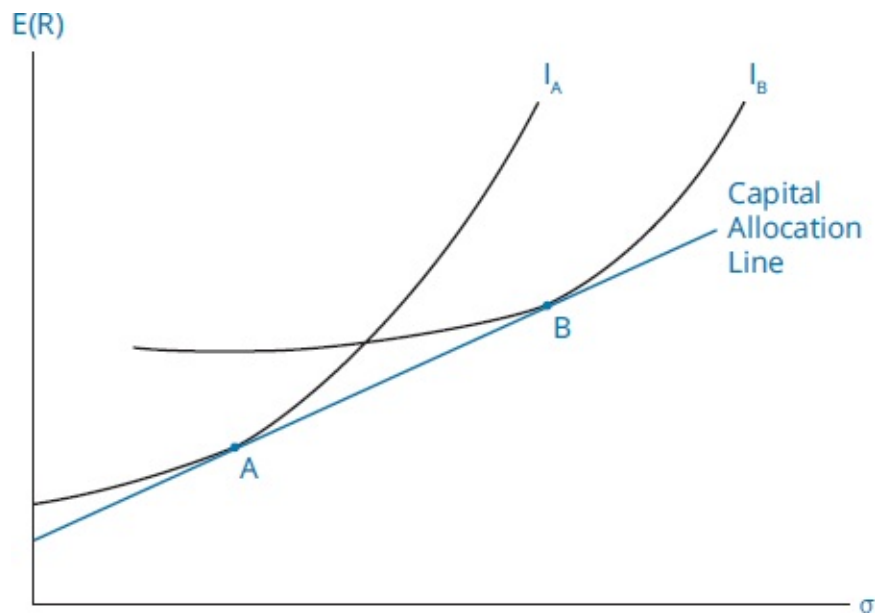
We can combine a CAL with indifference curves, representing an individual's preferences for risk and return, to illustrate the logic of selecting an optimal portfolio (i.e., one that maximizes the investor's expected utility). In Figure 8.8, we can see that Investor A, with preferences represented by indifference curves I_1 , I_2 , and I_3 , can reach the level of expected utility on I_2 by selecting Portfolio X. This is the optimal portfolio for this investor, as any portfolio that lies on I_2 is preferred to all portfolios that lie on I_3 (and, in fact, to any portfolios that lie between I_2 and I_3). Portfolios on I_1 would be preferred to those on I_2 , but none of the portfolios that lie on I_1 are obtainable in the market.

Figure 8.8: Indifference Curves and a CAL



The result of our analysis here is not surprising: Investors who are less risk averse will select portfolios with more risk. Recall that the lower an investor's risk aversion, the flatter their indifference curves. As illustrated in Figure 8.9, the flatter indifference curve for Investor B (I_B) results in an optimal (tangency) portfolio that lies to the right of the one that results from a steeper indifference curve, such as that for Investor A (I_A). An investor who is less risk averse should optimally choose a portfolio with more invested in the risky asset and less invested in the risk-free asset.

Figure 8.9: Portfolio Choices Based on Two Investors' Indifference Curves



Capital Market Line

The **capital market line (CML)** is a special case of a CAL, in which the risky asset (tangency portfolio) is the **market portfolio**. The market portfolio consists of all risky assets weighted by their market values. The market portfolio is said to be **mean-variance efficient**, in that it has the highest Sharpe ratio of any possible portfolio. The **mutual fund theorem** posits that combining a risk-free asset with the market portfolio results in an efficient portfolio for all investors.

A major difference between a CAL and the CML is that a CAL assumes investors have heterogeneous expectations of assets' expected returns, variances, and covariances, while the CML assumes investors have homogeneous expectations. The implication is that each investor would have a unique CAL, depending on their expectations, but the CML is the same for all investors.

Clearly, a market portfolio that "contains all investable assets" is impossible to achieve. In practice, investors rely on broad stock market indexes to represent the market portfolio.

The CML can be written as follows:

$$E(r_P) = r_f + \frac{E(r_M) - r_f}{\sigma_M} \sigma_P$$

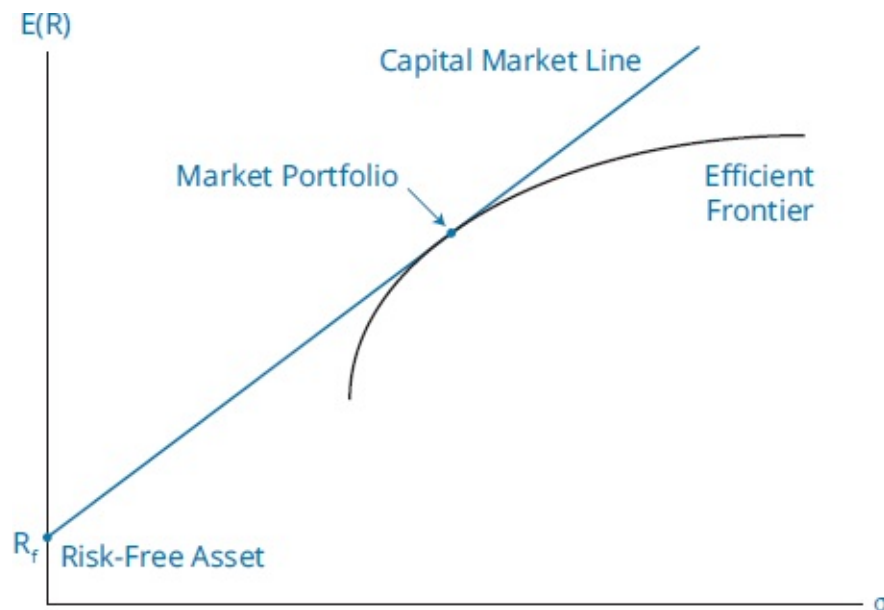
where:

r_M = expected return on the market portfolio

σ_M = standard deviation of the market portfolio

Figure 8.10 shows how the CML intercepts the vertical axis at the risk-free rate, which represents a 100% weight in the risk-free asset. As the investor takes on risk by allocating a portion to the market portfolio, the CML increases at a slope equal to the Sharpe ratio of the market portfolio. The market portfolio lies on the efficient frontier of risky asset portfolios.

Figure 8.10: The Capital Market Line



MODULE QUIZ 8.6

- According to the mutual fund theorem, an investor can achieve an efficient portfolio by combining a risk-free asset with:
 - any risky portfolio on the efficient frontier.
 - the market portfolio.
 - the minimum-variance portfolio.
- Portfolio manager Alex Pano has built his portfolio with a combination of the risk-free rate and a 78.3% allocation to a broad market index to achieve a target return of 10%. The market index has a standard deviation of 12.3%, and the current risk-free rate is 2%. The Sharpe ratio of his portfolio is *closest* to:
 - 0.51.
 - 0.65.
 - 0.83.

MODULE 8.7: THE CAPITAL ASSET PRICING MODEL

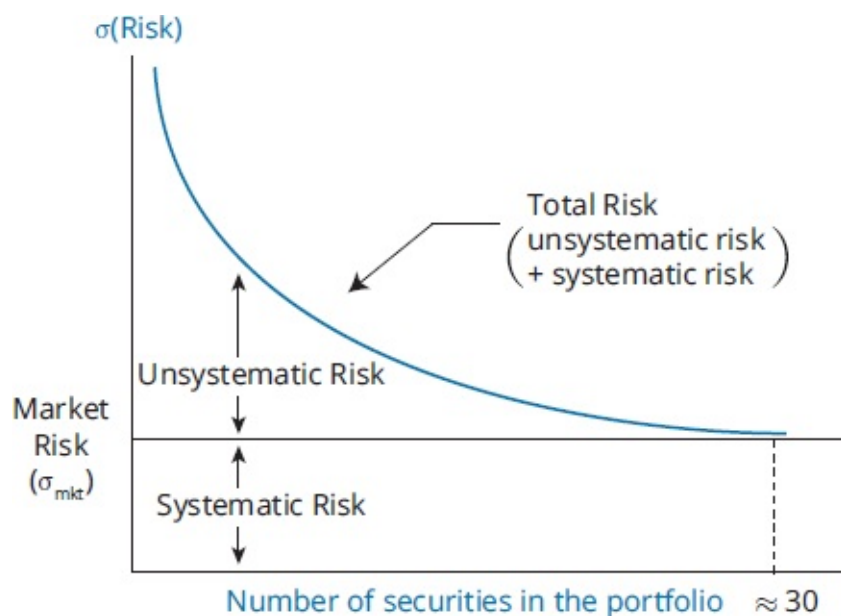
When an investor diversifies across assets that are not perfectly correlated, the portfolio's risk is less than the weighted average of the risks of the individual

securities in the portfolio. The risk that is eliminated by diversification is called **unsystematic risk** (also called *unique*, *diversifiable*, or *firm-specific risk*).

Because the market portfolio contains all risky assets, it must be a well-diversified portfolio. All the risk that can be diversified away has been. The risk that remains cannot be diversified away and is called **systematic risk** (also called *nondiversifiable risk* or *market risk*).

You don't actually have to buy all the securities in the market to diversify away unsystematic risk. As you increase the number of stocks in a portfolio, the portfolio's risk falls toward the level of market risk. After 30 or so securities, the portfolio standard deviation remains approximately constant. The remaining risk is systematic, or nondiversifiable, risk. Figure 8.11 provides a general representation of this concept.

Figure 8.11: Risk vs. Number of Portfolio Assets



Investment managers use **return-generating models** to estimate the expected returns of risky securities based on specific factors. Some models (which we will examine later in the Level I curriculum) use macroeconomic factors such as GDP growth, inflation, or consumer confidence, along with fundamental factors such as earnings, earnings growth, firm size, and research expenditures.

A widely used return generating model is the **capital asset pricing model (CAPM)**. This model estimates expected return based on a single factor, which is an asset's degree of exposure to systematic or nondiversifiable risk.

The CAPM makes the following assumptions:

- ▮ All investors have homogeneous expectations for assets' expected returns, standard deviation of returns, and returns correlations between assets.
- ▮ Investors choose the portfolio on the efficient frontier, based on their individual preferences, with the risk and return combination that maximizes their expected utility.
- ▮ Investors take the market price as given, and no investor can influence prices with their trades.
- ▮ Markets are in equilibrium (i.e., there are no mispriced assets).

Formally, the CAPM is stated as:

$$E(R_i) = R_f + \beta_i[E(R_{\text{market}}) - R_f]$$

An asset's exposure to systematic risk is represented by **beta** (β). Beta is expressed as:

$$\beta_i = \frac{\sigma_{i,m}}{\sigma_m^2}$$

where:

$\sigma_{i,m}$ = covariance between the returns of asset i and the return on the market portfolio

σ_m^2 = variance of the market portfolio's returns

We can also express beta in terms of the returns correlation between asset i and the market. Because $\sigma_{i,m}$ equals $\sigma_i \sigma_m \rho_{i,m}$:

$$\beta_i = \frac{\sigma_i \sigma_m \rho_{i,m}}{\sigma_m^2}; \text{ therefore, } \beta_i = \frac{\sigma_i}{\sigma_m} \rho_{i,m}$$

Beta reflects the sensitivity of an investment to the overall market risk. A beta greater than 1 indicates that an investment's expected return is greater than that of the market. When beta is less than 1, an investment has an expected return less than that of the market. The market portfolio has a beta of 1.



PROFESSOR'S NOTE

In our reading on linear regression, we will illustrate how to estimate an asset's beta using historical data.

An individual security's expected return in equilibrium can be shown on the **security market line (SML)**, which we express using the following formula:

$$E(r_i) = r_f + \frac{E(r_m) - r_f}{\sigma_m^2} \sigma_{i,m}$$



PROFESSOR'S NOTE

In the Portfolio Construction topic area, we will examine how analysts use the SML to identify overvalued or undervalued assets. The SML is a graphical representation of the CAPM.

The difference between the expected market return and the risk-free rate, $[E(r_m) - r_f]$, is known as the **market risk premium**. It reflects the additional return that an investor can expect by holding the market portfolio instead of the risk-free asset. Beta links the risk premium of an individual asset with that of the market.

EXAMPLE: Using the CAPM to calculate an asset's expected return

In a given market environment, the risk-free rate is 1%, and the expected market return is 12%. An asset's beta is 0.9. Calculate the asset's expected return according to the CAPM.

Answer:

$$\begin{aligned} E(r) &= r_f + \beta [E(r_m) - r_f] \\ &= 0.01 + 0.9(0.12 - 0.01) = 0.109 = 10.9\% \end{aligned}$$

Portfolio Theory Limitations

While the models we have introduced here provide insight into portfolio optimization, several of their underlying assumptions may not hold true in reality:

- | Evidence does not support the idea that financial asset returns follow a normal distribution; instead, they tend to show significant skewness and kurtosis.
- | Correlations among assets' returns do not remain stable over time—especially not at times of market stress.
- | Financial markets are less than perfectly efficient. Information asymmetries can result in mispriced assets.
- | The models are highly sensitive to estimation errors. This can result in incorrect weighting of assets in a portfolio.

- Future returns and variances may not reflect their past behavior because of changing market expectations or changes to the assets themselves.

Portfolio management also has practical constraints that are not incorporated into modern portfolio theory:

- Portfolio rebalancing strategies can be unrealistic if they do not account for transactions costs.
- Taxes can vary across portfolios due to both capital distributions and capital appreciation. Taxes can change the optimal composition of a portfolio significantly.
- After-tax and risk-adjusted realized returns will be affected if a portfolio manager takes advantage of capital losses to reduce capital gains.



MODULE QUIZ 8.7

1. If the standard deviation of market returns is 8%, which of the following investments has the highest beta?

Investment	Correlation With Market Returns	Standard Deviation of Asset Returns
1	0.75	12%
2	0.80	15%
3	0.70	20%

- A. Investment 1.
 - B. Investment 2.
 - C. Investment 3.
2. Using the security market line (SML) as a model, investor Carla Simmons believes Security XYZ, which has a beta of 1.2, will provide a 12% return over the next year. The risk-free rate is 4%, and the overall market is expected to return 8%. Is Simmons accurate in her estimate of the expected return of XYZ?
 - A. Yes, the expected return is 12%.
 - B. No, the expected return is less than 12%.
 - C. No, the expected return is greater than 12%.
 3. Which of the following is *least likely* an example of a limitation on traditional portfolio management theories?
 - A. Financial asset returns exhibit significant skewness.
 - B. Estimation errors can significantly affect model results.
 - C. In the real world, all available information is instantaneously reflected in asset prices.
 4. Which of the following is *least likely* a practical constraint that affects portfolio management in practice?
 - A. No proxy for a risk-free asset is available.

- B. Portfolio rebalancing increases transaction costs.
- C. Taxes on different forms of income may alter an investor's optimal portfolio.

KEY CONCEPTS

LOS 8.a

The historical return of a portfolio for a period is the weighted average of the realized returns for each asset. Over multiple periods, we must incorporate the cumulative return of the assets.

Portfolio drift may occur as some assets decline in value, while others increase. Over time, this can lead to style drift, where a portfolio becomes misaligned with its stated strategy.

Geometric average return with changing weights:

$$\bar{r}_{g,P} = \left[\prod_{t=1}^T \left(1 + \sum_{i=1}^N w_{i,t-1} \times r_{i,t} \right) \right]^{\frac{1}{T}} - 1$$

Variance of portfolio returns:

$$\sigma_P^2 = \sum_{i=1}^N \sum_{j=1}^N w_i w_j \sigma_{ij}$$

where σ_{ij} = covariance between the returns for Assets i and j

$$\sigma_{ij} = \sigma_i \sigma_j \rho_{ij}$$

where ρ_{ij} = the correlation of returns between Assets i and j

If asset returns are less than perfectly positively correlated, combining them in a portfolio creates a diversification benefit that can reduce overall risk. A correlation matrix can help investors identify which assets have the lowest returns correlations.

Expected variance:

$$\sigma_P^2 = \sum_{m=1}^M P(m) (r_{p,m} - E[r_p])^2$$

where m = a scenario in the model

Diversification reduces unsystematic risk, which is specific to individual assets, but it does not reduce systematic risk of the broader market. A portfolio of 20 – 30 assets should capture most of the benefits of diversification.

LOS 8.b

Portfolios that have the lowest risk for a given level of expected return form the minimum-variance frontier. Portfolios that have the highest expected return for a given level of risk form the efficient frontier, which is the segment of the minimum-variance frontier that only includes efficient portfolios.

For a two-asset portfolio, we can calculate the weights of each asset that will form the minimum-variance portfolio:

$$w_A = \frac{\sigma_B^2 - \sigma_{A,B}}{\sigma_A^2 + \sigma_B^2 - 2\sigma_{A,B}}, \text{ and } w_B = 1 - w_A.$$

If a risk-free asset is available, an investor can combine it with an optimal risky portfolio (tangency portfolio) to achieve an optimal risk-return tradeoff. The standard deviation of the combined portfolio is simply the weight of the risky portfolio times its standard deviation of returns. The tangency portfolio is the efficient portfolio with the greatest Sharpe ratio, or excess expected return per unit of risk.

$$\text{Sharpe ratio} = \frac{E(r_P) - r_f}{\sigma_P}$$

Given a target expected rate of return r_P , the weight of the risky portfolio that is consistent with that rate of return is:

$$w_{\text{risky}} = \frac{r_P - r_f}{r_{\text{risky}} - r_f}, \text{ and } w_{\text{risk-free}} = 1 - w_{\text{risky}}$$

Return-constrained optimization minimizes portfolio risk for a given level of return. Risk-constrained optimization finds the highest expected return for a given level of risk. Risk-adjusted return optimization maximizes the Sharpe ratio.

LOS 8.c

Risk tolerance is the extent to which an investor is willing to accept risk to achieve greater expected return. Risk tolerance is the inverse of risk aversion; the lower the risk aversion, the higher the risk tolerance. A more steeply sloped indifference curve between risk and expected return implies a greater degree of risk aversion.

We can state the expected return on a portfolio of a risky asset and a risk-free asset as a straight-line relationship with the portfolio's risk:

$$E(r_P) = r_f + \frac{E(R_{\text{risky asset}}) - r_f}{\sigma_{\text{risky asset}}} \sigma_P$$

The line representing these possible combinations of risk-free assets and a risky asset is referred to as the risky asset's capital allocation line (CAL). An investor's optimal portfolio is tangent to the CAL on the investor's highest attainable indifference curve. Investors who are less risk averse (more risk tolerant) will select portfolios with more risk.

The capital market line (CML) is a special case of a CAL in which the risky asset is the market portfolio. The market portfolio consists of all risky assets weighted by their market values. The mutual fund theorem posits that combining a risk-free asset with the market portfolio results in an efficient portfolio for all investors.

The CML can be written as:

$$E(r_P) = r_f + \frac{E(r_M) - r_f}{\sigma_M} \sigma_P$$

where:

r_M = expected return on the market portfolio

σ_M = standard deviation of the market portfolio

The capital asset pricing model (CAPM) estimates expected return based on a single factor, which is an asset's degree of exposure to systematic risk. The CAPM is stated as:

$$E(R_i) = R_f + \beta_i [E(R_{\text{market}}) - R_f]$$

where:

$$\beta_i = \frac{\sigma_{i,m}}{\sigma_m^2}$$

Beta greater than one indicates that an investment's expected return is greater than that of the market. Beta less than one indicates that an investment's expected return is less than that of the market.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 8.1

. A The geometric average return can be calculated as follows:

$$= [(1 + 0.11) \times (1 + 0.06) \times (1 + (-0.03))]^{1/3} - 1 = 4.5\%.$$

(LOS 8.a)

- . **A** Style drift can occur when an asset's characteristics change over time. Portfolio drift occurs naturally when asset weights are adjusted due to underperformance or outperformance in the market. (LOS 8.a)

Module Quiz 8.2

- . **A** $\sigma_{AB}^2 = \sigma_A \sigma_B \rho_{AB} = 0.16 \times 0.08 \times 0.40 = 0.00512$.

(LOS 8.a)

- . **C** The first step in the calculation is to determine the values and weightings at the end of Period 1:

$$\text{Asset 1 value} = 100 \times 1.063 = 106.30$$

$$\text{Asset 2 value} = 100 \times 1.021 = 102.10$$

$$\text{Portfolio value} = 106.3 + 102.10 = 208.40$$

$$\text{Asset 1 weighting at the end of Period 1} = 106.30 / 208.40 = 51.01\%$$

$$\text{Asset 2 weighting at the end of Period 1} = 102.10 / 208.40 = 48.99\%$$

The next step is to apply these weights to Period 2:

$$\text{Asset 1 return} = 106.30 \times 1.085 = 115.34$$

$$\text{Asset 2 return} = 102.10 \times 1.027 = 104.86$$

$$\text{Portfolio value} = 115.34 + 104.86 = \$220.20$$

$$\text{Weight of Asset 1} = 115.34 / 220.20 = 52.38\%$$

$$\text{Weight of Asset 2} = 104.86 / 220.20 = 47.62\%$$

(LOS 8.a)

Module Quiz 8.3

- . **C** Unsystematic risk can be reduced by adding assets to a portfolio. Systematic risk is the overall market risk that cannot be diversified away. Style drift is not related to the correlation between assets. (LOS 8.a)

- . **C** Approximate portfolio variance = $0.07 \times 0.20 = 0.014$

$$\text{Standard deviation} = \sqrt{0.014} = 0.1183 = 11.8\%$$

(LOS 8.a)

- . **A** First, we need to calculate the expected return:

Growth: $(0.30 \times 12\%) + (0.60 \times 8\%) + (0.10 \times 14\%) = 9.8\%$

Flat: $(0.30 \times 4\%) + (0.60 \times 2\%) + (0.10 \times 6\%) = 3.0\%$

Recession: $(0.30 \times -3\%) + (0.60 \times 0\%) + (0.10 \times -6\%) = -1.5\%$

Expected return = $(0.50 \times 9.8\%) + (0.35 \times 3.0\%) + (0.15 \times -1.5\%) = 5.725\%$

Next, we can calculate the probability-weighted deviations from the expected return:

Growth: $(9.8\% - 5.725\%)^2 \times 0.50 = 0.00083$

Flat: $(3.0\% - 5.725\%)^2 \times 0.35 = 0.00026$

Recession: $(-1.5\% - 5.725\%)^2 \times 0.15 = 0.00078$

Variance = $0.00083 + 0.00026 + 0.00078 = 0.00187$

Standard deviation = $= \sqrt{0.00187} = 0.0432 = 4.32\%$.

(LOS 8.a)

Module Quiz 8.4

- . **B** A portfolio above the efficient frontier (higher return with the same risk) cannot be created with the current assets. A less efficient portfolio (lower expected return with the same risk) is attainable, and a portfolio with both a higher expected return and more risk may be attainable. (LOS 8.b)
- . **A** The tangency portfolio is the optimal risky portfolio on the efficient frontier with the highest Sharpe ratio. It represents the optimal combination of assets to achieve the highest risk-adjusted return. (LOS 8.b)
- . **C** The three key inputs for the portfolio optimization process are expected returns for each asset, volatility for each asset, and correlations of returns among the pairs of assets. (LOS 8.b)

Module Quiz 8.5

- . **B** Utility for Investment P = $8\% - (1/2 \times 2 \times 4\%) = 4\%$

Utility for Investment Q = $6\% - (1/2 \times 2 \times 1\%) = 5\%$

Utility for Investment R = $5\% - (1/2 \times 2 \times 2\%) = 3\%$

Investment Q provides the highest utility level for the client. (LOS 8.c)

- . **A** A risk-averse investor has convex indifference curves and requires higher returns to accept increased risk. (LOS 8.c)

- . C Optimal portfolio selection involves maximizing the investor's satisfaction, with the highest risk-return tradeoff given the investor's preferences. Utility is another term for the investor's satisfaction. (LOS 8.c)

Module Quiz 8.6

- . B The mutual fund theorem states that all investors can achieve an efficient portfolio by combining a risk-free asset with the market portfolio. (LOS 8.c)
- . C The Sharpe ratio is: $\frac{r_A - r_f}{\sigma_A}$

Because the risk-free asset has a zero standard deviation, this is the standard deviation of the portfolio:

$$\begin{aligned}\sigma_P &= (1 - w_f) \times \sigma_r \\ &= 78.3\% \times 0.123 = 0.096 = 9.6\%\end{aligned}$$

Sharpe ratio = $(10\% - 2\%) / 9.6\% = 0.83$.

(LOS 8.c)

Module Quiz 8.7

- . C The formula for beta using correlation is: $\beta_i = \frac{\sigma_i}{\sigma_m} \rho_{i,m}$

Investment 1: $\beta = 0.75 \times 0.12 / 0.08 = 1.125$

Investment 2 = $0.80 \times 0.15 / 0.08 = 1.50$

Investment 3 = $0.70 \times 0.20 / 0.08 = 1.75$

(LOS 8.c)

- . B The SML equation is:

$$E[r] = r_f + \beta [E(r_m) - r_f]$$

Given the information, the expected return on Security XYZ is:

$$E[r] = 4\% + 1.2(8\% - 4\%) = 8.8\%.$$

(LOS 8.c)

- . C In the real world, assets may be mispriced due to non-efficient markets and information asymmetries. (LOS 8.c)
- . A In most markets, a proxy for a risk-free asset (e.g., short-term government debt) should be available. Transactions costs and taxes are practical

constraints that affect portfolio management. (LOS 8.c)

READING 9

SIMULATION OF FINANCIAL ASSET PRICES AND RETURNS

MODULE 9.1: HISTORICAL SIMULATION

LOS 9.a: Describe historical simulation and explain how it can be used in investment applications.

A **simulation** is a predictive modeling technique used to estimate potential outcomes that involve complex systems or calculations. The process involves modeling assumptions based on controlled variables to provide insight into future performance. The technique is invaluable in evaluating portfolio optimization, risk and returns, and cash flow and liquidity modeling.

The process of setting up a simulation is as follows:

- Step 1: Set parameters.* Establish the goal, set the output variables to be tested, and define the required calculations to link the inputs to the desired outputs.
- Step 2: Generate scenarios.* These may be complex with multiple distributions, or simple enough to calculate manually.
- Step 3: Evaluate scenarios.* Run the model through the calculation engine to generate multiple potential paths.
- Step 4: Compile and interpret results.* Aggregate the data and interpret the outcomes.

A **historical simulation** incorporates historical data, such as past asset prices and returns, into scenario models to assess potential outcomes. Using observed market conditions and events, it predicts how a system might respond to similar circumstances and identifies potential risks.

Historical simulation offers several advantages:

- | It is easy to explain. Describing the methodology and building narratives around the results is straightforward.
- | The approach is nonparametric. With no need to force the data into a distributional assumption (e.g., normal, log-normal), it is particularly useful for modeling assets with nonnormal return distributions, such as those with excess kurtosis or skewness. It is better than parametric models at predicting the impact of extreme market events and fat-tailed distributions.
- | It captures volatility. It accounts naturally for **volatility clustering**, the tendency for periods of low or high volatility to persist.

Historical simulation assumes implicitly that past performance reliably predicts future prices and risks, and that historical data represent the population of outcomes. However, these may not always be the case. Relying on past assumptions may result in less accurate predictions when faced with unprecedented circumstances or regime changes.

In addition, unreliable or incomplete data may distort results. Missing data may result from the following:

- | Financial instruments that are thinly traded, or have low trading volumes, may have gaps in their data.
- | Global markets are asynchronous when it comes to nontrading days, such as holidays, technology outages, or other halted trading.
- | Securities that did not exist during the simulation period, such as a company formed post-simulation, will not have historical price data.

One way to compensate for data deficiencies is to use a proxy, or substitute financial instrument, for the security with sparse data. An analyst can create a proxy by weighting financial instruments that have data available over the period studied.

Using the steps we outlined earlier, a historical simulation runs as follows:

Step 1: Set parameters. Select the dates and the inputs, or valuation and risk factors, then assign the historical change that occurred under those parameters. Sometimes, simplifying assumptions are required to align returns to the same time frames. For equities, the parameters tend to be logarithmic or continuously compounded price returns, though percentage returns are also used. For fixed income, simulated yields are

used to discount future cash flows.

Step 2: Generate scenarios. Use prior observations to determine the scenarios.

Step 3: Evaluate scenarios. Run the model for each scenario identified.

Step 4: Compile and interpret results. Compare the performance under the scenarios that occurred previously.

By applying historical returns to current positions, a historical simulation can determine the profit and loss distribution of a long-only portfolio, one that purchases assets with the expectation their value will appreciate. A key metric is **value at risk (VaR)**, or the maximum potential loss over specified time frame at a given level of confidence. Because VaR focuses on the losses of the distribution, it is largely unaffected by a few positive returns. For a two-asset portfolio, the process would be as follows:

Step 1: Set parameters. Calculate daily logarithmic returns over a specified prior period for each asset in the portfolio.

Step 2: Generate scenarios. Calculate the profit or loss for each position under each scenario.

Step 3: Evaluate scenarios. Create a P&L distribution, which can be represented as a histogram.

Step 4: Compile and interpret results. Calculate the VaR at a 95% confidence level by ranking the P&L estimates from lowest to highest. The VaR is the fifth percentile of the values observed (where 5% of the worst outcomes sit). If that percentile falls between two observations, analysts may use the value above or below, or a weighted average of the two.

VaR can be calculated on a parametric basis if returns are normally distributed, or on a historical basis using actual data. While parametric models require assumptions about the shape of the distribution, historical simulations are nonparametric and require no assumptions. This allows the historical simulation to capture “drift” and “fat tails” (extreme events) that a normal distribution often underestimates. The two methods can be compared by calculating the expected standard deviation of the portfolio.

When calculating daily VaR, analysts typically assume a mean return of zero for two reasons:

The null hypothesis. It is difficult to statistically disprove that the market has no meaningful directional move on any single given day.

Volatility dominance. The daily movement of returns measured by standard deviation, or volatility, is usually significantly larger than any trend.

Fixed-income instruments require a special methodology because historical price changes for a specific security can be misleading. For example, a 10-year zero-coupon bond analyzed five years ago does not have the same risk profile as it does today. Five years ago, that bond had a duration, or interest rate sensitivity, of 10 years; today, its duration is only 5 years.



PROFESSOR'S NOTE

We will examine the concept of duration for bonds in the Fixed Income topic area.

As a fixed-income instrument approaches maturity, its price volatility decreases. As a result, to evaluate fixed-income instruments using historical simulation, analysts focus on changes in benchmark yields or prices, rather than those of the actual security in question. Analysts tend to use the implied price changes for zero-coupon bonds and evaluate how corresponding changes would affect the cash flows of fixed-income instruments.



MODULE QUIZ 9.1

1. The primary advantage of using historical simulation over a parametric approach, when dealing with assets that exhibit high excess kurtosis, is that a historical model:
 - A. reduces computational time and costs by using approximations.
 - B. uses a lookback period to eliminate the impact of regime shifts.
 - C. does not require data that fits specific distributional assumptions.
2. In a historical simulation, an analyst uses historical prices for a 10-year zero-coupon bond that was issued eight years ago. Is using these historical prices appropriate?
 - A. Yes.
 - B. No, because zero-coupon bonds are thinly traded.
 - C. No, because the bond's interest rate sensitivity changes during its life.

MODULE 9.2: BOOTSTRAP RESAMPLING

LOS 9.b: Describe bootstrap resampling, and explain how it can be used in investment applications.

A **bootstrap simulation** is a nonparametric simulation method that takes historical data, resamples it to create new datasets, and replaces the data each time they are used. This last step introduces randomness by allowing repeated sampling of the same data points. Compared to historical simulation, this more flexible approach to scenario generation reduces the risk of overfitting to specific historical events by preserving the empirical distribution of the original data. **Resampling** involves repeatedly taking samples from observed data to infer population parameters.

Bootstrapping uses resampling with replacement to generate distributions. Each data point is selected randomly, analyzed, and returned to the dataset to be reused. The selection probability for each point in the dataset with a sample size of n is $1/n$ for each draw. The ability to resample multiple times makes it easy to simulate a sampling distribution even when the original dataset is small. Generating a random variable from a distribution is known as **pulling**.

The fundamental difference between bootstrapping and historical simulation is the treatment of data. Historical simulation uses the past dataset directly, a true population, and relies on the events that took place to simulate possible future outcomes. By contrast, bootstrapping creates a simulated sample by resampling with replacement and treats this observed sample as a proxy for the population. That makes bootstrapping useful if a population distribution is unknown but easily inferred, as often occurs in finance.

With bootstrapping, there is no need for distributional assumptions, and it is easy to assess the stability of an estimate. A bootstrapping simulation is also relatively easy to run, although replacing the data requires a bit more computation compared to historical simulation. As with historical simulation, bootstrapping does not require theoretical models or external data.

One application of the bootstrapping method is valuing European-style call options, financial instruments that give the purchaser the right, but not the obligation, to buy the underlying asset. A European-style option can only be exercised at maturity and only has value if the price of the underlying asset is above the exercise price.

Theoretical option pricing models, such as the Black-Scholes-Merton formula (which is addressed in the Level II CFA curriculum), are often used to determine option values. However, these models are based on several assumptions that may not hold in the real world, including that volatility remains constant and that

returns follow a normal distribution. By contrast, bootstrapping circumvents these limiting assumptions with a nonparametric approach: resampling historical returns to simulate future price paths.

The steps in bootstrapping to simulate future price paths for a European-style call option are the same ones that apply to all simulations:

- Step 1: Set parameters.* In this case, the objective is to calculate the option value at maturity. The variables are the exercise price, current stock price, interest rate, and investment period.
- Step 2: Generate scenarios.* To construct a scenario, randomly sample the data by selecting a number of discrete points. For example, if there are fifteen data points, pull five random numbers between 1 and 15 (random sampling), look up the return that corresponds to that number, then repeat the process.
- Step 3: Evaluate scenarios.* The calculations include return, price, and option value. Multiple simulations are run to generate a distribution of potential outcomes with the simulated price paths, the mean price path, and confidence intervals.
- Step 4: Compile and interpret results.* An option value distribution may be more skewed than a share price distribution, reflecting the leveraged exposure to stock prices that options provide while limiting downside risk.

Another example of bootstrap simulation is to estimate portfolio volatility for target date funds, which shift their asset mix toward less risky investments as the investor's retirement date approaches. The steps for analyzing target date fund volatility using bootstrapping are as follows:

- Step 1: Set parameters.* Review the monthly returns over the period under observation and the fund's net asset value for each period.
- Step 2: Generate scenarios.* Create a predetermined number of simulated return scenarios using bootstrapping. This involves random sampling with replacement, assigning returns, and repeating the process.
- Step 3: Evaluate scenarios.* Calculate the standard deviation of each of the simulated returns to measure volatility and rank the simulated returns from lowest to highest.

Step 4: Compile and interpret results. The result is a range for the average monthly volatility of the target date fund. Comparing the volatility of different target date funds enables an investor to determine whether the fund is more aggressive (higher volatility) or less aggressive (lower volatility) than its peers.

Bootstrap simulation complements analytical methods that provide more detail on the direct relationships and their underpinnings. By eliminating the need for distributional assumptions, bootstrapping can be used in situations when the underlying data are complex and cannot be modeled with parametric distributions. This can be the case when distributions exhibit time-varying volatility, skewness, or fat tails. Bootstrapping is also a preferred technique when historical patterns in the data are expected to persist.

The strengths and weaknesses of bootstrapping simulation are outlined in Figure 9.1.

Figure 9.1: Bootstrapping Strengths and Weaknesses

Feature	Strengths	Weaknesses
Data requirements	<i>Minimal assumptions:</i> does not assume a specific underlying distribution (like normality), making it highly flexible for distributions that are unknown or hard to model	<i>Sensitive to sample quality:</i> relies entirely on the original sample being representative; if the initial data are biased or unrepresentative, the bootstrap will amplify those flaws
Implementation	<i>Simplicity:</i> easy to execute by resampling with replacement, bypassing the need for complex analytical or parametric models	<i>Computationally intensive:</i> generating thousands of resamples can be demanding and time-consuming, especially when dealing with larger datasets
Statistical power	<i>Statistical robustness:</i> highly effective for small datasets, providing reliable estimates of the variability of a statistic	<i>No analytical insight:</i> provides estimates but offers no theoretical explanation for the underlying behavior of the population
Estimation quality	<i>Robust variability estimates:</i> delivers reliable standard error estimates and confidence intervals while reducing the influence of outliers compared to historical simulations	<i>Limited extrapolation:</i> cannot simulate outcomes outside the range of the observed data, making it less effective for predicting extreme tail-risk events
Versatility	<i>Adaptable:</i> can be applied to a wide range of statistics, such as mean, median, and correlation when data are complex or nonparametric	<i>Not ideal for autocorrelated data:</i> unsuitable for time-series data without specialized modifications because bootstrapping breaks time-sequence dependencies
Context requirements	<i>No need for large external datasets:</i> works directly with the sample at hand, which is ideal when data are scarce	



MODULE QUIZ 9.2

- Which of the following is *most likely* a strength of bootstrapping?
 - It represents the statistical features of a population.
 - It provides only statistical estimates, not exact results.
 - Inputs may be limited by the distribution of actual outcomes.
- Which of the following *best* describes the fundamental difference between the scenario generation step in historical and bootstrap simulation? Historical simulation:
 - assumes a parametric distribution of returns, while bootstrapping relies on empirical observations from the sample.
 - relies on past events, while bootstrapping uses resampling with replacement to create new data combinations.

- C. is limited to the range of observed data, while bootstrapping can simulate outcomes outside the range of observed data.

MODULE 9.3: MONTE CARLO SIMULATION

LOS 9.c: Describe Monte Carlo simulation and explain how it can be used in investment applications.

A **Monte Carlo simulation** estimates outcomes using a random sampling technique. Instead of using raw data points, it relies on distributional assumptions (e.g., normal, log-normal) for input variables. By running several thousands of simulated trials, it produces a simulated frequency distribution of prices or returns, which is then used to generate risk and performance measures. An analyst should run as many simulations as is practical, given the tradeoff between the time needed to run them and the increased precision more simulations provide.

One objective of Monte Carlo simulation is to estimate portfolio performance over a specific future time frame. By drawing random observations from a simulated frequency distribution of portfolio returns, analysts can calculate the expected return and standard deviation to a specified degree of confidence. Because the underlying distribution assumptions are controlled, it is possible to track how sensitive the output is to changes in these assumptions.

Monte Carlo simulation is also used to value financial instruments, such as complex securities and derivatives that lack an **analytic pricing formula**—one that provides an answer through a closed-form expression without iteration or approximation. Conducting a Monte Carlo simulation is basically the same for securities that have an analytic pricing formula and those that do not.

When valuing European call options, a Monte Carlo simulation should produce the same results as the Black-Scholes-Merton formula. The process for conducting a Monte Carlo simulation is as follows:

Step 1: Set parameters. Define parameters such as the time horizon, the risk-free rate, and the option value. Assumptions specific to a Monte Carlo simulation include the price distribution (in this case, log-normal); the annualized volatility, or the dispersion of stock prices over the horizon; and the expected return.

Step 2: Generate scenarios. The calculation engine pulls a random number

from the specified distribution to create a potential price path.

Step 3: Evaluate scenarios. The calculation engine runs thousands of simulations, and the payoff is calculated for each path. Run the model for each scenario identified.

Step 4: Compile and interpret results. The distribution mean is calculated, then discounted back to present at the risk-free rate. The more simulations that are run on a single price path, the closer the value obtained will be to the result from the Black-Scholes-Merton formula.

One example of path-dependent options is Asian-style options, which realize a payoff only if the underlying stock price at maturity is greater than the average value of the stock over a specified period. Traditional models cannot handle the complexity of valuing path-dependent options, but Monte Carlo simulation can do so by simulating multiple price paths.

Monte Carlo simulations can also be used to value other complex financial instruments such as mortgage-backed securities, which are financial instruments with payments from pools of underlying mortgage loans. As we will see in the Fixed Income topic area, because mortgage borrowers have the right to prepay their loans, mortgage-backed securities behave as if they contain an embedded call option. These embedded options make it difficult to value these instruments by other means. Similarly, convertible bonds have embedded options and can be valued using Monte Carlo simulation.

Some circumstances require simulating multiple variables at the same time to capture complex interactions, correlations, and compound effects. For simulations that involve multiple variables, such as portfolios or options dependent on multiple assets, a **multivariate Monte Carlo simulation** is required. This process involves generating normal variables and transforming them using the **Cholesky decomposition**, which captures how the variables are related by mapping uncorrelated normal variables into correlated normal variables. A Cholesky decomposition uses a matrix factorization methodology to decompose the covariance or correlation matrix and generate correlated, random normally distributed numbers.

Steps to conduct a multivariate Monte Carlo simulation are as follows:

Step 1: Set parameters. Define the stochastic process for the assets. Under the usual log-normal price model, log-returns are assumed to follow a joint

normal distribution. Required parameters include each asset's volatility and the correlation between their returns.

Step 2: Generate scenarios. Transform two independent standard variables into correlated variables that reflect the specified correlation structure using the Cholesky decomposition. Generate two independent standard normal variables with a mean of 0 and standard deviations of 1, designated as Z_1 and Z_2 . Then, construct the correlated variables. Define the first variable as $X_1 = \sigma_{X_1} Z_1$ and link the two variables such that

$X_2 = \sigma_{X_2}(\rho Z_1 + \sqrt{1 - \rho^2} Z_2)$. The correlated variables are used as the inputs for each simulated scenario.

Step 3: Evaluate scenarios. Simulate thousands of full price paths, applying the correlated random variables at each step, then compute the payoff for each scenario identified.

Step 4: Compile and interpret results. Estimate the expected payoff by averaging across simulations. Discount the expected payoff back to the present value at the risk-free rate.

Monte Carlo simulation provides a statistical estimate rather than an exact result. To determine cause-and-effect relationships, analytical methods are more useful. Usually, Monte Carlo simulation is run alongside analytical methods rather than as a replacement, but it can be used as a valuation tool when analytical methods cannot be applied. Under these circumstances, an analyst can test how robust the analysis is by changing the assumptions and rerunning the estimates to evaluate the impact of the incremental changes.

The strengths and weaknesses of Monte Carlo simulation are summarized in Figure 9.2.

Figure 9.2: Monte Carlo Strengths and Weaknesses

Feature	Strengths	Weaknesses
Asset and portfolio modeling	<i>Wide applicability:</i> models assets, risk factors, and portfolios; handles nonlinear relationships, time-varying volatility, and fat-tailed distributions	<i>Complexity:</i> requires advanced modeling expertise, and complex, time-consuming setup, especially for multi-asset portfolios
Statistical distribution	<i>Flexibility for nonnormal distributions:</i> adapts to nonnormal distributions, such as those that exhibit skewness and excess kurtosis	<i>Model dependency:</i> accuracy depends on the reliability of the initial assumptions made about distributions and volatilities
Resource requirements	<i>Precision:</i> provides highly accurate estimates of expected outcomes, given enough simulations	<i>Computational intensity:</i> needs significant computational power to run many thousands of scenarios, making it slow and costly for real-time assessments
Relationship modeling	<i>Correlation handling:</i> models interdependencies among different assets, which is critical for portfolio risk	<i>Data requirements:</i> requires extensive historical data to estimate parameters like correlation and volatility
Market scenarios	<i>Forward-looking:</i> can simulate hypothetical or extreme future conditions that have never occurred in historical data	

Figure 9.3 compares the three simulation methods we have discussed based on their key features.

Figure 9.3: Comparison of Historical Simulation, Bootstrapping, and Monte Carlo Simulation¹

Feature	Historical Simulation	Bootstrapping	Monte Carlo Simulation
Source of data	Historical dataset	Observed sample	Assumed or modeled distribution
Sampling method	Sampling from historical data	Resampling with replacement from the sample	Random sampling from a predefined probability distribution
Assumptions	Assumes past data reflect future	Assumes sample represents population	Assumes the process generating outcomes can be modeled statistically
Illustrative use cases	Risk management, financial stress tests	Estimating confidence intervals, standard errors	Modeling complex systems, simulating long-term outcomes
Flexibility	Limited to historical patterns	Limited by sample size	Highly flexible, accommodates a wide range of scenarios
Uncertainty modeling	Captures only historically observed uncertainty	Captures uncertainty based on resampling variability	Captures hypothetical uncertainty based on modeled inputs
Computational intensity	Low to moderate	Moderate	High, depending on the complexity of the model



MODULE QUIZ 9.3

- Which of the following is *least likely* to be a limitation of Monte Carlo analysis?
 - Monte Carlo simulation is a statistical, rather than an analytic, method.
 - Results of the analysis are no better than the assumptions used to generate it.
 - Monte Carlo simulation is unable to provide answers to “what if” questions.
- In the scenario generation step of a multivariate simulation, what specific role does Cholesky decomposition perform regarding normally distributed variables?
 - It serves as the calculation engine for discounting future payoffs.
 - It decomposes the frequency distribution into a log-normal price path.
 - It maps uncorrelated normal variables into correlated normal variables.

KEY CONCEPTS

LOS 9.a

Historical simulation is a nonparametric method of assessing potential investment outcomes by applying actual past price or return changes to current portfolio positions. It makes no assumptions about the distribution of the data,

which enables it to capture extreme market events. It also inherently accounts for volatility clustering, recognizing that periods of low or high volatility persist over time. The method is highly valued for its simplicity and low computational cost, as it requires no complex calculations or the generation of synthetic scenarios.

LOS 9.b

Bootstrapping creates new datasets by repeatedly resampling from a historical dataset. Each data point is replaced once it has been selected, so it may appear multiple times in a single simulation. Bootstrapping is nonparametric in that it does not require prior assumptions about the population distribution. By using the empirical distribution of the actual data, it is flexible and works well with small sample sizes or complex data.

LOS 9.c

Monte Carlo simulation estimates outcomes by using a random sampling technique based on distributional assumptions, such as normal or log-normal distributions. Instead of relying on raw historical data, it runs several thousand simulated trials to produce a frequency distribution for prices or returns. While Monte Carlo simulation provides high precision and flexibility for nonnormal distributions, it is computationally intensive. Analysts must balance the need for increased precision against the time and computational power required to process those simulations.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 9.1

- . C Because historical simulation is nonparametric, it uses actual observed data points rather than forcing a normal distribution curve, which often underestimates the frequency of extreme outliers. Missing the impact of events that did not occur during the lookback period is a weakness of historical simulation. (LOS 9.a)
- . C The closer a bond is to maturity, the less sensitive it is to interest rate changes, so this bond's price volatility today is lower than it was eight years ago. For this reason, using its past price data in a simulation may produce misleading results. (LOS 9.a)

Module Quiz 9.2

- . **A** One of the strengths of bootstrapping is that it offers a good representation of the statistical features of a population. However, this method does not provide exact results, and the inputs can be limited by the distribution of actual outcomes. (LOS 9.b)
- . **B** Historical simulation relies on past events, while bootstrapping introduces randomness by allowing any data point to be drawn multiple times. Neither technique requires assumptions about the distribution of the data, but both are limited to the range of observed data. (LOS 9.b)

Module Quiz 9.3

- . **C** The ability to address “what if” questions (i.e., using input data outside the range that has been observed historically) is an advantage of Monte Carlo simulation. (LOS 9.c)
- . **C** Cholesky decomposition captures how the variables are related by mapping uncorrelated normal variables into correlated normal variables. (LOS 9.c)

¹ Reproduced from the Level I CFA curriculum learning module, “Simulation of Financial Asset Prices and Returns,” with permission from CFA Institute.

READING 10

APPLICATIONS OF SIMPLE LINEAR REGRESSION IN FINANCE

MODULE 10.1: LINEAR REGRESSION BASICS

LOS 10.a: Describe, interpret, and explain simple linear regression, including coefficient estimation using the least squares criterion.

The purpose of **simple linear regression** is to explain the variation in a dependent variable in terms of the variation in a single independent variable. Here, the term *variation* is interpreted as the degree to which a variable differs from its mean value. Don't confuse *variation* with *variance*—they are related, but they are not the same:

$$\text{variation in } Y = \sum_1^n (Y_i - \bar{Y})^2$$

- | The **dependent variable** (usually denoted as Y) is the variable whose variation is explained by the independent variable. We are interested in answering the question, “What explains fluctuations in the dependent variable?” The dependent variable is also referred to using the terms *explained variable*, *endogenous variable*, or *predicted variable*.
- | The **independent variable** (usually denoted as X) is the variable used to explain the variation of the dependent variable. The independent variable is also referred to as the terms *explanatory variable*, *exogenous variable*, or *predicting variable*.

EXAMPLE: Dependent vs. independent variables

Suppose you want to predict stock returns with GDP growth. Which variable is the independent variable?

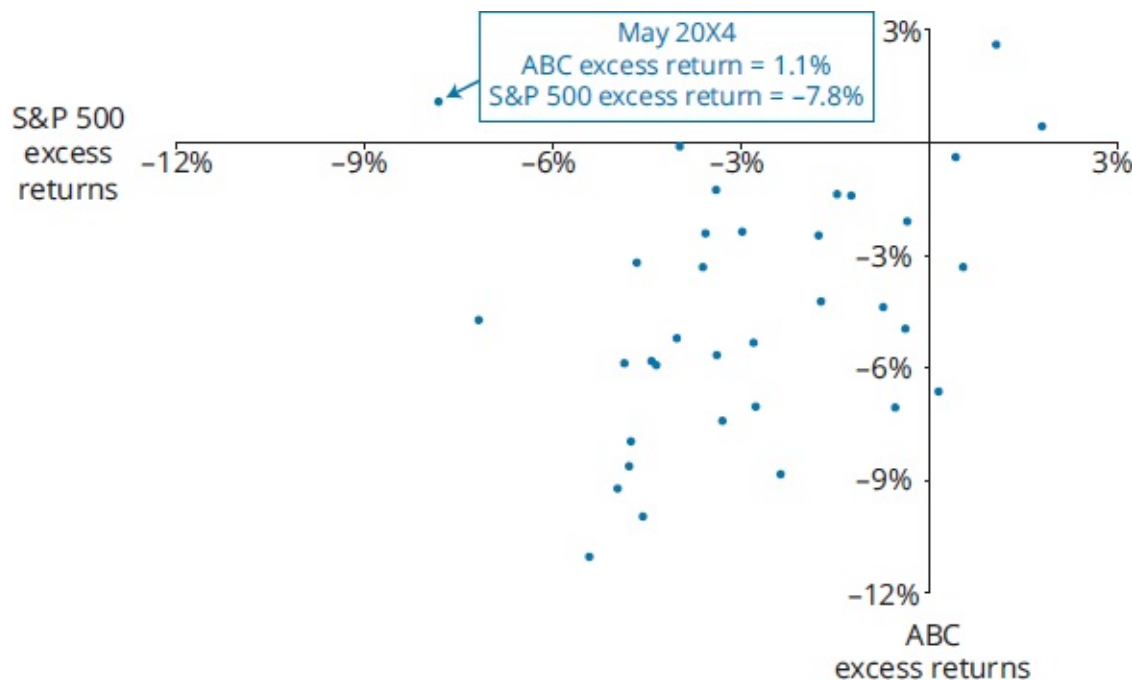
Answer:

Because GDP is going to be used as a predictor of stock returns, stock returns are being *explained* by GDP. Hence, stock returns are the dependent (explained) variable, and GDP is the independent (explanatory) variable.

Suppose we want to use excess returns on the S&P 500 (the independent variable) to explain the variation in excess returns on ABC common stock (the dependent variable). For this model, we define excess return as the difference between the actual return and the return on 1-month Treasury bills.

We would start by creating a scatter plot with ABC excess returns on the vertical axis and S&P 500 excess returns on the horizontal axis. Monthly excess returns for both variables from June 20X2 to May 20X5 are plotted in Figure 10.1. For example, look at the point labeled May 20X4. In that month, the excess return on the S&P 500 was -7.8% , and the excess return on ABC was 1.1% .

Figure 10.1: Scatter Plot of ABC Excess Returns vs. S&P 500 Index Excess Returns



The two variables in Figure 10.1 appear to be positively correlated: excess ABC returns tended to be positive (negative) in the same month that S&P 500 excess returns were positive (negative). This is not the case for all the observations, however (e.g., May 20X4). In fact, the correlation between the variables in this data set is approximately 0.40.

Simple Linear Regression Model

The following linear regression model is used to describe the relationship between two variables, X and Y :

$$Y_i = b_0 + b_1 X_i + \epsilon_i, \dots i = 1, \dots, n$$

where:

Y_i = i th observation of the dependent variable, Y

X_i = i th observation of the independent variable, X

b_0 = regression intercept term

b_1 = regression slope coefficient

ϵ_i = **residual** for the i th observation (also referred to as the disturbance term or **error term**). With this, a residual is the difference between the actual value of the dependent variable and the value forecast by the regression model, for a given value of the independent variable.

Based on this regression model, the regression process estimates an equation for a line through a scatter plot of the data that “best” explains the observed values for Y in terms of the observed values for X .

The linear equation, often called the line of best fit or **regression line**, takes the following form:

$$\hat{Y}_i = \hat{b}_0 + \hat{b}_1 X_i, i = 1, 2, 3, \dots, n$$

where:

$\hat{Y}_i$ = estimated value of Y_i given X_i

$\hat{b}_0$ = estimated intercept term

$\hat{b}_1$ = estimated slope coefficient



PROFESSOR'S NOTE

The hat “^” above a variable or parameter indicates an estimated value.

The regression line is just one of the many possible lines that can be drawn through the scatter plot of X and Y . The criteria used to estimate this line is the essence of linear regression. The regression line is the line that minimizes the sum of the squared differences (vertical distances) between the Y values predicted by the regression equation ($\hat{Y}_i = \hat{b}_0 + \hat{b}_1 X_i$) values, Y_i . The sum of the squared vertical distances between the estimated and actual Y values is referred to as the **sum of squared errors (SSE)**.

Thus, the regression line is the line that minimizes the SSE. This explains why simple linear regression is frequently referred to as **ordinary least squares**

(OLS) regression, and the values determined by the estimated regression equation, $\hat{Y}_i$ are called least-squares estimates.

The estimated **slope coefficient** ($\hat{b}_1$) for the regression line describes the change in Y for a one-unit change in X . It can be positive, negative, or zero, depending on the relationship between the regression variables. The slope term is calculated as follows:

$$\hat{b}_1 = \frac{\text{Cov}_{XY}}{\sigma_X^2}$$

The intercept term ($\hat{b}_0$) is the line's intersection with the Y -axis at $X = 0$. It can be positive, negative, or zero. A property of the least-squares method is that the intercept term may be expressed as follows:

$$\hat{b}_0 = \bar{Y} - \hat{b}_1 \bar{X}$$

where:

$\bar{Y}$ = mean of Y

$\bar{X}$ = mean of X

The intercept equation highlights the fact that the regression line passes through a point with coordinates equal to the mean of the independent and dependent variables (i.e., the point $\bar{X}, \bar{Y}$).

EXAMPLE: Computing the slope coefficient and intercept term

Compute the slope coefficient and intercept term using the following information:

$\text{Cov}(\text{S\&P 500}, \text{ABC}) = 0.000336$ Mean return, S\&P 500 = -2.70%

$\text{Var}(\text{S\&P 500}) = 0.000522$ Mean return, ABC = -4.05%

Answer:

The slope coefficient is calculated as $\hat{b}_1 = 0.000336 / 0.000522 = 0.64$.

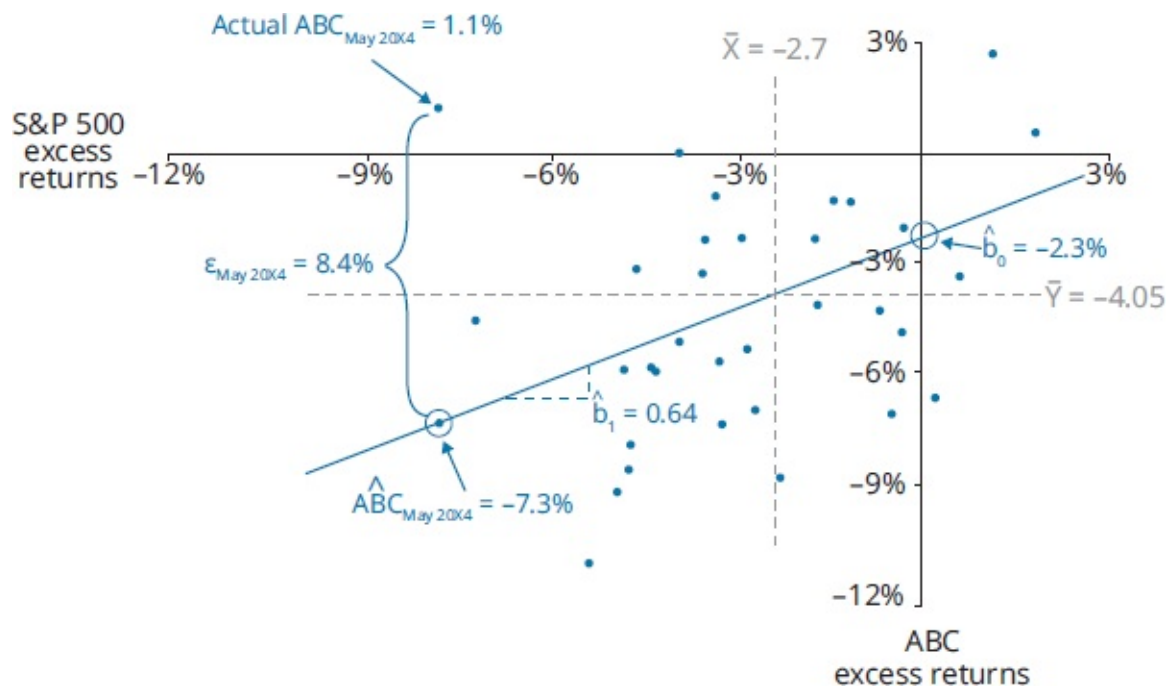
The intercept term is calculated as follows:

$$\bar{b}_0 = \overline{\text{ABC}} - \hat{b}_1 \overline{\text{S\&P 500}} = -4.05\% - 0.64(-2.70\%) = -2.3\%.$$

The estimated regression line that minimizes the SSE in our ABC stock return example is shown in Figure 10.2.

This regression line has an intercept of -2.3% and a slope of 0.64 . The model predicts that if the S&P 500 excess return is -7.8% (May 20X4 value), then the ABC excess return would be $-2.3\% + (0.64)(-7.8\%) = -7.3\%$. The residual (error) for the May 20X4 ABC prediction is 8.4% —the difference between the actual ABC excess return of 1.1% and the predicted return of -7.3% .

Figure 10.2: Estimated Regression Equation for ABC vs. S&P 500 Excess Returns



Interpreting a Regression Coefficient

The estimated intercept represents the value of the dependent variable at the point of intersection of the regression line and the axis of the dependent variable (usually, the vertical axis). In other words, the intercept is an estimate of the dependent variable when the independent variable is zero.

We also mentioned earlier that the estimated slope coefficient is interpreted as the expected change in the dependent variable for a one-unit change in the independent variable. For example, an estimated slope coefficient of 2 would indicate that the dependent variable is expected to change by two units for every one-unit change in the independent variable.

EXAMPLE: Interpreting regression coefficients

In the previous example, the estimated slope coefficient was 0.64, and the estimated intercept term was -2.3% . Interpret each coefficient estimate.

Answer:

The slope coefficient of 0.64 can be interpreted to mean that when excess S&P 500 returns increase (decrease) by 1%, ABC excess returns are expected to increase (decrease) by 0.64%.

The intercept term of -2.3% can be interpreted to mean that when the excess return on the S&P 500 is zero, the expected return on ABC stock is -2.3% .



PROFESSOR'S NOTE

The slope coefficient in a regression of the excess returns of an individual security (the y variable) on the return on the market (the x variable) is called the stock's beta, which is an estimate of systematic risk. Later in this reading we will demonstrate how to estimate a security's beta using linear regression.

Keep in mind, however, that any conclusions regarding the importance of an independent variable in explaining a dependent variable are based on the statistical significance of the slope coefficient. The magnitude of the slope coefficient tells us nothing about the strength of the linear relationship between the dependent and independent variables. A hypothesis test must be conducted, or a confidence interval must be formed, to assess the explanatory power of the independent variable. Later in this reading, we will perform these hypothesis tests.

Time Series, Cross-Sectional Data, and Panel Data

Data types used in regression analysis include time series, cross-sectional data, and panel data.

A **time series** is a set of observations taken periodically, most often at equal intervals over time. Daily closing prices of a stock over the past year and quarterly earnings per share of a company over a five-year period are examples of time series data.

Cross-sectional data refers to a set of comparable observations, all taken at one specific point in time. Today's closing prices of the 30 stocks in the Dow Jones Industrial Average (DJIA) and fourth-quarter earnings per share for 10 health care companies are examples of cross-sectional data.

Time series and cross-sectional data may be combined to form **panel data**. Panel data are often presented in tables, where each row (or column) represents cross-sectional data and each column (or row) represents time series data.



MODULE QUIZ 10.1

1. What is the *most appropriate* interpretation of a slope coefficient estimate equal to 10.0?
 - A. The predicted value of the dependent variable when the independent variable is 0 is 10.0.
 - B. For every 1-unit change in the independent variable, the model predicts that the dependent variable will change by 10 units.
 - C. For every 1-unit change in the independent variable, the model predicts that the dependent variable will change by 0.1 units.
2. The value of the dependent variable when the independent variable equals zero is described by a regression's:
 - A. error term.
 - B. intercept term.
 - C. slope coefficient.
3. A simple linear regression produces a slope coefficient of -2.7 and an intercept term of 34.8 . If the independent variable equals 8.4 , the predicted value for the dependent variable is *closest* to:
 - A. 12.
 - B. 58.
 - C. 290.

MODULE 10.2: ANALYSIS OF VARIANCE (ANOVA) AND GOODNESS OF FIT

LOS 10.b: Describe and compare the assumptions of simple linear regression, identify violations through analyzing residuals, evaluate the estimated model's goodness-of-fit and regression coefficients, and results of ANOVA estimates.

Linear regression is based on numerous assumptions. Most of the major assumptions pertain to the regression model's residual term (ϵ). Linear regression assumes the following:

A linear relationship exists between the dependent and the independent variables.

The variance of the residual term is constant for all observations (homoskedasticity).

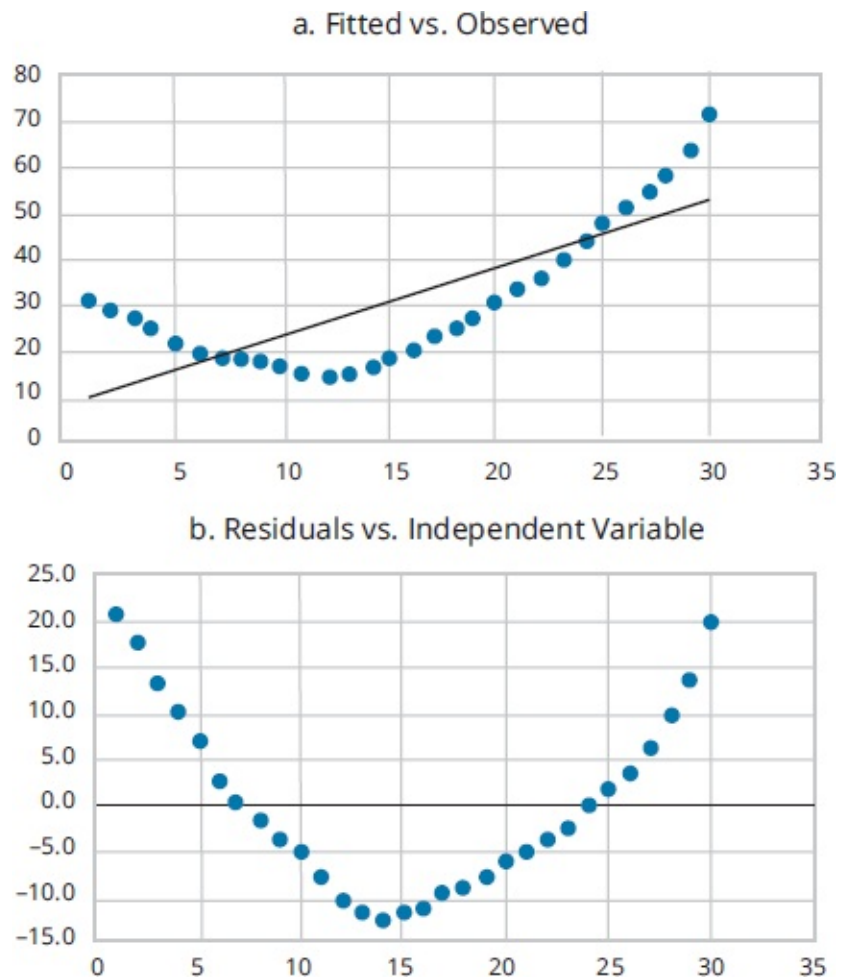
The residual term is independently distributed; that is, the residual for one observation is not correlated with that of another observation (or, the paired x and y observations are independent of each other).

The residual term is normally distributed.

Linear Relationship

A linear regression model is not appropriate when the underlying relationship between X and Y is nonlinear. In Panel A of Figure 10.3, we illustrate a regression line fitted to a nonlinear relationship. Note that the prediction errors (vertical distances from the dots to the line) are positive for low values of X , then increasingly negative for higher values of X , and then turning positive for still-greater values of X . One way of checking for linearity is to examine the model residuals (prediction errors) in relation to the independent regression variable. In Panel B, we show the pattern of residuals over the range of the independent variable: positive, negative, then positive. In a well-fitting regression model, the residuals should be random and show no pattern.

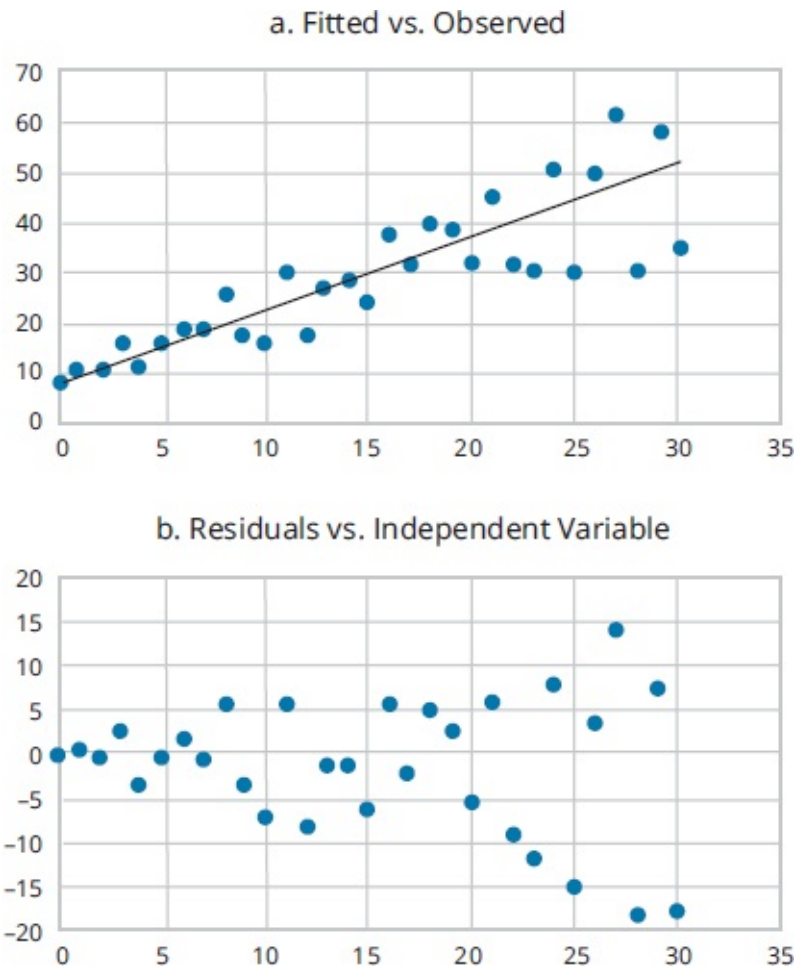
Figure 10.3: Nonlinear Relationship



Homoskedasticity

Homoskedasticity refers to the case where prediction errors all have the same variance. **Heteroskedasticity** refers to the situation when the assumption of homoskedasticity is violated. In Figure 10.4, Panel A shows a scatter plot of observations around a fitted regression line where the residuals (prediction errors) increase in magnitude with larger values of the independent variable X . Panel B shows the residuals plotted versus the value of the independent variable, and it illustrates that the variance of the error terms is not likely constant for all observations.

Figure 10.4: Heteroskedasticity



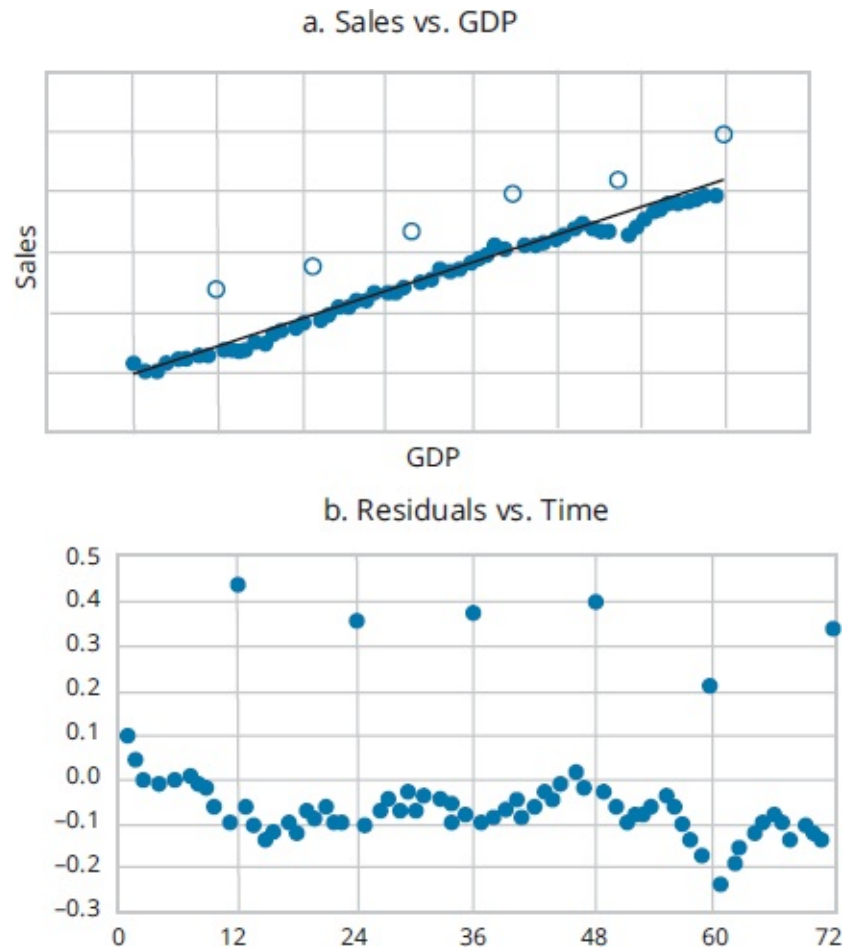
Another type of heteroskedasticity results if the variance of the error term changes over time (rather than with the magnitude of the independent variable). We could observe this by plotting the residuals from a linear regression model versus the dates of each observation and finding that the magnitude of the errors exhibits a pattern of changing over time.

Independence

Suppose we collect a company's monthly sales and plot them against monthly GDP, as in Figure 10.5, Panel A—and suppose we observe that some prediction errors (the unfilled dots) are noticeably larger than others. To investigate this, we plot the residuals versus time, as in Panel B. The residuals plot illustrates that large prediction errors occur every 12 months (in December). This suggests that there is seasonality in sales such that December sales (the unfilled dots in Figure 10.5) are noticeably farther from their predicted values than sales for the other months. If the relationship between X and Y is not independent, the residuals are

not independent, and our estimates of the model parameters' variances will not be correct. A regression is said to exhibit **autocorrelation** if its residuals are correlated.

Figure 10.5: Non-Independence



Normality

When the residuals (prediction errors) are normally distributed, we can conduct hypothesis testing to evaluate the goodness of fit of the model (discussed later). With a large sample size, based on the central limit theorem, our parameter estimates may be valid—even when the residuals are not normally distributed.

Outliers are observations (one or a few) that are far from our regression line (have large prediction errors or X values that are far from the others). Outliers will influence our parameter estimates so that the OLS model will not fit the other observations well.

Analysis of variance (ANOVA) is a statistical procedure for analyzing the total variability of the dependent variable. Let's define some terms before we discuss ANOVA tables:

- | The **total sum of squares (SST)** measures the total variation in the dependent variable. SST is equal to the sum of the squared differences between the actual Y values and the mean of Y :

$$SST = \sum_{i=1}^n (Y_i - \bar{Y})^2$$

- | The **sum of squares regression (SSR)** measures the variation in the dependent variable that is explained by the regression model. SSR is the sum of the squared distances between the predicted Y values and the mean of Y :

$$SSR = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2$$

- | The **mean square regression (MSR)** is the SSR divided by the number of independent variables. A simple linear regression has only one independent variable, so in this case, $MSR = SSR$.



PROFESSOR'S NOTE

Multiple regression (i.e., with more than one independent variable) is addressed in the Level II CFA curriculum.

- | The **sum of squared errors (SSE)** measures the unexplained variation in the dependent variable. It's also known as the sum of squared residuals or the residual sum of squares. SSE is the sum of the squared vertical distances between the actual Y values and the predicted Y values on the regression line:

$$SSE = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

- | The **mean squared error (MSE)** is the SSE divided by the degrees of freedom, which are $n - k - 1$, where k is the number of independent variables. A simple linear regression has only one independent variable, so in this case, degrees of freedom are $n - 2$.

You probably will not be surprised to learn the following:

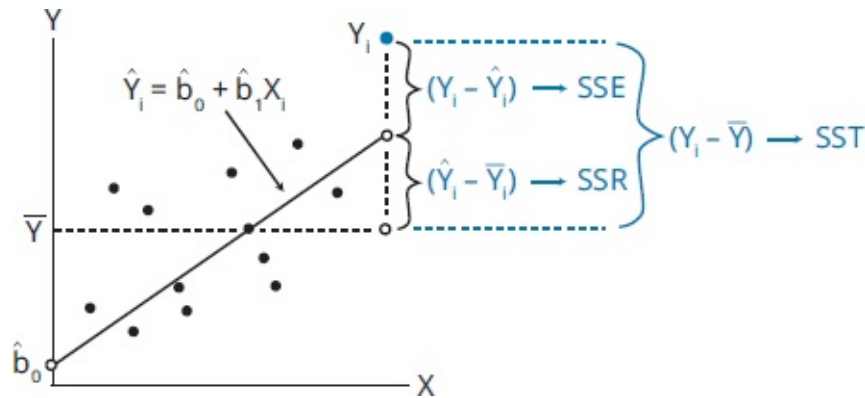
total variation = explained variation + unexplained variation

or:

$$SST = SSR + SSE$$

Figure 10.6 illustrates how the total variation in the dependent variable (SST) comprises SSR and SSE.

Figure 10.6: Components of Total Variation



The output of the ANOVA procedure is an ANOVA table, which is a summary of the variation in the dependent variable. ANOVA tables are included in the regression output of many statistical software packages. You can think of the ANOVA table as data source for many of the regression concepts discussed in this reading. A generic ANOVA table for a simple linear regression (one independent variable) is presented in Figure 10.7.

Figure 10.7: ANOVA Table for a Simple Linear Regression

Source of Variation	Degrees of Freedom	Sum of Squares	Mean Sum of Squares
Regression (explained)	1	SSR	$MSR = \frac{SSR}{k} = \frac{SSR}{1} = SSR$
Error (unexplained)	$n - 2$	SSE	$MSE = \frac{SSE}{n - 2}$
Total	$n - 1$	SST	

Standard Error of Estimate (SEE)

The **standard error of estimate (SEE)**, or **root mean squared error**, for a regression is the standard deviation of its residuals. The lower the SEE, the better the model fit:

$$SEE = \sqrt{MSE}$$

Coefficient of Determination (R^2)

The **coefficient of determination** (R^2) is defined as the percentage of the total variation in the dependent variable explained by the independent variable. For example, an R^2 of 0.63 indicates that the variation of the independent variable explains 63% of the variation in the dependent variable:

$$R^2 = SSR / SST$$



PROFESSOR'S NOTE

For simple linear regression (i.e., with one independent variable), the coefficient of determination, R^2 , may be computed by simply squaring the correlation coefficient, r . In other words, $R^2 = r^2$ for a regression with one independent variable.

EXAMPLE: Using the ANOVA table

Given the following ANOVA table based on 36 observations, calculate the R^2 and the SEE.

Completed ANOVA Table for ABC Regression

Source of Variation	Degrees of Freedom	Sum of Squares	Mean Sum of Squares
Regression (explained)	1	0.0076	0.0076
Error (unexplained)	34	0.0406	0.0012
Total	35	0.0482	

Answer:

$$R^2 = \frac{\text{explained variation (SSR)}}{\text{total variation (SST)}} = \frac{0.0076}{0.0482} = 0.158 \text{ or } 15.8\%$$

$$SEE = \sqrt{MSE} = \sqrt{0.0012} = 0.035$$

The *F*-Statistic

An *F*-test assesses how well a set of independent variables, as a group, explains the variation in the dependent variable.

The *F*-statistic is calculated as follows:

$$F = \frac{MSR}{MSE} = \frac{SSR / k}{SSE / (n - k - 1)}$$

where:

MSR = mean regression sum of squares

MSE = mean squared error

In multiple linear regression, the F -statistic is used to test the hypothesis that all the slope coefficients equal zero. For a simple linear regression, there is only one independent variable, so the F -test is equivalent to a t -test of the statistical significance of the slope coefficient:

$$H_0: b_1 = 0 \text{ versus } H_a: b_1 \neq 0$$

To determine whether b_1 is statistically significant using the F -test, the calculated F -statistic is compared with the critical F -value, F_c , at the appropriate level of significance. The degrees of freedom for the numerator and denominator with one independent variable are as follows:

$$\begin{aligned} df_{\text{numerator}} &= k = 1 \\ df_{\text{denominator}} &= n - k - 1 = n - 2 \end{aligned}$$

where:

n = number of observations

The decision rule for the F -test is to reject H_0 if $F > F_c$.

Rejecting the null hypothesis that the value of the slope coefficient equals zero at a stated level of significance indicates that the independent variable and the dependent variable have a significant linear relationship.

EXAMPLE: Calculating and interpreting the F -statistic

Use the ANOVA table from the previous example to calculate the F -statistic. Test the null hypothesis at the 5% significance level that the slope coefficient is equal to 0.

Answer:

$$F = \frac{MSR}{MSE} = \frac{0.0076}{0.0012} = 6.33$$

$$df_{\text{numerator}} = k = 1$$

$$df_{\text{denominator}} = n - k - 1 = 36 - 1 - 1 = 34$$

The null and alternative hypotheses are $h_0: b_1 = 0$ versus $h_a: b_1 \neq 0$. The critical F -value for 1 and 34 degrees of freedom at a 5% significance level is

approximately 4.1. (Remember, it's a one-tailed test, so we use the 5% F -table.) Therefore, we can reject the null hypothesis and conclude that the slope coefficient is significantly different than zero.

Hypothesis Test of a Regression Coefficient

A t -test may also be used to test the hypothesis that a true regression coefficient (b_0 or b_1 for a simple linear regression) is equal to a hypothesized value. For example, letting $\hat{b}_1$ be the point estimate for b_1 , the appropriate test statistic with $n - k - 1$ degrees of freedom is:

$$t_{b_1} = \frac{\hat{b}_1 - b_1}{s_{\hat{b}_1}}$$

The decision rule for tests of significance for regression coefficients is:

Reject H_0 if $t > +t_{\text{critical}}$ or $t < -t_{\text{critical}}$

Rejection of the null supports the alternative hypothesis that the slope coefficient is different from the hypothesized value of b_1 . To test whether an independent variable explains the variation in the dependent variable (i.e., it is statistically significant), the null hypothesis is that the true slope is zero ($b_1 = 0$). The appropriate test structure for the null and alternative hypotheses is:

$H_0: b_1 = 0$ versus $H_a: b_1 \neq 0$

EXAMPLE: Hypothesis test for significance of regression coefficients

The estimated slope coefficient from our earlier ABC example was 0.64 with a standard error equal to 0.26. Assuming that the sample has 36 observations, determine if the estimated slope coefficient is significantly different than zero at a 5% level of significance.

Answer:

The calculated test statistic is:

$$t = \frac{\hat{b}_1 - b_1}{s_{\hat{b}_1}} = \frac{0.064 - 0}{0.26} = 2.46$$

The critical two-tailed t -values are ± 2.03 (from the t -table with $df = 36 - 2 = 34$). Because $t > t_{\text{critical}}$ (i.e., $2.46 > 2.03$), we reject the null hypothesis and conclude that the slope is different from zero.

Note that the t -test for a simple linear regression is equivalent to a t -test for the correlation coefficient between x and y :

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

Indicator Variables

An **indicator variable** or **dummy variable** is a time series that takes on a value of 1 in periods when some condition holds, and 0 in periods when that condition does not hold. In a simple regression, an analyst can use an indicator as the independent variable to test whether a condition has a significant effect on the dependent variable.

For example, let's say an analyst wants to decide whether to classify a company's quarterly earnings as cyclical. One way to do so is with an indicator variable for economic contractions, giving it a value of 1 in quarters when the economy was in recession and 0 in all other quarters. A simple linear regression, with the company's quarterly earnings as the dependent variable and this indicator (*recession*) as the independent variable, is as follows:

$$\text{earnings}_i = b_0 + b_1(\text{recession}_i) + \epsilon_i$$

The analyst can test the hypothesis that the slope, b_1 , is equal to zero. If the analyst can reject the hypothesis, this would suggest the company's earnings are significantly different in recessionary quarters than they are in expansionary quarters.



PROFESSOR'S NOTE

In this example, we would expect the slope coefficient to be negative, because earnings would likely be lower in recessionary periods if the business cycle affects them.

This approach is equivalent to grouping the quarterly earnings into recessionary and nonrecessionary periods and performing a difference-in-means test. In fact, the slope coefficient b_1 should equal the difference in means. It represents how different earnings are, on average, in recessionary periods.



MODULE QUIZ 10.2

1. Which of the following is *least likely* a necessary assumption of simple linear regression analysis?

- A. The residuals are normally distributed.
 - B. There is a constant variance of the error term.
 - C. The dependent variable is uncorrelated with the residuals.
2. What is the appropriate alternative hypothesis to test the statistical significance of the intercept term in the following regression?
- $$Y = a_1 + a_2(X) + \varepsilon$$
- A. $H_A: a_1 \neq 0$.
 - B. $H_A: a_1 > 0$.
 - C. $H_A: a_2 \neq 0$.
3. The variation in the dependent variable explained by the independent variable is measured by the:
- A. mean squared error (MSE).
 - B. sum of squared errors (SSE).
 - C. regression sum of squares.

MODULE 10.3: PREDICTED VALUES AND FUNCTIONAL FORMS OF REGRESSION

LOS 10.c: Calculate and interpret predicted values, the standard error of the estimate, and prediction intervals for the dependent variable in a simple linear regression model and describe different functional forms.

Predicted values are values of the dependent variable based on the estimated regression coefficients and a prediction about the value of the independent variable. They are the values that are predicted by the regression equation, given an estimate of the independent variable.

For a simple regression, this is the predicted (or forecast) value of Y :

$$\hat{Y} = \hat{b}_0 + \hat{b}_1 X_p$$

where:

$\hat{Y}$ = predicted value of the dependent variable

X_p = forecast value of the independent variable

EXAMPLE: Predicting the dependent variable

Here is the following ABC regression equation:

$$\widehat{ABC} = -2.3\% + (0.64)(\widehat{S\&P\ 500})$$

Calculate the predicted value of ABC excess returns if forecast S&P 500 excess returns are 10%.

Answer:

The predicted value for ABC excess returns is determined as follows:

$$\widehat{ABC} = -2.3\% + (0.64)(10\%) = 4.1\%$$

Confidence Intervals for Predicted Values

This is the equation for the confidence interval for a predicted value of Y :

$$\hat{Y} \pm (t_c \times s_f) \Rightarrow [\hat{Y} - (t_c \times s_f) < Y < \hat{Y} + (t_c \times s_f)]$$

where:

t_c = two-tailed critical t -value at the desired level of significance with $df = n - 2$

s_f = standard error of the forecast

The challenge with computing a confidence interval for a predicted value is calculating s_f . On the Level I exam, it's highly unlikely that you will have to calculate the standard error of the forecast (it will probably be provided if you need to compute a confidence interval for the dependent variable). However, if you do need to calculate s_f , it can be done with the following formula for the variance of the forecast:

$$s_f^2 = SEE^2 \left[1 + \frac{1}{n} + \frac{(X - \bar{X})^2}{(n - 1)s_x^2} \right]$$

where:

SEE^2 = variance of the residuals = the square of the SEE

s_x^2 = variance of the independent variable

$\bar{X}$ = value of the independent variable for which the forecast was made

From this formula, we can identify the factors that affect the standard error of the forecast. Any of the following conditions will result in a smaller standard error of the forecast:

- | A smaller SEE
- | A larger number of observations
- | A predicted value of the independent variable that is closer to its mean
- | A greater variance in the independent variable

EXAMPLE: Confidence interval for a predicted value

Calculate a 95% confidence interval on the predicted value of ABC excess returns from the previous example. Assume the standard error of the forecast is 3.67%, and the forecast value of S&P 500 excess returns is 10%.

Answer:

This is the predicted value for ABC excess returns:

$$\widehat{ABC} = -2.3\% + (0.64)(10\%) = 4.1\%$$

The 5% two-tailed critical t -value with 34 degrees of freedom is 2.03. This is the confidence interval at the 95% confidence level:

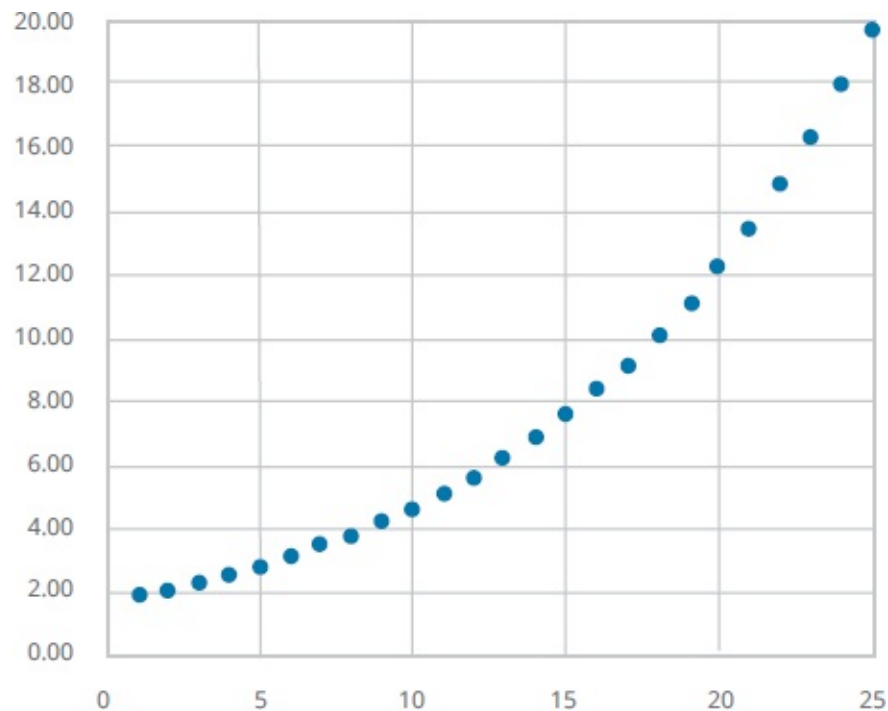
$$\begin{aligned}\widehat{ABC} &= (t_c \times s_f) \rightarrow [4.1\% \pm (2.03 \times 3.67\%)] = 4.1\% \pm 7.5\% \\ &= -3.4\% \text{ to } 11.6\%\end{aligned}$$

We can interpret this range to mean that, given a forecast value for S&P 500 excess returns of 10%, we can be 95% confident that the ABC excess returns will be between -3.4% and 11.6%.

Functional Forms of Simple Linear Regression

One of the assumptions of linear regression is that the relationship between X and Y is linear. What if that assumption is violated? Consider $Y = \text{EPS}$ for a company and $X = \text{time index}$. Suppose that EPS is growing at approximately 10% annually. Figure 10.8 shows the plot of actual EPS versus time.

Figure 10.8: Nonlinear Relationship



In such a situation, transforming one or both variables can produce a linear relationship. The appropriate transformation depends on the relationship between the two variables. One often-used transformation is to take the natural log of one or both variables. Here are some examples:

- | **Log-lin model.** This is if the dependent variable is logarithmic, while the independent variable is linear.
- | **Lin-log model.** This is if the dependent variable is linear, while the independent variable is logarithmic.
- | **Log-log model.** Both the dependent variable and the independent variable are logarithmic.

Selecting the correct functional form involves determining the nature of the variables and evaluating the goodness-of-fit measures (e.g., R^2 , SEE, F -stat).

Log-Lin Model

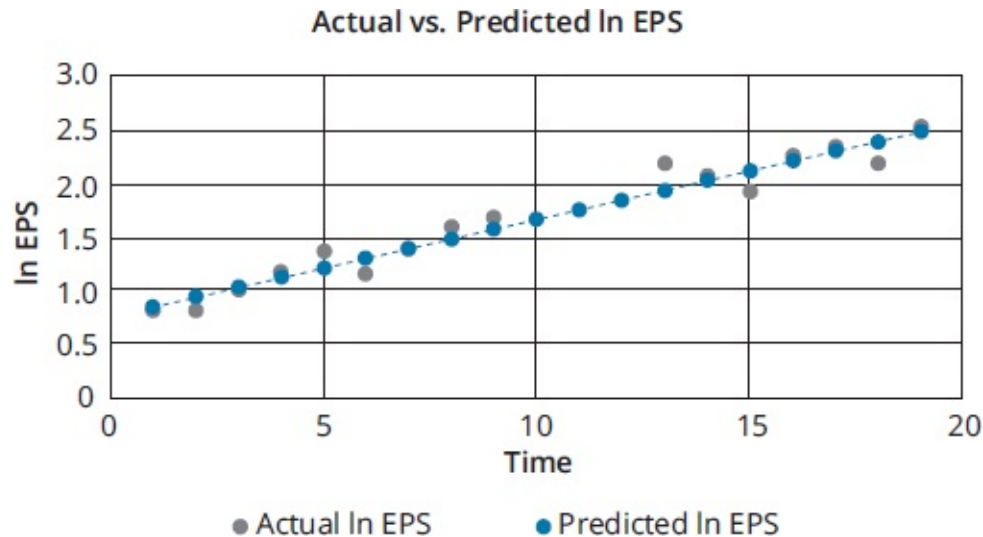
Taking the natural logarithm of the dependent variable, our model now becomes this:

$$\ln Y_i = b_0 + b_1 X_i + \epsilon_i$$

In this model, the slope coefficient is interpreted as the *relative* change in the dependent variable for an absolute change in the independent variable. Figure

10.9 shows the results after taking the natural log of EPS and fitting that data using a log-lin model.

Figure 10.9: Log-Lin Model, EPS Data



Lin-Log Model

Taking the natural logarithm of the independent variable, our model now becomes this:

$$Y_i = b_0 + b_1 \ln(X)_i + \varepsilon_i$$

In this model, the slope coefficient is interpreted as the *absolute* change in dependent variable for a *relative* change in the independent variable.

Log-Log Model

Taking the natural logarithm of both variables, our model now becomes this:

$$\ln Y_i = b_0 + b_1 \ln(X)_i + \varepsilon_i$$

In this model, the slope coefficient is interpreted as the relative change in the dependent variable for a relative change in the independent variable.



MODULE QUIZ 10.3

1. For a regression model of $Y = 5 + 3.5X$, the analysis (based on a large data sample) provides the standard error of the forecast as 2.5, and the standard error of the slope coefficient as 0.8. A 90% confidence interval for the estimate of Y when the value of the independent variable is 10 is *closest* to:
 - A. 35.1 to 44.9.
 - B. 35.6 to 44.4.
 - C. 35.9 to 44.1.

2. The appropriate regression model for a linear relationship between the relative change in an independent variable and the absolute change in the dependent variable is a:
- A. log-lin model.
 - B. lin-log model.
 - C. lin-lin model.

MODULE 10.4: ESTIMATING CAPM VALUES WITH SIMPLE LINEAR REGRESSION

LOS 10.d: Calculate and interpret the variable estimates of the capital asset pricing model (CAPM).

As we saw in our reading on The Return and Risk of a Financial Portfolio, the **capital asset pricing model (CAPM)** can be stated as:

$$E(R_i) = r_f + \beta[E(R_m) - R_f],$$

where:

$E(R_i)$ = expected return on asset i

R_f = risk-free interest rate

$E(R_m) - R_f$ = expected market risk premium, the difference between the market rate of return R_m and the risk-free rate

Because the only variable on the right side of the equation is the market risk premium, we describe the CAPM as a **one-factor risk model**. This feature allows us to use simple linear regression to estimate the expected return on an asset from observed data. Doing so assumes we can use past observations of the model's inputs as proxies for their expected values.



PROFESSOR'S NOTE

Later in the Level I curriculum, we will introduce multifactor risk models. Estimating these models requires the technique of multiple regression, which is addressed at Level II.

Typically, we begin by restating the model such that we are estimating an asset's expected excess return (i.e., above the risk-free rate) as a function of its degree of exposure (beta) to systematic risk as reflected in the market risk premium:

$$E(R_i) - r_f = \beta[E(R_m) - R_f]$$

The steps in estimating an asset's expected return from historical data using the CAPM are as follows:

Step 1: Collect data on the asset's returns, the returns on a market index (to use as a proxy for the market rate of return), and interest rates on short-term government debt (to use as a proxy for the risk-free rate). Analysts generally consider a minimum of 30 observations to be a statistically valid sample, such as a year of weekly data (1–3 years is typical) or three years of monthly data (3–5 years is typical).



PROFESSOR'S NOTE

The appropriate time period and frequency of observations depend on the nature and purpose of the analysis. For example, a speculator might use daily data for the most recent few months to form a short-term expectation for an asset's return, while a buy-and-hold investor might use monthly or quarterly data over a span of years that includes at least one full business cycle to estimate an asset's long-term expected return.

Step 2: Calculate the asset's excess returns ($R_i - R_f$) and the market risk premium ($R_m - R_f$) for each period in the sample.

Step 3: Estimate a least-squares regression using the asset's excess returns as the dependent variable and the market risk premium as the independent variable. If we let Z_i represent the asset's excess returns in period i and $Z_{m,i}$ represent the market risk premium in period i , we can state the regression as follows:

$$Z_i = b_0 + b_1(Z_{m,i}) + \varepsilon_i$$

Step 4: Finally, using the regression estimate of the asset's beta ($\hat{b}_1$) and expected values of the market risk premium, estimate the asset's expected return according to the CAPM as originally stated:

$$E(r) = r_f + \hat{b}_1[E(r_m) - r_f]$$

EXAMPLE: Estimating an asset's beta

An analyst performs a linear regression using a stock's excess returns as the dependent variable and the market risk premium as the independent variable. The regression returns the following statistics:

	Coefficient	Standard Error	<i>t</i> -Statistic
Intercept	−0.498	0.113	−4.369
Slope	0.857	0.024	35.841

1. Determine and interpret the asset's estimated beta.
2. Assuming a sufficiently large sample size to use critical values from a *z*-distribution, can the analyst reject the hypothesis that the asset's beta equals zero, at a 5% significance level?
3. Calculate a 95% confidence interval for the asset's beta.

Answer:

1. The estimated beta is the slope coefficient, 0.857. An asset with a beta less than one has excess returns that are relatively insensitive to the market's excess returns.
2. Because the *t*-statistic for the slope coefficient is greater than the critical value of 1.96, the analyst can reject the hypothesis that the asset's beta equals zero.
3. 95% confidence interval = $0.857 \pm 1.96(0.024) = 0.810$ to 0.904 .



MODULE QUIZ 10.4

1. In a typical ordinary least squares regression estimate of the CAPM, the dependent variable is an asset's:
 - A. price.
 - B. return.
 - C. excess return.
2. An asset has an estimated beta of 1.2. If the risk-free rate is expected to increase by 0.5% and the market risk premium is expected to decrease by 0.5%, according to the CAPM the expected return on this asset:
 - A. increases.
 - B. decreases.
 - C. remains the same.

KEY CONCEPTS

LOS 10.a

Linear regression provides an estimate of the linear relationship between an independent variable (the explanatory variable) and a dependent variable (the predicted variable).

The general form of a simple linear regression model is as follows:

$$Y_i = b_0 + b_1X_i + \varepsilon_i$$

The least-squares model minimizes the sum of squared errors (SSE):

$$\begin{aligned} \hat{b}_0 &= \text{fitted intercept} = \bar{Y} - \hat{b}_1\bar{X} \\ \hat{b}_1 &= \text{fitted slope coefficient} = \frac{\text{Cov}(X,Y)}{\text{variance of } X} \end{aligned}$$

The estimated intercept, $\hat{b}_0$, represents the value of the dependent variable at the point of intersection of the regression line and the axis of the dependent variable (usually, the vertical axis). The estimated slope coefficient, $\hat{b}_1$, is interpreted as the change in the dependent variable for a one-unit change in the independent variable.

LOS 10.b

Assumptions made regarding simple linear regression include the following:

A linear relationship exists between the dependent and the independent variable.

The variance of the residual term is constant (homoskedasticity).

The residual term is independently distributed (residuals are uncorrelated).

The residual term is normally distributed.

The total sum of squares (SST) measures the total variation in the dependent variable and equals the sum of the squared differences between its actual values and its mean.

The sum of squares regression (SSR) measures the variation in the dependent variable that is explained by the independent variable, and it equals the sum of the squared distances between the predicted values and the mean of the dependent variable.

The mean square regression (MSR) is the SSR divided by the number of independent variables. For a simple linear regression, $MSR = SSR$.

The sum of squared errors (SSE) measures the unexplained variation in the dependent variable and is the sum of the squared vertical distances between the actual and the predicted values of the dependent variable.

The mean squared error (MSE) is the SSE divided by the degrees of freedom ($n - 2$ for a simple linear regression).

The coefficient of determination, R^2 , is the proportion of the total variation of the dependent variable explained by the regression:

$$R^2 = \frac{SSR}{SST} = \frac{SST - SSE}{SST}$$

In simple linear regression, because there is only one independent variable ($k = 1$), the F -test tests the same null hypothesis as testing the statistical significance of b_1 using the t -test: $H_0: b_1 = 0$ versus $H_a: b_1 \neq 0$. With only one independent variable, F is calculated as follows:

$$F\text{-stat} = \frac{MSR}{MSE} \text{ with } 1 \text{ and } n - 2 \text{ degrees of freedom}$$

An indicator variable is a time series that has a value of 1 in periods when a specified condition holds, and 0 in all other periods. These can be used to test whether a dependent variable is significantly different in periods when the condition holds.

ANOVA Table for Simple Linear Regression ($k = 1$)

Source of Variation	Degrees of Freedom (df)	Sum of Squares	Mean Sum of Squares
Regression (explained)	1	SSR	$MSR = \frac{SSR}{k} = \frac{SSR}{1} = SSR$
Error (unexplained)	$n - 2$	SSE	$MSE = \frac{SSE}{n - 2}$
Total	$n - 1$	SST	

The standard error of the estimate in a simple linear regression is calculated as follows:

$$SEE = \sqrt{\frac{SSE}{n - 2}}$$

LOS 10.c

A predicted value of the dependent variable, $\hat{Y}$, is determined by inserting the predicted value of the independent variable, X_p , in the regression equation and calculating $\hat{Y}_p = \hat{b}_0 + \hat{b}_1 X_p$.

The confidence interval for a predicted Y -value is $[\hat{Y} - (t_c \times s_f) < Y < \hat{Y} + (t_c \times s_f)]$ where s_f is the standard error of the forecast.

Dependent Variable	Independent Variable	Model	Slope Interpretation
Logarithmic	Linear	Log-lin	<i>Relative</i> change in dependent variable for an <i>absolute</i> change in the independent variable
Linear	Logarithmic	Lin-log	<i>Absolute</i> change in dependent variable for a <i>relative</i> change in the independent variable
Logarithmic	Logarithmic	Log-log	<i>Relative</i> change in dependent variable for a <i>relative</i> change in the independent variable

LOS 10.d

The capital asset pricing model (CAPM) is a one-factor model according to which the expected return on an asset is a function of its degree of exposure to market risk. We can use ordinary least squares regression to estimate an asset's beta (exposure to market risk) from historical observations, and in turn, to estimate its expected return based on expected values of the market return.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 10.1

- . **B** The slope coefficient is best interpreted as the predicted change in the dependent variable for a 1-unit change in the independent variable; if the slope coefficient estimate is 10.0 and the independent variable changes by 1 unit, the dependent variable is expected to change by 10 units. The intercept term is best interpreted as the value of the dependent variable when the independent variable is equal to zero. (LOS 10.a)
- . **B** The intercept term is the value of the dependent variable when the independent variable equals zero. The slope coefficient is the change in the dependent variable for a one-unit change in the independent variable. (LOS 10.a)
- . **A** $\hat{Y} = 34.8 - 2.7(8.4) = 12.12$. (LOS 10.a)

Module Quiz 10.2

- . **C** The model does not assume that the dependent variable is uncorrelated with the residuals. It does assume that the independent variable is uncorrelated

with the residuals. (LOS 10.b)

- . **A** In this regression, a_1 is the intercept term. To test the statistical significance means to test the null hypothesis that a_1 is equal to zero, versus the alternative that a_1 is not equal to zero. (LOS 10.b)
- . **C** The regression sum of squares measures the amount of variation in the dependent variable explained by the independent variable (i.e., the explained variation). The SSE measures the variation in the dependent variable not explained by the independent variable. The MSE is equal to the SSE divided by its degrees of freedom. (LOS 10.b)

Module Quiz 10.3

- . **C** The estimate of Y , given $X = 10$, is $Y = 5 + 3.5(10) = 40$. The critical value for a 90% confidence interval with a large sample size (z-statistic) is approximately 1.65. Given the standard error of the forecast of 2.5, the confidence interval for the estimated value of Y is $40 \pm 1.65(2.5) = 35.875$ to 44.125 . (LOS 10.c)
- . **B** The appropriate model would be a lin-log model, in which the values of the dependent variable (Y) are regressed on the natural logarithms of the independent variable (X): $Y = b_0 + b_1 \ln(X)$. (LOS 10.c)

Module Quiz 10.4

- . **C** Typically, a regression estimate of the CAPM uses an asset's excess return over the risk-free rate as the dependent variable, and the market risk premium as the independent variable. (LOS 10.d)

- . **B** Expected return using CAPM: $E(r) = r_f + \hat{b}_1[E(r_m) - r_f]$

If r_f increases by 0.5% and $[E(r_m) - r_f]$ decreases by 0.5%, the change in this asset's expected return is $0.5\% + 1.2(-0.5\%) = -0.1\%$.

(LOS 10.d)

READING 11

INTRODUCTION TO FINANCIAL DATA SCIENCE

MODULE 11.1: INTRODUCTION TO FINANCIAL DATA SCIENCE

LOS 11.a: Describe how big data, machine learning, and artificial intelligence are used in financial data science, fintech, and investment management.

The objective of **financial data science** is to use quantitative and qualitative analysis to produce insights into a specific financial matter, such as consumer behavior patterns or predictions of stock market movements. Three of the main uses of data analysis techniques in the finance world are (1) evaluation of investment opportunities, (2) portfolio optimization, and (3) risk mitigation.

The term **fintech** refers to the meeting of the finance and technology worlds. In this reading, we focus on fintech developments that aid quantitative analysis within the investment industry.



PROFESSOR'S NOTE

The Level I CFA curriculum recognizes that there are alternative definitions of *fintech*, including referring to developments in technology that can be applied to the financial services industry. Companies that are in the business of developing technologies for the finance industry are also often referred to as fintech companies. However, for the Level I exam, stick to the definition just listed.

Big Data

The foundation of any form of financial data science is data. **Big data** is a widely used expression that refers to all the potentially useful information that is

generated. The data collected can be classified as structured, unstructured, or semi-structured.

- | **Structured data** are organized in tables with rows and columns, and typically stored in a database, such as balance sheet data for companies or records of market movements.
- | **Unstructured data** are not as organized (e.g., data from social media, voice recordings, pictures, and sensors), making it tricky for a human to effectively and efficiently deal with the information without assistance in the form of machine learning algorithms or AI. This often requires specialized computer programs or code.
- | **Semi-structured data** are somewhere in the middle. While they cannot be stored in a table, they can be tagged and stored in an organized fashion (e.g., HTML code).

Regardless of the data format, the first step in analysis is to obtain the relevant data. Data can come from traditional sources such as financial markets, company financial reports, and government economic statistics, or from alternative sources.

Alternative data sources can be categorized as follows:

- | *Individuals*. This is social media activity, website activity (such as clicks, time spent on page, searches), and personal data. The quantity of alternative data coming from individuals is growing (most of it unstructured).
- | *Business processes*. Transaction and corporate data are typically structured, such as sales information and banking records. Using this data appropriately can help to give leading or real-time indicators of performance—unlike traditional business sources, such as quarterly earnings reports, that provide a lagging view.
- | *Sensors*. Devices such as smartphones, smart buildings, connected vehicles, and satellites can provide unstructured data at vastly higher volumes than either of the two previous categories. The broad network of such devices is referred to as the **Internet of Things (IoT)**.

Given the exponentially growing quantity of data produced from nontraditional sources and the complex process required to convert this information into a usable format, businesses have emerged that collect, aggregate, and sell this

alternative data. Sellers and buyers need to be mindful of legal and ethical implications of dealing with this information. Using methods such as web scraping—the automated extraction of data from websites using specialized tools—may result in collecting information that is not in the public domain. One of the key risks is that some of the information captured may be personal and protected under privacy regulations.

Key characteristics of big data are its volume, velocity, and variety (the “3 Vs”):

- | The *volume* of data continues to grow by orders of magnitude. The units in which data can be measured have increased from megabytes (MB) and gigabytes (GB) to terabytes (TB) (1,000 gigabytes) and even petabytes (PB) (1,000 terabytes).
- | *Velocity* refers to how quickly data are communicated. Real-time data such as stock market price feeds are said to have low **latency**. Data that are only communicated periodically, or with a lag, have high latency. Generally speaking, the higher the velocity, the larger the volume of data collected, increasing from batch collections (typically measured in MB), to periodic (GB), near real time (TB) and finally, real time (PB).
- | The *variety* of data refers to the varying structures in which data may exist. The more complex the structure, the more storage volume it requires. Variety ranges from structured forms (tables, typically measured in MB) through to databases (GB); semi-structured forms like photos, web page code, or audio files (TB); and unstructured forms like social media, mobile, and video content (PB).

A fourth “V” is also important when considering financial big data: *veracity*. This considers whether the underlying information is reliable and credible. If not, a user must work to improve these factors before carrying out any analysis.

Taking advantage of big data presents numerous challenges. Analysts must ensure that the data they use are of high quality, accounting for the possibilities of outliers, bad or missing data, or sampling biases. The volume of data collected must be sufficient and appropriate for its intended use.

Data Science

The field of **data science** concerns how we extract information from big data. Data science describes methods for processing and visualizing data. Processing

methods include the following:

- | *Capture*. This is collecting data and transforming data into usable forms. If users require low latency, the data need to be presented in real time; otherwise, slower access may be acceptable.
- | *Curation*. This is assuring data quality by adjusting for errors or missing data.
- | *Storage*. This is archiving and accessing data. To assess the most appropriate form of storage, one must consider whether the data are structured, semi-structured, or unstructured, as well as their latency. Data can be stored in **data lakes**, which are centralized repositories capable of storing all three types of data in their original format.
- | *Search*. This is examining stored data to find needed information.
- | *Transfer*. This is moving data from their source or a storage medium to the analytical tool that will use the data. **Distributed computing frameworks** are systems that enable data to be processed and analyzed across multiple machines.

Visualization techniques can help present data in understandable ways. These techniques include the familiar charts and graphs that display structured data. Visualizing less structured data requires other methods, such as **word clouds** that illustrate the frequency with which words appear in a sample of text; heat maps; 3D graphics; or **mind maps** that display logical relations between concepts.

The typical information analyzed in **financial data science** includes stock prices and trading data, interest and exchange rates, asset returns, accounting data (e.g., financial statement analysis and corporate disclosures), and macroeconomic data. Some of these are high-frequency data (e.g., **high-frequency trading**, where data are updated on a millisecond or microsecond basis), while others are much slower to update, such as corporate disclosures. Financial data science requires expert knowledge not only in how to handle data, but also in finance to understand the inputs and outputs of the analysis. Much of the data used is sequential and time dependent—even cyclical. Methods used to analyze the data must be able to identify potential issues in the data such as noise, nonstationarity, interdependencies, or nonlinearity. Some financial data, such as stock performance, are prone to high probabilities of extreme events (i.e., fat tails). Other financial data may be harder to analyze due to the low volume available, such as data originating from emerging markets.

Data mining is one of the main types of data science, focusing on identifying patterns within large datasets that can then be used to predict outputs. Typically, this is used on structured data, but as preprocessing methods become more sophisticated, unstructured data can potentially be converted into a usable format for this purpose. Pattern identification methods are based on conditional probabilities (such as Bayes' theorem) of patterns emerging. One key risk of data mining is **overfitting**, where the model misinterprets noise in the data as important when identifying and extrapolating patterns, rather than excluding it.

Those who deal with financial data must comply with any financial regulatory requirements. Banks and financial service providers are regulated firms and must adhere to strict regulations concerning data management, risk assessment, and model validation. This ensures protection for clients and consumers, alongside upholding market integrity. Without this, mishandling of the data could, in extreme circumstances, lead to financial crises.

Even nonregulated firms must ensure they adhere to business law and any contractual obligations, though they have more flexibility in how they handle data. As the use of programs and AI grows, users increasingly focus on thoroughly developing appropriate models, with high levels of testing and verification. Data privacy is paramount, as is human review and interaction to ensure any decisions made by models are appropriate.

Machine Learning

In a world with increasing amounts of data available, data science covers the complex process of collecting, cleaning, modeling, and interpreting the data. Methods used need to be reliable and allow informed decisions to be made on a useful time scale, sometimes in real time. Techniques and systems such as machine learning and AI are growing in use to improve efficiency while maintaining or improving accuracy compared to human-led analysis.

One of the most prominent subcategories of AI is **machine learning (ML)**. In ML, a computer algorithm is given inputs of source data, with no assumptions about their probability distributions, and it may be given outputs of target data. The algorithm is designed to learn, without human assistance, how to model the output data based on the input data, or to learn how to detect and recognize patterns in the input data. The goal is to be able to identify statistically significant patterns within the data and create predictions as the output.

Interpreting these outputs requires expertise in the subject, whether it is fraud detection, portfolio optimization, or stock price forecasts.

- | In **supervised learning**, the input and output data are labeled, and the machine learns to model the outputs from the inputs. Then, the machine is given new data on which to use the model, with the intention of predicting outcomes. Financial data science is most likely to apply this in uses such as predicting stock and company performance. Assuming the underlying data are reliable, this should give accurate results if trained on sufficiently large enough datasets, but it might not adapt well to new scenarios.
- | In **unsupervised learning**, the input data are not labeled, and the machine learns to describe the structure of the data and identify patterns. No output data are provided. This can be useful in financial data science for activities like customer segmentation and anomaly detection. Unsupervised learning is widely used and understood. It tends to reduce the complexity (dimensionality) of data, so it can be helpful for exploring data that are otherwise difficult to analyze.
- | **Reinforcement learning** models learn through trial and error and are improved by providing feedback on their output in the form of rewards or penalties, which the models use to adjust their analytical techniques. The downside of this type of ML is that it can take significant training and resources to design a rewards system that effectively improves the model. Reinforcement learning has been used well for optimizing portfolios, as it is helpful in sequential decision-making where it can learn from its past decisions to improve on its next ones.
- | **Deep learning**, sometimes referred to as deep learning nets, is a branch of ML that uses layers of neural networks (described later in this reading) to identify patterns, beginning with simple patterns and advancing to more complex ones. Deep learning may employ supervised, unsupervised, or semi-supervised learning. Applications of deep learning include image and speech recognition and natural language processing.

Underfitting is a risk in ML. Underfitting occurs when the machine fails to identify actual patterns and relationships, treating true parameters as noise. This means that the model is not complex enough to describe the data. A further challenge with ML is that its results can be thought of as opaque or a “black

box,” producing outcomes based on relationships that are not readily explainable.

ML typically requires vast amounts of data. A typical process begins with a *training dataset*, in which the model looks for relationships. A *validation dataset* is then used to refine these connections, which can be applied to a *test dataset* to analyze its predictive ability. The training set is the largest of the three datasets, including around 60%–80% of the total data available (typically, for larger datasets, the percentage will be toward the upper end of this range; for smaller datasets, it will be toward the lower end). The validation and test datasets split the remaining 20%–40% of the data evenly.

Data are allocated randomly among the three datasets to reduce bias and the risk of overfitting, and to help prevent **data leakage**, which is information from outside the training dataset influencing the model’s learning. However, with time series data that rely on chronological order to be interpreted accurately, allocating randomly among the three datasets would lose underlying patterns and dependencies within the data. This also reopens the algorithm to data leakage if a later piece of data (which has only occurred because of earlier data) ends up in the training dataset. Two methods of allocating time series data are time-based splits and rolling-window validations.

Time-based split. The data are simply split in chronological order. The earliest 60%–80% of data is used in the training dataset, the next 10%–20% in the validation dataset, and the most recent 10%–20% in the testing dataset.

Rolling-window validation. This first trains the model on an initial portion of the period covered (e.g., January to July), validates it on a later period (e.g., September and October), and then goes back toward the beginning and trains on another dataset (e.g., February to August), thus “rolling the window” along.

Regardless of which method is chosen to allocate time series data, the period covered across the validation and test datasets should match the intended forecast horizon. For example, if an analyst wishes to forecast for the next two years, the algorithm needs to have been able to learn what patterns are presented over a two-year period.

Human expertise is required to ensure an appropriate model is selected and to ensure that the input data are reliable, have been cleaned (fixing or removing any errors in the original data), and minimize any biases. Expertise is also required to interpret and appropriately act on the model’s predictions and suggestions.

Neural Networks

For nonlinear relationships with many interdependencies, **neural networks** may be the most suitable form of ML. These are inspired by biological neural systems, and they have many layers of interconnected nodes (known as neurons) through which the data are processed to find patterns.

Neural networks begin with an initial or input layer that receives raw data, much like the other types of ML models, and they can engage in supervised or unsupervised learning. They have many hidden, or intermediate, layers that process this information and produce neurons to represent each different variable or characteristic, with the idea that the input data may link to these variables through several layers. Each layer has weights assigned to the different neurons, depending on their importance.

Neural networks can be trained through **backpropagation**, a form of supervised learning. During training, the data move through each layer, and an initial prediction is made. This prediction is compared to the actual outcome, a process known as a **forward pass**. Next, the neural network quantifies the error and calculates the loss. The objective of backpropagation is to minimize this loss. Once the loss has been calculated, the network tests to see how much a change in weights of the different neurons affects the error, known as the **loss gradient**. These weights are adjusted gradually (known as the **gradient descent**) to reduce the loss gradient as much as possible. This process repeats many times to improve the accuracy of the prediction. The model's task is to identify all these different links and apply appropriate weights before reaching the output, or final, layer where a prediction can be made.

More complex types of neural networks include **generative adversarial networks (GANs)** and **variational autoencoders (VAEs)**.

- | GANs combine two separate neural networks, a generator network and a discriminator network, to produce realistic synthetic data in a comparable way to how an analyst might use a spreadsheet to produce scenarios to stress test. The training is performed on historic market data, with the aim of producing plausible future scenarios. Though this can be helpful, one must review the use of any output of GANs carefully because they can be used to create highly realistic fake material.

- | VAEs start by compressing data into a compact form, then trying to reconstruct the data. This is often used in finance to reduce dimensionality and improve efficiency in analyzing large sets of financial data without compromising on pattern recognition, and it can assist in identifying anomalies. This is a form of probabilistic generative model, using encoder and decoder networks.

A more traditional way of analyzing nonlinear relationships is using **copulas**. These are mathematical functions, such as the Cholesky decomposition, which use marginal probability distributions to identify dependencies between random variables.

Neural networks are expected to increase in use within finance as technical capabilities improve and become more accessible, reducing the associated cost.

Natural Language Processing and Large Language Models

AI was initially created to perform tasks such as pattern recognition and decision-making in a faster and more accurate way than humans. In finance, this can be used to analyze qualitative as well as quantitative data.

Natural language processing (NLP) refers to the use of computers and AI to interpret human language. Speech recognition and language translation are among the uses of NLP. Possible applications in finance could be to check for regulatory compliance in examining employee communications, or to evaluate large volumes of research reports to detect more subtle changes in sentiment than can be discerned from analysts' recommendations alone.

For text analytics, the input is typically unstructured and in the form of writing (company filings, written reports) or voice recordings (quarterly earnings calls, surveys). NLP can translate these into a usable format much faster than humanly possible, while looking for patterns. **Lexical analysis** may also be used to analyze characteristics, such as word frequency.

Large language models (LLMs) are a form of generative AI that generate real-world scenarios in a human-like format. LLMs use neural networks to identify context and meaning within qualitative input data, such as a collection of market and company information. Nonfinancial input data, such as information on political polls for an upcoming election, can also have a significant impact on the finance-focused qualitative output. The output is human-like text covering the relevant topic, such as a summary of the input data or a performance forecast

based on the tone and sentiment of the input data. This is typically achieved through **self-supervised learning**, a form of unsupervised learning where the model creates its own labels for unlabeled input data and corrects its assigned weights as it learns. Training must first take place on a vast dataset to ensure the model learns the appropriate linguistics, such as semantics and grammar, as well as accurately interpreting finance-specific terminology.

One of the risks of LLMs is that, while they are well practiced in producing human-like text and predicting word patterns, they are not designed to assess or identify errors in financial data. Analysts must review the output with a skeptical, expert eye. To ensure accuracy, analysts should use LLMs alongside more traditional data-driven models.

Programming languages used in financial data science include the following:

- | *Python*. This is the basis for many fintech apps.
- | *R*. This is used for statistical analysis.
- | *Java*. This is used for many internet apps and can be run from different computers, servers, or operating systems concurrently.
- | *C/C++*. This is a specialized, high-speed programming language that is used for algorithmic and high-frequency trading.

Some of the databases used are *SQL* (structured, on a server), *SQLite* (structured, not on a server, most frequently used for mobile apps that require access to data), and *NoSQL* (for unstructured data).



MODULE QUIZ 11.1

1. In contrast with traditional data sources used in finance, which of the following is *most likely* classified as an alternative data source? A company's:
 - A. annual report.
 - B. banking records.
 - C. regulatory filings.
2. In machine learning, which of the following datasets is *most likely* to use the largest proportion of the underlying data?
 - A. Test dataset.
 - B. Training dataset.
 - C. Validation dataset.
3. Underfitting is *most accurately* described as treating:
 - A. noise as noise.
 - B. noise as true parameters.

- C. true parameters as noise.
- 4. In neural networks, the purpose of backpropagation is to:
 - A. train the model by using a system of rewards and penalties.
 - B. ensure data privacy is protected and other regulations are upheld.
 - C. minimize the difference between the predicted and actual outcome.

KEY CONCEPTS

LOS 11.a

Financial data science is the quantitative and qualitative analysis of financial information. Fintech refers to developments in technology that can be applied to the financial services industry.

Big data refers to the vast amount of information produced that could be useful for analysis, including data from traditional and alternative sources.

Characteristics of financial big data include its volume, velocity, variety, and veracity.

Data science includes appropriate approaches to the capture, curation, storage, search, and transfer of data to the appropriate model being used for analysis, as well as consideration of how the results should be presented.

Data mining focuses on identifying patterns from large datasets, which can then be used to predict outputs.

AI refers to computer systems that can be programmed to simulate human cognition. Machine learning (ML) and neural networks are examples of AI.

ML is programming that gives a computer system the ability to improve its performance of a task over time and is often used to detect patterns in large sets of data. The model can be trained using supervised or unsupervised learning. Some models will use reinforcement learning to improve their output.

ML splits data into three datasets: training (the largest), validation, and test. Data can be allocated randomly among these three, but with time-series data, more sophisticated allocation techniques such as time-based split and rolling-window validations should be used.

Key risks of data mining and ML are overfitting, where a model misinterprets noise as important, and underfitting, where a model misinterprets important information as noise.

Neural networks are used for nonlinear relationships. Starting with an input layer, there are then several hidden layers which link the variables to each other in a non-linear fashion, before reaching the output layer. Backpropagation is used to improve the accuracy of the output by calculating the loss and loss gradients, then changing the weights of each neuron to minimize the loss.

Applications of fintech to investment management include text analytics, natural language processing, risk governance, and algorithmic trading.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 11.1

- . **B** Data generated from business processes, such as banking and sales records, are considered alternative data. Financial reports and regulatory filings are traditional sources of data. (LOS 11.a)
- . **B** The training dataset comprises 60%–80% of the data available. The validation and test datasets each comprise 10%–20%. (LOS 11.a)
- . **C** Underfitting occurs when the machine fails to identify actual patterns and relationships, treating true parameters as noise. Treating noise as true parameters is overfitting. (LOS 11.a)
- . **C** Neural networks can be trained through backpropagation, where an initial prediction is compared to the actual outcome. The neural network will quantify this error and calculate the loss (the loss gradient) before changing the weights of different neurons to reduce the difference as much as possible. Reinforcement learning trains the model through a system of rewards and penalties. While it is important that data privacy is protected and other regulations are upheld, this is not related to backpropagation. (LOS 11.a)

TOPIC QUIZ: QUANTITATIVE METHODS

You have now finished the Quantitative Methods topic section. Please log into your Schweser online dashboard and take the Topic Quiz on this section. The Topic Quiz provides immediate feedback on how effective your study has been for this material. Questions are more exam-like than typical Module Quiz or QBank questions; a score of less than 70% indicates that your study likely needs improvement. These tests are best taken timed; allow 1.5 minutes per question.

READING 12

THE FIRM AND MARKET STRUCTURES

MODULE 12.1: BREAKEVEN, SHUTDOWN, AND SCALE

LOS 12.a: Determine and interpret breakeven and shutdown points of production, as well as how economies and diseconomies of scale affect costs under perfect and imperfect competition.

In economics, we define the **short run** for a firm as the time period over which some factors of production are fixed. Typically, we assume that capital is fixed in the short run so that a firm cannot change its scale of operations (plant and equipment) over the short run. All factors of production (costs) are variable in the **long run**. The firm can let its leases expire and sell its equipment, thereby avoiding costs that are fixed in the short run.

Shutdown and Breakeven Under Perfect Competition

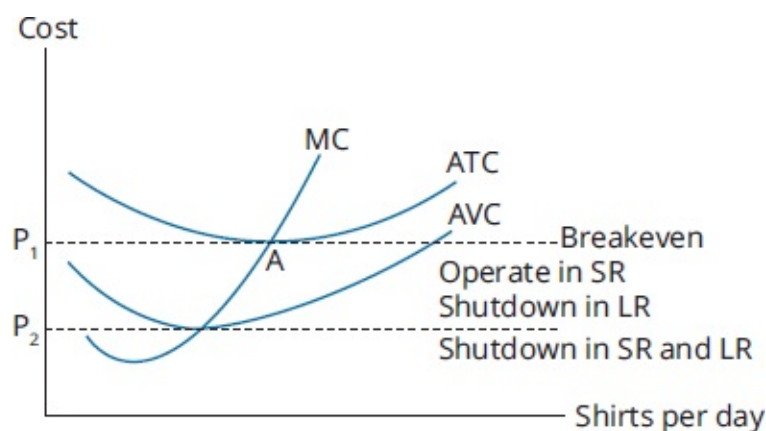
As a simple example of shutdown and breakeven analysis, consider a retail store with a one-year lease (fixed cost) and one employee (quasi-fixed cost), so that variable costs are simply the store's cost of merchandise. If the total sales (total revenue) just cover both fixed and variable costs, price equals both average revenue and average total cost—so we are at the breakeven output quantity, and economic profit equals zero.

During the period of the lease (the short run), as long as items are being sold for more than their variable cost, the store should continue to operate to minimize losses. If items are being sold for less than their average variable cost, losses would be reduced by shutting down the business in the short run.

In the long run, a firm should shut down if the price is less than average total cost, regardless of the relation between price and average variable cost.

For a firm under perfect competition (a price-taker), we can use a graph of cost functions to examine the profitability of the firm at different output prices. In Figure 12.1, at price P_1 , price and average revenue equal average total cost. At the output level of Point A, the firm is making an economic profit of zero. At a price above P_1 , economic profit is positive, and at prices less than P_1 , economic profit is negative (the firm has economic losses).

Figure 12.1: Shutdown and Breakeven



Because some costs are fixed in the short run, it will be better for the firm to continue production in the short run as long as average revenue is greater than average variable costs. At prices between P_1 and P_2 in Figure 12.1, the firm has losses, but the losses are smaller than would occur if all production were stopped. As long as total revenue is greater than total variable cost, at least some of the firm's fixed costs are covered by continuing to produce and sell its product. If the firm were to shut down, losses would be equal to the fixed costs that still must be paid. As long as price is greater than average variable costs, the firm will minimize its losses in the short run by continuing in business.

If average revenue is less than average variable cost, the firm's losses are greater than its fixed costs, and it will minimize its losses by shutting down production in the short run. In this case (a price less than P_2 in Figure 12.1), the loss from continuing to operate is greater than the loss (total fixed costs) if the firm is shut down.

In the long run, all costs are variable, so a firm can avoid its (short-run) fixed costs by shutting down. For this reason, if price is expected to remain below minimum average total cost (Point A in Figure 12.1) in the long run, the firm will shut down rather than continue to generate losses.

To sum up, if average revenue is less than average variable cost in the short run, the firm should shut down. This is its **short-run shutdown point**. If average revenue is greater than average variable cost in the short run, the firm should continue to operate, even if it has losses. In the long run, the firm should shut down if average revenue is less than average total cost. This is the **long-run shutdown point**. If average revenue is just equal to average total cost, total revenue is just equal to total (economic) cost, and this is the firm's **breakeven point**.

- | If $AR \geq ATC$, the firm should stay in the market in both the short and long run.
- | If $AR \geq AVC$, but $AR < ATC$, the firm should stay in the market in the short run but will exit the market in the long run.
- | If $AR < AVC$, the firm should shut down in the short run and exit the market in the long run.

Shutdown and Breakeven Under Imperfect Competition

For price-searcher firms (those that face downward-sloping demand curves), we could compare average revenue to ATC and AVC, just as we did for price-taker firms, to identify shutdown and breakeven points. However, marginal revenue is no longer equal to price.

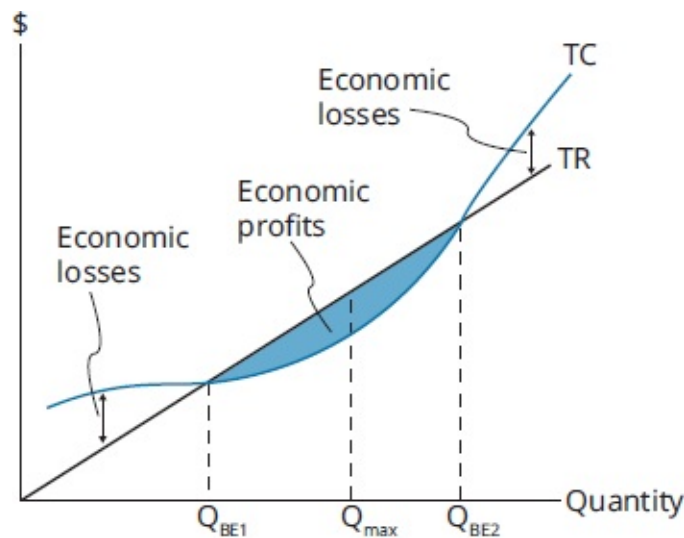
We can, however, still identify the conditions under which a firm is breaking even, should shut down in the short run, and should shut down in the long run in terms of total costs and total revenue. These conditions are as follows:

- | $TR = TC$: break even
- | $TC > TR > TVC$: firm should continue to operate in the short run but shut down in the long run
- | $TR < TVC$: firm should shut down in the short run and the long run

Because price does not equal marginal revenue for a firm in imperfect competition, analysis based on total costs and revenues is better suited for examining breakeven and shutdown points.

The previously described relations hold for both price-taker and price-searcher firms. We illustrate these relations in Figure 12.2 for a price-taker firm (TR increases at a constant rate with quantity). Total cost equals total revenue at the breakeven quantities Q_{BE1} and Q_{BE2} . The quantity for which economic profit is maximized is shown as Q_{max} .

Figure 12.2: Breakeven Point Using the Total Revenue/Total Cost Approach



If the entire TC curve exceeds TR (i.e., no breakeven point), the firm will want to minimize the economic loss in the short run by operating at the quantity corresponding to the smallest (negative) value of $TR - TC$.

EXAMPLE: Short-run shutdown decision

For the last fiscal year, Legion Gaming reported total revenue of \$700,000, total variable costs of \$800,000, and total fixed costs of \$400,000. Should the firm continue to operate in the short run?

Answer:

The firm should shut down. Total revenue of \$700,000 is less than total costs of \$1,200,000, and it is also less than total variable costs of \$800,000. By shutting down, the firm will lose an amount equal to fixed costs of \$400,000. This is less than the loss of operating, which is $TR - TC = \$500,000$.

EXAMPLE: Long-run shutdown decision

Suppose, instead, that Legion Gaming reported total revenue of \$850,000. Should the firm continue to operate in the short run? Should it continue to operate in the long run?

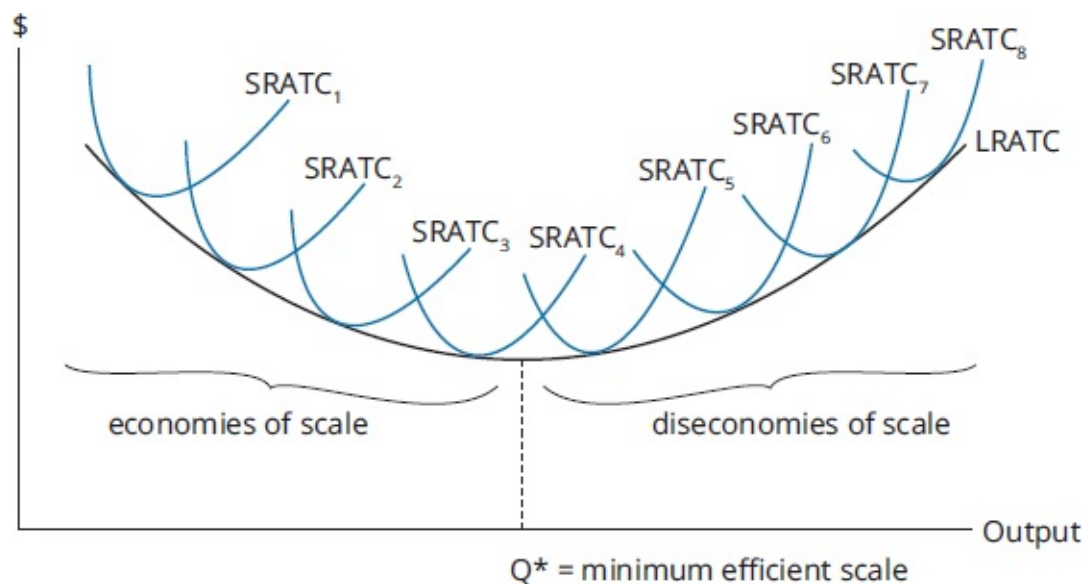
Answer:

In the short run, $TR > TVC$, and the firm should continue operating. The firm should consider exiting the market in the long run, as TR is not sufficient to cover all of the fixed costs and variable costs.

Economies and Diseconomies of Scale

While plant size is fixed in the short run, in the long run, firms can choose their most profitable scale of operations. Because the long-run average total cost (LRATC) curve is drawn for many different plant sizes or scales of operation, each point along the curve represents the minimum ATC for a given plant size or scale of operations. In Figure 12.3, we show a firm's LRATC curve along with short-run average total cost (SRATC) curves for many different plant sizes, with $SRATC_{n+1}$ representing a larger scale of operations than $SRATC_n$.

Figure 12.3: Economies and Diseconomies of Scale



We draw the LRATC curve as U-shaped. Average total costs first decrease with larger scale, but eventually begin to increase with larger scale. The lowest point on the LRATC corresponds to the scale or plant size at which the average total cost of production is at a minimum. This scale is sometimes called the **minimum efficient scale**. Under perfect competition, firms must operate at minimum

efficient scale in long-run equilibrium, and LRATC will equal the market price. Recall that under perfect competition, firms earn zero economic profit in long-run equilibrium. Firms that have chosen a different scale of operations with higher average total costs will have economic losses and must either leave the industry or change to the minimum efficient scale.

The downward-sloping segment of the LRATC curve presented in **economies of scale** (or *increasing returns to scale*) are present. Economies of scale result from factors such as labor specialization, mass production, and investment in more efficient equipment and technology. In addition, the firm may be able to negotiate lower input prices with suppliers as it increases in size and purchases more resources. A firm operating with economies of scale can increase its competitiveness by expanding production and reducing costs.

The upward-sloping segment of the LRATC curve indicates that **diseconomies of scale** are present. Diseconomies of scale may result as the increasing bureaucracy of larger firms leads to inefficiency, problems with motivating a larger workforce, and greater barriers to innovation and entrepreneurial activity. A firm operating under diseconomies of scale will want to decrease output and move back toward the minimum efficient scale. The US auto industry is an example of an industry that has exhibited diseconomies of scale.

There may be a relatively flat portion at the bottom of the LRATC curve that exhibits *constant returns to scale*, or relatively constant costs across a range of plant sizes.



MODULE QUIZ 12.1

1. In a purely competitive market, economic losses indicate that:
 - A. price is below average total costs.
 - B. collusion is occurring in the marketplace.
 - C. firms need to expand output to reduce costs.
2. A firm is likely to operate in the short run as long as price is at least as great as:
 - A. marginal cost.
 - B. average total cost.
 - C. average variable cost.
3. A firm's average revenue is greater than its average variable cost and less than its average total cost. If this situation is expected to persist, the firm should:
 - A. shut down in the short run and in the long run.
 - B. shut down in the short run, but operate in the long run.
 - C. operate in the short run, but shut down in the long run.

4. If a firm increases its plant size by 10% and its minimum average total cost increases by 10%, the firm is experiencing:
- A. constant returns to scale.
 - B. diseconomies of scale.
 - C. economies of scale.

MODULE 12.2: CHARACTERISTICS OF MARKET STRUCTURES

LOS 12.b: Describe characteristics of perfect competition, monopolistic competition, oligopoly, and pure monopoly.

Perfect competition results in firm demand that is horizontal (perfectly elastic) at the market price. The firm demand curves for the three other market structures we discuss are all downward sloping. When a firm's demand curve slopes downward, marginal revenue (MR) is less than price. For both horizontal and downward-sloping demand curves, a firm will maximize profits by producing the quantity for which MR is just equal to marginal cost.

While it may not be true in every case, it may be useful to think of firms under pure monopoly as having the steepest demand curves. Firms under monopolistic competition typically have relatively elastic downward-sloping demand curves, while firms in an oligopoly market will face downward-sloping demand curves somewhere between these two extremes.

We can analyze where a market falls along the spectrum from perfect competition to pure monopoly by examining five factors:

- Number of firms and their relative sizes
- Degree to which firms differentiate their products
- Bargaining power of firms with respect to pricing
- Barriers to entry into or exit from the industry
- Degree to which firms compete on factors other than price

Perfect competition refers to a market in which many firms produce identical products, barriers to entry into the market are very low, and firms compete for sales only on the basis of price. Firms face perfectly elastic (horizontal) demand curves at the price determined in the market because no firm has a large enough portion of the overall market to affect the market price of the good. The market

for wheat in a region is a good approximation of such a market. Overall market supply and demand determine the price of wheat, and each producer can sell all that they choose to at that price.

Monopolistic competition differs from perfect competition in that products are not identical. Each firm differentiates its product(s) from those of other firms through some combination of differences in product quality, product features, and marketing. The demand curve faced by each firm is downward sloping (i.e., neither perfectly elastic nor perfectly inelastic). Prices that producers charge are not identical because of perceived differences among their products, and typically, barriers to entry are low. The market for toothpaste is a good example of monopolistic competition. Firms differentiate their products through features and marketing with claims of more attractiveness, whiter teeth, fresher breath, and even actually cleaning your teeth and preventing decay. If the price of your personal favorite increases, you are not likely to immediately switch to another brand as we assume under perfect competition. Some customers may switch brands in response to a 10% increase in price, and some may not. This is why firm demand is downward sloping rather than perfectly elastic.

The most important characteristic of an **oligopoly** market is that only a few firms are in the industry. In such a market, each firm must consider the actions and responses of other firms in setting price and business strategy. We say that such firms are *interdependent*. While the firms' products are typically good substitutes for each other, they may be either quite similar or differentiated through features, branding, marketing, and quality. Barriers to entry are typically high, often because of economies of scale in production or marketing, which accounts for the existence of a small number of firms with relatively large market shares. Demand can be more or less elastic than for firms in monopolistic competition. The automobile market is dominated by a small number of large firms and can be characterized as an oligopoly. The product and pricing decisions of Toyota certainly affect those of Ford, and vice versa. Automobile makers compete based on price, but they also compete through marketing, product features, and quality, which are often signaled strongly through the brand name. The oil industry also has a few firms with large market shares, but their products are, in most cases, good substitutes for each other.

A **monopoly** market is characterized by a single seller of a product with no good substitutes. This fact alone means that the firm faces a downward-sloping demand curve (the market demand curve) and has the power to choose the price

at which it sells its product. High barriers to entry protect a monopoly producer from competition. One source of monopoly power is the protection offered by copyrights and patents. Another possible source of monopoly power is control over a resource specifically needed to produce the product. Most frequently, monopoly power is supported by specific laws or government regulation (e.g., a local electric utility).

Figure 12.4 shows the key features of each market structure.

Figure 12.4: Characteristics of Market Structures

	Perfect Competition	Monopolistic Competition	Oligopoly	Monopoly
Number of sellers	Many firms	Many firms	Few firms	Single firm
Barriers to entry	Very low	Low	High	Very high
Nature of substitute products	Very good substitutes	Good substitutes, but differentiated	Good substitutes or differentiated	No good substitutes
Nature of competition	Price only	Price, marketing, features	Price, marketing, features	Advertising
Pricing power	None	Some	Some to significant	Significant

LOS 12.c: Explain supply and demand relationships under monopolistic competition, including the optimal price and output for firms as well as pricing strategy.

Monopolistic competition has the following market characteristics:

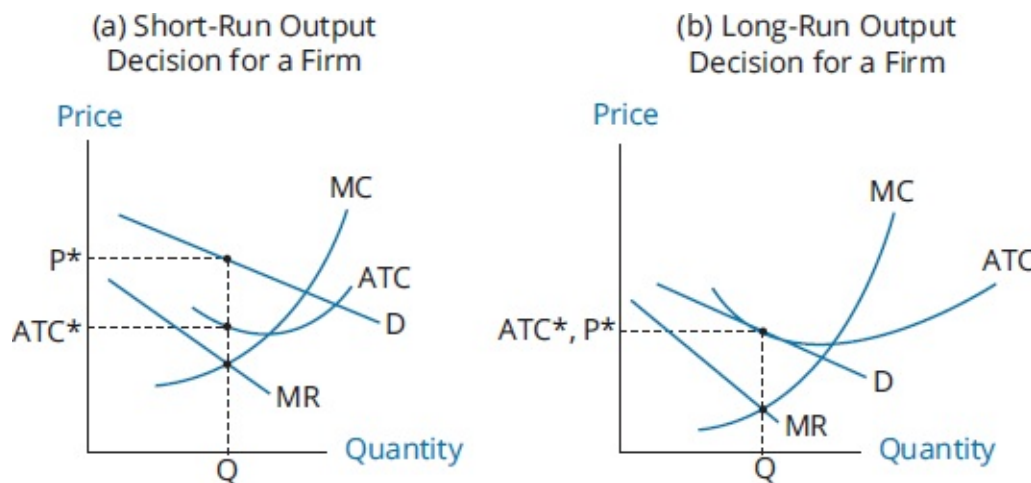
- | *A large number of independent sellers.* (1) Each firm has a relatively small market share, so no individual firm has any significant power over price. (2) Firms only need to pay attention to average market price, not the prices of individual competitors. (3) There are too many firms in the industry for collusion (price-fixing) to be possible.
- | *Differentiated products.* Each producer has a product that is, in some way, different from those of its competitors (in the minds of consumers). The competing products are considered close substitutes for one another.

- ▮ *Firms compete less on price and more on marketing, perceived quality, and differences in features.* Firms must make price and output decisions because they face downward-sloping demand curves.
- ▮ *Low barriers to entry.* The cost of entering the market and exiting the market are relatively low.

Think about the market for toothpaste. Brands of toothpaste are quite similar, and it is reasonable to assume that toothpaste is not too difficult or costly to produce. But brands are differentiated based on specific features, on influential advertising and marketing, and on the reputations of the producers.

The price/output decision for monopolistic competition is illustrated in Figure 12.5. Panel A of Figure 12.5 illustrates the short-run price/output characteristics of monopolistic competition for a single firm. As indicated, firms in monopolistic competition maximize economic profits by producing where marginal revenue (MR) equals marginal cost (MC), and by charging the price for that quantity from the demand curve, D . Here, the firm earns positive economic profits because price, P^* , exceeds average total cost, ATC^* . Due to low barriers to entry, competitors can enter the market in pursuit of these economic profits.

Figure 12.5: Short-Run and Long-Run Output Under Monopolistic Competition

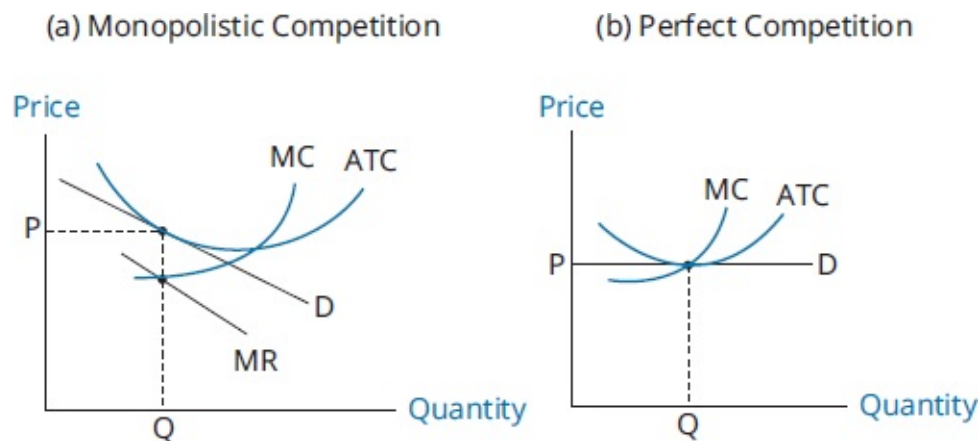


Panel B of Figure 12.5 illustrates long-run equilibrium for a *representative* firm after new firms have entered the market. As indicated, the entry of new firms shifts the demand curve faced by each individual firm down to the point where price equals average total cost ($P^* = ATC^*$), such that economic profit is zero. At this point, there is no longer an incentive for new firms to enter the market, and long-run market equilibrium is established. A firm in monopolistic

competition continues to produce at the quantity where $MR = MC$, but it no longer earns positive economic profits. We can get to a similar long-run equilibrium even without entry of new firms if each firm increases its marketing spending (a component of ATC) to either increase or defend its market share until ATC has increased to that shown in Panel B. If all firms compete in this way, each firm will produce Q^* and sell at P^* but earn no economic profit because marketing costs have increased ATC to ATC^* , which is equal to price. Advertising expenses are often relatively high for firms in monopolistic competition.

Figure 12.6 illustrates the differences between long-run equilibrium in markets with monopolistic competition and markets with perfect competition. Note that with monopolistic competition, average total cost is not at a minimum for the quantity produced (suggesting excess capacity, or an inefficient scale of production), and the price is slightly higher than under perfect competition. The point to consider here, however, is that perfect competition is characterized by no product differentiation. The question of the efficiency of monopolistic competition becomes, “Is there an economically efficient amount of product differentiation?”

Figure 12.6: Firm Output Under Monopolistic and Perfect Competition



In a world with only one brand of toothpaste, clearly, average production costs would be lower. That fact alone probably does not mean that a world with only one brand or type of toothpaste would be a better world. While product differentiation has costs, it also has benefits to consumers. Consider the market for a pharmaceutical that reduces blood pressure to prolong life. There may be several competing drugs that are more or less effective for, or well or poorly

tolerated by, different groups of patients. In this case, we may find that firm demand curves are relatively steep compared to those for brands of toothpaste, indicating that the competing drugs are not considered good substitutes for many patients.

LOS 12.d: Explain supply and demand relationships under oligopoly, including the optimal price and output for firms as well as pricing strategy.

Compared to monopolistic competition, an oligopoly market has higher barriers to entry and fewer firms. The other key difference is that the firms are interdependent; a price change by one firm can be expected to be met by a price change by its competitors in response. This means that the actions of another firm will directly affect a given firm's demand curve for the product. Given this complicating fact, models of oligopoly pricing and profits must make numerous important assumptions. In the following, we describe four of these models and their implications for price and quantity:

Kinked demand curve model

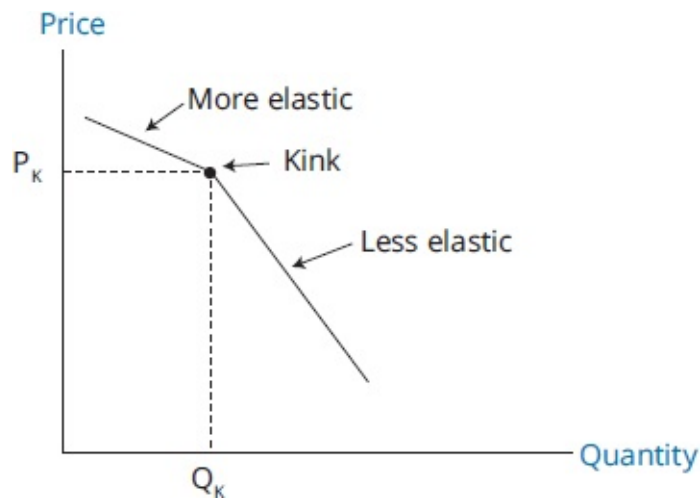
Cournot duopoly model

Nash equilibrium model

Stackelberg dominant firm model

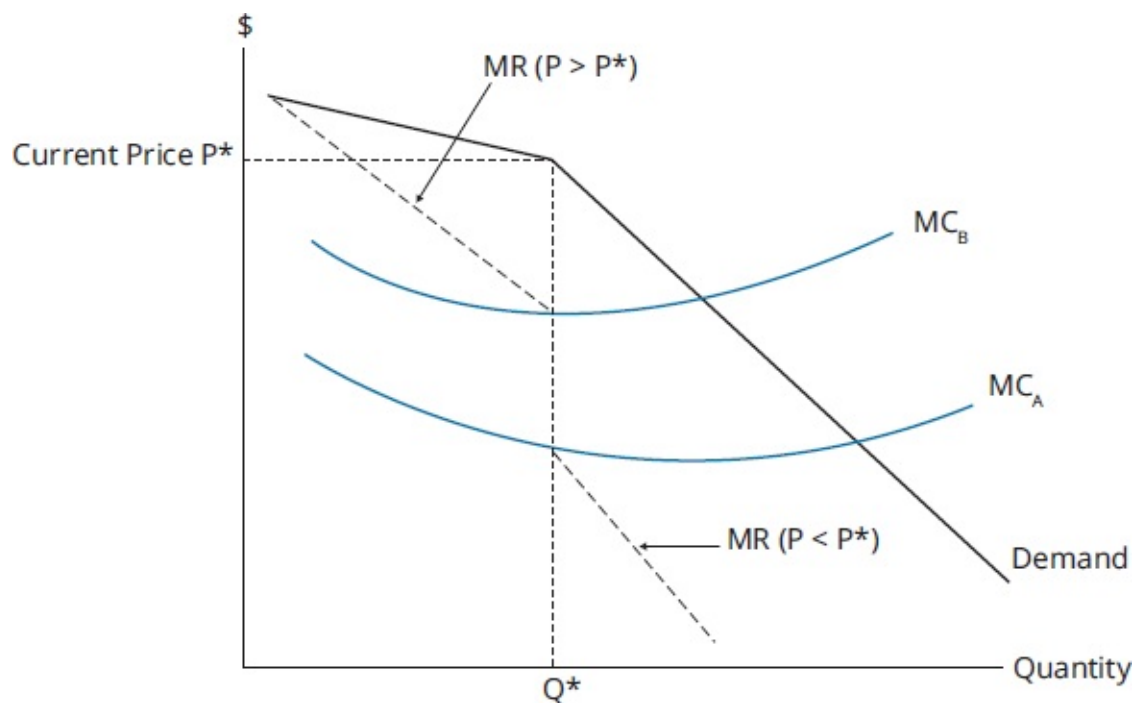
One traditional model of oligopoly, the **kinked demand curve model**, is based on the assumption that competitors are unlikely to match a price increase by a competitor, but very likely to match a price decrease by a competitor. This results in a kink in the demand curves faced by each producer, at the current market price. Each firm believes that it faces a demand curve that is more elastic (flatter) above the current price (the kink in the demand curve) than it is below the given price. The kinked demand curve model is illustrated in Figure 12.7. The kink price is at price P_K , where a firm produces Q_K . A firm believes that if it raises its price above P_K , its competitors will remain at P_K , and it will lose market share because it has the highest price. Above P_K , the demand curve is considered to be relatively elastic (i.e., a small price increase will result in a large decrease in demand). On the other hand, if a firm decreases its price below P_K , other firms will match the price cut, and all firms will experience a relatively small increase in sales relative to any price reduction. Therefore, Q_K is the profit-maximizing level of output.

Figure 12.7: Kinked Demand Curve Model



With a kink in the demand curve, we also get a gap in the associated MR curve, as shown in Figure 12.8. For any firm with a MC curve passing through this gap, the price where the kink is located is the firm's profit-maximizing price.

Figure 12.8: Marginal Revenue With Kinked Demand Curve



We say that the decisions of firms in an oligopoly are interdependent; that is, the pricing decision of one firm depends on the pricing decisions of other firms. Some models of market price equilibrium have a set of rules for the actions of oligopolists. These rules assume they choose prices based on the choices of the

other firms. By specifying the decision rules that each firm follows, we can design a model that allows us to determine the equilibrium prices and quantities for firms operating in an oligopoly market.

An early model of oligopoly pricing decisions is the **Cournot model**. In Cournot's duopoly model, two firms with identical MC curves each choose their preferred selling price based on the price the other firm chose in the previous period. Firms assume that the competitor's price will not change. The long-run equilibrium for an oligopoly with two firms (duopoly), in the Cournot model, is for both firms to sell the same quantity, dividing the market equally at the equilibrium price. The equilibrium price is less than the price that a single monopolist would charge, but greater than the equilibrium price that would result under perfect competition. With a greater number of producers, the long-run market equilibrium price moves toward the competitive price.

Another model, the **Stackelberg model**, uses a different set of rules and produces a different result. While the Cournot model assumes the competitors choose price simultaneously each period, the Stackelberg model assumes pricing decisions are made sequentially. One firm, the "leader," chooses its price first, and the other firm chooses a price based on the leader's price. In long-run equilibrium, under these rules, the leader charges a higher price and receives a greater proportion of the firms' total profits.

These models are early versions of *rules-based models*, which fall under the heading of what are now generally termed *strategic games*. Strategic games comprise decision models in which the best choice for a firm depends on the expected actions (reactions) of other firms.

A more general model of strategic games was developed by Nobel Prize winner John Nash, who developed the concept of a **Nash equilibrium**. A Nash equilibrium is reached when the choices of all firms are such that there is no other choice that makes any firm better off (increases profits or decreases losses). The Cournot model results in a Nash equilibrium. In equilibrium, neither competitor can increase profits by changing the price they charge.

The concept of a Nash equilibrium can be illustrated with the situation presented in Figure 12.9, which shows the choices and resulting profits for two firms. Each firm can charge either a high price or a low price. If both firms charge a high price, Firm A earns 1,000 and Firm B earns 600. While Firm A would not charge the low price (it would earn less regardless of Firm B's decision), Firm B can

increase profits to 700 by charging a low price. With Firm A charging a high price and Firm B charging a low price, neither firm can increase profits by unilaterally changing its price strategy. Thus, we can identify the Nash equilibrium in this scenario as Firm B charging a low price and Firm A charging a high price.

Figure 12.9: Nash Equilibrium

	Firm B high price	Firm B low price
Firm A high price	A earns 1,000 B earns 600	A earns 600 B earns 700
Firm A low price	A earns 160 B earns 0	A earns 100 B earns 140

Thus far, we have assumed that firms act competitively to maximize their individual profits. We can illustrate the potential of cooperative behavior among producers to increase the total profits of all firms in the market. **Collusion** refers to competitors making a joint agreement to charge a given price—or, alternatively, to agree to specific levels of output. In Figure 12.9, the greatest joint profits (1,600) are earned when both firms charge a high price. If Firm A offers to pay Firm B 200 for charging a high price, Firm A's profits increase from 600 to 1,000. After paying 200 to Firm B, Firm A still gains 200. Firm B's profits (including the payment of 200) increase from 700 to 800. Collusion, in this case, increases the profits of both firms, compared to the Nash equilibrium. If firms can enter into and enforce an agreement regarding pricing and output, often they can all benefit. Such agreements among producers are illegal in many countries because they reduce competition.

An example of a collusive agreement is the OPEC **cartel**. Cartel-member countries agree to restrict their oil production to increase the world price of oil. Members sometimes choose to “cheat” on the cartel agreement by producing more than the amount of oil they have agreed to produce. If members of a cartel do not adhere to the agreement (taking advantage of the higher market price but failing to restrict output to the agreed-upon amount), the agreement can quickly break down.

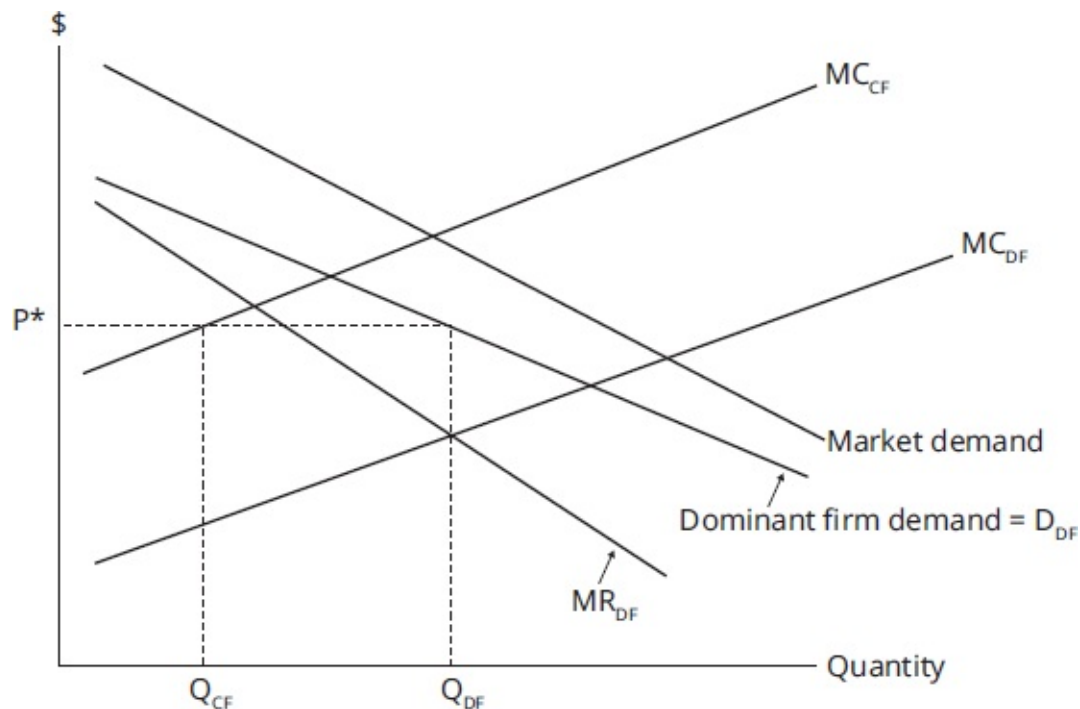
In general, collusive agreements to increase price in an oligopoly market will be more successful (have less cheating) under the following conditions:

- There are fewer firms.
- Products are more similar (less differentiated).
- Cost structures are more similar.
- Purchases are relatively small and frequent.
- Retaliation by other firms for cheating is more certain and more severe.
- There is less actual or potential competition from firms outside the cartel.

A final model of oligopoly behavior to consider is the **dominant firm model**. In this model, a single firm has a significantly large market share because of its greater scale and lower cost structure—the dominant firm (DF). In such a model, the market price is essentially determined by the DF, and the other competitive firms (CFs) take this market price as given.

The DF believes that the quantity supplied by the other firms decreases at lower prices, so that the DF's demand curve is related to the market demand curve, as shown in Figure 12.10. Based on this demand curve (D_{DF}) and its associated marginal revenue (MR_{DF}) curve, the firm will maximize profits at a price of P^* . The CFs maximize profits by producing the quantity for which their marginal cost (MC_{CF}) equals P^* , quantity Q_{CF} .

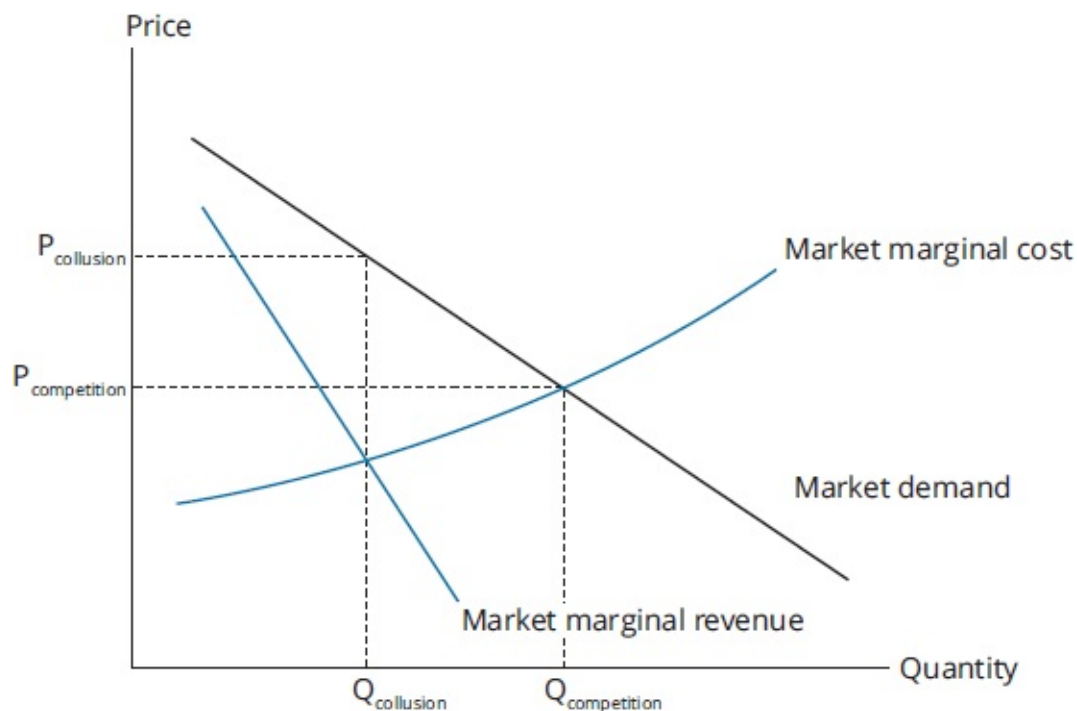
Figure 12.10: Dominant Firm Oligopoly



A price decrease by one of the CFs, which increases Q_{CF} in the short run, will lead to a decrease in price by the DF, and CFs will decrease output or exit the industry in the long run. The long-run result of such a price decrease by competitors below P^* would then be to decrease the overall market share of competitor firms and increase the market share of the DF.

Clearly, oligopoly markets exhibit many possible outcomes that depend on the characteristics of the firms and of the market itself. The important point is that the firms' decisions are interdependent so that the expected reactions of other firms are an important consideration. Overall, the resulting price will be somewhere between the price based on perfect collusion that would maximize total profits to all firms in the market (which is actually the monopoly price), and the price that would result from perfect competition and generate zero economic profits in the long run. These two limiting outcomes are illustrated in Figure 12.11 as $P_{\text{collusion}}$ with $Q_{\text{collusion}}$ for perfect collusion, and $P_{\text{competition}}$ and $Q_{\text{competition}}$ for perfect competition.

Figure 12.11: Collusion vs. Perfect Competition



MODULE QUIZ 12.2

1. The demand for products from monopolistic competitors is relatively elastic due to:
 - A. high barriers to entry.
 - B. the availability of many close substitutes.

- C. the availability of many complementary goods.
- 2. Compared to a perfectly competitive industry, in an industry characterized by monopolistic competition:
 - A. both price and quantity are likely to be lower.
 - B. price is likely to be higher, and quantity is likely to be lower.
 - C. quantity is likely to be higher, and price is likely to be lower.
- 3. A firm will *most likely* maximize profits at the quantity of output for which:
 - A. price equals marginal cost.
 - B. price equals marginal revenue.
 - C. marginal cost equals marginal revenue.
- 4. An oligopolistic industry has:
 - A. few barriers to entry.
 - B. few economies of scale.
 - C. a great deal of interdependence among firms.
- 5. Consider a firm in an oligopoly market that believes the demand curve for its product is more elastic above a certain price than below this price. This belief fits *most appropriately* to which of the following models?
 - A. Cournot model.
 - B. Dominant firm model.
 - C. Kinked demand model.
- 6. Consider an agreement between France and Germany that will restrict wine production so that maximum economic profit can be realized. The possible outcomes of the agreement are presented in the following table.

	Germany complies	Germany defaults
France complies	France gets €8 billion Germany gets €8 billion	France gets €2 billion Germany gets €10 billion
France defaults	France gets €10 billion Germany gets €2 billion	France gets €4 billion Germany gets €4 billion

Based on the concept of a Nash equilibrium, the *most likely* strategy followed by the two countries with respect to whether they comply with or default on the agreement will be:

- A. both countries will default.
- B. both countries will comply.
- C. one country will default and the other will comply.

MODULE 12.3: IDENTIFYING MARKET STRUCTURES

LOS 12.e: Identify the type of market structure within which a firm operates and describe the use and limitations of concentration measures.

We can use the characteristics we outlined earlier to identify the type of market structure within which a firm is operating. Our earlier table is repeated here in Figure 12.12. For an analyst attempting to determine the degree of pricing power that firms in the industry have, the focus is on the number of firms in the industry, its barriers to entry, the nature of substitute products, and the nature of industry competition. Significant interdependence among firm pricing and output decisions is a characteristic of all oligopoly markets, although some interdependence may be present under monopolistic competition—even with many more firms than for an oligopoly structure.

The following table illustrates the differences in characteristics among the various market structures.

Figure 12.12: Characteristics of Market Structures

	Perfect Competition	Monopolistic Competition	Oligopoly	Monopoly
Number of sellers	Many firms	Many firms	Few firms	Single firm
Barriers to entry	Very low	Low	High	Very high
Nature of substitute products	Very good substitutes	Good substitutes, but differentiated	Good substitutes or differentiated	No good substitutes
Nature of competition	Price only	Price, marketing, features	Price, marketing, features	Advertising
Pricing power	None	Some	Some to significant	Significant

Given the differences in cost structures, suitability of substitutes, and the degree of differentiation, simply classifying the primary characteristics of a market does not tell us the degree of pricing power for individual firms or the magnitude of the difference between market prices and the prices implied by perfect competition. What we would really like (especially for the regulation of markets) is to be able to measure the elasticity of firm demand directly, but that is difficult and subject to estimation error.

Consequently, regulators often use market shares (percentages of market sales) to measure the degree of monopoly or market power of firms in an industry. Often, mergers or acquisitions of companies in the same industry or market are not permitted by government authorities when they determine that the market

share of the combined firms will be too high and, therefore, detrimental to the economy.

Market or industry **concentration measures** are often used as an indicator of market power. One concentration measure is the **N-firm concentration ratio**, which is calculated as the sum of the percentage market shares of the largest N firms in a market. While this measure is simple to calculate and understand, it does not directly measure market power or elasticity of demand.

One limitation of the N-firm concentration ratio is that it may be relatively insensitive to mergers of firms within an industry. This problem is reduced by using an alternative measure of market concentration, the **Herfindahl-Hirschman Index (HHI)**. The HHI is calculated as the sum of the squares of the market shares of the largest firms in the market. The following example illustrates this difference between the two measures and their calculation.

EXAMPLE: 4-firm concentration ratio and 4-firm HHI

Given the market shares of the following firms, calculate the 4-firm concentration ratio and the 4-firm HHI, both before and after a merger of Acme and Blake.

Firm	Market Share
Acme	25%
Blake	15%
Curtis	15%
Dent	10%
Erie	5%
Federal	5%

Answer:

Before the merger, the 4-firm concentration ratio for the market is $25 + 15 + 15 + 10 = 65\%$. After the merger, the Acme + Blake firm has 40% of the market, and the 4-firm concentration ratio is $40 + 15 + 10 + 5 = 70\%$. Although the 4-firm concentration ratio has only increased slightly, the market power of the largest firm in the industry has increased significantly, from 25% to 40%.

Before the merger, the 4-firm HHI is $0.25^2 + 0.15^2 + 0.15^2 + 0.10^2 = 0.1175$. After the merger, the 4-firm HHI is $0.40^2 + 0.15^2 + 0.10^2 + 0.05^2 = 0.1950$, a significant increase.

A limitation that applies to both of our simple concentration measures is that barriers to entry are not considered in either case. Even a firm with high market share may not have much pricing power if barriers to entry are low and there is *potential competition*. With low barriers to entry, it may be the case that other firms stand ready to enter the market if firms currently in the market attempt to increase prices significantly. In this case, the elasticity of demand for existing firms may be high, even though they have relatively high market shares and industry concentration measures.



MODULE QUIZ 12.3

1. Which of the following is *most likely* an advantage of the Herfindahl-Hirschman Index (HHI) relative to the *N*-firm concentration ratio?
 - A. The HHI is simpler to calculate.
 - B. The HHI considers barriers to entry.
 - C. The HHI is more sensitive to mergers.
2. A market characterized by low barriers to entry, good substitutes, limited pricing power, and marketing of product features is *best* characterized as:
 - A. oligopoly.
 - B. perfect competition.
 - C. monopolistic competition.

KEY CONCEPTS

LOS 12.a

The breakeven quantity of production is the quantity for which price (P) = average total cost (ATC), and total revenue (TR) = total cost (TC).

A firm should shut down in the long run if $P < ATC$ so that $TR < TC$. A firm should shut down in the short run (and the long run) if $P < \text{average variable cost (AVC)}$ so that $TR < \text{total variable cost (TVC)}$.

The long-run average total cost (LRATC) curve shows the minimum average total cost for each level of output, assuming that the plant size (scale of the firm) can be adjusted. A downward-sloping segment of an LRATC curve indicates economies of scale (increasing returns to scale). Over such a segment, increasing the scale of the firm reduces ATC. An upward-sloping segment of an LRATC curve indicates diseconomies of scale, where average unit costs will rise as the scale of the business (and long-run output) increases.

LOS 12.b

Perfect competition is characterized by the following:

- | Many firms, each small relative to the market
- | Very low barriers to entry into or exit from the industry
- | Homogeneous products that are perfect substitutes; no advertising or branding
- | No pricing power

Monopoly is characterized by the following:

- | A single firm that comprises the whole market
- | Very high barriers to entry into or exit from the industry
- | Advertising used to compete with substitute products
- | Significant pricing power

Monopolistic competition is characterized by the following:

- | Many firms
- | Low barriers to entry into or exit from the industry
- | Differentiated products; heavy advertising and marketing expenditure
- | Some pricing power

Oligopoly markets are characterized by the following:

- | Few sellers
- | High barriers to entry into or exit from the industry
- | Products that may be homogeneous or differentiated by branding and advertising
- | Firms that may have significant pricing power

LOS 12.c

Under monopolistic competition, firms face downward-sloping demand curves so that marginal revenue is less than price, and the price from the demand curve at the profit-maximizing quantity is the price at the optimal (profit-maximizing) level of output. Resources expended on product differentiation may increase ATC so that there are no economic profits at long-run equilibrium. There is no well-defined firm supply curve.

Monopolistic competition:

- | Price > marginal revenue = marginal cost (in equilibrium)
- | Zero economic profit in long-run equilibrium

LOS 12.d

Under oligopoly, the pricing strategy is not clear. Because firm decisions are interdependent, the optimal pricing and output strategy depends on assumptions made about other firms' cost structures and about competitors' responses to a firm's price changes.

Models of oligopoly pricing include the following:

- | *Cournot*. Firms with a given MC that make pricing decisions simultaneously share the market equally in LR equilibrium.
- | *Stackelberg*. When firm pricing decisions are sequential, the first firm to set its price (price leader) will maintain a larger share of the market, even in the long run.
- | *Dominant firm*. A firm with a significantly lower cost of production will gain a disproportionate share of the market and essentially set the price that other competitors can charge.

In general, in the absence of collusion, the long-run oligopoly equilibrium is Nash equilibrium, one in which no competitor can unilaterally change price to increase its profits. There is no well-defined firm supply curve.

Oligopoly:

- | Price > marginal revenue = marginal cost (in equilibrium)
- | May have positive economic profit in long-run equilibrium, but may move toward zero economic profit over time

LOS 12.e

To identify the market structure in which a firm is operating, we can examine the number of firms in its industry, the prevalence of nonprice competition, and barriers to entry, then compare these to the characteristics that define each market structure. However, none of this provides us with the elasticity of firm demand.

Because direct estimation of the elasticity of firm demand is statistically difficult and imprecise, industry concentration measures are often used as a proxy for the degree of competition. A concentration ratio for N firms is calculated as the

percentage of market sales accounted for by the N largest firms in the industry and is used as a simple measure of market structure and market power.

The Herfindahl-Hirschman Index measure of concentration is calculated as the sum of the squared market shares of the largest N firms in an industry and better reflects the effect of mergers on industry concentration.

Neither measure indicates market power directly. Both can be misleading when potential competition restricts pricing power.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 12.1

- . **A** In a purely competitive market, economic losses indicate that firms are overproducing, causing prices to fall below average total costs. This can occur in the short run. In the long run, however, market supply will decrease as firms exit the industry, and prices will rise to the point where economic profits are zero. (LOS 12.a)
- . **C** If price is greater than average variable cost, a firm will continue to operate in the short run because it is covering at least some of its fixed costs. (LOS 12.a)
- . **C** If a firm is generating sufficient revenue to cover its variable costs and part of its fixed costs, it should continue to operate in the short run. If average revenue is likely to remain below average total costs in the long run, the firm should shut down. (LOS 12.a)
- . **B** If minimum average total costs increase as plant size is increased, the firm is experiencing diseconomies of scale. (LOS 12.a)

Module Quiz 12.2

- . **B** The demand for products from firms competing in monopolistic competition is relatively elastic due to the availability of many close substitutes. If a firm increases its product price, it will lose customers to firms selling substitute products at lower prices. (LOS 12.c)
- . **B** Monopolistic competition is likely to result in a higher price and lower quantity of output compared to perfect competition. (LOS 12.c)

- . C The profit-maximizing output is the quantity at which marginal revenue equals marginal cost. In a price-searcher industry structure (i.e., any structure that is not perfect competition), price is greater than marginal revenue. (LOS 12.c)
- . C An oligopolistic industry has a great deal of interdependence among firms. One firm's pricing decisions or advertising activities will affect the other firms. (LOS 12.d)
- . C The kinked demand model assumes that each firm in a market believes that at some price, demand is more elastic for a price increase than for a price decrease. (LOS 12.d)
- . A The Nash equilibrium results when each nation pursues the strategy that is best, given the strategy that is pursued by the other nation.
 - Given that Germany complies with the agreement: France will get €8 billion if it complies, but €10 billion if it defaults. Therefore, France should default.
 - Given that Germany defaults: France will get €2 billion if it complies, but €4 billion if it defaults. Therefore, France should default.

Because France is better off in either case by defaulting, France will default. Germany will follow the same logic and reach the same conclusion. (LOS 12.d)

Module Quiz 12.3

- . C Although the N -firm concentration ratio is simple to calculate, it can be relatively insensitive to mergers between companies with large market shares. Neither the HHI nor the N -firm concentration ratio consider barriers to entry. (LOS 12.e)
- . C These characteristics are associated with a market structure of monopolistic competition. Firms in perfect competition do not compete on product features. Oligopolistic markets have high barriers to entry. (LOS 12.e)

READING 13

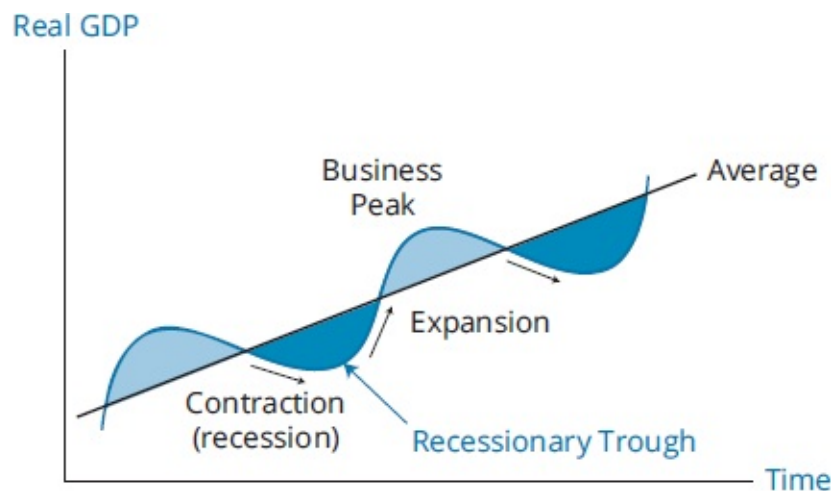
UNDERSTANDING BUSINESS CYCLES

MODULE 13.1: BUSINESS CYCLES

LOS 13.a: Describe the business cycle and its phases.

The **business cycle** is characterized by fluctuations in economic activity. The business cycle has four phases: **expansion** (real GDP is increasing), **peak** (real GDP stops increasing and begins decreasing), **contraction** or **recession** (real GDP is decreasing), and **trough** (real GDP stops decreasing and begins increasing). The phases are illustrated in Figure 13.1.

Figure 13.1: Business Cycle



There are alternative ways to show business cycles. The curve in the figure illustrates the **classical cycle**, which is based on real GDP relative to a beginning value. The **growth cycle** refers to changes in the percentage difference between real GDP and its longer-term trend or potential value (shown as the “average” line in the figure). The **growth rate cycle** refers to changes in the annualized percentage growth rate from one period to the next and tends to show both peaks

and troughs earlier than the other two measures. The growth rate cycle—which, like the growth cycle, shows GDP relative to a trend rate—is the preferred measure for economists and practitioners.

An expansion features growth in most sectors of the economy, with increasing employment, consumer spending, and business investment. As an expansion approaches its peak, the rates of increase in spending, investment, and employment slow but remain positive, while inflation accelerates.

A contraction or recession is associated with declines in most sectors, with inflation typically decreasing. When a contraction reaches a trough and the economy begins a new expansion or **recovery**, economic growth becomes positive again and inflation is typically moderate, but employment growth might not start to increase until the expansion has taken hold convincingly.

Economists commonly consider two consecutive quarters of growth in real GDP to be the beginning of an expansion, and two consecutive quarters of declining real GDP to be the beginning of a contraction. Statistical agencies that date expansions and recessions, such as the National Bureau of Economic Research in the United States, look at a wider variety of additional economic data—especially unemployment, industrial production, and inflation—to identify turning points in the business cycle.

A key aspect of business cycles is that they recur, but not at regular intervals. Past business cycles have been as short as a year, or longer than a decade. The idea of a business cycle applies to economies that consist mainly of businesses. For economies that are mostly subsistence agriculture or dominated by state planning, fluctuations in activity are not really “business cycles” in the sense we are discussing here.

LOS 13.b: Describe credit cycles.

Credit cycles refer to cyclical fluctuations in interest rates and the availability of loans (credit). Typically, lenders are more willing to lend, and tend to offer lower interest rates, during economic expansions. Conversely, they are less willing to lend, and require higher interest rates, when the economy is slowing (contracting).

Credit cycles may amplify business cycles. Widely available or “loose” credit conditions during expansions can lead to “bubbles” (prices based on implausible

expectations) in the markets for some assets, such as subprime mortgages in the period leading up to the financial crisis of 2007–2009. Some research suggests that expansions tend to be stronger, and contractions deeper and longer lasting, when they coincide with credit cycles. They do not always coincide, however, as historical data suggest credit cycles have been longer in duration than business cycles on average.

LOS 13.c: Describe how resource use, consumer and business activity, housing sector activity, and external trade sector activity vary over the business cycle and describe their measurement using economic indicators.

Business Cycles and Resource Use Fluctuation

Inventories are an important business cycle indicator. Firms try to keep enough inventory on hand to meet sales demand but do not want to keep too much of their capital tied up in inventory. As a result, the **inventory-sales ratio** in many industries tends toward a normal level in times of steady economic growth.

When an expansion is approaching its peak, sales growth begins to slow, and unsold inventories accumulate. This can be seen in an increase in the inventory-sales ratio above its normal level. Firms respond to this unplanned increase in inventory by reducing production, which is one of the causes of the subsequent contraction in the economy. An increase in inventories is counted in the GDP statistics as economic output whether the increase is planned or unplanned. An analyst who looks only at GDP growth, rather than the inventory-sales ratio, might see economic strength rather than the beginning of weakness.

The opposite occurs when a contraction reaches its trough. Having reduced their production levels to adjust for lower sales demand, firms find their inventories becoming depleted more quickly once sales growth begins to accelerate. This causes the inventory-sales ratio to decrease below its normal level. To meet the increase in demand, firms will increase output, and the inventory-sales ratio will increase back toward normal levels.

One of the ways that firms react to fluctuations in business activity is by adjusting their use of labor and physical capital. Adding and subtracting workers in lockstep with changes in economic growth would be costly for firms, in terms of both direct expenses and the damage it would do to employee morale and loyalty. Instead, firms typically begin by changing how they use their current

workers, producing less or more output per hour or adjusting the hours they work by adding or removing overtime. Only when an expansion or contraction appears likely to persist will they hire or lay off workers.

Similarly, because it is costly to adjust production levels by frequently buying and selling plant and equipment, firms first adjust their production levels by using their existing physical capital more or less intensively. As an expansion persists, firms will increase their production capacity by investing more in plant and equipment. During contractions, however, firms will not necessarily sell plant and equipment outright. They can reduce their physical capacity by spending less on maintenance or by delaying the replacement of equipment that is nearing the end of its useful life.

Consumer Sector Activity

Consumer spending, the largest component of gross domestic product, depends on the level of consumers' current incomes and their expectations about their future incomes. As a result, consumer spending increases during expansions and decreases during contractions.

Consumer spending in some sectors is more sensitive to business cycle phases than spending in other sectors. Spending on **durable goods** is highly cyclical because they are often higher-value purchases. Consumers are more willing to purchase high-value durable goods (e.g., appliances, furniture, automobiles) during expansions, when incomes are increasing and economic confidence is high. During contractions (and sometimes extending into the early stages of expansions), consumers often postpone durable goods purchases until they are more confident about their employment status and prospects for income growth.

Consumer spending on **services** is also positively correlated with business cycle phases, but not to the same extent as durable goods spending. Services include some discretionary spending, such as for travel, lodging, and restaurant meals, but they also include spending that is less discretionary, such as for telecommunications, health care, and insurance. Spending on **nondurable goods**, such as food at home or household products for everyday use, remains relatively stable over the business cycle.

Housing Sector Activity

Although the housing sector is a smaller part of the economy relative to overall consumer spending, cyclical swings in activity in the housing market can be

large, so that the effect on overall economic activity is greater than it otherwise would be. Important determinants of the level of economic activity in the housing sector are as follows:

Mortgage rates. Low interest rates tend to increase home buying and construction, while high interest rates tend to reduce home buying and construction.

Housing costs relative to income. When incomes are cyclically high (low) relative to home costs, including mortgage financing costs, home buying and construction tend to increase (decrease). Housing activity can decrease even when incomes are rising if home prices are rising faster than incomes, as often occurs late in expansions.

Speculative activity. As we saw in the housing sector in 2007–08 in many economies, rising home prices can lead to purchases based on expectations of further gains. Higher prices led to more construction and eventually excess building. This resulted in falling prices that decreased or eliminated speculative demand and led to dramatic decreases in house prices and in housing activity overall.

Demographic factors. The proportion of the population in the 25- to 40-year-old segment is positively related to activity in the housing sector because these are the ages of greatest household formation. Strong population shifts from rural areas to cities, as may occur in newly industrializing economies, may require large increases in construction of new housing to accommodate those needs.

External Trade Sector Activity

The most important factors determining the level of a country's imports and exports are domestic GDP growth, GDP growth of trading partners, and currency exchange rates. Increasing growth of domestic GDP leads to increases in purchases of foreign goods (imports), while decreasing domestic GDP growth reduces imports. Exports depend on the growth rates of GDP of other economies (especially those of important trading partners). Increasing foreign incomes increase sales to foreigners (exports), and decreasing economic growth in foreign countries decreases domestic exports.

An increase in the value of a country's currency makes its goods more expensive to foreign buyers and foreign goods less expensive to domestic buyers, which

tends to decrease exports and increase imports. A decrease in the value of a country's currency has the opposite effect (increasing exports and decreasing imports). Currencies affect import and export volumes over time in response to persistent trends in foreign exchange rates, rather than in response to short-term changes, which can be quite volatile.

Currency effects on imports and exports can differ in direction from GDP growth effects and respond to a complex set of variables. The effects of changes in GDP levels and growth rates are more direct and immediate.

Typical business cycle characteristics may be summarized as follows:

| *Trough:*

- The GDP growth rate changes from negative to positive.
- There is a high unemployment rate, and an increasing use of overtime and temporary workers.
- Spending on consumer durable goods and housing may increase.
- There is a moderate or decreasing inflation rate.

| *Expansion:*

- The GDP growth rate increases.
- The unemployment rate decreases as hiring accelerates.
- There are investment increases in producers' equipment and home construction.
- The inflation rate may increase.
- Imports increase as domestic income growth accelerates.

| *Peak:*

- The GDP growth rate decreases.
- The unemployment rate decreases, but hiring slows.
- Consumer spending and business investments grow at slower rates.
- The inflation rate increases.

| *Contraction/recession:*

- The GDP growth rate is negative.
- Hours worked decrease; unemployment rate increases.
- Consumer spending, home construction, and business investments decrease.

- The inflation rate decreases with a lag.
- Imports decrease as domestic income growth slows.

Economic Indicators

Economic indicators can be classified into three categories: **leading indicators** that have been known to change direction before peaks or troughs in the business cycle, **coincident indicators** that change direction at roughly the same time as peaks or troughs, and **lagging indicators** that tend not to change direction until after expansions or contractions are already underway.

Figure 13.2 lists the index components for leading, coincident, and lagging indicators in the United States. Other major economies use indexes of their own that have different components.



PROFESSOR'S NOTE

Because these indexes differ by country, we do not believe their specific components are fair game for the exam.

Figure 13.2: US Leading, Lagging, and Coincident Indicators

Leading Indicators	Coincident Indicators	Lagging Indicators
<i>Financial components:</i> ■ Leading Credit Index™ ■ S&P 500 Stock Index ■ Interest rate spread, 10-year T-bonds less Fed Funds	■ Employees on nonagricultural payrolls ■ Personal income less transfer payments ■ Manufacturing and trade sales ■ Index of industrial production	■ Average duration of unemployment ■ Ratio, manufacturing and trade inventories to sales ■ Change in labor cost per unit output, manufacturing ■ Average prime rate charged by banks ■ Commercial and industrial loans outstanding ■ Ratio, consumer installment credit to personal income ■ Change in CPI for services
<i>Non-financial components:</i> ■ Average consumer expectations for business conditions ■ ISM® new orders index ■ Building permits, private housing ■ Average weekly hours, manufacturing ■ Manufacturers' new orders, nondefense capital goods excluding aircraft ■ Manufacturers' new orders, consumer goods and materials ■ Average weekly initial claims for unemployment insurance		

Source: The Conference Board, www.conference-board.org



MODULE QUIZ 13.1

- In the early part of an economic expansion, inventory-sales ratios are *most likely* to:
 - increase because sales are unexpectedly low.
 - increase because businesses plan for expansion.
 - decrease because of unexpected increases in sales.
- The contraction phase of the business cycle is *least likely* accompanied by decreasing:
 - unemployment.
 - inflation pressure.
 - economic output.
- Which economic sector would *most likely* correlate strongly and positively with credit cycles?

- A. Exports.
 - B. Food retail.
 - C. Construction.
4. An economic indicator that has turning points that tend to occur after the turning points in the business cycle is classified as a:
- A. lagging indicator.
 - B. leading indicator.
 - C. trailing indicator.
5. When they recognize the beginning of a recession, companies are *most likely* to adjust their stock of physical capital by:
- A. selling physical assets.
 - B. deferring maintenance of equipment.
 - C. canceling orders for new construction equipment.

KEY CONCEPTS

LOS 13.a

The business cycle has four phases:

Expansion. Real GDP is increasing (or GDP growth relative to trend is increasing).

Peak. Real GDP stops increasing/begins decreasing (GDP growth relative to trend peaks).

Contraction. Real GDP is decreasing (or GDP growth relative to trend is decreasing).

Trough. Real GDP stops decreasing/begins increasing (GDP growth relative to trend reaches a low).

Expansions feature increasing output, employment, consumption, investment, and inflation. Contractions are characterized by decreases in these indicators.

Business cycles are recurring, but they do not occur at regular intervals, can differ in strength or severity, and do not persist for specific lengths of time.

LOS 13.b

Credit cycles are cyclical fluctuations in interest rates and credit availability. Credit cycles may amplify business cycles and cause bubbles in the markets for some assets.

LOS 13.c

Inventory to sales ratios typically increase late in expansions when sales slow, and decrease near the end of contractions when sales begin to accelerate. Firms decrease or increase production to restore their inventory-sales ratios to their desired levels.

Because hiring and laying off employees have high costs, firms prefer to adjust their use of current employees. As a result, firms are slow to lay off employees early in contractions and slow to add employees early in expansions.

Firms use their physical capital more intensively during expansions, investing in new capacity only if they believe the expansion is likely to continue. They use physical capital less intensively during contractions, but they are more likely to reduce capacity by deferring maintenance and not replacing equipment than by selling their physical capital.

Consumer spending fluctuates with the business cycle. Durable goods spending is highly sensitive to business cycles, and spending on services is somewhat sensitive, but spending on nondurable goods is relatively less sensitive to business cycles.

The level of activity in the housing sector is affected by mortgage rates, demographic changes, the ratio of income to housing prices, and investment or speculative demand for homes resulting from recent price trends.

Domestic imports tend to rise with increases in GDP growth and domestic currency appreciation, while increases in foreign incomes and domestic currency depreciation tend to increase domestic export volumes.

Leading indicators have turning points that tend to precede those of the business cycle.

Coincident indicators have turning points that tend to coincide with those of the business cycle.

Lagging indicators have turning points that tend to occur after those of the business cycle.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 13.1

- . **C** Early in an expansion, inventory-sales ratios typically decrease below their normal levels as accelerating sales draw down inventories of produced goods. (LOS 13.c)
- . **A** An economic contraction is likely to feature increasing unemployment (i.e., decreasing employment), along with declining economic output and decreasing inflation pressure. (LOS 13.a)
- . **C** Credit cycles are associated with interest rates and the availability of credit, which is important in the financing of construction and the purchase of property. (LOS 13.b)
- . **A** Lagging indicators have turning points that occur after business cycle turning points. (LOS 13.c)
- . **B** Physical capital adjustments to downturns typically are made through aging of equipment plus postponing maintenance. (LOS 13.c)

READING 14

FISCAL POLICY

MODULE 14.1: FISCAL POLICY OBJECTIVES

LOS 14.a: Compare monetary and fiscal policy.

Fiscal policy refers to a government's use of spending and taxation to influence economic activity. The budget is said to be *balanced* when tax revenues equal government expenditures. A **budget surplus** occurs when government tax revenues exceed expenditures, and a **budget deficit** occurs when government expenditures exceed tax revenues. An increase in the deficit (or a decrease in a surplus) is considered expansionary in that it tends to increase GDP. A decrease in a deficit (or increase in a surplus) is considered contractionary in that it tends to decrease GDP.

Monetary policy refers to the central bank's actions that affect the quantity of money and credit in an economy to influence economic activity. Monetary policy is said to be expansionary (or accommodative or easy) when the central bank increases the quantity of money and credit in an economy. Conversely, when the central bank is reducing the quantity of money and credit in an economy, monetary policy is said to be contractionary (or restrictive or tight).

Policymakers use both monetary and fiscal policies with the goals of maintaining stable prices and producing positive economic growth. Fiscal policy can also be used as a tool for redistribution of income and wealth.

LOS 14.b: Describe roles and objectives of fiscal policy as well as arguments as to whether the size of a national debt relative to GDP matters.

Objectives of fiscal policy may include the following:

- ▮ Influencing the level of economic activity and aggregate demand
- ▮ Redistributing wealth and income among segments of the population

- | Allocating resources among economic agents and sectors in the economy

In general, decreased taxes and increased government spending both *increase* a budget deficit, overall demand, economic growth, and employment; increased taxes and decreased government spending *decrease* these. Budget deficits are increased in response to recessions, and budget deficits are decreased to slow growth when inflation is too high.

Keynesian economists believe that fiscal policy, through its effect on aggregate demand, can have a strong effect on economic growth when the economy is operating at less than full employment. Monetarists believe that the effect of fiscal stimulus is only temporary, and that monetary policy should be used to increase or decrease inflationary pressures over time. Monetarists do not believe that monetary policy should be used in an attempt to influence aggregate demand to counter cyclical movements in the economy.

Discretionary fiscal policy refers to the spending and taxing decisions of a national government that are intended to stabilize the economy. In contrast, **automatic stabilizers** are built-in fiscal devices triggered by the state of the economy. For example, during a recession, tax receipts will fall, and government expenditures on unemployment insurance payments will increase. Both of these tend to increase budget deficits and are expansionary. Similarly, during boom times, higher tax revenues coupled with lower outflows for social programs tend to decrease budget deficits and are contractionary.

When a government runs fiscal deficits, it incurs debt that needs to be repaid as well as ongoing interest expense. Total deficits, annual deficits, and interest expense can all be evaluated relative to annual GDP. When these ratios increase beyond certain levels, it may be a cause for concern, and the solvency of the country may be questioned.

A country's **debt ratio** is the ratio of aggregate debt to GDP. Because taxes are linked to GDP, when an economy grows in real terms, so will tax revenues. If the real interest rate on the government's debt is higher than the real growth rate of the economy, then the debt ratio will increase over time (keeping tax rates constant). Similarly, if the real interest rate on the government's debt is lower than real growth in GDP, the debt ratio will decrease (i.e., improve) over time.

Arguments *for* being concerned with the size of a fiscal deficit are as follows:

- | Higher deficits lead to higher future taxes. Higher future taxes will lead to disincentives to work and entrepreneurship. This leads to lower long-term economic growth.
- | If markets lose confidence in the government, investors may not be willing to refinance the debt. This can lead to the government defaulting (if debt is in a foreign currency) or having to simply print money (if the debt is in local currency). Printing money would ultimately lead to higher inflation.
- | Increased government borrowing will tend to increase interest rates, and firms may reduce their borrowing and investment spending as a result, thus decreasing the impact on aggregate demand of deficit spending. This is referred to as the **crowding-out effect** because government borrowing is taking the place of private-sector borrowing.

Arguments *against* being concerned with the size of a fiscal deficit are as follows:

- | If the debt is primarily held by domestic citizens, the scale of the problem is overstated.
- | If the debt is used to finance productive capital investment, future economic gains will be sufficient to repay the debt.
- | Fiscal deficits may prompt needed tax reform.
- | Deficits would not matter if **Ricardian equivalence** holds, which means private-sector savings in anticipation of future tax liabilities just offset the government deficit.
- | If the economy is operating at less than full capacity, deficits do not divert capital away from productive uses and can aid in increasing GDP and employment.



MODULE QUIZ 14.1

1. Both monetary and fiscal policy are used to:
 - A. balance the budget.
 - B. achieve economic targets.
 - C. redistribute income and wealth.
2. Roles and objectives of fiscal policy *most likely* include:
 - A. controlling the money supply to limit inflation.
 - B. adjusting tax rates to influence aggregate demand.
 - C. using government spending to control interest rates.

MODULE 14.2: FISCAL POLICY TOOLS AND IMPLEMENTATION

LOS 14.c: Describe tools of fiscal policy, including their advantages and disadvantages.

Fiscal policy tools include spending tools and revenue tools.

Spending Tools

Transfer payments, also known as entitlement programs, redistribute wealth, taxing some and making payments to others. Examples include government-run retirement income plans (such as Social Security in the United States) and unemployment insurance benefits. Transfer payments are not included in GDP computations.

Current spending refers to government purchases of goods and services on an ongoing and routine basis.

Capital spending refers to government spending on infrastructure, such as roads, schools, bridges, and hospitals. Capital spending is expected to boost future productivity of the economy.

Spending tools are commonly justified as means of pursuing the following goals:

- | Provide services such as national defense that benefit all the residents in a country.
- | Invest in infrastructure to enhance economic growth.
- | Support the country's growth and unemployment targets by directly affecting aggregate demand.
- | Provide a minimum standard of living.
- | Subsidize investment in research and development for certain high-risk ventures consistent with future economic growth or other goals (e.g., green technology).

Revenue Tools

Direct taxes are levied on income or wealth. These include income taxes, taxes on income for national insurance, wealth taxes, estate taxes, corporate taxes,

capital gains taxes, and Social Security taxes. Some progressive taxes (such as income taxes and wealth taxes) collect revenue for wealth and income redistributing.

Indirect taxes are levied on goods and services. These include sales taxes, value-added taxes (VATs), and excise taxes. Indirect taxes can be used to reduce consumption of some goods and services (e.g., alcohol, tobacco, gambling).

Desirable attributes of tax policy are as follows:

- | *Simplicity* to use and enforce
- | *Efficiency*, defined here as minimizing interference with market forces and not acting as a deterrent to working
- | *Fairness* is quite subjective, but two of the commonly held beliefs are **horizontal equality** (people in similar situations should pay similar taxes) and **vertical equality** (richer people should pay more in taxes)
- | *Sufficiency*, in that taxes should generate enough revenues to meet the spending needs of the government

Advantages of fiscal policy tools include:

- | Social policies (e.g., discouraging tobacco use) can be implemented quickly via indirect taxes.
- | Quick implementation of indirect taxes also means that government revenues can be increased without significant additional costs.

Disadvantages of fiscal policy tools include:

- | Direct taxes and transfer payments take time to implement, delaying the impact of fiscal policy.
- | Capital spending also takes a long time to implement; the economy may have recovered by the time its impact is felt.

Announcing a change in fiscal policy may have significant effects on expectations. For example, an announcement of a future increase in taxes may immediately reduce current consumption, rapidly producing the desired goal of reducing aggregate demand.

Not all fiscal policy tools affect economic activity equally. Spending tools are most effective in increasing aggregate demand. Tax reductions are somewhat

less effective, as people may not spend the entire amount of the tax savings. Tax reductions for those with low incomes will be more effective in increasing aggregate demand, as those with lower incomes tend to spend a larger proportion of income on consumption; that is, they save a smaller proportion of income and have a higher **marginal propensity to consume (MPC)**.

Fiscal Multiplier

Changes in government spending have magnified effects on aggregate demand because those whose incomes increase from increased government spending will, in turn, increase their spending, which increases the incomes and spending of others. The magnitude of the *multiplier effect* depends on the tax rate and on the marginal propensity to consume.

To understand the calculation of the multiplier effect, consider an increase in government spending of \$100 when the MPC is 80% and the tax rate is 25%. The increase in spending increases incomes by \$100, but \$25 (100×0.25) of that will be paid in taxes. **Disposable income** is equal to income after taxes, so disposable income increases by $\$100 \times (1 - 0.25) = \75 . With an MPC of 80%, additional spending by those who receive the original \$100 increase is $\$75 \times 0.8 = \60 .

This additional spending will increase others' incomes by \$60 and disposable incomes by $\$60 \times 0.75 = \45 , from which they will spend $\$45 \times 0.8 = \36 .

Because each iteration of this process reduces the amount of additional spending, the effect reaches a limit. The **fiscal multiplier** determines the potential increase in aggregate demand resulting from an increase in government spending:

$$\text{fiscal multiplier} = \frac{1}{1 - \text{MPC}(1 - t)}$$

Here, with a tax rate of 25% and an MPC of 80%, the fiscal multiplier is $1 / [1 - 0.8(1 - 0.25)] = 2.5$, and the increase of \$100 in government spending has the potential to increase aggregate demand by \$250.

The fiscal multiplier is inversely related to the tax rate (higher tax rate decreases the multiplier) and is directly related to the marginal propensity to consume (higher MPC increases the multiplier).

Balanced Budget Multiplier

To balance the budget, the government could increase taxes by \$100 to just offset a \$100 increase in spending. Changes in taxes also have a magnified effect on aggregate demand. An increase in taxes will decrease disposable income and consumption expenditures, thereby decreasing aggregate demand. The initial decrease in spending from a tax increase of \$100 is $100 \times \text{MPC} = 100 \times 0.8 = \80 ; beyond that, the multiplier effect is the same as we described for a direct increase in government spending, and the overall decrease in aggregate demand for a \$100 tax increase is $100(\text{MPC}) \times \text{fiscal multiplier}$ —or, for our example, $100(0.8)(2.5) = \$200$.

Combining the total increase in aggregate demand from a \$100 increase in government spending with the total decrease in aggregate demand from a \$100 tax increase shows that the net effect on aggregate demand of both is an increase of $\$250 - \$200 = \$50$, so we can say that the balanced budget multiplier is positive.

If, instead of a \$100 increase in taxes, we increased taxes by $100 / \text{MPC} = 100 / 0.8 = \125 and increased government spending by \$100, the net effect on aggregate demand would be zero.

Ricardian Equivalence

Increases in the current deficit mean greater taxes in the future. To maintain their preferred pattern of consumption over time, taxpayers may increase current savings (reduce current consumption) to offset the expected cost of higher future taxes. If taxpayers reduce current consumption and increase current saving by just enough to repay the principal and interest on the debt the government issued to fund the increased deficit, there is no effect on aggregate demand. This is known as Ricardian equivalence, after economist David Ricardo. If taxpayers underestimate their future liability for servicing and repaying the debt so that aggregate demand is increased by equal spending and tax increases, Ricardian equivalence does not hold. Whether it does is an open question.

LOS 14.d: Explain the implementation of fiscal policy and difficulties of implementation as well as whether a fiscal policy is expansionary or contractionary.

Fiscal policy is implemented through changes in taxes and spending. This is called **discretionary fiscal policy** (as opposed to automatic stabilizers, discussed

previously). Discretionary fiscal policy would be designed to be expansionary when the economy is operating below full employment. Fiscal policy aims to stabilize aggregate demand. During recessions, actions can be taken to increase government spending or decrease taxes. Either change tends to strengthen the economy by increasing aggregate demand, putting more money in the hands of corporations and consumers to invest and spend. During inflationary economic booms, actions can be taken to decrease government spending or increase taxes. Either change tends to slow the economy by decreasing aggregate demand, taking money out of the hands of corporations and consumers and causing both investment and consumption spending to decrease.

Discretionary fiscal policy is not an exact science. First, economic forecasts might be wrong, leading to incorrect policy decisions. Second, complications arise in practice that delay both the implementation of discretionary fiscal policy and the impact of policy changes on the economy. The lag between recessionary or inflationary conditions in the economy and the impact on the economy of fiscal policy changes can be divided into three types:

Recognition lag. Discretionary fiscal policy decisions are made by a political process. The state of the economy is complex, and it may take policymakers time to recognize the nature and extent of the economic problems.

Action lag. Governments take time to discuss, vote on, and enact fiscal policy changes.

Impact lag. Time elapses between the enactment of fiscal policy changes and when the impact of the changes on the economy actually takes place. It takes time for corporations and individuals to act on the fiscal policy changes, and fiscal multiplier effects occur only over time as well.

These lags can actually make fiscal policy counterproductive. For example, if the economy is in a recession phase, fiscal stimulus may be deemed appropriate. However, by the time fiscal stimulus is implemented and has its full impact, the economy may already be on a path to a recovery driven by the private sector.

Additional macroeconomic issues may hinder the usefulness of fiscal policy:

- | *Misreading economic statistics.* The full employment level for an economy is not precisely measurable. If the government relies on expansionary fiscal policy mistakenly at a time when the economy is already at full capacity, it will simply drive inflation higher.

- | *Crowding-out effect.* Expansionary fiscal policy may crowd out private investment. Greater government borrowing tends to increase interest rates, which decreases private investments (some investments will be uneconomic with higher borrowing costs). This crowding-out effect can reduce the impact of expansionary fiscal policy on aggregate demand. Opinions vary on the magnitude of this effect.
- | *Supply shortages.* If economic activity is slow due to resource constraints (low availability of labor or other resources) and not due to low demand, expansionary fiscal policy will fail to achieve its objective and will probably lead to higher inflation.
- | *Limits to deficits.* There is a limit to expansionary fiscal policy. If the markets perceive that the deficit is already too high as a proportion of GDP, funding the deficit will be problematic. This could lead to higher interest rates and actually make the situation worse.
- | *Multiple targets.* If the economy has high unemployment coupled with high inflation, fiscal policy cannot address both problems simultaneously.

To determine if fiscal policy is expansionary or contractionary, economists often focus on *changes* in the budget surplus or deficit. An increase (decrease) in surplus is indicative of a contractionary (expansionary) fiscal policy. Similarly, an increase (decrease) in deficit is indicative of an expansionary (contractionary) fiscal policy.



PROFESSOR'S NOTE

For the exam, an increase (decrease) in a revenue item (e.g., sales tax) should be considered contractionary (expansionary), and an increase (decrease) in a spending item (e.g., construction of highways) should be considered expansionary (contractionary).

A government's intended fiscal policy is not necessarily obvious from just examining the current deficit. Consider an economy that is in recession so that transfer payments are increased and tax revenue is decreased, leading to a deficit. This does not necessarily indicate that fiscal policy is expansionary as, at least to some extent, the deficit is a natural outcome of the recession without any explicit action of the government. Economists often use a measure called the **structural budget deficit** (or **cyclically adjusted budget deficit**) to gauge fiscal policy. This is the deficit that would occur based on current policies if the economy were at full employment.



MODULE QUIZ 14.2

1. A government enacts a program to subsidize farmers with an expansive spending program of \$10 billion. At the same time, the government enacts a \$10 billion tax increase over the same period. Which of the following statements *best* describes the impact on aggregate demand?
 - A. Lower growth, because the tax increase will have a greater effect.
 - B. No effect, because the tax and spending effects just offset each other.
 - C. Higher growth, because the spending increase will have a greater effect.
2. A government reduces spending by \$50 million. The tax rate is 30%, and consumers exhibit a marginal propensity to consume of 80%. The change in aggregate demand caused by the change in government spending is *closest* to:
 - A. -\$66 million.
 - B. -\$114 million.
 - C. -\$250 million.
3. The size of a national debt is *most likely* to be a concern for policymakers if:
 - A. Ricardian equivalence holds.
 - B. a crowding-out effect occurs.
 - C. debt is used to finance capital growth.
4. A government is concerned about the timing of fiscal policy changes and is considering requiring the compilation and reporting of economic statistics weekly, rather than quarterly. The new reporting frequency is intended to decrease the:
 - A. action lag.
 - B. impact lag.
 - C. recognition lag.
5. Fiscal policy is *most likely* to be expansionary if tax rates:
 - A. and government spending both decrease.
 - B. decrease and government spending increases.
 - C. increase and government spending decreases.

KEY CONCEPTS

LOS 14.a

Fiscal policy is a government's use of taxation and spending to influence the economy. Monetary policy deals with determining the quantity of money supplied by the central bank. Both policies aim to achieve economic growth with price level stability, although governments use fiscal policy for social and political reasons as well.

LOS 14.b

Objectives of fiscal policy can include (1) influencing the level of economic activity, (2) redistributing wealth or income, and (3) allocating resources among

industries.

There are arguments for being concerned with the size of a fiscal deficit:

- | Higher future taxes lead to disincentives to work, negatively affecting long-term economic growth.
- | Fiscal deficits may not be financed by the market when debt levels are high.
- | There is a crowding-out effect as government borrowing increases interest rates and decreases the private-sector investment.

There are arguments against being concerned with the size of a fiscal deficit:

- | Debt may be financed by domestic citizens.
- | Deficits for capital spending can boost the productive capacity of the economy.
- | Fiscal deficits may prompt needed tax reform.
- | Ricardian equivalence may prevail: private savings rise in anticipation of the need to repay principal on government debt.
- | When the economy is operating below full employment, deficits do not crowd out private investment.

LOS 14.c

Fiscal policy tools include spending tools and revenue tools. Spending tools include transfer payments, current spending (goods and services used by government), and capital spending (investment projects funded by government). Revenue tools include direct and indirect taxation.

An advantage of fiscal policy is that indirect taxes can be used to quickly implement social policies and can also be used to quickly raise revenues at a low cost.

Disadvantages of fiscal policy include time lags for implementing changes in direct taxes and time lags for capital spending changes to have an impact.

LOS 14.d

Fiscal policy is implemented by government changes in taxing and spending policies. Delays in realizing the effects of fiscal policy changes limit their usefulness. Delays can be caused by the following:

- | *Recognition lag.* Policymakers may not immediately recognize when fiscal policy changes are needed.

- | *Action lag.* Governments take time to enact needed fiscal policy changes.
- | *Impact lag.* Fiscal policy changes take time to affect economic activity.

In general, fiscal policies that increase the government deficit are often considered expansionary, and policies that decrease the deficit are considered contractionary. However, because a deficit will naturally increase during a recession and decrease during an expansion, the impact of a current deficit is often judged relative to a structural (cyclically adjusted) deficit amount.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 14.1

- . **B** Both monetary and fiscal policies primarily strive to achieve economic targets such as inflation and GDP growth. Balancing the budget is not a goal for monetary policy and is a potential outcome of fiscal policy. Fiscal policy (but not monetary policy) may secondarily be used as a tool to redistribute income and wealth. (LOS 14.a)
- . **B** Influencing the level of aggregate demand through taxation and government spending is an objective of fiscal policy. Controlling inflation and interest rates are typical objectives of monetary policy. (LOS 14.b)

Module Quiz 14.2

- . **C** The amount of the spending program exactly offsets the amount of the tax increase, leaving the budget unaffected. The multiplier for government spending is greater than the multiplier for a tax increase. Therefore, the balanced budget multiplier is positive. All of the government spending enters the economy as increased expenditure, whereas spending is reduced by only a portion of the tax increase. (LOS 14.c)
- . **B** $\text{fiscal multiplier} = 1 / [1 - \text{MPC}(1 - T)] = 1 / [1 - 0.80(1 - 0.3)] = 2.27$
 change in government spending = -\$50 million
 change in aggregate demand = $-(50 \times 2.27) = -\$113.64$ million
 (LOS 14.c)
- . **B** Crowding out refers to the possibility that government borrowing causes interest rates to increase and private investment to decrease. If government debt is financing the growth of productive capital, this should increase

future economic growth and tax receipts to repay the debt. Ricardian equivalence is the theory that if government debt increases, private citizens will increase savings in anticipation of higher future taxes, and it is an argument against being concerned about the size of government debt and budget deficits. (LOS 14.d)

- . **C** More frequent and current economic data would make it easier for authorities to monitor the economy and to recognize problems. The reduction in the time between economic reports should reduce the recognition lag. (LOS 14.d)
- . **B** Increases in government spending and decreases in taxes are expansionary fiscal policy. Decreases in spending and increases in taxes are contractionary fiscal policy. (LOS 14.d)

READING 15

MONETARY POLICY

MODULE 15.1: CENTRAL BANK OBJECTIVES AND TOOLS

LOS 15.a: Describe the roles and objectives of central banks.

There are several key **roles of central banks**:

Sole supplier of currency. Central banks have the sole authority to supply money. Traditionally, such money was backed by gold; the central bank stood ready to convert the money into a prespecified quantity of gold. Later on, the gold backing was removed, and money supplied by the central bank was deemed **legal tender** by law. Money not backed by any tangible value is termed **fiat money**. As long as fiat money holds its value over time and is acceptable for transactions, it can continue to serve as a medium of exchange.

Banker to the government and other banks. Central banks provide banking services to the government and other banks in the economy.

Regulator and supervisor of payments system. In many countries, central banks may regulate the banking system by imposing standards of risk-taking allowed and reserve requirements of banks under its jurisdiction. Central banks also oversee the payments system to ensure smooth clearing operations domestically and in conjunction with other central banks for international transactions.

Lender of last resort. Central banks' ability to print money allows them to supply money to banks that are experiencing shortages. This government backing tends to prevent runs on banks (i.e., large-scale withdrawals) by assuring depositors that their funds are secure.

Holder of gold and foreign exchange reserves. Central banks are often the repositories of the nation's gold and reserves of foreign currencies.

Conductor of monetary policy. Central banks control or influence the quantity of money supplied in an economy and growth of money supply over time.

The primary objective of a central bank is to *control inflation* so as to promote price stability. High inflation is not conducive to a stable economic environment. High inflation leads to **menu costs** (i.e., cost to businesses of constantly having to change their prices) and **shoe leather costs** (i.e., costs to individuals of making frequent trips to the bank so as to minimize their holdings of cash that are depreciating in value due to inflation).

In addition to price stability, some central banks have other stated goals:

- | Stability in exchange rates with foreign currencies
- | Full employment
- | Sustainable positive economic growth
- | Moderate long-term interest rates

The target inflation rate in most developed countries is a range around 2% to 3%. A target of zero inflation is not used because that increases the risk of deflation, which can be disruptive for an economy.

While most developed countries have an explicit target inflation rate, the US Fed and the Bank of Japan do not. In the United States, this is because the Fed has the additional goals of maximum employment and moderate long-term interest rates. In Japan, it is because deflation, rather than inflation, has been a persistent problem in recent years.

Some developed countries, and several developing countries, choose a target level for the exchange rate of their currency with that of another country, primarily the US dollar. This is referred to as **pegging** their exchange rate with the dollar. If their currency appreciates (i.e., becomes relatively more valuable), they can sell their domestic currency reserves for dollars to reduce the exchange rate. While such actions may be effective in the short run, for stability of the exchange rate over time, the monetary authorities in the pegging country must manage interest rates and economic activity to achieve their goal. This can lead to increased volatility of their money supply and interest rates. The pegging country essentially commits to a policy intended to make its inflation rate equal to the inflation rate of the country to which it pegs its currency.

LOS 15.b: Describe tools used to implement monetary policy tools and the monetary transmission mechanism, and explain the relationships between monetary policy and economic growth, inflation, interest, and exchange

rates.

Monetary policy is implemented using the **monetary policy tools** of the central bank. The three main policy tools of central banks are as follows:

Policy rate. In the United States, banks can borrow funds from the Fed if they have temporary shortfalls in reserves. The rate at which banks can borrow reserves from the Fed is called the *discount rate*. For the European Central Bank (ECB), it is called the *refinancing rate*.

One way to lend money to banks is through a *repurchase agreement*. The central bank purchases securities from banks that, in turn, agree to repurchase the securities at a higher price in the future. The percentage difference between the purchase price and the repurchase price is effectively the rate at which the central bank is lending to member banks. The Bank of England uses this method, and its policy rate is called the *two-week repo (repurchase) rate*. A lower rate reduces banks' cost of funds, encourages lending, and tends to decrease interest rates overall. A higher policy rate has the opposite effect, decreasing lending and increasing interest rates.

In the United States, the *federal funds rate* is the rate that banks charge each other on overnight loans of reserves. The Fed sets a target for this market-determined rate and uses open market operations to influence it toward the target rate.

Reserve requirements. By increasing the reserve requirement (the percentage of deposits that banks are required to retain as reserves), the central bank effectively decreases the funds that are available for lending and thereby decreases the money supply, which will tend to increase interest rates. A decrease in the reserve requirement will increase the funds available for lending and the money supply, which will tend to decrease interest rates. This tool only works well to increase the money supply if banks are willing to lend and customers are willing to borrow.

Open market operations. Buying and selling of securities by the central bank is referred to as open market operations. When the central bank buys securities, cash replaces securities in investor accounts, banks have excess reserves, more funds are available for lending, the money supply increases, and interest rates decrease. Sales of securities by the central bank have the opposite effect,

reducing cash in investor accounts, excess reserves, funds available for lending, and the money supply, which will tend to cause interest rates to increase. In the United States, open market operations are the Fed's most commonly used tool and are important in achieving the federal funds target rate.

Monetary Transmission Mechanism

The **monetary transmission mechanism** refers to the ways in which a change in monetary policy, specifically the central bank's policy rate, affects the price level and inflation. A change in the policy rates that the monetary authorities control directly is transmitted to prices through four channels: other short-term rates, asset values, currency exchange rates, and expectations.

We can examine the transmission mechanism in more detail by considering the effects of a change to a contractionary monetary policy implemented through an increase in the policy rate:

- | Banks' *short-term lending rates will increase* in line with the increase in the policy rate. The higher rates will decrease aggregate demand as consumers reduce credit purchases and businesses cut back on investment in new projects.
- | Bond prices, equity prices, and *asset prices in general will decrease* as the discount rates applied to future expected cash flows are increased. This may have a wealth effect because a decrease in the value of households' assets may increase the savings rate and decrease consumption.
- | Both consumers and businesses may decrease their expenditures because their *expectations for future economic growth decrease*.
- | The increase in interest rates may attract foreign investment in debt securities, leading to an *appreciation of the domestic currency relative to foreign currencies*. An appreciation of the domestic currency increases the foreign currency prices of exports and can reduce demand for the country's export goods.

Taken together, these effects act to decrease aggregate demand and put downward pressure on the price level. A decrease in the policy rate would affect the price level through the same channels, but in the opposite direction.

Monetary Policy Relation With Economic Growth, Inflation, Interest, and Exchange Rates

If money neutrality holds, changes in monetary policy and the policy rate will have no effect on real output. In the short run, however, changes in monetary policy can affect real economic growth as well as interest rates, inflation, and foreign exchange rates. The effects of a change to a more expansionary monetary policy may include any or all of the following:

- | The central bank buys securities, which increases bank reserves.
- | The interbank lending rate decreases as banks are more willing to lend each other reserves.
- | Other short-term rates decrease as the increase in the supply of loanable funds decreases the equilibrium rate for loans.
- | Longer-term interest rates also decrease.
- | The decrease in real interest rates causes the currency to depreciate in the foreign exchange market.
- | The decrease in long-term interest rates increases business investments in plant and equipment.
- | Lower interest rates cause consumers to increase their purchases of houses, autos, and durable goods.
- | Depreciation of the currency increases foreign demand for domestic goods.
- | The increases in consumption, investment, and net exports all increase aggregate demand.
- | The increase in aggregate demand increases inflation, employment, and real GDP.



MODULE QUIZ 15.1

1. A central bank's policy goals *least likely* include:
 - A. price stability.
 - B. minimizing long-term interest rates.
 - C. maximizing the sustainable growth rate of the economy.
2. A country that targets a stable exchange rate with another country's currency *least likely*:
 - A. accepts the inflation rate of the other country.
 - B. will sell its currency if its foreign exchange value rises.
 - C. must also match the money supply growth rate of the other country.
3. If a country's inflation rate is below the central bank's target rate, the central bank is *most likely* to:
 - A. sell government securities.
 - B. increase the reserve requirement.

- C. decrease the overnight lending rate.
- 4. A central bank conducts monetary policy primarily by altering the:
 - A. policy rate.
 - B. inflation rate.
 - C. long-term interest rate.
- 5. Purchases of securities in the open market by the monetary authorities are *least likely* to increase:
 - A. excess reserves.
 - B. cash in investor accounts.
 - C. the interbank lending rate.
- 6. An increase in the policy rate will *most likely* lead to an increase in:
 - A. business investments in fixed assets.
 - B. consumer spending on durable goods.
 - C. the foreign exchange value of the domestic currency.

MODULE 15.2: MONETARY POLICY EFFECTS AND LIMITATIONS

LOS 15.c: Describe qualities of effective central banks; contrast their use of inflation, interest rate, and exchange rate targeting in expansionary or contractionary monetary policy; and describe the limitations of monetary policy.

For a central bank to succeed in its inflation targeting policies, it should have three essential qualities:

Independence. For a central bank to be effective in achieving its goals, it should be free from political interference. Reducing the money supply to reduce inflation can also be expected to decrease economic growth and employment. The political party in power has an incentive to boost economic activity and reduce unemployment before elections. For this reason, politicians may interfere with the central bank's activities, compromising its ability to manage inflation. Independence should be thought of in relative terms (degrees of independence) rather than absolute terms. Even in the case of relatively independent central banks, the heads of the banks may be appointed by politicians.

Independence can be evaluated based on both **operational independence** and **target independence**. Operational independence means that the central bank

is allowed to independently determine the policy rate. Target independence means the central bank also defines how inflation is computed, sets the target inflation level, and determines the horizon over which the target is to be achieved. The ECB has both target and operational independence, while most other central banks have only operational independence.

Credibility. To be effective, central banks should follow through on their stated intentions. If a government with large debts, instead of a central bank, sets an inflation target, the target would not be credible because the government has an incentive to allow inflation to exceed the target level. On the other hand, a credible central bank's targets can become self-fulfilling prophecies. If the market believes that a central bank is serious about achieving a target inflation rate of 3%, wages and other nominal contracts will be based on 3% inflation, and actual inflation will then be close to that level.

Transparency. Transparency on the part of central banks aids their credibility. Transparency means that central banks periodically disclose the state of the economic environment by issuing **inflation reports**. Transparent central banks periodically report their views on the economic indicators and other factors they consider in their interest rate setting policy. When a central bank makes clear the economic indicators it uses in establishing monetary policy and how they will be used, it not only gains credibility, but also makes policy changes easier to anticipate and implement.

Inflation, Interest Rate, and Exchange Rate Targeting by Central Banks

Central banks have used various economic variables and indicators over the years to make monetary policy decisions. In the past, some have used **interest rate targeting**, increasing the money supply when specific interest rates rose above the target band and decreasing the money supply (or the rate of money supply growth) when interest rates fell below the target band. Currently, **inflation targeting** is the most widely used tool for making monetary policy decisions, and it is the method required by law in some countries. Central banks that currently use inflation targeting include the United Kingdom, Brazil, Canada, Australia, Mexico, and the ECB.

The **neutral interest rate** of an economy is the growth rate of the money supply that neither increases nor decreases the economic growth rate:

neutral interest rate = real trend rate of economic growth + inflation target

When the policy rate is above the neutral rate, monetary policy is said to be contractionary; when the policy rate is below the neutral rate, monetary policy is said to be expansionary.

The most common inflation rate target is 2%, with a permitted deviation of $\pm 1\%$ so the target band is 1% to 3%. The reason the inflation target is not 0% is that variations around that rate would allow for negative inflation (i.e., deflation), which is considered disruptive to the smooth functioning of an economy. Central banks are not necessarily targeting current inflation, which is the result of prior policy and events, but inflation in the range of two years in the future.

Some countries, especially developing countries, use **exchange rate targeting**. That is, they target a foreign exchange rate between their currency and another (often the US dollar), rather than targeting inflation. As an example, consider a country that has targeted an exchange rate for its currency versus the US dollar. If the foreign exchange value of the domestic currency falls relative to the US dollar, the monetary authority must use foreign reserves to purchase their domestic currency (which will reduce money supply growth and increase interest rates) to reach the target exchange rate. Conversely, an increase in the foreign exchange value of the domestic currency above the target rate will require sale of the domestic currency in currency markets to reduce its value (increasing the domestic money supply and decreasing interest rates) to move toward the target exchange rate. One result of exchange rate targeting may be greater volatility of the money supply because domestic monetary policy must adapt to the necessity of maintaining a stable foreign exchange rate.

Over the short term, the targeting country can purchase or sell its currency in the foreign exchange markets to influence the exchange rate. There are limits, however, on how much influence currency purchases or sales can have on exchange rates over time. For example, a country may run out of foreign reserves with which to purchase its currency when the exchange value of its currency is still below the target exchange rate.

The net effect of exchange rate targeting is that the targeting country will have the same inflation rate as the country with the targeted currency, and the targeting country will need to follow monetary policy and accept interest rates that are consistent with this goal regardless of domestic economic circumstances.

Limitations of Monetary Policy

This transmission mechanism for monetary policy previously described does not always produce the intended results. In particular, long-term rates may not rise and fall with short-term rates because of the effect of monetary policy changes on expected inflation.

If individuals and businesses believe that a decrease in the money supply intended to reduce inflation will be successful, they will expect lower future inflation rates. Because long-term bond yields include a premium for expected inflation, long-term rates could fall (tending to increase economic growth), even while the central bank has increased short-term rates to slow economic activity. Conversely, increasing the money supply to stimulate economic activity could lead to an increase in expected inflation rates and long-term bond yields, even as short-term rates fall.

From a different perspective, monetary tightening may be viewed as too extreme—increasing the probability of a recession, making long-term bonds more attractive, and reducing long-term interest rates. If money supply growth is seen as inflationary, higher expected future asset prices will make long-term bonds relatively less attractive and will increase long-term interest rates. Bond market participants that act in this way have been called **bond market vigilantes**. When the central bank's policy is credible and investors believe that the inflation target rate will be maintained over time, this effect on long-term rates will be small.

Another situation in which the transmission mechanism may not perform as expected is if demand for money becomes very elastic and individuals willingly hold more money, even without a decrease in short-term rates. Such a situation is called a **liquidity trap**. Increasing the growth of the money supply will not decrease short-term rates under these conditions because individuals hold the money in cash balances instead of investing in interest-bearing securities. If an economy is experiencing deflation even though monetary policy has been expansionary, liquidity trap conditions may be present.

Compared to inflation, deflation is more difficult for central banks to reverse. In a deflationary environment, monetary policy needs to be expansionary. However, the central bank is limited to reducing the nominal policy rate to zero. Once it reaches zero, the central bank has a limited ability to further stimulate the economy.

Another reason standard tools for increasing the money supply might not increase economic activity is that even with increasing excess reserves, banks may not be willing to lend. When what has become known as the *credit bubble* collapsed in 2008, banks around the world lost equity capital and desired to rebuild it. For this reason, they decreased their lending even as money supplies were increased, and short-term rates fell. With short-term rates near zero, economic growth still poor, and a threat of deflation, central banks began a policy termed **quantitative easing**.

In the United Kingdom, quantitative easing entailed large purchases of British government bonds in the maturity range of three to five years. The intent was to reduce interest rates to encourage borrowing and to generate excess reserves in the banking system to encourage lending. Uncertainty about the economy's future caused banks to behave quite conservatively and willingly hold more excess reserves, rather than make loans.

In the United States, billions of dollars were made available for the Fed to buy assets other than short-term Treasury securities. Large amounts of mortgage securities were purchased from banks to encourage bank lending and to reduce mortgage rates in an attempt to revive the housing market, which had collapsed. When this program did not have the desired effect, a second round of quantitative easing (QE2) was initiated. The Fed purchased long-term Treasury bonds in large quantities (hundreds of billions of dollars) with the goal of bringing down longer-term interest rates and generating excess reserves to increase lending and economic growth. The Fed has also purchased securities with credit risk as part of its quantitative easing, improving banks' balance sheets but perhaps shifting risk from the private sector to the public sector.

Monetary Policy in Developing Economies

Developing countries face problems in successfully implementing monetary policy. Without a liquid market in their government debt, interest rate information may be distorted and open market operations difficult to implement. In a very rapidly developing economy, it may be quite difficult to determine the neutral rate of interest for policy purposes. Rapid financial innovation may change the demand to hold monetary aggregates. Central banks may lack credibility because of past failure(s) to maintain inflation rates in a target band and might not be given independence by the political authority.

LOS 15.d: Explain the interaction of monetary and fiscal policy.

Monetary policy and fiscal policy may each be either expansionary or contractionary, so there are four possible scenarios:

Expansionary fiscal and monetary policy. In this case, the impact will be highly expansionary, taken together. Interest rates will usually be lower (due to monetary policy), and the private and public sectors will both expand.

Contractionary fiscal and monetary policy. In this case, aggregate demand and GDP would be lower, and interest rates would be higher due to tight monetary policy. Both the private and public sectors would contract.

Expansionary fiscal policy and contractionary monetary policy. In this case, aggregate demand will likely be higher (due to fiscal policy), while interest rates will be higher (due to increased government borrowing and tight monetary policy). Government spending as a proportion of GDP will increase.

Contractionary fiscal policy and expansionary monetary policy. In this case, interest rates will fall from decreased government borrowing and from the expansion of the money supply, increasing both private consumption and output. Government spending as a proportion of GDP will decrease due to contractionary fiscal policy. The private sector would grow as a result of lower interest rates.

Not surprisingly, the fiscal multipliers for different types of fiscal stimulus differ, and the effects of expansionary fiscal policy are greater when it is combined with expansionary monetary policy. The fiscal multiplier for direct government spending increases has been much higher than the fiscal multiplier for increases in transfers to individuals or tax reductions for workers. Transfer payments to the poor have the greatest relative impact, followed by tax cuts for workers, and broader-based transfers to individuals (not targeted). For all types of fiscal stimulus, the impact is greater when the fiscal actions are combined with expansionary monetary policy. This may reflect the impact of greater inflation, falling real interest rates, and the resulting increase in business investments.



MODULE QUIZ 15.2

1. Qualities of effective central banks include:
 - A. credibility and verifiability.
 - B. comparability and relevance.
 - C. independence and transparency.

2. Monetary policy is likely to be *least* responsive to domestic economic conditions if policymakers employ:
 - A. inflation targeting.
 - B. interest rate targeting.
 - C. exchange rate targeting.
3. Monetary policy is *most likely* to fail to achieve its objectives when the economy is:
 - A. growing rapidly.
 - B. experiencing deflation.
 - C. experiencing disinflation.

KEY CONCEPTS

LOS 15.a

Central bank roles include supplying currency, acting as a banker to the government and to other banks, regulating and supervising the payments system, acting as a lender of last resort, holding the nation's gold and foreign currency reserves, and conducting monetary policy.

Central banks have the objective of controlling inflation, and some have additional goals of maintaining currency stability, full employment, positive sustainable economic growth, or moderate interest rates.

LOS 15.b

Policy tools available to central banks include the policy rate, reserve requirements, and open market operations. The policy rate is called the discount rate in the United States, the refinancing rate by the ECB, and the two-week repo rate in the United Kingdom.

Decreasing the policy rate, decreasing reserve requirements, and making open market purchases of securities are all expansionary. Increasing the policy rate, increasing reserve requirements, and making open market sales of securities are all contractionary.

The transmission mechanism for changes in the central bank's policy rate through to prices and inflation includes one or more of the following:

- | Short-term bank lending rates
- | Asset prices
- | Expectations for economic activity and future policy rate changes
- | Exchange rates with foreign currencies

A contractionary monetary policy (increase in policy rate) will tend to decrease economic growth, increase market interest rates, decrease inflation, and lead to appreciation of the domestic currency in foreign exchange markets. An expansionary monetary policy (decrease in policy rate) will have opposite effects, tending to increase economic growth, decrease market interest rates, increase inflation, and reduce the value of the currency in foreign exchange markets.

LOS 15.c

Effective central banks exhibit independence, credibility, and transparency.

- | *Independence.* The central bank is free from political interference.
- | *Credibility.* The central bank follows through on its stated policy intentions.
- | *Transparency.* The central bank makes it clear what economic indicators it uses and reports on the state of those indicators.

Most central banks set target inflation rates, typically 2%–3%, rather than targeting interest rates, as was once common. When inflation is expected to rise above (fall below) the target band, the money supply is decreased (increased) to reduce (increase) economic activity.

Developing economies sometimes target a stable exchange rate for their currency relative to that of a developed economy, selling their currency when its value rises above the target rate and buying their currency with foreign reserves when the rate falls below the target. The developing country must follow a monetary policy that supports the target exchange rate and essentially commits to having the same inflation rate as the developed country.

Reasons that monetary policy may not work as intended:

- | Monetary policy changes may affect inflation expectations to such an extent that long-term interest rates move opposite to short-term interest rates.
- | Individuals may be willing to hold greater cash balances without a change in short-term rates (liquidity trap).
- | Banks may be unwilling to lend greater amounts, even when they have increased excess reserves.
- | Short-term rates cannot be reduced below zero.
- | Developing economies face unique challenges in using monetary policy due to undeveloped financial markets, rapid financial innovation, and lack of

credibility of the monetary authority.

LOS 15.d

Interaction of monetary and fiscal policies:

Monetary Policy	Fiscal Policy	Interest Rates	Output	Private Sector Spending	Public Sector Spending
Tight	Tight	Higher	Lower	Lower	Lower
Easy	Easy	Lower	Higher	Higher	Higher
Tight	Easy	Higher	Higher	Lower	Higher
Easy	Tight	Lower	Varies	Higher	Lower

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 15.1

- . **B** Central bank goals often include the following: maximum employment, which is interpreted as the maximum sustainable growth rate of the economy; stable prices; and *moderate* (not minimum) long-term interest rates. (LOS 15.a)
- . **C** The money supply growth rate may need to be adjusted to keep the exchange rate within acceptable bounds, but it is not necessarily the same as that of the other country. The other two statements are true. (LOS 15.a)
- . **C** Decreasing the overnight lending rate would add reserves to the banking system, which would encourage bank lending, expand the money supply, reduce interest rates, and allow GDP growth and the rate of inflation to increase. Selling government securities or increasing the reserve requirement would have the opposite effect, reducing the money supply and decreasing the inflation rate. (LOS 15.b)
- . **A** The primary method by which a central bank conducts monetary policy is through changes in the target short-term rate or policy rate. (LOS 15.b)
- . **C** Open market purchases by monetary authorities decrease the interbank lending rate by increasing excess reserves that banks can lend to one another—and therefore, increasing their willingness to lend. (LOS 15.b)

- . **C** An increase in the policy rate is likely to increase longer-term interest rates, causing decreases in consumption spending on durable goods and business investments in plant and equipment. The increase in rates, however, makes investment in the domestic economy more attractive to foreign investors, increasing demand for the domestic currency and causing the currency to appreciate. (LOS 15.b)

Module Quiz 15.2

- . **C** The three qualities of effective central banks are independence, credibility, and transparency. (LOS 15.c)
- . **C** Exchange rate targeting requires monetary policy to be consistent with the goal of a stable exchange rate with the targeted currency, regardless of domestic economic conditions. (LOS 15.c)
- . **B** Monetary policy has a limited ability to act effectively against deflation because the policy rate cannot be reduced below zero, and demand for money may be highly elastic (liquidity trap). (LOS 15.c)

READING 16

INTRODUCTION TO GEOPOLITICS

MODULE 16.1: GEOPOLITICS

LOS 16.a: Describe geopolitics from a cooperation versus competition perspective.

Geopolitics refers to interactions among nations, including the actions of **state actors** (national governments) and **nonstate actors** (corporations, nongovernment organizations, and individuals).

Geopolitics also refers to the study of how geography affects interactions among nations and their citizens. For example, firms located in coastal countries naturally tend to be the dominant participants in international shipping.

One way to examine geopolitics is through analysis of the extent to which individual countries cooperate with one another. Potential areas for cooperation include diplomatic and military matters, and economic and cultural interactions. In terms of economics, areas of cooperation include freedom of movement across borders for goods, services, and capital; agreements to harmonize tariffs; international standardization of rules; and transfers of information and technology.

While a country that engages with other countries on these matters may be considered cooperative and one that does not may be considered noncooperative, the extent of cooperation actually varies along a spectrum. A country might be more cooperative on some issues and less cooperative on others, and its degree of cooperation can change over time or with the outcomes of the country's domestic politics. A country's current decision makers and the length of its political cycle are factors to consider when analyzing geopolitics.

A country will typically cooperate with other countries when doing so advances its national interests. For example, a country may cooperate with its neighbors in

a military alliance if doing so will further its interests in protecting its citizens from foreign invaders.

We can analyze a country's national interests as a hierarchy, with its top priorities being those that ensure its survival. A country's **geophysical resource endowment** may influence its priorities. For example, a country that has mineral resources but lacks arable land needs to trade minerals for food; therefore, it has an interest in cooperating with other countries to keep international trade lanes open.

Nonstate actors often have interests in cooperating across borders. Individuals and firms seek to direct their resources to their highest-valued uses, and some of those uses may be in other countries. To facilitate the flow of resources, state and nonstate actors may cooperate on **standardization** of regulations and processes. One key example of standardization among countries is International Financial Reporting Standards for firms presenting their accounting data to the public, which we will examine in the Financial Statement Analysis topic area.

Cultural factors, such as historical emigration patterns or a shared language, can be another influence on a country's level of cooperation. Among these cultural factors are a country's formal and informal **institutions**, such as laws, public and private organizations, or distinct customs and habits. Strong and stable institutions can make cooperation easier for state and nonstate actors. For example, countries that produce and export large amounts of cultural content tend to be those with legal and ethical institutions that protect intellectual property. Cultural exchange is one means through which a country may exercise **soft power**, which is the ability to influence other countries without using or threatening force.

LOS 16.b: Describe geopolitics and its relationship with globalization.

Globalization refers to the long-term trend toward worldwide integration of economic activity and cultures. Data from the World Bank suggest economic openness, as measured by international trade as a percentage of total output, increased steadily from about 25% in the early 1970s to about 60% before the 2008 financial crisis, and has remained near that level since then. We may contrast globalization with **nationalism**—which, in this context, refers to a nation pursuing its own economic interests independently of, or in competition with, the economic interests of other countries.



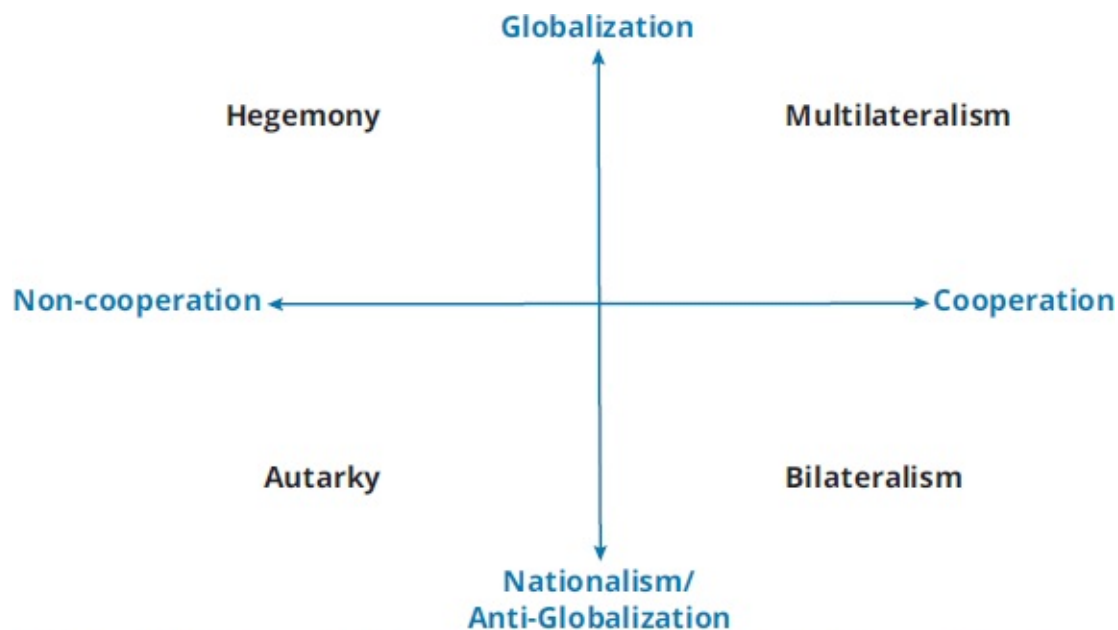
PROFESSOR'S NOTE

Debate about what the word *nationalism* means is beyond the scope of the CFA curriculum. We use it here only in the sense of opposition or resistance to globalization.

As we did with cooperation versus noncooperation, we can think of countries' actions along a spectrum from globalization to nationalism. In general, countries that are closer to the globalization end of the spectrum are those that more actively import and export goods and services, permit freer movement of capital across borders and exchange of currencies, and are more open to cultural interaction.

In Figure 16.1, we draw each spectrum as an axis. This creates four quadrants, each of which we can associate with a type of behavior by countries. While individual countries rarely fit neatly into one of these categories, this gives us a general framework within which we can describe geopolitical actions.

Figure 16.1: Archetypes of Globalization and Cooperation



Source: Reproduced from Level I CFA Curriculum learning module, "Introduction to Geopolitics," with permission from CFA Institute.

Characteristics we may associate with each of these categories are as follows:

- ▮ **Autarky** (noncooperation and nationalism) refers to a goal of national self-reliance, including producing most or all necessary goods and services

domestically. Autarky is often associated with a state-dominated society in general, with attributes such as government control of industry and media.

- | **Hegemony** (noncooperation and globalization) refers to countries that are open to globalization but have the size and scale to influence other countries without necessarily cooperating.
- | **Bilateralism** (cooperation and nationalism) refers to cooperation between two countries. A country that engages in bilateralism may have many such relationships with other countries while tending not to involve itself in multicountry arrangements.
- | **Multilateralism** (cooperation and globalization) refers to countries that engage extensively in international trade and other forms of cooperation with many other countries. Some countries may exhibit **regionalism**, cooperating multilaterally with nearby countries but less so with the world at large.

Some of the nonstate actors within a country may be more oriented toward globalization than their governments. Businesses may look outside their home country for opportunities to increase profits, reduce costs, and sell to new markets. Investors may seek higher returns or diversification by investing outside their home country. Nonstate actors might buy and sell foreign securities (**portfolio investment flows**) or own physical production capacity in other countries (**foreign direct investment**).

LOS 16.c: Describe functions and objectives of the international organizations that facilitate trade, including the World Bank, the International Monetary Fund, and the World Trade Organization.

Perhaps the best way to understand the roles of the organizations designed to facilitate trade is to examine their own statements.

The following is according to the **International Monetary Fund (IMF)** (more available at www.IMF.org):

Article I of the Articles of Agreement sets out the IMF's main goals:

- | Promoting international monetary cooperation
- | Facilitating the expansion and balanced growth of international trade
- | Promoting exchange stability
- | Assisting in the establishment of a multilateral system of payments

- Making resources available (with adequate safeguards) to members experiencing balance of payments difficulties

The following is according to the **World Bank** (www.worldbank.org):

The World Bank is a vital source of financial and technical assistance to developing countries around the world. Our mission is to fight poverty with passion and professionalism for lasting results and to help people help themselves and their environment by providing resources, sharing knowledge, building capacity, and forging partnerships in the public and private sectors.

We are not a bank in the common sense; we are made up of two unique development institutions owned by 187 member countries: the International Bank for Reconstruction and Development (IBRD) and the International Development Association (IDA).

Each institution plays a different but collaborative role in advancing the vision of inclusive and sustainable globalization. The IBRD aims to reduce poverty in middle-income and creditworthy poorer countries, while IDA focuses on the world's poorest countries.

[. . .] Together, we provide low-interest loans, interest-free credits, and grants to developing countries for a wide array of purposes that include investments in education, health, public administration, infrastructure, financial and private sector development, agriculture, and environmental and natural resource management.

The following is according to the **World Trade Organization (WTO)** (more available at www.WTO.org):

The World Trade Organization (WTO) is the only international organization dealing with the global rules of trade between nations. Its main function is to ensure that trade flows as smoothly, predictably, and freely as possible.

[. . .] Trade friction is channeled into the WTO's dispute settlement process where the focus is on interpreting agreements and commitments, and how to ensure that countries' trade policies conform with them. That way, the risk of disputes spilling over into political or military conflict is reduced.

[. . .] At the heart of the system—known as the multilateral trading system—are the WTO's agreements, negotiated and signed by a large majority of the world's trading nations, and ratified in their parliaments. These agreements are the legal ground-rules for international commerce. Essentially, they are contracts, guaranteeing member countries important trade rights. They also bind governments to keep their trade policies within agreed limits to everybody's benefit.

LOS 16.d: Describe geopolitical risk.

Geopolitical risk is the possibility of events that interrupt peaceful international relations. We can classify geopolitical risk into three types:

Event risk refers to events about which we know the timing but not the outcome, such as national elections.

Exogenous risk refers to unanticipated events, such as outbreaks of war or rebellion.

Thematic risk refers to known factors that have effects over long periods, such as human migration patterns or cyber risks.

Geopolitical risk affects investment values by increasing or decreasing the risk premium investors require to hold assets in a country or region. To forecast the effect on investments of a geopolitical risk, we need to consider its probability (*likelihood*), the magnitude of its effects on investment outcomes (*impact*), and how quickly investment values would reflect these effects (*velocity*).

We can use our framework of cooperation and globalization to help estimate the **likelihood of geopolitical risk**. Countries that are more cooperative and globalized tend to have less likelihood of some geopolitical risks, such as armed conflict, but may have greater likelihood of other risks, such as the supply chain disruptions that followed the COVID-19 pandemic in 2020 and 2021.

To analyze the **velocity of geopolitical risk**, we can classify risks as high velocity (short term), medium velocity, or low velocity (long term). Exogenous risks often have high-velocity effects on financial markets and investment values. **Black swan risk** is a term for the risk of low-likelihood exogenous events that have substantial short-term effects. Investors with longer time horizons typically do not need to react to these kinds of events, but investors with shorter horizons might find it necessary to react.

Medium-velocity risks can potentially damage specific companies or industries by increasing their costs or disrupting their production processes, while low-velocity risks tend to affect them in the “environmental, social, and governance” realm. Analyzing these kinds of risk is important for investors with long time horizons.

LOS 16.e: Describe tools of geopolitics and their impact on regions and economies.

We can consider **tools of geopolitics**, the means by which (primarily) state actors advance their interests in the world, as falling into three broad categories of national security, economic, and financial.

National security tools may include armed conflict, espionage, or bilateral or multilateral agreements designed to reinforce or prevent armed conflict. We can say a national security tool is *active* if a country is currently using it or *threatened* if a country is not currently using it but appears likely to do so. Armed conflict affects regions and economies by destroying productive capital and causing migration away from areas of conflict.

Economic tools can be cooperative or noncooperative. Examples of cooperative economic tools include free trade areas, common markets, and economic and monetary unions (each of which we describe in our reading on international trade and capital flows). Examples of noncooperative economic tools include domestic content requirements, voluntary export restraints, and nationalization (i.e., the state taking control) of companies or industries.

Financial tools include foreign investment and the exchange of currencies. We can view countries as using these tools cooperatively if they allow foreign investment and the free exchange of currencies, or noncooperatively when they restrict these activities. **Sanctions**, or restrictions on a specific geopolitical actor's financial interests, are a financial tool that state actors may use alongside national security tools.

LOS 16.f: Describe the impact of geopolitical risk on investments.

Because analyzing geopolitical risks requires effort, time, and resources, investors should consider whether the **impact of geopolitical risk** is likely to be high or low, and focus their analysis on risks that could have a high impact. With regard to those risks, investors should determine whether they are likely to have *discrete impacts* on a company or industry or *broad impacts* on a country, a region, or the world. Business cycles can affect the impact of geopolitical risk, in that these risks may have greater impacts on investment values when an economy is in recession than they would have during an expansion.

Investors can use qualitative or quantitative **scenario analysis** to gauge the potential effects of geopolitical risks on their portfolios. To help identify geopolitical risks over time, investors may identify **signposts**, or data that can signal when the likelihood of an event is increasing or decreasing, such as volatility indicators in financial markets.



MODULE QUIZ 16.1

1. A state actor that is generally cooperative with other countries and primarily nationalist in pursuing its objectives is *most accurately* said to exhibit:
 - A. autarky.
 - B. hegemony.
 - C. bilateralism.
2. Which of the following tools of geopolitics is *best* described as a noncooperative economic tool?
 - A. Voluntary export restraints.

- B. Regional free trade agreements.
 - C. Restrictions on conversion of currencies.
3. When investing for a long time horizon, a portfolio manager should *most likely* devote resources to analyzing:
- A. event risks.
 - B. thematic risks.
 - C. exogenous risks.
4. Which international organization is primarily concerned with providing economic assistance to developing countries?
- A. World Bank.
 - B. World Trade Organization.
 - C. International Monetary Fund.

KEY CONCEPTS

LOS 16.a

Geopolitics refers to interactions among nations. On various issues ranging from diplomacy and military force to economic or cultural openness, countries lie along a spectrum from cooperative to noncooperative.

LOS 16.b

Globalization refers to integration of economic activity and cultures among countries, and can be contrasted with nationalism, which refers to a country pursuing its own interests independently of other countries. Analysts should view geopolitical actions as being on a spectrum from nationalism to globalization.

We may describe geopolitics and its relationship with globalization using the following four broad categories: autarky (noncooperation and nationalism), hegemony (noncooperation and globalization), bilateralism (cooperation and nationalism), and multilateralism (cooperation and globalization).

LOS 16.c

The International Monetary Fund facilitates trade by promoting international monetary cooperation and exchange rate stability, assists in setting up international payments systems, and makes resources available to member countries with balance of payments problems.

The World Bank provides low-interest loans, interest-free credits, and grants to developing countries for many specific purposes. It also provides resources and

knowledge and helps form private/public partnerships with the overall goal of fighting poverty.

The World Trade Organization has the goal of ensuring that trade flows freely and works smoothly. Its main focus is on instituting, interpreting, and enforcing numerous multilateral trade agreements that detail global trade policies for a large majority of the world's trading nations.

LOS 16.d

Categories of geopolitical risk are event risk (when the timing is known), exogenous risk (unanticipated events), and thematic risk (known factors that have long-term effects).

LOS 16.e

Tools of geopolitics include national security tools, economic tools, and financial tools.

National security tools may include armed conflict, espionage, or bilateral or multilateral national security agreements.

Cooperative economic tools include free trade areas, common markets, and economic and monetary unions. Noncooperative economic tools include domestic content requirements, voluntary export restraints, and nationalization.

Financial tools include foreign investment, exchange of currencies, and sanctions.

LOS 16.f

Investors should analyze the likelihood of a geopolitical risk, the impact on investment values of an event if it occurs, and the velocity with which it would affect investment values.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 16.1

- . C Bilateralism is characterized by nationalism (as contrasted with globalization) and cooperation. Both autarky and hegemony are characterized by noncooperation. (LOS 16.b)

- . **A** Voluntary export restraints (exporting less of a good than the global market demands) are an example of a noncooperative economic tool. Restrictions on the exchange of currencies are a financial tool. Free trade agreements are a cooperative economic tool. (LOS 16.e)
- . **B** Thematic risks are those that have effects over the long term. Event risks and exogenous risks are more likely to have high-velocity impacts on investment values, but they are less of a focus for investors with longer time horizons. (LOS 16.d)
- . **A** The World Bank provides technical and financial assistance to economically developing countries. The World Trade Organization is primarily concerned with settling disputes among countries concerning international trade. The International Monetary Fund promotes international trade and exchange rate stability and assists member countries that experience balance of payments trouble. (LOS 16.c)

READING 17

INTERNATIONAL TRADE

MODULE 17.1: INTERNATIONAL TRADE

LOS 17.a: Describe the benefits and costs of international trade.

Historically, economic models of trade have focused on the gains that result when countries with lower relative cost of (a comparative advantage in) the production of a good specialize in producing that good and export it, importing goods for which other countries have a lower relative cost or production. This increases the total output of goods and the wealth of both countries. Comparative advantage results from differences in technology and resource endowments across countries.

Newer models of trade emphasize gains from economies of scale that reduce costs of export goods, an increased variety of goods produced, decreasing costs and improved quality from additional competition, and more efficient allocation of productive resources.

Free trade can also benefit consumers by reducing the pricing power of domestic monopolies. Countries that produce the same good can also operate in market of monopolistic competition with differentiated products. Consider the global market for automobiles. Several countries export one type of autos and import other types. Consumers benefit from the greater variety offered, as well as the reduced costs from specialization.

While domestic consumers of imported goods and domestic producers of exported goods both gain, international trade imposes costs as well. The most cited costs of free trade are the loss of domestic jobs in an importing industry and increased economic inequality. Consider a country with a comparative advantage in the production of steel that exports to another country with a higher relative cost of steel production. When free trade is permitted, consumers of steel in the importing country gain and workers and producers of steel lose. In the

exporting country, increased demand for steel production may increase prices to domestic consumers, while workers and companies in the steel industry will gain, at least in the short run.

In a situation where labor costs are lower in the exporting country, free trade can decrease wages and employment in the domestic economy. Consider textile production in a country with a relatively high cost of labor. For a labor-intensive industry such as some textile production, importing textiles from a lower-cost country will result in job and wage losses in the domestic industry, possibly increasing income inequality.

When we speak of the gains from trade in general, it is based on economic analysis suggesting that, overall, the gains from trade are greater than the losses—especially in the long run. That is, the gainers could, at least in theory, compensate the losers, with net gains shared by both importing and exporting countries. While free trade surely imposes costs on the workers and companies in an industry facing competition from imports in the short run, it is argued that in the long run—after workers receive additional training and find work in other industries—the short-run costs of free trade are mitigated to some significant degree, or even reversed.

LOS 17.b: Compare types of trade restrictions, such as tariffs, quotas, and export subsidies, and their economic implications.

There are many reasons (at least stated reasons) given by governments that impose trade restrictions. Some have support among economists as conceivably valid in terms of increasing a country's welfare, while others get little or no support from economic theory. Some of the reasons for trade restrictions that have support from economists are as follows:

- | *Infant industry.* Protection from foreign competition is given to new industries to give them an opportunity to grow to an internationally competitive scale and get up the learning curve in terms of efficient production methods.
- | *National security.* Even if imports are cheaper, it may be in the country's best interest to protect producers of goods crucial to the country's national defense so that those goods are available domestically in the event of conflict.

Other arguments for trade restrictions that have little support in theory are as follows:

- | *Protecting domestic jobs.* While some jobs are certainly lost, and some groups and regions are negatively affected by free trade, other jobs (in export industries or growing domestic goods and services industries) will be created, and prices for domestic consumers will be less without import restrictions.
- | *Protecting domestic industries.* Industry firms often use political influence to get protection from foreign competition—usually to the detriment of consumers, who pay higher prices.

Other arguments include retaliation for foreign trade restrictions, government collection of tariffs (like taxes on imported goods), countering the effects of government subsidies paid to foreign producers, and preventing foreign exports at less than their cost of production (*dumping*).

Types of trade restrictions include the following:

- | **Tariffs.** These are taxes on imported goods collected by the government.
- | **Quotas.** These are limits on the amount of imports allowed over some period.
- | **Export subsidies.** These are government payments to firms that export goods.
- | **Minimum domestic content.** This is the requirement that some percentage of product content must be from the domestic country.
- | **Voluntary export restraint.** A country voluntarily restricts the amount of a good that can be exported, often in the hope of avoiding tariffs or quotas imposed by its trading partners.

Economic Implications of Trade Restrictions

We will now examine the effects of the primary types of trade restrictions, tariffs, and subsidies.

A **tariff** placed on an imported good increases its domestic price, decreases the quantity imported, and increases the quantity supplied domestically. Domestic producers gain, foreign exporters lose, and the domestic government gains by the amount of the tariff revenues.

A **quota** restricts the quantity of a good imported to the quota amount. Domestic producers gain, and domestic consumers lose from an increase in the domestic price. The right to export a specific quantity to the domestic country is granted by the domestic government, which may or may not charge for the import

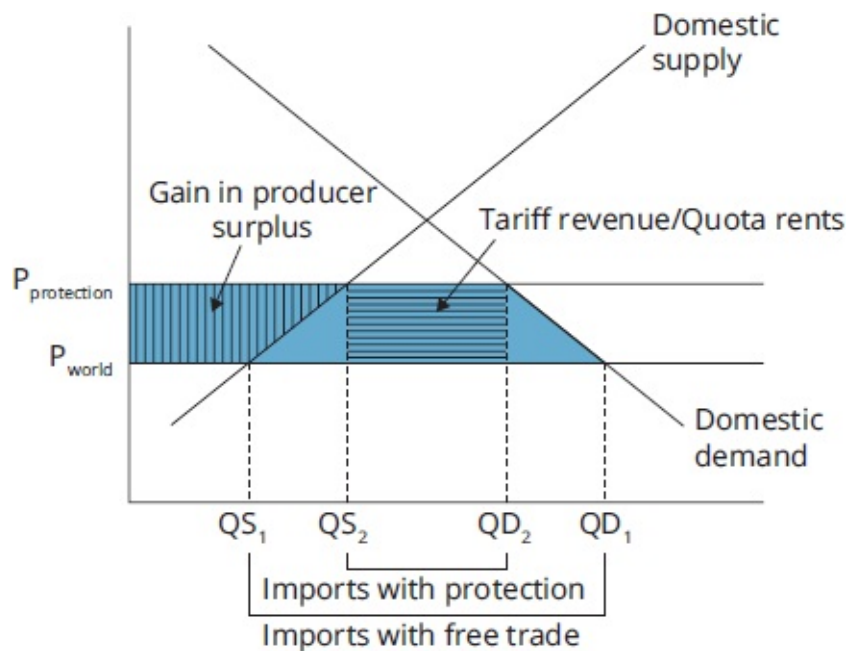
licenses to foreign countries. If the import licenses are sold, the domestic government gains the revenue.

We illustrate the overall welfare effects of quotas and tariffs for a small country in Figure 17.1. We define a quota that is equivalent to a given tariff as a quota that will result in the same decrease in the quantity of a good imported as the tariff. Defined this way, a tariff and an equivalent quota both increase the domestic price from P_{world} , the price that prevails with no trade restriction, to $P_{\text{protection}}$.

At P_{world} , before any restriction, the domestic quantity supplied is QS_1 , and the domestic quantity demanded is QD_1 , with the difference equal to the quantity imported, $QD_1 - QS_1$. Placing a tariff on imports increases the domestic price to $P_{\text{protection}}$, increases the domestic quantity supplied to QS_2 , and decreases the domestic quantity demanded to QD_2 . The difference is the new quantity imported. An equivalent quota will have the same effect, decreasing the quantity imported to $QD_2 - QS_2$.

The entire shaded area in Figure 17.1 represents the loss of consumer surplus in the domestic economy. The portion with vertical lines, the area to the left of the domestic supply curve between $P_{\text{protection}}$ and P_{world} , represents the gain in the producer surplus of domestic producers. The portion with horizontal lines, the area bounded by $QD_2 - QS_2$ and $P_{\text{protection}} - P_{\text{world}}$, represents the gain to the domestic government from tariff revenue. The two remaining triangular areas are the deadweight loss from the restriction on free trade.

Figure 17.1: Effects of Tariffs and Quotas



In the case of a quota, if the domestic government collects the full value of the import licenses, the result is the same as for a tariff. If the domestic government does not charge for the import licenses, this amount is a gain to those foreign exporters who receive the import licenses under the quota and are termed **quota rents**.

In terms of overall economic gains from trade, the deadweight loss is the amount of lost welfare from the imposition of a quota or tariff. From the viewpoint of the domestic country, the loss in consumer surplus is only partially offset by the gains in domestic producer surplus and the collection of tariff revenue.

If none of the quota rents are captured by the domestic government, the overall welfare loss to the domestic economy is greater by the amount of the quota rents. It is the entire difference between the gain in producer surplus and the loss of consumer surplus.

A **voluntary export restraint (VER)** is just as it sounds. It refers to a voluntary agreement by a government to limit the quantity of a good that can be exported. VERs are another way of protecting the domestic producers in the importing country. They result in a welfare loss to the importing country equal to that of an equivalent quota with no government charge for the import licenses—that is, no capture of the quota rents.

Export subsidies are payments by a government to its country's exporters. Export subsidies benefit producers (exporters) of the good but increase prices

and reduce consumer surplus in the exporting country. In a small country, the price will increase by the amount of the subsidy to equal the world price plus the subsidy. In the case of a large exporter of the good, the world price decreases and some benefits from the subsidy accrue to foreign consumers, while foreign producers are negatively affected.

Most of the effects of all four of these protectionist policies are the same. With respect to the domestic (importing) country, import quotas, tariffs, and VERs all do the following:

- | Reduce imports
- | Increase price
- | Decrease consumer surplus
- | Increase domestic quantity supplied
- | Increase producer surplus

With one exception, all will decrease national welfare. Quotas and tariffs in a large country could increase national welfare under a specific set of assumptions, primarily because for a country that imports a large amount of the good, setting a quota or tariff could reduce the world price for the good.

Capital Restrictions

Some countries impose **capital restrictions** on the flow of financial capital across borders. Restrictions include outright prohibition of investment in the domestic country by foreigners, prohibition of or taxes on the income earned on foreign investments by domestic citizens, prohibition of foreign investments in certain domestic industries, and restrictions on repatriation of earnings of foreign entities operating in a country.

Overall, capital restrictions are thought to decrease economic welfare. However, over the short term, they have helped developing countries avoid the impact of great inflows of foreign capital during periods of optimistic expansion and the impact of large outflows of foreign capital during periods of correction and market unease, or outright panic. Even these short-term benefits may not offset longer-term costs if the country is excluded from international markets for financial capital flows.

LOS 17.c: Explain motivations for and advantages of trading blocs, common markets, and economic unions.

There are various types of agreements among countries with respect to trade policy. The essence of all of them is to reduce trade barriers among the countries. Reductions in trade restrictions among countries have some, by now familiar, positive and negative effects on economic welfare. The positive effects result from increased trade according to comparative advantage, as well as increased competition among firms in member countries. The negative effects result because some firms, some industries, and some groups of workers will see their wealth and incomes decrease. Workers in affected industries may need to learn new skills to get new jobs.

On balance, economic welfare is improved by reducing or eliminating trade restrictions. Note, however, that to the extent that a trade agreement increases trade restrictions on imports from nonmember countries, economic welfare gains are reduced and, in an extreme case, could be outweighed by the costs such restrictions impose. This could result if restrictions on trade with nonmember countries increase a country's (unrestricted) imports from a member that has higher prices than the country's previous imports from a nonmember.

We list these types of agreements, generally referred to as **trading blocs** or **regional trading agreements (RTAs)**, in order of their degrees of integration.

Free Trade Areas

All barriers to import and export of goods and services among member countries are removed.

Customs Union

All barriers to import and export of goods and services among member countries are removed.

All countries adopt a common set of trade restrictions with nonmembers.

Common Market

All barriers to import and export of goods and services among the countries are removed.

All countries adopt a common set of trade restrictions with nonmembers.

All barriers to the movement of labor and capital goods among member countries are removed.

Economic Union

All barriers to import and export of goods and services among the countries are removed.

All countries adopt a common set of trade restrictions with nonmembers.

All barriers to the movement of labor and capital goods among member countries are removed.

Member countries establish common institutions and economic policy for the union.

Monetary Union

All barriers to import and export of goods and services among the countries are removed.

All countries adopt a common set of trade restrictions with nonmembers.

All barriers to the movement of labor and capital goods among member countries are removed.

Member countries establish common institutions and economic policy for the union.

Member countries adopt a single currency.

The North American Free Trade Agreement (NAFTA) is an example of a free trade area, the European Union (EU) is an example of an economic union, and the Eurozone is an example of a monetary union.



MODULE QUIZ 17.1

1. Which of the following effects is *most likely* to occur in a country that increases its openness to international trade?
 - A. Increased prices of consumer goods.
 - B. Greater specialization in domestic output.
 - C. Decreased employment in exporting industries.
2. An agreement with another country to limit the volume of goods and services sold to them is *best* described as a:
 - A. quota.
 - B. voluntary export restraint.
 - C. minimum domestic content rule.

3. Which of the following groups would be *most likely* to suffer losses from the imposition of a tariff on steel imports?
 - A. Domestic steel producers.
 - B. Workers in the domestic auto industry.
 - C. Workers in the domestic steel industry.
4. The *most likely* motivation for establishing a trading bloc is to:
 - A. increase economic welfare in the member countries.
 - B. increase tariff revenue for the member governments.
 - C. protect domestic industries in the member economies.
5. In which type of regional trade agreement are economic policies conducted independently by the member countries, while labor and capital are free to move among member countries?
 - A. Free trade area.
 - B. Common market.
 - C. Economic union.

KEY CONCEPTS

LOS 17.a

Free trade among countries increases overall economic welfare. Countries can benefit from trade because one country can specialize in the production of an export good and benefit from economies of scale. Economic welfare can also be increased by greater product variety, more competition, and a more efficient allocation of resources.

Costs of free trade are primarily losses to those in domestic industries that lose business to foreign competition, especially less efficient producers who leave an industry. While other domestic industries will benefit from freer trade policies, unemployment may increase over the period in which workers are retrained for jobs in the expanding industries. Some argue that greater income inequality may result, but overall, the gains from liberalization of trade policies are thought to exceed the costs, so that the winners could conceivably compensate the losers and still be better off.

LOS 17.b

Types of trade restrictions include the following:

- | *Tariffs*. These are taxes on imported goods collected by the government.
- | *Quotas*. These are limits on the amount of imports allowed over some period.

- | *Minimum domestic content.* This is the requirement that some percentage of product content must be from the domestic country.
- | *Voluntary export restraints.* A country voluntarily restricts the amount of a good that can be exported, often in the hope of avoiding tariffs or quotas imposed by their trading partners.

Within each importing country, all of these restrictions will tend to do the following:

- | Increase prices of imports and decrease quantities of imports
- | Increase quantity supplied of domestically produced goods
- | Increase producer's surplus and decrease consumer surplus

Export subsidies decrease export prices and benefit importing countries at the expense of the government of the exporting country.

Restrictions on the flow of financial capital across borders include outright prohibition of investment in the domestic country by foreigners, prohibition of or taxes on the income earned on foreign investments by domestic citizens, prohibition of foreign investments in certain domestic industries, and restrictions on repatriation of earnings of foreign entities operating in a country.

LOS 17.c

Trade agreements, which increase economic welfare by facilitating trade among member countries, take the following forms:

- | *Free trade area.* All barriers to the import and export of goods and services among member countries are removed.
- | *Customs union.* Member countries *also* adopt a common set of trade restrictions with nonmembers.
- | *Common market.* Member countries *also* remove all barriers to the movement of labor and capital goods among members.
- | *Economic union.* Member countries *also* establish common institutions and economic policy for the union.
- | *Monetary union.* Member countries *also* adopt a single currency to reduce the cost of cross-border trade.

Module Quiz 17.1

ANSWER KEY FOR MODULE QUIZZES

- . **B** Openness to international trade increases specialization as production shifts to those products in which domestic producers have a comparative advantage. Greater competition from imports will tend to decrease prices for consumer goods. Increasing international trade is likely to increase profitability and employment in exporting industries, but it may decrease profitability and employment in industries that compete with imported goods. (LOS 17.a)
- . **B** Voluntary export restraints are agreements to limit the volume of goods and services exported to another country. Minimum domestic content rules are limitations imposed by a government on its domestic firms. Import quotas are limitations on imports, not on exports. (LOS 17.b)
- . **B** Imposing a tariff on steel imports benefits domestic steel producers and workers by increasing the domestic price of steel and benefits the national government by increasing tax (tariff) revenue. However, the increase in the domestic price of steel would increase costs in industries that use significant amounts of steel, such as the automobile industry. The resulting increase in the price of automobiles reduces the quantity of automobiles demanded, and it ultimately reduces employment in that industry. (LOS 17.b)
- . **A** The motivation for trading blocs is to increase economic welfare in the member countries by eliminating barriers to trade. Joining a trading bloc may have negative consequences for some domestic industries and may decrease tariff revenue for the government. (LOS 17.c)
- . **B** These characteristics describe a common market. In a free trade area, member countries remove restrictions on goods and services and trade with one another but may still restrict movement of labor and capital among member countries. In an economic union, member countries also coordinate their economic policies and institutions. (LOS 17.c)

READING 18

CAPITAL FLOWS AND THE FX MARKET

MODULE 18.1: THE FOREIGN EXCHANGE MARKET

LOS 18.a: Describe the foreign exchange market, including its functions and participants, distinguish between nominal and real exchange rates, and calculate and interpret the percentage change in a currency relative to another currency.

Foreign currency markets serve companies and individuals that purchase or sell foreign goods and services denominated in foreign currencies. An even larger market, however, exists for capital flows. Foreign currencies are needed to purchase foreign physical assets as well as foreign financial securities.

Many companies have foreign exchange risk arising from their cross-border transactions. A Japanese company that expects to receive 10 million euros when a transaction is completed in 90 days has yen/euro exchange rate risk as a result. By entering into a **forward currency contract** to sell 10 million euros in 90 days for a specific quantity of yen, the firm can reduce or eliminate the foreign exchange risk associated with the transaction. When a firm takes a position in the foreign exchange market to reduce an existing risk, we say the firm is **hedging** its risk.

Alternatively, when a transaction in the foreign exchange markets increases currency risk, we describe the transaction or position as **speculating**. Investors, companies, and financial institutions, such as banks and investment funds, all regularly enter into speculative foreign currency transactions.

The primary dealers in foreign exchange (FX) and originators of forward FX contracts are large multinational banks. This part of the FX market is often

called the **sell side**. On the other hand, the **buy side** consists of the many buyers of foreign currencies and forward FX contracts. These buyers include the following:

- | *Corporations* regularly engage in cross-border transactions, purchase and sell foreign currencies as a result, and enter into FX forward contracts to hedge the risk of expected future receipts and payments denominated in foreign currencies.
- | *Investment accounts* of many types transact in foreign currencies, hold foreign securities, and may both speculate and hedge with currency derivatives. **Real money accounts** refer to mutual funds, pension funds, insurance companies, and other institutional accounts that do not use derivatives. **Leveraged accounts** refer to the various types of investment firms that use derivatives, including hedge funds, firms that trade for their own accounts, and other trading firms of various types.
- | *Governments and government entities*, including sovereign wealth funds and pension funds, acquire foreign exchange for transactional needs, investment, or speculation. Central banks sometimes engage in FX transactions to affect exchange rates in the short term in accordance with government policy.
- | The **retail FX market** refers to FX transactions by households and relatively small institutions, and it may be for tourism, cross-border investment, or speculative trading.

Types of Exchange Rates

An **exchange rate** is simply the price or cost of units of one currency in terms of another. In this book we state exchange rates in the form 1.416 USD/EUR, to mean that each euro costs \$1.416. If you read the “/” as *per*, you will have no trouble with the notation. We say the exchange rate is \$1.416 per euro.



PROFESSOR'S NOTE

This is the way exchange rates are quoted in the Level I CFA curriculum. Foreign exchange traders typically use the inverse (an indirect quotation).

In a foreign currency quotation, we have the price of one currency in units of another currency. These are often referred to as the **base currency** and the **price currency**. In the quotation 1.25 USD/EUR, the USD is the price currency and

the EUR is the base currency. The price of one euro (base currency) is 1.25 USD (the price currency), so 1.25 is the price of one unit of the base currency in terms of the other. It may help to remember that the euro, in this example, is in the bottom or “base” of the exchange rate given in terms of USD/EUR.

An exchange rate expressed as price currency/base currency is referred to as a **direct quote** from the point of view of an investor in the price currency country and an **indirect quote** from the point of view of an investor in the base currency country. For example, a quote of 1.17 USD/EUR would be a direct quote for a USD-based investor and an indirect quote for a EUR-based investor. Conversely, a quote of $1 / 1.17 = 0.855$ EUR/USD would be a direct quote for a EUR-based investor and an indirect quote for a USD-based investor.

The exchange rate at a point in time is referred to as a **nominal exchange rate**. If this rate (price/base) increases, the cost of a unit of the base currency in terms of the price currency has increased, so that the purchasing power of the price currency has decreased. If the USD/EUR exchange rate increases from 1.10 to 1.15, the cost of 100 euros increases from \$110 to \$115. The purchasing power of the dollar has decreased relative to the euro because the cost of 100 euros' worth of goods to a consumer in the United States has increased over the period.

The purchasing power of one currency relative to another is also affected by changes in the price levels of the two countries. The **real exchange rate** between two currencies refers to the purchasing power of one currency in terms of the amount of goods priced in another currency, relative to an earlier (base) period.

Consider a situation in which the nominal USD/EUR exchange rate is unchanged at 1.00 over a period and the price level in the United States is unchanged, while prices in the Eurozone have increased by 5%. Eurozone goods that cost 100 euros at the beginning of the period cost 105 euros at the end of the period. With the nominal exchange rate unchanged, the purchasing power of the USD in the Eurozone has decreased, because exchanging 100 USD for 100 EUR will now buy only $100 / 105 = 95.2\%$ of the goods 100 EUR could buy at the beginning of the period.

Here is a summary:

- | An increase in the *nominal* USD/EUR rate decreases the purchasing power of the USD in the Eurozone (and increases the purchasing power of the EUR in the United States); the *real* USD/EUR exchange rate has increased.

- | A decrease in the *nominal* USD/EUR rate increases the purchasing power of the USD in the Eurozone (and decreases the purchasing power of the EUR in the United States); the *real* USD/EUR exchange rate has decreased.
- | An increase in the Eurozone price level, relative to the price level in the United States, will increase the *real* USD/EUR exchange rate, decreasing the purchasing power of the USD in the Eurozone (and increasing the purchasing power of the EUR in the United States).
- | A decrease in the Eurozone price level, relative to the price level in the United States, will decrease the *real* USD/EUR exchange rate, increasing the purchasing power of the USD in the Eurozone (and decreasing the purchasing power of the EUR in the United States).

The end-of-period real P/B exchange rate can be calculated as follows:

$$\text{real P/B exchange rate} = \text{nominal P/B exchange rate} \times \left(\frac{\text{CPI}_{\text{base currency}}}{\text{CPI}_{\text{price currency}}} \right)$$

where the CPI values are relative to base period values of 100.

We can see from the formula the following:

- | An increase (decrease) in the nominal exchange rate over the period increases (decreases) the end-of period real exchange rate, and the purchasing power of the price currency decreases (increases).
- | An increase in the price level in the price currency country relative to the price level in the base currency country will decrease the real exchange rate, increasing the purchasing power of the price currency in terms of base country goods.
- | Conversely, a decrease in the price level in the price currency country relative to the price level in the base currency country will increase the real exchange rate, decreasing the purchasing power of the price currency in terms of base country goods.

In the following example, we calculate the end-of-period real \$/£ exchange rate when the nominal \$/£ exchange rate has decreased over the period (which tends to decrease the real exchange rate and increase the purchasing power of the price currency)—and when the price level in the United Kingdom has increased by more than the price level in the United States over the period (which tends to increase the real exchange rate and decrease the purchasing power of the price

currency). The relative increase in UK prices has reduced the effects of the decrease in the nominal exchange rate on the increase in the purchasing power of the USD.

EXAMPLE: Real exchange rate

At a base period, the CPIs of the United States and United Kingdom are both 100, and the exchange rate is \$1.70/£. Three years later, the exchange rate is \$1.60/£, and the CPI has risen to 110 in the United States and 112 in the United Kingdom. What is the real exchange rate at the end of the three-year period?

Answer:

The real exchange rate is $\$1.60 / \text{£} \times 112 / 110 = \$1.629 / \text{£}$, which means that US goods and services that cost \$1.70 at the base period now cost only \$1.629 (in real terms) if purchased in the United Kingdom and the real exchange rate, \$/£, has fallen. The decrease in the real exchange rate (and the increase in the purchasing power of the USD in terms of UK goods) over the period is less than it would have been if the relative prices between the two countries had not changed.

A **spot exchange rate** is the currency exchange rate for immediate delivery, which for most currencies means the exchange of currencies takes place two days after the trade.

A **forward exchange rate** is a currency exchange rate for an exchange to be done in the future. Forward rates are quoted for various future dates (e.g., 30 days, 60 days, 90 days, or one year). A forward is actually an agreement to exchange a specific amount of one currency for a specific amount of another on a future date specified in the forward agreement.

A French firm that will receive 10 million GBP from a British firm six months from now has uncertainty about the amount of euros that payment will be equivalent to six months from now. By entering into a forward agreement covering 10 million GBP at the 6-month forward rate of 1.192 EUR/GBP, the French firm has agreed to exchange 10 million GBP for 11.92 million euros in six months.

Calculating the Percentage Change in the FX Value of a Currency

Consider a USD/EUR exchange rate that has changed from 1.42 to 1.39 USD/EUR. The percentage change in the dollar price of a euro is simply $1.39 / 1.42 - 1 = -0.0211 = -2.11\%$. Because the dollar price of a euro has fallen, the euro has *depreciated* relative to the dollar, and a euro now buys 2.11% fewer US dollars. It is correct to say that the euro has depreciated by 2.11% relative to the dollar.

On the other hand, it is *not* correct to say that the dollar has appreciated by 2.11%. To calculate the percentage appreciation of the dollar, we need to convert the quotes to EUR/USD. So, our beginning quote of 1.42 USD/EUR becomes $1 / 1.42 = 0.7042$ EUR/USD, and our ending quote of 1.39 USD/EUR becomes $1 / 1.39 = 0.7194$ EUR/USD. Using these exchange rates, we can calculate the change in the euro price of a dollar as $0.7194 / 0.7042 - 1 = 0.0216 = 2.16\%$. In this case, it is correct to say that the dollar has appreciated 2.16% with respect to the euro. Note that for the given exchange rate quotes, the percentage appreciation of the dollar is not the same as the percentage depreciation in the euro.



MODULE QUIZ 18.1

1. One year ago, the nominal exchange rate for USD/EUR was 1.300. Since then, the real exchange rate has increased by 3%. This *most likely* implies that:
 - A. the nominal exchange rate is less than USD/EUR 1.235.
 - B. the purchasing power of the euro has increased approximately 3% in terms of US goods.
 - C. inflation in the euro zone was approximately 3% higher than inflation in the United States.
2. Sell-side participants in the foreign exchange market are *most likely* to include:
 - A. banks.
 - B. hedge funds.
 - C. insurance companies.
3. Suppose that the quote for British pounds (GBP) in New York is USD/GBP 1.3110. What is the quote for US dollars (USD) in London (GBP/USD)?
 - A. 0.3110.
 - B. 0.7628.
 - C. 1.3110.
4. The Canadian dollar (CAD) exchange rate with the Japanese yen (JPY) changes from JPY/CAD 75 to JPY/CAD 78. The CAD has:
 - A. depreciated by 3.8%, and the JPY has appreciated by 4.0%.
 - B. appreciated by 3.8%, and the JPY has depreciated by 4.0%.
 - C. appreciated by 4.0%, and the JPY has depreciated by 3.8%.

MODULE 18.2: MANAGING EXCHANGE RATES

LOS 18.b: Describe exchange rate regimes and explain the effects of exchange rates on countries' international trade and capital flows.

The IMF categorizes **exchange rate regimes** into the following types, which include two for countries that do not issue their own currencies and seven for countries that issue their own currencies.

Countries That Do Not Have Their Own Currency

A country can use the currency of another country (**formal dollarization**). The country cannot have its own monetary policy, as it does not create money or issue currency.

A country can be a member of a **monetary union** in which several countries use a common currency. Within the European Union, for example, most countries use the euro. While individual countries give up the ability to set domestic monetary policy, they all participate in determining the monetary policy of the European Central Bank.

Countries That Have Their Own Currency

A **currency board arrangement** is an explicit commitment to exchange domestic currency for a specified foreign currency at a fixed exchange rate. A notable example of such an arrangement is Hong Kong. In Hong Kong, currency is (and may be) only issued when fully backed by holdings of an equivalent amount of US dollars. The Hong Kong Monetary Authority can earn interest on its US dollar balances. With dollarization, there is no such income, as the income is earned by the US Federal Reserve when it buys interest-bearing assets with the US currency it issues. While the monetary authority gives up the ability to conduct independent monetary policy and essentially imports the inflation rate of the outside currency, there may be some latitude to affect interest rates over the short term.

In a **conventional fixed peg arrangement**, a country pegs its currency within margins of $\pm 1\%$ versus another currency or a basket that includes the currencies of its major trading or financial partners. The monetary authority can maintain exchange rates within the band by purchasing or selling foreign currencies in the foreign exchange markets (*direct intervention*). In addition, the country can use

indirect intervention, including changes in interest rate policy, regulation of foreign exchange transactions, and convincing people to constrain foreign exchange activity. The monetary authority retains more flexibility to conduct monetary policy than with dollarization, a monetary union, or a currency board. However, changes in policy are constrained by the requirements of the peg.

In a system of pegged exchange rates within horizontal bands or a **target zone**, the permitted fluctuations in currency value relative to another currency or basket of currencies are wider (e.g., $\pm 2\%$). Compared to a conventional peg, the monetary authority has more policy discretion because the bands are wider.

With a **crawling peg**, the exchange rate is adjusted periodically, typically to adjust for higher inflation versus the currency used in the peg. This is termed a *passive crawling peg* as opposed to an *active crawling peg*, in which a series of exchange rate adjustments over time is announced and implemented. An active crawling peg can influence inflation expectations, adding some predictability to domestic inflation. Monetary policy is restricted in much the same way it is with a fixed peg arrangement.

With **management of exchange rates within crawling bands**, the width of the bands that identify permissible exchange rates is increased over time. This method can be used to transition from a fixed peg to a floating rate when the monetary authority's lack of credibility makes an immediate change to floating rates impractical. Again, the degree of monetary policy flexibility increases with the width of the bands.

With a system of **managed floating exchange rates**, the monetary authority attempts to influence the exchange rate in response to specific indicators such as the balance of payments, inflation rates, or employment without any specific target exchange rate or predetermined exchange rate path. Intervention may be direct or indirect. Such management of exchange rates may induce trading partners to respond in ways that reduce stability.

When a currency is **independently floating**, the exchange rate is market determined, and foreign exchange market intervention is used only to slow the rate of change and reduce short-term fluctuations—not to keep exchange rates at a target level.

Changes in Exchange Rates

Changes in exchange rates can affect both imports and exports. If the USD/EUR exchange rate decreases, the USD has appreciated relative to the EUR. In this case, the USD prices of imports from the Eurozone have decreased, and it takes more euros to purchase a US good at the new exchange rate. As a result, Eurozone imports from the United States (US exports to the Eurozone) will decrease, and US imports of Eurozone goods (Eurozone exports to the United States) will increase.

The effects on the goods market (imports and exports of merchandise) occur more slowly than the effects on capital flows between countries. This is reflected in a country's **balance of payments**. Essentially, the balance of payments refers to the fact that capital flows must offset any imbalance between the value of a country's exports to and imports from another country.

Consider two countries that have an imbalance between the value of their exports to and imports from another country, such as between China and the United States. Typically, US imports from China have a greater value than US exports to China. We say that the United States has a goods and services **trade deficit** with China and China has a trade surplus with the United States. Capital flows must offset this difference.

Capital flows result primarily from purchases of assets (both physical and financial) in one country by investors or governments in other countries. A large portion of the capital flows between the United States and China are purchases of US debt securities by China. China has a **capital account deficit** that offsets its trade surplus with the United States.

The relation between the balance of trade and capital flows is expressed by the following identity:

$$(\text{exports} - \text{imports}) \equiv (\text{private savings} - \text{investment in physical capital}) \\ + (\text{tax revenue} - \text{government spending})$$

or:

$$(X - M) \equiv (S - I) + (T - G)$$

The intuition is that a trade deficit ($X - M < 0$) means that the right-hand side must also be negative, which implies that total domestic savings (private savings + government savings) are less than domestic investment in physical capital. The additional amount to fund domestic investment must come from foreigners, which results in a surplus in the capital account to offset the deficit in the trade account. Another thing we can see from this identity is that any government

deficit not funded by an excess of domestic saving over domestic investment is consistent with a trade deficit (imports > exports), which is offset by an inflow of foreign capital (a surplus in the capital account).

Capital flows adjust more rapidly than spending, savings, and asset prices, so they are the primary determinant of exchange rates in the short and intermediate term. Trade flows are more important in determining exchange rates in the long term as asset prices and saving/investment decisions adjust over time.

LOS 18.c: Describe common objectives of capital restrictions imposed by governments.

Governments sometimes place restrictions on the flow of investment capital into their country, out of their country, or both. Commonly cited objectives of capital flow restrictions include the following:

- | *Reduce the volatility of domestic asset prices.* In times of macroeconomic crisis, capital flows out of the country can drive down asset prices drastically, especially with prices of liquid assets such as stocks and bonds. With no restrictions on inflows or outflows of foreign investment capital, the asset markets of countries with economies that are small relative to the amount of foreign investment can be quite volatile over a country's economic cycle.
- | *Maintain fixed exchange rates.* For countries with fixed exchange rate targets, limiting flows of foreign investment capital makes it easier to meet the exchange rate target—and, therefore, to be able to use monetary and fiscal policy to pursue only the economic goals for the domestic economy.
- | *Keep domestic interest rates low.* By restricting the outflow of investment capital, countries can keep their domestic interest rates low and manage the domestic economy with monetary policy, as investors cannot pursue higher rates in foreign countries. China is an example of a country with a fixed exchange rate regime where restrictions on capital flows allow policymakers to maintain the target exchange rate as well as to pursue a monetary policy independent of concerns about its effect on currency exchange rates.
- | *Protect strategic industries.* Governments sometimes prohibit investment by foreign entities in industries considered to be important for national security, such as the telecommunications and defense industries.

MODULE QUIZ 18.2



1. The monetary authority of the Stoddard Islands will exchange its currency for US dollars at a one-for-one ratio. As a result, the exchange rate of the Stoddard Islands currency with the US dollar is 1.00, and many businesses in the Islands will accept US dollars in transactions. This exchange rate regime is *best* described as:
 - A. a fixed peg.
 - B. dollarization.
 - C. a currency board.
2. Other things equal, which of the following is *most likely* to decrease a country's trade deficit?
 - A. Increase the capital account surplus.
 - B. Decrease expenditures relative to income.
 - C. Decrease domestic saving relative to domestic investment.
3. The goal of a government that imposes restrictions on foreign capital flows is *most likely* to:
 - A. stimulate domestic interest rates.
 - B. decrease domestic asset price volatility.
 - C. encourage competition with domestic industries.

KEY CONCEPTS

LOS 18.a

Currency exchange rates are given as the price of one unit of currency in terms of another. A nominal exchange rate of 1.44 USD/EUR is interpreted as \$1.44 per euro. We refer to the USD as the price currency and the EUR as the base currency.

An increase (decrease) in an exchange rate represents an appreciation (depreciation) of the base currency relative to the price currency.

A spot exchange rate is the rate for immediate delivery. A forward exchange rate is a rate for exchange of currencies at some future date.

A real exchange rate measures changes in relative purchasing power over time:

$$\text{real exchange rate} = \text{nominal exchange rate} \times \left(\frac{\text{CPI}_{\text{base currency}}}{\text{CPI}_{\text{price currency}}} \right)$$

For a change in an exchange rate, we can calculate the percentage appreciation (price goes up) or depreciation (price goes down) of the base currency. For example, a decrease in the USD/EUR exchange rate from 1.44 to 1.42 represents a depreciation of the EUR relative to the USD of 1.39% ($1.42 / 1.44 - 1 = -0.0139$) because the value of a euro has fallen 1.39%.

Given the depreciation of the EUR (base currency) of 1.39%, we can calculate the appreciation of the price currency as $\frac{1}{(1 - 0.0139)} - 1 = 0.0141 = 1.41\%$.

The market for foreign exchange is the largest financial market in terms of the value of daily transactions and has various participants, including large multinational banks (the sell side) and corporations, investment fund managers, hedge fund managers, investors, governments, and central banks (the buy side).

Participants in the foreign exchange markets are referred to as hedgers if they enter into transactions that decrease an existing foreign exchange risk, and as speculators if they enter into transactions that increase their foreign exchange risk.

LOS 18.b

Exchange rate regimes for countries that do not have their own currency:

- | With *formal dollarization*, a country uses the currency of another country.
- | In a *monetary union*, several countries use a common currency.

Exchange rate regimes for countries that have their own currency:

- | A *currency board arrangement* is an explicit commitment to exchange domestic currency for a specified foreign currency at a fixed exchange rate.
- | In a *conventional fixed peg arrangement*, a country pegs its currency within margins of $\pm 1\%$ versus another currency.
- | In a system of *pegged exchange rates within horizontal bands* or a *target zone*, the permitted fluctuations in currency value relative to another currency or basket of currencies are wider (e.g., $\pm 2\%$).
- | With a *crawling peg*, the exchange rate is adjusted periodically, typically to adjust for higher inflation versus the currency used in the peg.
- | With *management of exchange rates within crawling bands*, the width of the bands that identify permissible exchange rates is increased over time.
- | With a system of *managed floating exchange rates*, the monetary authority attempts to influence the exchange rate in response to specific indicators, such as the balance of payments, inflation rates, or employment, without any specific target exchange rate.
- | When a currency is *independently floating*, the exchange rate is market determined.

LOS 18.c

Commonly cited objectives of capital flow restrictions include the following:

- | Reducing the volatility of domestic asset prices
- | Maintaining fixed exchange rates
- | Keeping domestic interest rates low and enabling greater independence regarding monetary policy
- | Protecting strategic industries from foreign ownership

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 18.1

- . **B** An increase in the real exchange rate USD/EUR (the number of USD per one EUR) means that a euro is worth more in purchasing power (real) terms in the United States. Changes in a real exchange rate depend on the change in the nominal exchange rate relative to the difference in inflation. By itself, a real exchange rate does not indicate the directions or degrees of change in either the nominal exchange rate or the inflation difference. (LOS 18.a)
- . **A** Large multinational banks make up the sell side of the foreign exchange market. The buy side includes corporations, real money and leveraged investment accounts, governments and government entities, and retail purchasers of foreign currencies. (LOS 18.a)
- . **B** $1 / 1.311 = 0.7628$ GBP/USD
(LOS 18.a)
- . **C** The CAD has appreciated because it is worth a larger number of JPY. The percentage appreciation is $(78 - 75) / 75 = 4.0\%$. To calculate the percentage depreciation of the JPY against the CAD, convert the exchange rates to make JPY the base currency: $1 / 75 = 0.0133$ CAD/JPY and $1 / 78 = 0.0128$ CAD/JPY. Percentage depreciation = $(0.0128 - 0.0133) / 0.0133 = -3.8\%$. (LOS 18.a)

Module Quiz 18.2

- . **C** This exchange rate regime is a currency board arrangement. The country has not formally dollarized because it continues to issue a domestic

currency. A conventional fixed peg allows for a small degree of fluctuation around the target exchange rate. (LOS 18.b)

- . **B** An improvement in a trade deficit requires that domestic saving increase relative to domestic investment, which would decrease a capital account surplus. Decreasing expenditures relative to income means domestic savings increase. Decreasing domestic saving relative to domestic investment is consistent with a larger capital account surplus (an increase in net foreign borrowing) and a greater trade deficit. (LOS 18.b)
- . **B** Decreasing the volatility of domestic asset prices may be a goal of a government that imposes capital restrictions. Other typical goals include keeping domestic interest rates low and protecting certain domestic industries, such as the defense industry. (LOS 18.c)

READING 19

EXCHANGE RATE CALCULATIONS

MODULE 19.1: FOREIGN EXCHANGE RATES

LOS 19.a: Calculate and interpret currency cross-rates.

A **cross rate** is the exchange rate between two currencies implied by their exchange rates with a common third currency. Cross-rates are necessary when there is no active FX market in a currency pair. The rate must be computed from the exchange rates between each of these two currencies and a third currency, usually the USD or EUR.

Let's assume that we have the following quotations for Mexican pesos and Australian dollars: $\text{MXN/USD} = 10.70$ and $\text{USD/AUD} = 0.60$. The cross rate between Australian dollars and pesos (MXN/AUD) is as follows:

$$\text{MXN/AUD} = \text{USD/AUD} \times \text{MXN/USD} = 0.60 \times 10.70 = 6.42$$

So, our MXN/AUD cross rate is 6.42 pesos per Australian dollar. The key to calculating cross-rates is to note that the basis of the quotations must be such that we get the desired result algebraically. If we had started with an AUD/USD quotation of 1.67, we would have taken the inverse to get the quotation into USD/AUD terms. Another approach is to divide through, as is illustrated in the following example.

EXAMPLE: Cross rate calculation

The spot exchange rate between the Swiss franc (CHF) and the USD is $\text{CHF/USD} = 1.7799$, and the spot exchange rate between the New Zealand dollar (NZD) and the US dollar is $\text{NZD/USD} = 2.2529$. Calculate the CHF/NZD spot rate.

Answer:

The CHF/NZD cross rate is $(\text{CHF/USD}) / (\text{NZD/USD}) = 1.7799 / 2.2529 = 0.7900$.

LOS 19.b: Explain the arbitrage relationship between spot and forward exchange rates and interest rates, calculate a forward rate using points or in percentage terms, and interpret a forward discount or premium.

When currencies are freely traded and forward currency contracts exist, the percentage difference between forward and spot exchange rates is approximately equal to the difference between the two countries' interest rates. This is because there is an arbitrage trade with a riskless profit to be made when this relation does not hold.

We call this a no-arbitrage condition because if it doesn't hold, there is an opportunity to make a profit without risk. The possible arbitrage is as follows: borrow Currency A at Interest Rate A, convert it to Currency B at the spot rate and invest it to earn Interest Rate B, and sell the proceeds from this investment forward at the forward rate to turn it back into Currency A. If the forward rate does not correctly reflect the difference between interest rates, such an arbitrage could generate a profit to the extent that the return from investing Currency B and converting it back to Currency A with a forward contract is greater than the cost of borrowing Currency A for the period.

The no-arbitrage condition requires that “you cannot earn more than your domestic riskless rate of interest by borrowing your domestic currency, converting it to a foreign currency to invest at the foreign riskless rate, and exchanging back to your domestic currency.” So:

$$(1 + r_{\text{domestic}}) = \frac{1}{\text{spot}_{\$/\text{€}}} (1 + r_{\text{foreign}}) \text{forward}_{\$/\text{€}}$$

The three terms on the right-hand side are equivalent to units of foreign currency for one unit of the domestic currency, growth if invested at the foreign interest rate, and then exchanged back to domestic currency at the forward rate.

Equivalent relationships are as follows:

$$\frac{\text{forward}_{\$/\text{€}}}{\text{spot}_{\$/\text{€}}} = \frac{(1 + r_{\text{domestic}})}{(1 + r_{\text{foreign}})}$$

$$\text{forward}_{\$/\text{€}} = \frac{(1 + r_{\text{domestic}})}{(1 + r_{\text{foreign}})} \text{spot}_{\$/\text{€}}$$

$$\text{spot}_{\$/\text{€}} = \frac{(1 + r_{\text{foreign}})}{(1 + r_{\text{domestic}})} \text{forward}_{\$/\text{€}}$$

EXAMPLE: Calculating the arbitrage-free forward exchange rate

Consider two currencies, the ABE and the DUB. The spot ABE/DUB exchange rate is 4.5671, the 1-year riskless ABE rate is 5%, and the 1-year riskless DUB rate is 3%. What is the 1-year no-arbitrage forward exchange rate?

Answer:

$$\text{forward}_{A/D} = \text{spot}_{A/D} \left(\frac{1 + r_A}{1 + r_D} \right)$$
$$\text{forward}_{A/D} = 4.5671 \left(\frac{1.05}{1.03} \right) = 4.6558_{A/D}$$

Note that the forward rate is greater than the spot rate by $4.6558 / 4.5671 - 1 = 1.94\%$. This is approximately equal to the interest rate differential of $5\% - 3\% = 2\%$. The currency with the higher interest rate must depreciate over time by approximately the amount of the interest rate differential, to prevent arbitrage.

If we are calculating a 30-, 90-, or 180-day forward exchange rate, we need to use interest rates for 30-, 90-, and 180-day periods rather than annual rates. Note that these shorter-term rates are quoted as money market yields.

EXAMPLE: Calculating the arbitrage-free forward exchange rate with 90-day interest rates

The spot ABE/DUB exchange rate is 4.5671, the 90-day riskless ABE rate is 5%, and the 90-day riskless DUB rate is 3%. What is the 90-day forward exchange rate that will prevent arbitrage profits?

Answer:

$$\text{no-arbitrage forward} = 4.5671 \left[\frac{1 + 0.05 \left(\frac{90}{360} \right)}{1 + 0.03 \left(\frac{90}{360} \right)} \right] = 4.5671 \left(\frac{1.0125}{1.0075} \right)$$
$$= 4.5898 \text{ ABE/DUB}$$

In our previous example, we calculated the no-arbitrage one-year forward ABE/DUB exchange rate as 4.6558. Next, we illustrate the arbitrage profit that could be gained if the forward exchange rate differs from this no-arbitrage rate. Consider a forward rate of 4.6000 so that the depreciation in the ABE is less than

that implied by the no-arbitrage relationship. This makes the ABE attractive to a DUB investor who can earn a riskless profit as follows:

- | Borrow 1,000 DUB for one year at 3% to purchase ABE and get 4,567.1 ABE.
- | Invest the 4,567.1 ABE at the ABE rate of 5% to have $1.05(4,567.1) = 4,795.45$ ABE at the end of one year.
- | Enter into a currency forward contract to exchange 4,795.45 ABE in one year at the forward rate of 4.6000 ABE/DUB to receive $4,795.45 / 4.6000 = 1,042.49$ DUB.

The investor has ended the year with a 4.249% return on his 1,000 DUB investment, which is higher than the 3% 1-year DUB interest rate. After repaying the 1,000 DUB loan plus interest (1,030 DUB), the investor has a profit of $1,042.49 - 1,030 = 12.49$ DUB with no risk and no initial out-of-pocket investment (i.e., a pure arbitrage profit).

Arbitrageurs will pursue this opportunity, buying ABE (driving down the spot ABE/DUB exchange rate) and selling ABE forward (driving up the forward ABE/DUB exchange rate), until the interest rate parity relation is restored and arbitrage profits are no longer available.

Note that the no-arbitrage forward exchange rate is approximately proportional to the annual interest rate differential and the time period of the forward contract.

Calculating a Forward Exchange Rate From a Forward Quote in Points or in Percentage Terms

A forward exchange rate quote typically differs from the spot quotation and is expressed in terms of the difference between the spot exchange rate and the forward exchange rate. One way to indicate this is with points. The unit of points is the last decimal place in the spot rate quote. For a spot currency quote to four decimal places, such as 2.3481, each point is 0.0001, or 1/10,000th. A quote of +18.3 points for a 90-day forward exchange rate means that the forward rate is 0.00183 greater than the spot exchange rate.

EXAMPLE: Forward exchange rates in points

The AUD/EUR spot exchange rate is 0.7313 with the 1-year forward rate quoted at +3.5 points. What is the 1-year forward AUD/EUR exchange rate?

Answer:

The forward exchange rate is $0.7313 + 0.00035 = 0.73165$.

EXAMPLE: Forward exchange rates in percentage

The AUD/EUR spot rate is quoted at 0.7313, and the 120-day forward exchange rate is given as -0.062% . What is the 120-day forward AUD/EUR exchange rate?

Answer:

The forward exchange rate is $0.7313 (1 - 0.00062) = 0.7308$.

While forward rates are typically quoted as forward points, one application where percentage forward quotes are useful is if we are interpreting a forward rate as the expected future spot rate, s_{t+1} . We can write our no-arbitrage relation as follows:

$$\frac{s_{t+1}}{s_t} - 1 = \% \Delta s_{t+1} = \frac{r_f - r_d}{1 + r_d}$$

Analysis of capital markets suggests that an increase in a country's interest rate will attract foreign investment, which will lead to appreciation of the domestic currency (spot rate decreases). This is opposite to our conclusion from our no-arbitrage forward rate result that higher domestic rates are associated with a depreciation of the domestic currency. Historically, interest rate differences are poor predictors of future spot rates, although they may be unbiased and tend to get the direction of change correct.

It's best to think of the no-arbitrage forward rate as simply the forward rate at a point in time that prevents currency arbitrage, rather than the expected future spot rate. Our no-arbitrage relation holds at a point in time and does not address the question of how changes in interest rates affect spot exchange rates over time.

Interpreting a Forward Discount or Premium

The **forward discount** or **forward premium** for a currency is calculated relative to the spot exchange rate. The forward discount or premium *for the base currency* is the percentage difference between the forward price and the spot price.

Consider the following spot and forward exchange rates:

USD/EUR spot = \$1.312

USD/EUR 90-day forward = \$1.320

The (90-day) forward premium or discount on the euro = $\text{forward} / \text{spot} - 1 = 1.320 / 1.312 - 1 = 0.610\%$. Because this is positive, it is interpreted as a forward premium on the euro of 0.610%. Because we have the forward rate for three months, we could annualize the discount simply by multiplying by 4 (= 12 / 3).

Because the forward quote is greater than the spot quote, it will take more dollars to buy one euro 90 days from now, so the euro is expected to appreciate versus the dollar, and the dollar is expected to depreciate relative to the euro.

If the forward quote were less than the spot quote, the calculated amount would be negative, and we would interpret that as a forward discount for the euro relative to the US dollar.



MODULE QUIZ 19.1

1. Today's spot rate for the Indonesian rupiah (IDR) is IDR/USD 2,400.00, and the New Zealand dollar trades at NZD/USD 1.6000. The NZD/IDR cross rate is:
A. 0.00067.
B. 1,492.53.
C. 3,840.00.
2. The New Zealand dollar (NZD) is trading at USD/NZD 0.3500, and the Swedish krona (SEK) is trading at NZD/SEK 0.3100. The USD/SEK cross rate is:
A. 0.1085.
B. 8.8573.
C. 9.2166.
3. The spot Swiss franc/British pound (CHF/GBP) exchange rate is 1.3050. In the 180-day forward market, the CHF/GBP exchange rate is -42.5 points. The 180-day forward CHF/GBP exchange rate is *closest* to:
A. 1.2625.
B. 1.3008.
C. 1.3093.
4. The spot rate on the New Zealand dollar (NZD) is NZD/USD 1.4286, and the 180-day forward rate is NZD/USD 1.3889. This difference means:
A. interest rates are lower in the United States than in New Zealand.
B. interest rates are higher in the United States than in New Zealand.
C. it takes more NZD to buy one USD in the forward market than in the spot market.
5. The current spot rate for the British pound (GBP) in terms of US dollars is \$1.533 and the 180-day forward rate is \$1.508. Relative to the pound, the dollar is trading *closest* to a 180-day forward:

- A. discount of 1.63%.
 - B. premium of 1.66%.
 - C. discount of 1.66%.
6. The annual interest rates in the United States (USD) and Sweden (SEK) are 4% and 7% per year, respectively. If the current spot rate is SEK/USD 9.5238, then the 1-year forward rate in SEK/USD is:
- A. 9.2568.
 - B. 9.7985.
 - C. 10.2884.
7. The annual risk-free interest rate is 10% in the United States (USD) and 4% in Switzerland (CHF), and the 1-year forward rate is USD/CHF 0.80. Today's USD/CHF spot rate is *closest* to:
- A. 0.7564.
 - B. 0.8462.
 - C. 0.8888.

KEY CONCEPTS

LOS 19.a

Given two exchange rate quotes for three different currencies, we can calculate a currency cross rate. If the MXN/USD quote is 12.1 and the USD/EUR quote is 1.42, we can calculate the cross rate of MXN/EUR as $12.1 \times 1.42 = 17.18$.

LOS 19.b

The condition that must be met so that there is no arbitrage opportunity available is as follows:

$$\frac{\text{forward}}{\text{spot}} = \frac{(1 + i_{\text{price currency}})}{(1 + i_{\text{base currency}})} \text{ so that } \text{forward} = \text{spot} \times \frac{(1 + i_{\text{price currency}})}{(1 + i_{\text{base currency}})}$$

If the spot exchange rate for the euro is 1.25 USD/EUR, the euro interest rate is 4% per year, and the dollar interest rate is 3% per year, the no-arbitrage one-year forward rate can be calculated as follows:

$$1.25 \times (1.03 / 1.04) = 1.238 \text{ USD/EUR}$$

Points in a foreign currency quotation are in units of the last digit of the quotation. For example, a forward quote of +25.3 when the USD/EUR spot exchange rate is 1.4158 means that the forward exchange rate is $1.4158 + 0.00253 = 1.41833$ USD/EUR.

For a forward exchange rate quote given as a percentage, the percentage (change in the spot rate) is calculated as $\text{forward} / \text{spot} - 1$. A forward exchange rate

quote of +1.787%, when the spot USD/EUR exchange rate is 1.4158, means that the forward exchange rate is $1.4158 (1 + 0.01787) = 1.4411$ USD/EUR.

If a forward exchange rate does not correctly reflect the difference between the interest rates for two currencies, an arbitrage opportunity for a riskless profit exists. In this case, borrowing one currency, converting it to the other currency at the spot rate, investing the proceeds for the period, and converting the end-of-period amount back to the borrowed currency at the forward rate will produce more than enough to pay off the initial loan, with the remainder being a riskless profit on the arbitrage transaction.

To calculate a forward premium or discount for the base currency, use the following formula:

$$(\text{forward} / \text{spot}) - 1$$

Forward exchange rates are poor predictors of future exchange rates. Forward exchange rates should be interpreted as simply the no-arbitrage forward rates at a point in time, not as predictors of future spot rates.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 19.1

- . **A** Start with one NZD and exchange for $1 / 1.6 = 0.625$ USD. Exchange the USD for $0.625 \times 2,400 = 1,500$ IDR. We get a cross rate of 1,500 IDR/NZD, or $1 / 1,500 = 0.00067$ NZD/IDR. (LOS 19.a)
- . **A** $\text{USD/NZD } 0.3500 \times \text{NZD/SEK } 0.3100 = \text{USD/SEK } 0.1085$. Notice that the NZD term cancels in the multiplication. (LOS 19.a)
- . **B** The 180-day forward exchange rate is $1.3050 - 0.00425 = \text{CHF/GBP } 1.30075$. (LOS 19.b)
- . **B** Interest rates are higher in the United States than in New Zealand. It takes fewer NZD to buy one USD in the forward market than in the spot market. (LOS 19.b)
- . **B** To calculate a percentage forward premium or discount for the US dollar, we use the dollar to be the base currency. The spot and forward quotes given are US dollars per British pound (USD/GBP), so we must invert them to GBP/USD. The spot GBP/USD price is $1 / 1.533 = 0.6523$ and the

forward GBP/USD price is $1 / 1.508 = 0.6631$. Because the forward price is greater than the spot price, we say the dollar is at a forward premium of $0.6631 / 0.6523 - 1 = 1.66\%$. Alternatively, we can calculate this premium with the given quotes as spot / forward - 1 to get $1.533 / 1.508 - 1 = 1.66\%$. (LOS 19.b)

- . **B** The forward rate in SEK/USD is $9.5238 \left(\frac{1.07}{1.04} \right) = 9.7985$. Because the SEK interest rate is the higher of the two, the SEK must depreciate approximately 3%. (LOS 19.b)
- . **A** We can solve for the spot rate as follows. With the exchange rates quoted as USD/CHF, the spot is $0.80 \left(\frac{1.04}{1.10} \right) = 0.7564$. Because the interest rate is higher in the United States, it should take fewer USD to buy CHF in the spot market. In other words, the USD must be at a forward discount relative to the CHF. (LOS 19.b)

TOPIC QUIZ: ECONOMICS

You have now finished the Economics topic section. Please log into your Schweser online dashboard and take the Topic Quiz on this section. The Topic Quiz provides immediate feedback on how effective your study has been for this material. Questions are more exam-like than typical Module Quiz or QBank questions; a score of less than 70% indicates that your study likely needs improvement. These tests are best taken timed; allow 1.5 minutes per question.

READING 20

ORGANIZATIONAL FORMS, CORPORATE ISSUER FEATURES, AND OWNERSHIP

MODULE 20.1: FEATURES OF CORPORATE ISSUERS

LOS 20.a: Compare the organizational forms of businesses.

Organizational forms refer to how businesses are set up from a legal and organizational point of view. Key features of organizational forms include the following:

- | Whether the business is a separate legal entity from the owner(s)
- | Whether the owners of the business also operate the business, and if not, the nature of the relationship between its owners and operators
- | Whether the owners' liability for the actions and debts of the business is limited or unlimited
- | The tax treatment of profits or losses from the business
- | Access to additional capital to fund expansion and to distribute risk

To understand these features, we can compare them among four commonly used types of business structures: sole proprietorships, general partnerships, limited partnerships, and corporations.

A **sole proprietorship** is a business owned and operated by an individual. Legally, the business is an extension of the owner, who is personally responsible for claims against the business (i.e., unlimited liability) and receives all profits/losses.

Profits are then taxed as personal income of the owner. Sole proprietorships tend to be small in scale because they can only expand within the limits of the individual owner's ability to secure financing.

To do business on a scale that exceeds that of a sole proprietorship, two or more individuals can form a **general partnership**. In this structure, the **partnership agreement** specifies each partner's responsibilities for business operations and their shares of the partnership profits or losses. The agreement may be written, verbal, or even incidental through the actions of the partners. As with a sole proprietorship, the partners have unlimited liability for claims against the business, and profits from the business allocated to each partner are taxed as personal income.

A **limited partnership** involves two levels of partners. One or more general partners operate the business and have unlimited liability, as in a general partnership, but this structure also has **limited partners** who are liable only for the amount they invest in the partnership (i.e., **limited liability**) and have claims to its profits that are proportionate to their investments. Limited partners typically are not involved in appointing or removing general partners. How the profits are divided among the general and limited partners is specified in the partnership agreement. Because they are responsible for managing the business, the general partners typically receive a larger portion of profits than the limited partners. Profits allocated to the partners are taxed as personal income to each partner. Some jurisdictions allow a **limited liability partnership (LLP)** wherein a general partner is not required, and all the partners are limited partners. In the United States, LLPs are only allowed for providers of professional services such as law, accounting, and so on, and there are restrictions on the number of partners and the amount of equity investment.

The feature that distinguishes a **corporation**, or **limited company**, from the other business structures is that a corporation is a legal entity separate from its owners and managers. In this case, *all* of the corporation's shareholders have limited liability. An owner can lose his entire investment if the company goes bankrupt and the value of his shares goes to zero. Beyond their initial investments, shareholders are not responsible for claims against the corporation. A corporation may, but is not required to, distribute its profits to its owners. Most large firms are corporations because that structure gives them the greatest access to capital, both debt (**borrowed capital**) and equity (**ownership capital**).

Another distinguishing feature of corporations is the separation of its owners and managers. An investor who buys shares of a corporation does not directly influence the company's day-to-day operations. Instead, the owners appoint a **board of directors** that is responsible for hiring the senior managers to operate the company. The board and the managers it hires are responsible for acting in the interests of the shareholders.

A **public corporation** (or a **public limited company**) is one that has shares that are sold to the public and trade in an organized market (stock exchange). A **private limited company** is similar to a public company, but it has a limited number of shareholders and restrictions on transfer of shares.

Depending on the country, a corporation's profits may be subject to **double taxation** if the government taxes companies on their earnings *and* it taxes dividends (which are distributions of earnings to owners) as personal income.

LOS 20.b: Describe key features of corporate issuers.

A corporation's legal identity is separate from that of its owners and is formed by filing an **articles of incorporation** with a regulatory body. As a legal entity, a corporation has many of the rights and responsibilities of an individual, such as the right to hire employees, enter into contracts, borrow and lend money, and so on.

A corporation issues shares to the owners (shareholders), which allows it to raise large amounts of capital. Shareholders have **voting rights** that allow them to elect the board of directors. Shares are easily transferable if they are traded on an exchange. The board of directors may distribute a portion of the company's earnings to the shareholders as **dividends**.

The disadvantage of double taxation of corporate income is less for investors in companies that pay out a smaller fraction of their profits in dividends and reinvest the remaining, as illustrated in the following example.

EXAMPLE: Double taxation of dividends

ABC Corporation has pretax earnings of \$10 million and a corporate tax rate of 25%. Shareholders are taxed at 20% on their dividend income. Calculate the effective tax rate if:

(a) 100% of the profits are paid out in dividends, and

(b) 40% of the profits are paid out in dividends.

Answer:

	(a) 100% payout	(b) 40% payout
Earnings before tax	\$ 10,000,000	\$ 10,000,000
(-) Corporate income tax @ 25%	<u>2,500,000</u>	<u>2,500,000</u>
(=) After-tax income	7,500,000	7,500,000
Dividends	7,500,000	3,000,000
Tax on dividends @ 20%	1,500,000	600,000
Total tax paid	4,000,000	3,100,000
Effective tax rate		
= Total tax / earnings before tax	40%	31%

LOS 20.c: Compare publicly and privately owned corporate issuers.

Most public limited companies are **listed companies**, which means their shares are listed on an exchange, allowing investors to trade shares. A stock exchange is a rules-based open market, providing price and volume transparency.

Shareholders in a company can be individuals, other corporations, nonprofits, or government. Shares that are actively traded (i.e., not held by insiders, strategic investors, or sponsors) are called the company's **free float**. Free float is typically expressed as a percentage of total outstanding shares.

Public companies are subject to compliance and reporting requirements. For example, companies are required to file quarterly or annual financial reports with a regulatory body, and to disclose any material changes in the company's business or ownership.

Shares in private limited companies do not trade on an exchange. As a result, their value is not readily observable, and transfer between investors is difficult. Typically, investors in a private limited company have to wait until the company goes public or is sold to exit their investment. Private companies have fewer regulatory requirements and typically disclose less information than public companies. Also, with fewer investors, private companies can take a longer-term view of the business.

Private companies can raise equity capital through **private placements** of securities. Private placements are typically restricted to **accredited investors**

such as corporate and institutional investors or high net worth individuals.

Private companies can become public companies (“go public”) in one of three ways: initial public offering, direct listing, or acquisition by a special purpose entity.

- | A company can become public by issuing shares in an **initial public offering (IPO)**. To conduct an IPO, the company must meet exchange-specific requirements. Companies typically engage the services of an investment bank to underwrite the issue. Once the shares are listed on an exchange, owners can sell shares, and new owners can buy shares, without dealing directly with the company.
- | In a **direct listing**, a stock exchange agrees to list a private company’s existing shares. This differs from an IPO in that a direct listing does not raise any new capital for the company, but it has advantages in that it can be done more quickly than an IPO and without involving an underwriter.
- | A **special purpose acquisition company (SPAC)** is a corporate structure set up to acquire a private company in the future. The SPAC raises capital through an IPO and puts the funds into a trust that it must use to make an acquisition within a specified time. The acquired company does not have to be identified at the time of the IPO. For this reason, SPACs are also known as “blank check” companies.

Sometimes there can be a benefit to taking an underperforming public company private to restructure it and unlock its potential value. In such a case, an acquirer purchases all the outstanding shares of a public company, and the company is delisted from the exchange. The benefits of going private include a lower regulatory burden and the associated cost savings.



MODULE QUIZ 20.1

1. Which organizational form has the highest degree of separation between the owners and operators of a business?
 - A. Corporation.
 - B. Limited partnership.
 - C. General partnership.
2. A corporation is subject to a corporate income tax of 10%. Shareholders are taxed on dividends at a rate of 15%. Assume that the company pays out all its after-tax profits in dividends. The effective tax rate on the corporation's profit is:
 - A. 12.5%.
 - B. 23.5%.

- C. 25.0%.
3. A private company can become a public company through a:
- A. private placement.
 - B. leveraged buyout.
 - C. special purpose acquisition company.

KEY CONCEPTS

LOS 20.a

Feature	Sole Proprietorship	General Partnership	Limited Partnership	Corporation
Separate legal entity?	No	No	No	Yes
Managed by	Owner	Partners	GPs	Board and managers
Owner liability	Unlimited	Unlimited	GPs: unlimited LPs: limited	Limited
Access to capital	Limited	Limited	Limited	Virtually unlimited
Taxation of profits	Pass through to owner as personal income	Pass through to owner as personal income	Pass through to owner as personal income	Potential double taxation

LOS 20.b

Corporations incorporate as a separate legal entity by registering with a regulatory body. Owners are shareholders with a limited liability, but profits may be subject to double taxation. Shareholders vote for a board of directors who then hire managers and declare dividend payments to the shareholders.

LOS 20.c

Public limited companies list their shares on a stock exchange for trading. Free float, or actively traded shares, is typically quoted as a fraction of total shares outstanding. Exchange listing requires regulatory compliance and extensive disclosure requirements.

Private companies raise capital in private placements. A private company can become public by issuing shares in an initial public offering, carrying out a

direct listing on a stock exchange, or being acquired by a public company, which may be a special purpose acquisition company.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 20.1

- . **A** In a corporation, owners are most often not directly involved in operating the business. Both general partnerships and limited partnerships have general partners who operate the business. (LOS 20.a)
- . **B** The effective tax rate on profits distributed as dividends = $0.10 + 0.15 (1 - 0.10) = 23.5\%$. Alternatively:
 - Starting with a hypothetical \$100 of pretax profit, the corporate tax @10% is \$10.
 - Dividends = after-tax profit = \$90. Tax on dividends @ 15% = $0.15 \times 90 = \$13.50$.
 - Total tax = $10 + 13.50 = \$23.50$, or 23.5% of the pretax profit of \$100. (LOS 20.b)
- . **C** An IPO, direct listing, and special purpose acquisition company are methods for a private company to go public. Private placements allow a private company to raise capital but not take it public. Buyouts can result in a public company going private. (LOS 20.c)

READING 21

INVESTORS AND OTHER STAKEHOLDERS

MODULE 21.1: STAKEHOLDERS AND ESG FACTORS

LOS 21.a: Compare the financial claims and motivations of lenders and shareholders.

A company's lenders (debtholders) have a legal, contractual claim to the interest and principal payments the company has promised to make. Owners (equity holders) have a residual claim to the company's net assets (i.e., what remains after all other claims have been paid). That is, lenders have a higher priority of claims than equity owners. Because debt is less risky than equity, it is a less costly form of capital.

Both debtholders and equity holders can potentially lose their entire investment if a company fails, but their losses cannot exceed the amounts they have invested. A key difference between debt and equity investments is their upside potential. Regardless of a company's success, the best result debtholders can achieve is to receive the interest and principal payments promised by the company. Equity, on the other hand, has a theoretically unlimited upside if a company succeeds and grows over time.

The value of a company is the sum of the value of its debt and the value of its equity. As the company's value increases (assuming its value is greater than the value of its debt), the value of equity increases with it, while the value of debt is constant (i.e., no upside for debt investors from company growth). If the value of a company is less than the value of its debt, the value of its equity is zero, and the value of its debt is below the promised amount and moves with the value of the company (i.e., debt has downside exposure).

EXAMPLE: Impact of leverage on return on equity (ROE)

A company with revenues of \$1,000 and operating expenses of \$800 needs to invest \$1,000 in assets. The assets can be 100% equity or a mix of 50/50 debt and equity. The interest rate on debt is 10% (assume no taxes). Calculate the ROE under both financing scenarios.

Answer:

	100% Equity	50% Debt, 50% Equity
Revenues	\$1,000	\$1,000
Cash operating expense	\$800	\$800
Interest expense	\$0	\$50
Net income	\$200	\$150
Equity	\$1,000	\$500
ROE	20%	30%

As can be seen in the previous example, increasing leverage (and therefore, risk) can increase the return on equity as long as the *expected* rate of return on assets exceeds the cost of debt. If things work out, and the actual return on assets is equal to or exceeds the cost of debt, the equity investors benefit from taking the risk. Debt investors, however, receive only their promised 10% interest (and the return of principal when the debt matures).

EXAMPLE: Impact of a decrease in revenues

Continuing the previous example, calculate the company's ROE if its revenues decrease by 15%, assuming no change in cash operating expenses.

Answer:

	100% Equity	50% Debt, 50% Equity
Revenues	\$850	\$850
Cash operating expense	\$800	\$800
Interest expense	\$0	\$50
Net income	\$50	\$0
Equity	\$1,000	\$500
ROE	5%	0%

Assuming no change in cash operating expenses, a 15% decrease in revenues would reduce the ROE to 5% (100% equity scenario) or to 0% (50/50 debt and equity scenario). If the decrease in revenues exceeded 15%, revenues would not have been enough to cover operating expenses, and debt investors would receive less than their promised 10% interest. Again, debt investors have no upside—only potential downside.

Because of the difference in their risk profiles, the interests of debtholders may conflict with the interests of equity holders. Debtholders are primarily concerned with a company's ability to repay its obligations and less concerned with its growth prospects. Equity holders may favor actions that increase a company's potential growth, but also increase its risk level, such as adding financial leverage by issuing new debt. Issuing additional debt as opposed to equity also prevents **dilution** of shareholders' proportional ownership. A company's existing debtholders may oppose such actions because increasing the company's risk (and the probability of defaulting on its debts) does not increase their expected return. Therefore, debt investors usually limit borrowers' actions by including contractual provisions such as maximum leverage or a minimum interest coverage ratio (known as *covenants*, which we will examine in the Fixed Income topic area) in debt agreements.

LOS 21.b: Describe a company's stakeholder groups and compare their interests.

Under **shareholder theory**, the primary focus of corporate governance is the interests of the firm's shareholders, which is to maximize the market value of the firm's common equity. Under this theory, corporate governance is primarily

concerned with the conflict of interest between the firm's managers and its owners (shareholders).

The focus of corporate governance under **stakeholder theory** is broader. It considers conflicts among several groups that have an interest in the activities and performance of the firm.

Stakeholders of a Corporation

Shareholders have a *residual interest* in the corporation, in that they have claim to the net assets of the corporation after all liabilities have been settled.

Shareholders have voting rights to elect the board of directors and for other important corporate matters, which gives them control over the firm and its management. They have an interest in ongoing profitability and growth of the firm that will increase the value of their ownership shares.

Lenders can be public or private debtholders. Public debtholders or **bondholders** supply debt capital to the firm and are primarily owners of the firm's outstanding bonds, while **private debtholders** such as banks extend loans, credit facilities, and leases to the firm.

Private debtholders may have access to nonpublic information from company management, which decreases information asymmetry. For this reason, private debtholders are a critical source of financing for small-to-medium-sized businesses. In some cases, private debtholders hold equity in the firm, allowing them to take a more equity-like approach to evaluating the company and potentially making them more amenable to changes in terms of the loans (i.e., covenants). By contrast, bondholders rely only on public information and have little to no influence over an issuer's operations. The interests of both types of lenders are protected to varying degrees by covenants in their debt agreements with the firm.

The **board of directors** is responsible for protecting the interests of shareholders; hiring, firing, and setting the compensation of the firm's senior managers; establishing the strategic direction of the firm; and monitoring the company's financial performance and other aspects of its ongoing activities.

Board members include **inside directors** (e.g., senior executives, founders) and **independent directors** who have no material relationship with the company.

While inside directors may have conflicts of interest with shareholders, independent directors may better protect shareholders' interests.

In a **one-tier board structure**, both inside and independent directors serve on a single board. Major stock exchanges specify requirements for director independence, such as that most of the board should comprise independent directors. Other requirements may include diversity of backgrounds and competencies. In continental Europe, boards have a **two-tier structure** in which a **supervisory board** oversees a **management board** comprising inside directors.

Typically, board members are elected for a specified term (e.g., annual). In a **staggered board**, only a fraction of the board is elected each year. This decreases the power of shareholders to enact a major overhaul of the board. Firms with staggered boards justify them as providing continuity and allowing for a longer-term view of company strategy.

Senior managers typically receive compensation (remuneration) that is made up of a salary, a bonus based on some measure of company performance, and perks (e.g., expense accounts, use of company planes, special retirement benefits). Their interests can be expected to include continued employment and maximizing the total value of their compensation. Executive bonuses are typically tied to some measure of firm performance, giving senior managers a strong interest in the financial success of the firm.

Other employees are the **human capital** of the company, who provide their labor and skills. Employees also have an interest in the sustainability and success of the firm. They have interests in their rate of pay, opportunities for career advancement, training, and working conditions. Employees may hold equity in the company by participating in employee stock participation plans, which are designed to align their interests with those of the shareholders of the company. In some industries, employees join unions to negotiate the terms of their employment.

Suppliers of resources to the firm have an interest in preserving an ongoing relationship with the firm, in the profitability of their trade with the firm, and in the growth and ongoing stability of the firm. Because suppliers are typically short-term creditors of the firm, they also have an interest in the firm's solvency and ongoing financial strength.

Customers rely on the company to provide a good or service of high quality at a reasonable price. Customers may also have an interest in preserving an ongoing relationship to receive support and after-sale service. Maintaining the good will

of customers is critical for the long-term prospects of any company. Customers increasingly care about environmental and social responsibilities of corporations.

Governments rely on corporations for tax revenue, economic growth, social welfare, and employment creation. Regulators have an interest in ensuring compliance with various laws.

LOS 21.c: Describe environmental, social, and governance factors of corporate issuers considered by investors.

Some debt and equity investors who take a stakeholder perspective are interested in evaluating companies' environmental, social, and governance (ESG) factors. This is primarily due to three reasons:

- Government stakeholders increasingly prioritize climate change and social policies through regulatory changes.

- ESG factors can have a material impact on companies' results through potential loss of customer good will and financial losses due to fines and judgments.

- Poor corporate governance may lead to senior managers exploiting shareholders to advance personal interests.

- Many younger investors increasingly manage their wealth with ESG considerations in mind.

Negative externalities arise when a company or its investors do not bear the full cost of its actions (e.g., environmental damage). Increased government regulations and stakeholder awareness necessitate companies to recognize these costs, either explicitly in their financial statements or implicitly.

Environmental Factors

Environmental factors include company contributions to problems such as climate change, air and water pollution, deforestation, energy efficiency, waste management, or water scarcity. Material environmental factors can have a substantial impact on companies' operations or business models. Industries that are natural resource intensive have a direct impact on the environment, but other industries may have indirect impacts.

With respect to climate change, companies may face **physical risk** of adverse effects on assets or operations if severe weather increases in frequency, as well as **transition risk** as government regulations or consumer choices require

switching from high-carbon to low-carbon activities. **Stranded assets** are those that become unviable due to such changes. Poor safety policies or inadequate governance systems increase the risks for adverse events such as oil spills and contamination of groundwater. The costs of penalties, cleanup, litigation, and loss of reputation can be significant risks for a company's investors.

Social Factors

Social factors include the protection of customer privacy and information security, customer satisfaction, employee engagement, diversity and inclusion, labor relations, and community relations. Social factors contribute to a company's image in terms of how it treats its employees, customers, and the communities in which it operates. Taking measures to decrease social risk can reduce a company's costs through higher employee productivity, lower turnover, increased customer loyalty, and less risk of litigation.

Governance

Corporate governance encompasses factors such as the composition of the board and the internal audit committee, executive compensation, bribery and corruption, political contributions, and lobbying. Corporate governance systems should have adequate checks to ensure that managers act ethically, lawfully, and in the interests of shareholders.

Evaluation of ESG Risks

Analysts should identify and measure the ESG-related risks to which a company may be exposed and how they can affect the company's future cash flows. Equity investors bear the brunt of the risk from adverse outcomes. Debt investors have less exposure to adverse events unless they result in losses large enough to bring about default. Not all debtholders are equally exposed to ESG risks. Because some such risks may be delayed (e.g., a coal-fired electricity plant that is currently in compliance may become obsolete in the future), longer-maturity debt investors may have more exposure than short-term debt investors.



MODULE QUIZ 21.1

1. For a company that is financially sound, increasing the company's rate of growth is *most likely* to benefit:
 - A. equity holders.
 - B. debtholders.

- C. neither debtholders nor equity holders.
- 2. Which of the following board structures is *most likely* to be preferred by a minority shareholder?
 - A. Majority independent and staggered elections.
 - B. Majority independent and full board election.
 - C. Majority inside and staggered elections.
- 3. A company decides to shut down a production plant rather than retrofit it to comply with new environmental regulations. This is *best* described as an example of:
 - A. governance factors.
 - B. social factors.
 - C. stranded assets.

KEY CONCEPTS

LOS 21.a

A company's debtholders have a higher priority of claims than its equity holders. Debtholders have a legal claim to the promised interest and principal payments. Equity holders have a residual claim to the company's net assets after debtholders have been paid.

Debt has limited upside potential because the best result for debtholders is to receive the promised principal and interest payments. Equity has theoretically unlimited upside potential. This difference may create conflicts of interest between debtholders and equity holders.

LOS 21.b

The primary stakeholders of a corporation include shareholders, debtholders, the board of directors, senior management, employees, creditors, suppliers, and government. Stakeholder theory postulates that a company needs to balance the interests of all stakeholders.

LOS 21.c

Environmental factors include company contributions to climate change, air and water pollution, deforestation, energy efficiency, waste management, and water scarcity.

Social factors include the protection of customer privacy and information security, customer satisfaction, employee engagement, diversity and inclusion, labor relations, and community relations.

Corporate governance factors include the composition of the board, executive compensation, the internal audit function, bribery and corruption, political contributions, and lobbying.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 21.1

- . **A** If a company is financially sound, it is repaying interest and principal in full and on time. Debtholders have no additional claims to increased company profits. Equity holders benefit from the upside of a company's growth. (LOS 21.a)
- . **B** Minority shareholders prefer independent directors, as they would act in the best interests of the shareholders as opposed to acting in the interests of the management. Full board election would allow the shareholders to vote out the board if it was ineffective. Staggered elections delay changes to the board membership. (LOS 21.b)
- . **C** Stranded assets arise from obsolescence of existing assets that do not conform to new environmental standards. (LOS 21.c)

READING 22

CORPORATE GOVERNANCE: CONFLICTS, MECHANISMS, RISKS, AND BENEFITS

MODULE 22.1: CORPORATE GOVERNANCE

LOS 22.a: Describe the principal-agent relationship and conflicts that may arise between stakeholder groups.

When one party hires another to carry out a task, the two parties are said to have a **principal-agent relationship**. In such a relationship a **principal-agent conflict** has the potential to arise, because an agent is hired to act in the interests of the principal, but an agent's interests might not coincide exactly with those of the principal.

For example, consider an insurance agent who is paid a commission on policies written. It would be in the agent's interest to write insurance policies on people or property that are not good risks to maximize commission income. The principal (the owner of the insurance company) does not want to issue policies that are bad risks, as that is a money-losing proposition. Insurance companies mitigate this conflict by imposing underwriting standards for the policies they will issue and by continuing to work only with agents who consistently act in the company's best interest.

Agency costs are the costs of a principal-agent conflict. Agency costs can be direct, such as hiring employees to monitor the agent, or indirect, such as the opportunity cost of lost business.

Conflicts of Interest Between Shareholders and Managers or Directors

In the context of a corporation, shareholders are the principals (owners), and firm management and board members (directors) are their agents. Managers and directors may choose a lower level of business risk than shareholders would. Managers have their employment income tied to the firm and stand to lose it, should the firm fail. Shareholders can diversify their portfolios cheaply, and as such might prefer that individual companies take more risk.

Conflicts may arise if inside directors favor management interests at the expense of shareholder interests, or if directors favor one group of shareholders at the expense of another. Some conflicts result from **information asymmetry** between shareholders and managers, which refers to the fact that managers have more and better information about the functioning of the firm and its strategic direction than shareholders do. This decreases the ability of shareholders or nonexecutive directors to monitor and evaluate whether managers are acting in the best interests of shareholders. Information asymmetry is more acute for larger companies that operate in many businesses and geographical markets, companies that sell complex products, and companies with lower levels of institutional ownership and free float.

Common principal-agent conflicts arise in the following ways:

- | Managers may put in insufficient effort. This leads to poor evaluation of investment opportunities and risks taken, resulting in overall higher total costs.
- | Managers getting option grants may want to ratchet up risk because options do not have any downside risk. On the other hand, managers who are compensated primarily with cash may be inadequately motivated to take risk.
- | Manager compensation tied to company size creates incentives for *empire building* through poor or unnecessary acquisitions.
- | Managers may seek to entrench themselves by taking inadequate risk, mimicking competitors' actions rather than generating original ideas, or engaging in otherwise unprofitable projects that require their specific individual knowledge. Directors, in turn, might seek to entrench themselves by going along with management's decisions instead of questioning them.
- | Managers may undertake **self-dealing** by exploiting firm resources for personal benefit.

Conflicts Between Groups of Shareholders

A single shareholder or group of shareholders in a company may hold most of the votes (**controlling shareholders**) and act against the interests of the **minority shareholders**. A controlling shareholder may have concentrated ownership, in that a large proportion of his or her wealth might consist of shares in this company. In this situation a controlling shareholder might want the company to diversify into different businesses to mitigate risk. Minority shareholders, by contrast, might already hold diversified portfolios and do not want the company to squander resources by investing in a less desirable business just to diversify.

Some firms have a **dual-class structure** with different classes of common stock outstanding, some with more voting power than others. This can give a group of shareholders (e.g., the company's founders) effective control of the company even if they have claims to less than 50% of its earnings and assets. CFA Institute advocates against dual-class voting structures because they allow one group of shareholders to further their interests at the expense of other groups.

Conflicts of Interest Between Creditors and Shareholders

Shareholders may prefer more business risk than creditors do because creditors have a limited upside from good results compared to shareholders. Management actions that favor the interests of equity owners over the interests of creditors include issuing new debt that increases the default risk faced by existing debtholders, or increasing dividends at the cost of decreasing company assets as collateral and increasing the risk of default. This potential for conflict is a greater risk for long-term debtholders.

LOS 22.b: Describe corporate governance and mechanisms to manage stakeholder relationships and mitigate associated risks.

Corporate governance is the system of internal controls and procedures by which individual companies are managed. It includes a framework that defines the rights, roles, and responsibilities of various groups within an organization. The objective is to manage and minimize conflicts of interest between stakeholders of the company.

Stakeholder management refers to the management of company relations with stakeholders and is based on having a good understanding of stakeholder interests and maintaining effective communication with stakeholders. With

respect to the company's relationship with shareholders, there are standard practices. These practices are required by corporate laws and are similar in many jurisdictions, although there are some differences across countries.

Stakeholders gather information about the company from public reports filed by the company. Public companies release information in annual reports, proxy statements, and public notices. This reporting includes information about the financial performance and standing of the company, related-party transactions, executive remuneration, and governance structure. While reporting requirements for private companies are more limited, these companies typically provide information to their investors directly. Transparency in reporting reduces information asymmetry and allows stakeholders to evaluate whether the company's actions align with their interests.

Shareholder Mechanisms

Corporations typically hold an **annual general meeting** after the end of the firm's fiscal year. At the general meeting, company management provides shareholders with the audited financial statements for the year, addresses the company's performance and significant actions over the period, and answers shareholder questions.

Corporate laws dictate when the annual general meeting must occur and how the meeting must be communicated to shareholders. Typically, anyone owning shares is permitted to attend the annual general meeting, to speak or ask questions, and to vote their shares. A shareholder who does not attend the annual general meeting can vote her shares by **proxy**, meaning she assigns her right to vote to another person who will attend the meeting—often a director, a member of management, or the shareholder's investment advisor. A proxy may specify the shareholder's vote on specific issues or leave the vote to the discretion of the person to whom the proxy is assigned.

Ordinary resolutions, such as the approval of an auditor and the election of directors, require a simple majority of the votes cast. Other resolutions are addressed at **extraordinary general meetings**, which can be called any time a matter requires a shareholder vote. Examples include amendments to the company's bylaws, a merger or takeover, a shareholder-proposed special board election, or the liquidation of the firm.

Activist shareholders pressure companies in which they hold a significant number of shares for changes they believe will increase shareholder value. They may initiate shareholder lawsuits or seek representation on the board of directors. Other activist tactics include proposing shareholder resolutions for a vote and raising their issues to all shareholders or the public to gain wider support. Hedge funds have engaged increasingly in shareholder activism at firms in which they hold significant stakes.

A shareholder activist group may initiate a **proxy contest**, in which they seek the proxies of shareholders to vote in favor of their alternative proposals, or may make a **tender offer** for enough shares of a company to gain control. Both senior managers and boards of directors can be replaced by shareholders when they believe company performance is poor and would be improved by change. The threat of a **hostile takeover**, one not supported by the company's management, can act as an incentive for company managements and boards to pursue policies better aligned with the interests of shareholders. On the other hand, it might also cause the current management or board to adopt takeover defenses such as staggered board elections or **poison pill** provisions (low-price additional share offerings to current shareholders).

Creditor Mechanisms

When a company issues a bond, it specifies the rights of bondholders and the company's obligations in a legal document called a **bond indenture**. An indenture typically includes **covenants** that may require the company to take certain actions, or restrict it from taking certain actions. A bond can be backed by **collateral**, which is a specific asset pledged against which the bondholders will have a claim if the company defaults on the bond. A financial institution may act as a trustee to monitor the company's compliance with its bond covenants.

Creditor committees may form among bondholders to protect their interests when an issuer experiences financial distress. Some countries require such committees when a company files for bankruptcy. A group of bond investors may form an **ad hoc committee** when a company is struggling to meet its obligations. While the committee does not represent *all* the bondholders, their interests are generally aligned.

Board of Directors and Management Mechanisms

The **board of directors** is elected by shareholders to act in their interests. A board of directors typically has committees made up of board members with particular expertise. These committees report to the board, which retains the overall responsibility for the various board functions. The following are examples of typical board committees.

An **audit committee** is responsible for the following:

- | Oversight of the financial reporting function and implementation of accounting policies
- | Effectiveness of the company's internal controls and the internal audit function
- | Recommending an independent external auditor and its compensation
- | Proposing remedies based on their review of internal and external audits

A **nominating/governance committee** is responsible for the following:

- | Oversight of the company's corporate governance code including board elections
- | Setting policies for nomination of candidates for board membership
- | Implementing the company's code of ethics and policies regarding conflicts of interest
- | Monitoring changes in relevant laws and regulations
- | Ensuring that the company is in compliance with all applicable laws and regulations, as well as with the company's governance policies

A **compensation committee** or **remuneration committee** recommends to the board the amounts and types of compensation to be paid to directors and senior managers. This committee may also be responsible for oversight of employee benefit plans and evaluation of senior managers. Because managers should not be in a position to evaluate or compensate themselves, a compensation committee should (and in many countries is required to) be composed of independent directors only.

Other committees are industry specific. A risk committee (financial services industry) informs the board about appropriate risk policy and risk tolerance of the organization, and it oversees the enterprise-wide risk management processes of the organization. An investment committee (insurance industry) reviews and reports to the board prudent investment and capital management policies.

The number and size of board committees will depend on the size, complexity, and nature of the business. Regulations often require that firms have audit committees. The composition of a board committee is often based on its function, with audit committees, compensation committees, and governance committees often made up of only nonexecutive or independent directors.

Employee, Customer, and Supplier Mechanisms

Labor laws, employment contracts, and the right to form unions are the primary mechanisms for employees to manage relationships with employers. Some countries have laws that require boards of large companies to include employee representatives. **Employee stock ownership plans (ESOPs)** may help align company and employee interests. For customers and suppliers, contracts tend to be the mechanism through which they manage their relationships with companies. In recent years, customers and other stakeholders have increasingly used social media as a mechanism to influence company behavior.

Government Mechanisms

Governments enact and enforce regulations that govern companies' actions. They often do so by establishing agencies to regulate industries or sectors such as financial markets, or by monitoring specific issues such as workplace safety and environmental protection. In some countries, regulators develop corporate governance codes that companies must either adopt or explain why they have not done so. In some countries, corporate governance regulations are specified by stock exchanges as part of their listing requirements.

LOS 22.c: Describe potential risks of poor corporate governance and stakeholder management and benefits of effective corporate governance and stakeholder management.

Risks of Poor Governance and Stakeholder Management

When corporate governance is weak, the control functions of audits and board oversight may be weak as well. The risk is that some stakeholders can gain an advantage, to the disadvantage of other stakeholders. Accounting fraud, or simply poor recordkeeping, will have negative implications for company performance and value.

When governance is weak and managers are not monitored, they may serve their own interests by choosing less-than-optimal risk, reducing company value. Without proper monitoring and oversight, management may be given incentive compensation that allows them to pursue their own benefit rather than the company's interests. They may engage in related-party transactions that benefit their friends or family, to the detriment of shareholders.

Poor compliance procedures with respect to regulation and reporting can easily lead to legal and reputational risks. Violating stakeholder rights can lead to stakeholder lawsuits. Failure to comply with government regulations can damage company reputation. Failure to manage creditors' rights well can lead to debt default and bankruptcy.

Benefits of Effective Governance and Stakeholder Management

Effective corporate governance can improve operational efficiency by ensuring that management and board member incentives align their interests well with those of the shareholders. Effective governance implies effective control and monitoring. Just as weak control can lead to abuses, a strong system of controls and compliance with laws and regulations can avoid many legal and regulatory risks.

Formal policies regarding conflicts of interest and related-party transactions can also lead to better operating results. Proper governance with respect to the interests of creditors can reduce the risk of debt default or bankruptcy, thereby reducing the cost of debt financing. Alignment of management interests with those of shareholders leads to better financial performance and greater company value.



MODULE QUIZ 22.1

1. Between which two of a company's stakeholder groups does information asymmetry *most likely* make monitoring more difficult?
 - A. Suppliers and customers.
 - B. Employees and government.
 - C. Managers and shareholders.
2. The theory that deals with conflicts of interest between a company's owners and its creditors is *most appropriately* called:
 - A. governance theory.
 - B. stakeholder theory.
 - C. shareholder theory.

3. Benefits of effective corporate governance and stakeholder management *most likely* include:
- A. reduced risk of default.
 - B. more efficient related-party transactions.
 - C. greater control exercised by the most interested stakeholders.

KEY CONCEPTS

LOS 22.a

The principal-agent relationship refers to owners employing agents to act in their interests. Conflicts can arise because the agent's incentives may not align with those of the owner or, more generally, because the interests of one group within a corporation are not the same as those of other groups.

LOS 22.b

Corporate governance refers to the internal controls and procedures of a company that delineate the rights and responsibilities of various groups, and how conflicts of interest among the various groups are to be resolved.

Shareholders, creditors, boards of directors, employees, customers, suppliers, and governments have different mechanisms with which to manage their stakeholder relationships with companies.

Proxy voting is the primary shareholder mechanism. Shareholders can remove senior managers and boards of directors if they believe company performance would improve with a change. Activist shareholders may engage in proxy contests or hostile takeovers.

Creditor mechanisms include bond indentures and creditor committees. Employee mechanisms include labor laws and unions. Contracts are the primary mechanism for customers and suppliers. Governments may enact regulations or appoint regulatory agencies.

Duties of a board of directors include the following:

- | Selecting senior management, setting their compensation, and evaluating their performance
- | Setting the strategic direction for the company
- | Approving capital structure changes, significant acquisitions, and large investment expenditures

- | Reviewing company performance and implementing any necessary corrective steps
- | Planning for continuity of management and the succession of the CEO
- | Establishing, monitoring, and overseeing the firm's internal controls and risk management
- | Ensuring the quality of the firm's financial reporting and internal audit

LOS 22.c

The risks of poor governance include weak control systems, poor decision-making, legal risk, reputational risk, and default risk. Good corporate governance can improve operational efficiency and performance, reduce default risk, reduce the cost of debt, improve financial performance, and increase firm value.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 22.1

- . **C** Information asymmetry can exist between a company's shareholders and its managers because the company's managers may be much more knowledgeable about the company's functioning and strategic direction. This makes it more difficult for shareholders to monitor the firm's managers and determine whether they are acting in shareholders' interests. (LOS 22.a)
- . **B** Stakeholder theory focuses on the conflicts of interest among owners and several groups that have an interest in a company's activities, including creditors. (LOS 22.b)
- . **A** Reduced risk of default is among the benefits of effective corporate governance. Risks from poor corporate governance include related-party transactions by managers and opportunities for some stakeholder groups to gain an advantage at the expense of others. (LOS 22.c)

READING 23

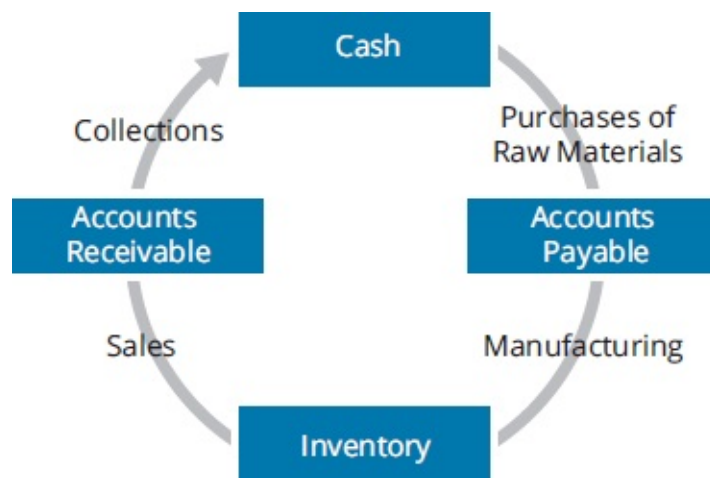
WORKING CAPITAL AND LIQUIDITY

MODULE 23.1: LIQUIDITY MEASURES AND MANAGEMENT

LOS 23.a: Explain the cash conversion cycle and compare issuers' cash conversion cycles.

The **cash conversion cycle (CCC)** measures the efficiency of a company's cash flow management. It represents the time it takes for a company to convert its investments in inventory and other resources into cash inflows from sales. In other words, it measures how quickly a company can convert its investments into cash and then use that cash to invest in new opportunities, as shown in Figure 23.1.

Figure 23.1: Cash Conversion Cycle



The CCC is calculated by adding the number of days it takes for a company to sell its inventory (**days of inventory on hand**), the number of days it takes for the company to collect payment from its customers (**days sales outstanding**),

and the number of days it takes for the company to pay its suppliers (**days payable outstanding**).

$$\text{cash conversion cycle} = \text{days of inventory on hand} + \text{days sales outstanding} \\ - \text{days payables outstanding}$$



PROFESSOR'S NOTE

The calculation of DOH, DSO, and DPO is covered in our Financial Statement Analysis reading on Financial Analysis Techniques.

A lower CCC is generally better as it indicates that a company can generate cash quickly and efficiently, meaning that less of the company's capital is devoted to working capital. A high CCC may indicate that a company is taking longer to convert its investments into cash, which could lead to cash flow problems and potentially limit the company's ability to invest in growth opportunities.

A company can decrease its CCC by decreasing its inventories and receivables or by increasing (extending) its payables. However, each of these has potential disadvantages:

- Reduced inventories of raw materials may create production bottlenecks due to supply chain disruptions. Reduced inventories of finished goods may mean an inability to meet spikes in customer demand.
- Reducing or tightening credit to customers may result in lost sales.
- We can think of accounts payable as an implicit source of credit from suppliers (as opposed to explicit sources such as bank loans). Suppliers offer payment terms in the form *a/b net c*, which means a percentage discount of *a* if the invoice is paid within *b* days, otherwise full payment is due within *c* days. Forgoing the discount for prompt payment amounts to borrowing money from the supplier for (*c* – *b*) days, at an effective annual rate we can calculate as follows:

$$\text{EAR of supplier financing} = \left(1 + \frac{a}{100 - a}\right)^{\left(\frac{365}{c-b}\right)} - 1$$

where:

a = percent discount

b = days until discount expires

c = days until full payment is due

Often, the company could find a lower EAR by borrowing from their lenders as opposed to forgoing the discount.

EXAMPLE: EAR of supplier financing

A supplier offers 2/10 net 30 terms. The bank interest rate is 8%. Which source of financing should the company prefer?

Answer:

Financing is for $30 - 10 = 20$ days.

EAR of supplier financing = $(1 + 0.02 / 0.98)^{(365/20)} - 1 = 0.446$, or 44.6%.

The cost of implicit supplier financing is much higher than the cost of explicit bank financing. The company should borrow from its bank at 8% to pay the invoice within 10 days.

Cash conversion cycles vary by industry. For example, pharmaceutical companies have long CCCs because they maintain inventories of high-margin drugs to meet unexpected surges in demand. By contrast, airlines have low CCCs because most of their sales are prepaid and they do not keep significant inventories. Analysts should use the CCC to compare companies within the same industry, or to track a company's performance over time.

In addition to the CCC, overall levels of working capital can indicate how efficiently a company manages its liquidity. To compare companies of different sizes, analysts calculate working capital as a proportion of sales. Because the ratio varies by industry, care should be taken to compare firms in similar lines of business. **Total working capital** is current assets less current liabilities. Analysts often prefer to measure **net working capital**, using only operating current assets and operating current liabilities, because it is closely linked to the CCC:

total working capital = current assets – current liabilities

net working capital = current assets (except cash and marketable securities)
– current liabilities (excluding short-term and current debt)

LOS 23.b: Explain liquidity and compare issuers' liquidity levels.

For an asset, **liquidity** refers to its nearness to cash. For a liability, liquidity refers to its nearness to settlement. Assets that can be quickly converted to cash (e.g., marketable securities) are considered highly liquid.

Inventory is less liquid than accounts receivable. Inventory may need processing before a sale, and once sold, inventory might be converted to accounts receivable, which needs to be collected to convert into cash.

For a corporate issuer, liquidity refers to availability of cash and other liquid assets to meet its short-term obligations. **Primary liquidity sources** include cash and marketable securities on hand, bank borrowings, and cash generated from the business. Long-term solvency of a company depends on its ability to generate sufficient cash from its business to service its liabilities. Analysts evaluate a company's liquidity management primarily by reviewing its statement of cash flows.

Companies generally rely on primary sources of liquidity. However, if needed, the company can rely on **secondary liquidity sources**, such as the following:

- | Cash saved by suspending dividends to shareholders
- | Delaying or reducing capital investments
- | Selling assets
- | Issuing additional equity
- | Restructuring debt to extend its maturity
- | Bankruptcy protection filing, which suspends the need to service the liabilities

Using secondary liquidity sources sends negative signals to the market, and they are generally more costly than primary liquidity sources.

A company's cash conversion cycle may vary by time of the year (i.e., seasonality) and as business conditions change (e.g., a sudden dip in demand for a company's product will increase inventory and increase DOH). Cash and marketable securities serve as a buffer to meet the company's obligations when there is a deviation from the normal CCC. This excess cash has a cost because it represents capital not invested in the business, but inadequate liquidity can result in having to rely on higher-cost secondary sources.

EXAMPLE: Cost of liquidity

Drake Corporation has a sudden need for liquidity to meet its payroll at the end of the week. The company's sources of liquidity and their costs are as shown here.

Source	\$ Fair Market Value (\$000s)	Liquidation Cost (%)
Cash and marketable securities	100	0
Inventory and receivables	200	15
Empty warehouse	<u>300</u>	30
Total	<u>600</u>	

Calculate the total proceeds from all sources, and the cost of liquidity in dollars and in percentage.

Answer:

The cost of liquidity is the discount to fair market value for which an asset is sold due to the need for immediate liquidity, rather than waiting for the normal time it takes to sell the asset.

Source	Fair Market Value (\$000s)	Liquidation Cost		Net Proceeds
Cash and marketable securities	\$100	0%	\$0	\$100
Inventory and receivables	\$200	15%	\$30	\$170
Empty warehouse	<u>\$300</u>	30%	<u>\$90</u>	<u>\$210</u>
Total	<u>\$600</u>		<u>\$120</u>	<u>\$480</u>

The cost of liquidity = \$120,000, or $120 / 600 = 20\%$.

Factors Affecting Liquidity

An increase in the CCC reduces an issuer's liquidity. The CCC can increase due to drags or pulls on liquidity. A **drag on liquidity** occurs when cash inflows lag. This can occur when excess inventory builds up or inventory becomes obsolete (DOH increases), or when collections are slow or receivables become uncollectible (DSO increases). A **pull on liquidity** occurs when cash outflows accelerate. This can occur when suppliers reduce credit lines or demand faster payments (DPO decreases).

Liquidity Ratios

The **current ratio** is the ratio of current assets to current liabilities. A current ratio greater than 1 indicates that the company has sufficient current assets to meet its current liabilities.

$$\text{current ratio} = \frac{\text{current assets}}{\text{current liabilities}}$$

Not all current assets are equally liquid. Because inventory is the least liquid current asset, the **quick ratio** excludes inventory from current assets and is a more stringent metric of liquidity.

$$\text{quick ratio} = \frac{(\text{cash and marketable securities} + \text{accounts receivable})}{\text{current liabilities}}$$

Finally, the **cash ratio** excludes both inventory and accounts receivables and is the most stringent measure of liquidity.

$$\text{cash ratio} = \frac{\text{cash and marketable securities}}{\text{current liabilities}}$$

EXAMPLE: Liquidity ratios

The following information is available for Drake Corporation for the past two years:

	20X1	20X2
Current Assets	(\$000s)	(\$000s)
Cash and marketable securities	\$ 98	\$ 54
Accounts receivable	\$ 120	\$ 110
Inventory	\$ 190	\$ 310
Prepaid accounts	\$ 24	\$ 18
Current Liabilities		
Accounts payable	\$ 85	\$ 100
Accrued expenses	\$ 34	\$ 39
Short-term notes payable	\$ 98	\$ 145

Calculate Drake's current ratio, quick ratio, and cash ratio for each of the two years and comment on the trend in the ratios.

Answer:

Current ratio = current assets / current liabilities

Current ratio (20X1) = $(98 + 120 + 190 + 24) / (85 + 34 + 98) = 1.99$

Current ratio (20X2) = $(54 + 110 + 310 + 18) / (100 + 39 + 145) = 1.73$

Quick ratio = cash and marketable securities + accounts receivable /
current liabilities

Quick ratio (20X1) = $(98 + 120) / (85 + 34 + 98) = 1.00$

Quick ratio (20X2) = $(54 + 110) / (100 + 39 + 145) = 0.58$

Cash ratio = cash and marketable securities / current liabilities

Cash ratio (20X1) = $98 / (85 + 34 + 98) = 0.45$

Cash ratio (20X2) = $54 / (100 + 39 + 145) = 0.19$

All three ratios are declining year-over-year from 20X1 to 20X2, indicating that Drake's liquidity is worsening.

LOS 23.c: Describe issuers' objectives and compare methods for managing working capital and liquidity.

Working capital management seeks to maximize firm profits while ensuring that sufficient liquidity is available to maintain the firm's operations and meet its obligations. Firms may choose to hold more short-term assets (which offer lower rates of return) to ensure that sufficient cash is available to service its obligations. The firm can also choose to finance its working capital using short-term loans, which are cheaper than sources such as long-term debt and equity. However, the firm must consider the risk of being able to roll over short-term debt at a reasonable cost.

In analyzing different approaches to working capital management, it is important to control for size differences between companies. We do this by calculating companies' relative amounts of long-term versus short-term assets as a proportion of sales.

A conservative approach to working capital management is for the company to hold higher amounts of short-term assets (relative to long-term assets) and finance the working capital using longer-term sources, such as long-term debt and equity. Benefits of a conservative approach include using more permanent capital with less need for rolling over; greater flexibility during market

disruptions; and a high probability of meeting short-term obligations. However, the conservative approach results in higher costs and lower profitability. Additionally, long-term lenders may impose operational constraints such as a minimum interest coverage ratio.

An aggressive approach would be to hold relatively lower levels of short-term assets and finance working capital using short-term debt. The benefit of an aggressive approach is lower costs, but its risks are failing to meet business obligations and vulnerability to market disruptions.

A moderate approach seeks to find a middle ground. Permanent current assets are funded using long-term sources of capital, while variable (seasonal) current assets are funded using short-term sources.

Short-Term Liquidity Sources

Firms should maintain a variety of alternative sources and evaluate the costs of each source, while securing sources ahead of time to meet spikes in liquidity needs.

Factors that affect a firm's approach to short-term funding include the following:

- | Company size (smaller firms have limited options)
- | Creditworthiness (which affects the interest rate on loans as well as operational restrictions imposed)
- | Legal systems (developed economies with well-defined protections for lenders offer more funding alternatives)
- | Regulatory concerns (firms in regulated industries, such as banks and utilities, have restrictions on funding sources as well as amounts raised)
- | Underlying assets (that serve as collateral on loans)



MODULE QUIZ 23.1

1. Compared to its industry peers, a company with a shorter cash conversion cycle *most likely*:
 - A. has more inventory.
 - B. has less accounts receivable.
 - C. pays its suppliers more promptly.
2. A company receives an invoice of \$150,000 for machine tools with terms of "1.5/15 net 40." The cost to the company of delaying payment of this invoice is *most* appropriately described as \$2,250 for the use of:
 - A. \$150,000 for 40 days.

- B. \$150,000 for 25 days.
 - C. \$147,750 for 25 days.
3. Which of the following actions is *most likely* to increase liquidity for a corporation?
- A. Selling inventory at a discount of 5%.
 - B. Availing of a discount of 10% by paying accounts payable early.
 - C. Extending customers' credit terms from 90 days to 120 days.
4. Which of the following is *least likely* a primary source of liquidity?
- A. Borrowings from banks.
 - B. Cash flow from operations.
 - C. Delaying capital expenditures.
5. Which of the following *most likely* represents conservative working capital management?
- A. Decreasing inventory on hand to reduce insurance costs.
 - B. Financing an increase in receivables by increasing long-term borrowing.
 - C. Selling marketable securities and using the proceeds to acquire real estate.

KEY CONCEPTS

LOS 23.a

The cash conversion cycle (CCC) represents the time it takes for a company to convert its investments in inventory and other resources into cash inflows from sales. The CCC increases with an increase in days of inventory on hand (DOH), days sales outstanding (DSO), and decreases with an increase in days payable outstanding (DPO).

$$\text{CCC} = \text{DOH} + \text{DSO} - \text{DPO}$$

$$\text{Total working capital} = \text{current assets} - \text{current liabilities}$$

$$\begin{aligned} \text{Net working capital} &= \text{current assets (except cash and marketable securities)} \\ &\quad - \text{current liabilities (excluding short-term and current debt)} \end{aligned}$$

LOS 23.b

Primary liquidity sources include cash and marketable securities on hand, bank borrowings, and cash generated from the business. Secondary sources of liquidity include forgoing dividends, delaying capital investments, selling assets, issuing equity, restructuring debt, and bankruptcy protection filing. Using secondary liquidity sources sends a negative signal to the market and incurs a cost of liquidity.

A drag on liquidity occurs when cash inflows lag (DOH or DSO increases), while a pull on liquidity occurs when cash outflows accelerate (DPO decreases).

Measures of a company's short-term liquidity include:

- | Current ratio = current assets / current liabilities.
- | Quick ratio = (cash + marketable securities + receivables) / current liabilities.
- | Cash ratio = (cash + marketable securities) / current liabilities.

LOS 23.c

A conservative approach to working capital management involves high levels of current assets financed with long-term debt and equity. Compared to a more aggressive approach, a conservative approach provides more liquidity and involves less financial risk, but has higher financing costs and reduces returns.

An aggressive approach to working capital management involves lower levels of current assets and financing working capital needs with short-term debt. Compared to a more conservative approach, an aggressive approach typically has lower financing costs and results in higher returns, but decreases liquidity and increases financial risk.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 23.1

- . **B** $CCC = DOH + DSO - DPO$. A company with a shorter CCC would have lower DOH (lower amount of cash tied in inventory), lower DSO (lower amount of cash tied in accounts receivables), or higher DPO (increased use of supplier credit). (LOS 23.a)
- . **C** The terms indicate that the company can pay $\$150,000(1 - 0.015) = \$147,750$ on day 15 (after the invoice date) or pay $\$150,000$ on day 40—effectively gaining the use of $\$147,750$ for 25 days at a cost of $\$2,250$. (LOS 23.a)
- . **A** Liquidity can be increased by reducing inventory. Extending repayment times would increase DSO, reducing liquidity. Paying suppliers earlier decreases DPO, thus reducing liquidity. (LOS 23.b)
- . **C** Primary sources of liquidity include cash and marketable securities on hand, bank borrowings, and cash generated from the business. Delaying capital expenditure is a secondary source of liquidity. (LOS 23.b)

- . **B** Financing an increase in a current asset with long-term borrowing is an example of conservative working capital management. The other choices describe decreases in current assets and therefore more likely represent aggressive working capital management. (LOS 23.c)

READING 24

CAPITAL INVESTMENTS AND CAPITAL ALLOCATION

MODULE 24.1: CAPITAL INVESTMENTS AND PROJECT MEASURES

LOS 24.a: Describe types of capital investments.

The four types of capital investments include going concern projects, regulatory/compliance projects, expansion projects, and other projects.

- | **Going concern projects** may be needed to maintain the business or reduce costs. Projects that maintain the business do not require detailed analysis. The only issues are whether the existing operations should continue and, if so, whether existing procedures or processes should be maintained. Projects to improve efficiency may involve determining if equipment that is obsolete, but still usable, should be replaced. To reduce financing risk, companies often use a **match funding** approach, financing projects with capital sources that are consistent with the project life. Analysts often use a company's annual depreciation expense as an estimate of the capital investment it needs for going concern projects.
- | **Regulatory/compliance projects** may be required by a government agency or insurance company and often involve safety-related or environmental concerns. These projects typically generate little to no revenue and require the company to evaluate alternative ways of carrying out the projects.
- | **Expansion projects** grow the business and require a complex decision-making process that includes forecasting future demand. Expansion projects can involve entering new markets or introducing new products within the same market. A detailed analysis including forecasting revenues and expenses is required.

- Other projects, such as new investments outside a company's existing lines of business, also entail a complex decision-making process with detailed analysis due to the uncertainty involved. Other projects are often similar to startups that explore a new technology or business idea. These projects can also involve buying out an existing company in a new industry, which involve risks that include overpaying.

LOS 24.b: Describe the capital allocation process, calculate net present value (NPV), internal rate of return (IRR), and return on invested capital (ROIC), and contrast their use in capital allocation.

The **capital allocation process** is identifying and evaluating capital projects (i.e., projects where the cash flows to the firm will be received over a period longer than a year). Any corporate decisions with an impact on future earnings can be examined using this framework. Decisions about buying a new machine, expanding business into another geographic area, moving the corporate headquarters to Cleveland, or replacing a delivery truck, to name a few, can be examined using a capital allocation analysis.

For a number of good reasons, capital allocation may be the most important responsibility that a financial manager has. First, because a capital allocation decision often involves the purchase of costly long-term assets with lives of many years, the decisions made may determine the future success of the firm. Second, the principles underlying the capital allocation process also apply to other corporate decisions, such as working capital management and making strategic mergers and acquisitions. Finally, making good capital allocation decisions is consistent with management's primary goal of maximizing shareholder value.

The capital allocation process has four administrative steps:

- Step 1: Idea generation.* The most important step in the capital allocation process is generating good project ideas. Ideas can come from a number of sources, including senior management, functional divisions, employees, or sources outside the company.
- Step 2: Analyzing project proposals.* Because the decision to accept or reject a capital project is based on the project's expected future cash flows, a cash flow forecast must be made for each project to determine its expected

profitability.

Step 3: Create the firm-wide capital budget. Firms must prioritize profitable projects according to the timing of the project's cash flows, available company resources, and the company's overall strategic plan. Many projects that are attractive individually may not make sense strategically.

Step 4: Monitoring decisions and conducting a post-audit. It is important to follow up on all capital allocation decisions. An analyst should compare the actual results to the projected results, and project managers should explain why projections did or did not match actual performance. Because the capital allocation process is only as good as the estimates of the inputs into the model used to forecast cash flows, a post-audit should be used to identify systematic errors in the forecasting process and improve company operations.

Net Present Value (NPV)

Net present value (NPV) is the sum of the present values of all the expected incremental cash flows if a project is undertaken. The discount rate used is the firm's cost of capital, adjusted for the risk level of the project. For a normal project, with an initial cash outflow followed by a series of expected after-tax cash inflows, the NPV is the present value of the expected inflows minus the initial cost of the project.

$$NPV = CF_0 + \frac{CF_1}{(1+k)^1} + \frac{CF_2}{(1+k)^2} + \dots + \frac{CF_n}{(1+k)^n} = \sum_{t=0}^n \frac{CF_t}{(1+k)^t}$$

where:

CF_0 = initial investment outlay (a negative cash flow)

CF_t = after-tax cash flow at time t (can be positive or negative)

k = required rate of return for project

A positive NPV project is expected to increase shareholder wealth, a negative NPV project is expected to decrease shareholder wealth, and a zero NPV project has no expected effect on shareholder wealth.

For *independent* projects, the *NPV decision rule* is simply to accept any project with a positive NPV and to reject any project with a negative NPV.

EXAMPLE: Net present value

Using the project cash flows presented in Figure 24.1, calculate the NPV of the project and determine whether it should be accepted or rejected. Assume that the cost of capital is 9%.

Figure 24.1: Expected Net After-Tax Cash Flows

Year	Cash Flow
0	-\$100
1	25
2	50
3	75

Answer:

$$\text{NPV} = -100 + \frac{25}{1.09} + \frac{50}{(1.09)^2} + \frac{75}{(1.09)^3} = 22.93$$

The project has a positive NPV, so it should be accepted.

You may calculate NPV directly by using the cash flow (CF) keys on your calculator. The process is illustrated in Figure 24.2.

Figure 24.2: Calculating NPV With the TI BA II Plus™

Keystrokes	Explanation	Display
[CF] [2nd] [CLR WORK]	Clear memory registers	CF0 = 0.0000
100 [+/-] [ENTER]	Initial cash outlay	CF0 = -100.0000
[↓] 25 [ENTER]	Period 1 cash flow	C01 = 25.0000
[↓]	Frequency of cash flow 1	F01 = 1.0000
[↓] 50 [ENTER]	Period 2 cash flow	C02 = 50.0000
[↓]	Frequency of cash flow 2	F02 = 1.0000
[↓] 75 [ENTER]	Period 3 cash flow	C03 = 75.0000
[↓]	Frequency of cash flow 3	F03 = 1.0000
[NPV] 9 [ENTER]	9% discount rate	I = 9.0000
[↓] [CPT]	Calculate NPV	NPV = 22.9335

Internal Rate of Return (IRR)

For a normal project, the **internal rate of return (IRR)** is the discount rate that makes the present value of the expected incremental after-tax cash inflows just equal to the initial cost of the project. More generally, the IRR is the discount rate that makes the present value of a project's estimated cash inflows equal to the present value of the project's estimated cash outflows. That is, IRR is the discount rate that makes the following relationship hold:

$$\text{PV inflows} = \text{PV outflows}$$

The IRR is also the discount rate for which the NPV of a project is equal to zero:

$$\text{NPV} = 0 = \text{CF}_0 + \frac{\text{CF}_1}{(1+\text{IRR})^1} + \frac{\text{CF}_2}{(1+\text{IRR})^2} + \dots + \frac{\text{CF}_n}{(1+\text{IRR})^n} = \sum_{t=0}^n \frac{\text{CF}_t}{(1+\text{IRR})^t}$$

IRR decision rule: If IRR is greater than the required rate of return, accept the project. If IRR is less than the required rate of return, reject the project.

The required rate of return for a given project is usually the firm's cost of capital. However, a project analyst may adjust the required rate of return above or below the firm's cost of capital to account for differences between the project's risk and the average risk of all of the firm's projects (which is reflected in the firm's current cost of capital).

For this reason, the minimum IRR above which a project will be accepted is often referred to as the **hurdle rate**. Projects with IRRs above this rate will be accepted, while those with IRRs below this rate will not be accepted.

EXAMPLE: Internal rate of return

Continuing with the cash flows presented in Figure 24.1 for the previous example, calculate the IRR of the project and determine whether it should be accepted or rejected. Assume that the required rate of return is 9%.

Answer:

$$0 = -100 + \frac{25}{(1+\text{IRR})} + \frac{50}{(1+\text{IRR})^2} + \frac{75}{(1+\text{IRR})^3}$$

The cash flows should be entered as in Figure 24.2 (if you haven't changed or cleared them, they are still there from the previous calculation of NPV).

With the TI calculator, the IRR can be calculated with [IRR] [CPT] to get 19.4377%.

The project should be accepted because its IRR is greater than the 9% required rate of return.

A project has a **conventional cash flow pattern** if the sign on the cash flows changes only once, with one or more cash outflows followed by one or more cash inflows. An **unconventional cash flow pattern** has more than one sign change. For example, a project might have an initial investment outflow, a series of cash inflows, and a cash outflow for asset retirement costs at the end of the project's life. Another example of unconventional cash flow is when the time intervals between cash flows differ, for example if the first cash flow occurs 1 year after project inception while the second cash flow occurs 1.5 years after the first. Spreadsheet software is most suitable for calculating NPV and IRR with unconventional cash flows.

Relative Advantages and Disadvantages of the NPV and IRR Methods

A key advantage of NPV is that it is a direct measure of the expected increase in the value of the firm. In theory, a positive NPV project should cause a proportionate increase in a company's stock price.

A key advantage of IRR is that it measures profitability as a percentage, showing the return on each dollar invested. The IRR provides information on the margin of safety that the NPV does not. From the IRR, we can tell how much below the IRR (estimated return) the actual project return could fall, in percentage terms, before the project becomes uneconomic (has a negative NPV).

The *disadvantages* of the IRR method are:

- | It assumes that project's cash flows are reinvested at the IRR while NPV assumes that those cash flows are reinvested at the project's required rate of return. It is more realistic to assume the latter.
- | For multiple sign changes, a project may have multiple IRRs that are difficult to interpret.

Return on Invested Capital

One way to examine whether a company is creating value for its shareholders is to compare the return on the company's investment in assets to its cost of capital. A company's **return on invested capital (ROIC)**, or simply **return on capital**, is defined as its **net operating profit after tax (NOPAT)**, or simply after-tax

operating profit, divided by the average book value of its total capital over the period:

$$\text{return on invested capital} = \frac{\text{net operating profit after tax}}{\text{average book value of invested capital}}$$

After-tax operating profit is net income plus after-tax interest expense. We use this because we want to measure the return to all sources of capital (both debt and equity). Invested capital includes long-term debt and equity, but excludes working capital.

We can rewrite the ROIC equation as:

$$\text{ROIC} = \left(\frac{\text{NOPAT}}{\text{Average Invested Capital}} \right) = \left(\frac{\text{NOPAT}}{\text{Sales}} \right) \times \left(\frac{\text{Sales}}{\text{Average Invested Capital}} \right)$$

We refer to the ratio of after-tax operating profit to sales as *operating margin after tax*, and we refer to the ratio of sales to invested capital as *capital turnover* or *asset turnover*. This means that ROIC is the product of operating margin after-tax and invested capital turnover. A company can increase its ROIC by improving its operating margin or by increasing its capital turnover.

Analysts compare ROIC to investors' required rate of return. In this context, the required rate of return is a blended rate for both debt and equity. If ROIC is greater than the required rate, the firm is adding value over time.

One attraction of ROIC is that it is based on accounting data, which are available to outside investors. While NPV and IRR are project-specific, ROIC is for the firm as a whole. ROIC is more relevant for outside investors because they typically cannot invest in a firm's individual projects.

There are three concerns with using ROIC:

Because accounting treatments differ, ROIC may not be comparable across companies.

ROIC is backward-looking and can be volatile from year to year.

Because ROIC is for the whole company, it may let profitable projects and good decisions mask unprofitable projects and poor decisions.



MODULE QUIZ 24.1

1. Which of the following is *most likely* a going concern project?
 - A. Opening a retail outlet in a new region.
 - B. Acquiring and merging with a supplier to secure a source for a key component.
 - C. Purchasing a new model of a factory machine that will decrease unit production costs.

2. In the capital allocation process, a post-audit is used to:
 - A. improve cash flow forecasts and stimulate management to improve operations and bring results into line with forecasts.
 - B. improve cash flow forecasts and eliminate potentially profitable but risky projects.
 - C. stimulate management to improve operations, bring results into line with forecasts, and eliminate potentially profitable but risky projects.
3. A company is considering the purchase of a copier that costs \$5,000. Assume a required rate of return of 10% and the following cash flow schedule:
 - Year 1: \$3,000.
 - Year 2: \$2,000.
 - Year 3: \$2,000.

The project's NPV is *closest* to:

- A. -\$309.
- B. +\$883.
- C. +\$1,523.

MODULE 24.2: CAPITAL ALLOCATION PRINCIPLES AND REAL OPTIONS

LOS 24.c: Describe principles of capital allocation and common capital allocation pitfalls.

Principles of Capital Allocation

The capital allocation process involves the following key principles:

- | *Decisions are based on after-tax cash flows, not accounting income.*
Accounting income is based on accruals and does not consider the timing of cash flows. The impact of taxes must be considered when analyzing all capital allocation projects. Firm value is based on cash flows firms get to keep, not those they send to the government. Any tax savings from non-cash tax deductions, such as depreciation and amortization, should also be included in the analysis.
- | *Incremental cash flows only.* Incremental cash flows are those that change if the project is undertaken. **Sunk costs** are costs that cannot be avoided even if the project is not undertaken. Because these costs are not affected by the accept/reject decision, they should not be included in the analysis. An example of a sunk cost is a consulting fee paid to a marketing research firm to estimate demand for a new product before making a decision on the project. Project

analysis should consider the cash flow impact on other parts of the business. A negative effect, called cannibalization, occurs when a new project replaces sales from an existing product. An example of cannibalization is when a soft drink company introduces a diet version of an existing beverage. A positive externality exists when doing the project would have a positive effect on sales of a firm's other product lines.

- | *The timing of cash flows is important.* Capital allocation decisions account for the time value of money, which means that cash flows received earlier are worth more than cash flows to be received later.

Common mistakes managers make when evaluating capital projects have two categories: cognitive errors (calculation errors) and behavioral biases (errors of judgment).

Cognitive Errors

- | *Poor forecasting.* Examples include incorrectly allocating overhead costs or neglecting to anticipate how competitors will respond to a project.
- | *Not considering the cost of internal funds.* Often, the cost of internally generated funds is not accounted for (it should be same as cost of equity because those funds would have been paid as dividends if not for the project). Companies that have an aversion to paying dividends are potentially using the retained earnings to fund poor projects.
- | *Incorrectly accounting for inflation.* Firms can analyze projects either in nominal or real terms. An analysis based on real cash flows must also use a real discount rate.

Behavioral Biases

- | *Pet projects of senior management.* Projects that have the personal backing of influential members of senior management may contain overly optimistic projections that make the project appear more profitable than it really is. In addition, the project might not be subjected to the same level of scrutiny as other projects.
- | *Inertia in setting the entire capital budget.* Capital budgets for many companies are similar from one year to the next. This indicates anchoring of capital budgets to the prior year rather than appropriately considering what opportunities exist every year. In a company with a culture of maximizing

shareholder value, managers will return excess funds to shareholders whenever there is a lack of positive NPV projects and make a case for expanding the budget when there are multiple positive NPV opportunities. Analysts should watch for companies with static or rising capital budgets coupled with declining returns as a sign of this bias.

- | *Basing investment decisions on EPS or ROE.* Managers whose incentive compensation is tied to increasing EPS or ROE may avoid positive long-term NPV investments that are expected to reduce EPS or ROE in the short run.
- | *Failure to generate alternative investment ideas.* Generating investment ideas is a crucial step in the capital allocation process. However, once a manager comes up with a “good” idea, he or she may go with it rather than searching for an idea that is “better.”

LOS 24.d: Describe types of real options relevant to capital investments.

Real options are future actions that a firm can take, given that they invest in a project today. Real options are similar to financial options (put and call options) in that they give the option holder the right, but not the obligation, to take a future action. The value of real options could enhance a project’s NPV. Options never have negative values because if, in the future, the specified action will have a negative value, the option holder will not take the action (i.e., not exercise the option).

Types of real options include the following:

- | **Timing options** allow a company to delay making an investment because it expects to have better information in the future.
- | **Abandonment options** allow management to abandon a project if the present value of the incremental cash flows from exiting a project exceeds the present value of the incremental cash flows from continuing the project.
- | **Expansion options** or **growth options** allow a company to make additional investments in future projects if the company decides they will create value.
- | **Flexibility options** give managers choices regarding the operational aspects of a project. The two main forms are price-setting and production-flexibility options.
 - **Price-setting options** allow the company to change the price of a product. For example, the company may raise prices if demand for a product is high in

order to benefit from that demand without increasing production.

- **Production-flexibility options** may include paying workers overtime, using different materials as inputs, or producing a different variety of product.
- **Fundamental options** are projects that are options themselves because the payoffs depend on the price of an underlying asset. For example, the payoff for a copper mine is dependent on the market price for copper. If copper prices are low, it may not make sense to open a copper mine, but if copper prices are high, opening the copper mine could be very profitable. The operator has the option to close the mine when prices are low and open it when prices are high.

One way companies can include real options in their project analysis is to estimate their value and add it to the NPV (while subtracting any extra cost to acquire the real option). This can incorporate available techniques such as option pricing models or decision trees. Another approach is to simply consider the NPV without real options to be the project's minimum value.



MODULE QUIZ 24.2

1. Which of the following statements concerning the principles underlying the capital allocation process is *most accurate*?
 - A. Cash flows should include tax benefits of non-cash expense deductions.
 - B. The net income for a project is essential for making a correct capital allocation decision.
 - C. Cash flows should be project specific, ignoring sunk costs and impacts on other parts of the business.
2. A manufacturer of clothes washing machines decides to add matching clothes dryers to its product line. In this case, it is *most likely* important in the project analysis to consider:
 - A. sunk costs.
 - B. negative impact on other parts of the business.
 - C. positive impact on other parts of the business.
3. An analyst is estimating the NPV of a project to introduce a new spicier version of its well-known barbeque sauce into its product line. A cost that should *most likely* be excluded from his analysis is:
 - A. \$200,000 to develop a recipe for the new sauce.
 - B. \$100,000 for a marketing survey that was conducted to determine demand for a spicier sauce.
 - C. a \$150,000 decrease in sales of its current sauce as some current customers switch to the spicier sauce.
4. Albert Duffy, a project manager at Crane Plastics, is considering taking on a new capital project. When presenting the project, Duffy shows members of Crane's executive management team that, because the company has the ability to have employees work overtime, the project makes sense. The project Duffy is taking on would be *best*

described as having a(n):

- A. flexibility option.
- B. expansion option.
- C. fundamental option.

KEY CONCEPTS

LOS 24.a

Capital investments include going concern projects to maintain a business or to reduce costs, required regulatory/compliance projects, expansion projects, and other projects that increase the size and scope of a company.

LOS 24.b

Capital allocation is the process of evaluating capital projects (i.e., projects with cash flows over a period longer than one year).

Steps of the capital allocation process are: (1) generate investment ideas; (2) analyze project ideas; (3) create a firm-wide capital budget; and (4) monitor decisions and conduct a post-audit.

NPV is the sum of the present values of a project's expected cash flows and represents the change in firm value from undertaking a project. Positive NPV projects should be undertaken, but negative NPV projects should not because they are expected to decrease the value of the firm.

An IRR is the discount rate at which the present values of a project's expected cash inflows and cash outflows are equal (i.e., the discount rate for which the NPV of a project is zero). A project for which the IRR is greater (less) than the appropriate discount rate for the project will have an NPV that is positive (negative) and should be accepted (not accepted).

Return on invested capital can be compared to a company's required rate of return to indicate whether the company has increased or decreased firm value over time.

LOS 24.c

Capital allocation decisions should be based on after-tax cash flows, ignore sunk costs, and capture any spillover effects on other parts of the business. Timing of cash flows is important.

Common mistakes in the capital allocation process include the following:

- | Having overly optimistic assumptions for pet projects of senior management
- | Basing long-term investment decisions on short-term EPS or ROE considerations
- | Poor cash flow estimation, misestimating overhead costs, and improper accounting for inflation
- | Not accounting for cost of internally generated funds in the estimate of required rate of return
- | Static capital budgets without regard to actual investment opportunities
- | Failure to generate alternative investment ideas

LOS 24.d

Real options allow managers to make future decisions that change the value of capital allocation decisions made today.

- | Timing options allow a company to delay making an investment.
- | Abandonment options allow management to abandon a project if the present value of the incremental cash flows from exiting a project exceeds the present value of the incremental cash flows from continuing a project.
- | Expansion options allow a company to make additional investments in a project if doing so creates value.
- | Flexibility options give managers choices regarding the operational aspects of a project. The two main forms are price-setting and production-flexibility options.
- | Fundamental options are projects that are options themselves because the payoffs depend on the price of an underlying asset.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 24.1

- . **C** Going concern projects are those to maintain the business or to increase the efficiency of existing operations. The other two projects are business growth investments that increase the size of the company. (LOS 24.a)
- . **A** A post-audit identifies what went right and what went wrong. It is used to improve forecasting and operations. (LOS 24.b)

- . **B** $CF_0 = -5,000$; $CF_1 = 3,000$; $CF_2 = 2,000$; $CF_3 = 2,000$; $I / Y = 10$; $NPV = \$883$.

(LOS 24.b)

Module Quiz 24.2

- . **A** Cash flows should be after-tax and include any tax savings from non-cash deductions (e.g., depreciation and amortization). While sunk costs should be ignored, the impact on other parts of the business (positive or negative) should be accounted for by adjusting cash flows. Accounting net income, which includes non-cash expenses, is irrelevant. Incremental cash flows are essential for making correct capital allocation decisions. (LOS 24.c)
- . **C** It is quite possible that offering a matching dryer will increase sales of their washers because some consumers will prefer a matching set. The increased sales of their washers is a positive impact, and those incremental profits should be considered in the analysis. A negative impact on another part of the business would be a consideration if introducing dryers could be expected to decrease washer sales. Sunk costs should not be considered in project analysis. (LOS 24.c)
- . **B** The cost of the marketing survey should not be included because it is a sunk cost; it will be incurred whether they decide to do the project or not. The decrease in sales of their current sauce if the spicier version is introduced (cannibalization) should be considered in the analysis. The cost of recipe development should be included because it will only be incurred if they decide to go ahead with the introduction of the new spicier sauce. (LOS 24.c)
- . **A** The project described has production flexibility regarding the level of production. Other flexibility options might be to produce a different product or to use different inputs at some future date. Including the value of real options can improve the NPV estimates for individual projects. (LOS 24.d)

READING 25

CAPITAL STRUCTURE

MODULE 25.1: WEIGHTED-AVERAGE COST OF CAPITAL

LOS 25.a: Calculate and interpret the weighted-average cost of capital for a company.

An issuer's **weighted-average cost of capital (WACC)** is a blended rate that includes its cost of debt and equity:

$$\text{WACC} = [\text{weight of debt} \times \text{pretax cost of debt} \times (1 - \text{tax rate})] + (\text{weight of equity} \times \text{cost of equity})$$

The cost of debt is lower than the cost of equity because debt has priority of claims over equity. Because most jurisdictions allow tax deductions for corporate interest expense, we adjust the pretax cost of debt to its after-tax value in the WACC formula.

The weights for debt and equity can be target weights or market value weights. Analysts often estimate a company's target weights based on book value of debt and equity. Market value weights are appropriate for estimating the current opportunity cost of capital as they reflect current market conditions.

EXAMPLE: WACC

ABC, Inc.'s, capital structure is 50% debt and 50% equity. ABC's cost of debt is 8%, while the cost of equity is 11%. ABC's corporate tax rate is 30%. Calculate the WACC for ABC.

Answer:

$$\text{WACC} = [0.50 \times 0.08 \times (1 - 0.30)] + (0.50 \times 0.11) = 0.083, \text{ or } 8.3\%$$

LOS 25.b: Explain factors affecting capital structure and the weighted-average cost of capital.

Capital structures vary considerably among companies. Typically a company will target a capital structure that minimizes its WACC, while also considering the nature of its assets (i.e., long-lived or shorter-term) when choosing the duration of its financing sources.

We may view factors that affect capital structures as those that determine a company's capacity to service debt. These factors may be internal or external to a firm. Internal factors include the characteristics of the business or industry, a company's life cycle stage, a company's existing debt level, and the corporate tax rate. External factors include market and business cycle conditions, regulation, and industry norms.

Company characteristics that influence the proportion of debt in a company's capital structure include the following:

- | *Growth and stability of revenue.* High growth of revenue or stability of significant revenue suggest a continuing ability to service debt.
- | *Growth and predictability of cash flow.* Growing cash flow increases the ability to service debt. Significant and stable cash flows indicate a continuing ability to service debt.
- | *Amount of business risk.* More business risk (operational risk and demand risk) means greater variability of earnings and cash flows, which decreases the ability to service debt.
- | *Amount and liquidity of company assets.* Assets can be pledged as collateral to make a company's debt more attractive. When assets are more liquid (easier to turn into cash, values more stable), they can be pledged more readily.
- | *Cost and availability of debt financing.* Companies find debt relatively more attractive when the cost of debt is lower and investors are more willing to lend to the company.

In general, the more stable, predictable, and recurring are a company's revenues and cash flows, the higher proportion of debt it can have in its capital structure. With this in mind, we can say, other things equal:

- | Companies in noncyclical industries are better able to support high proportions of debt than companies in cyclical industries.
- | Companies with low fixed operating costs as a proportion of total costs (i.e., low operating leverage) are better able to support high proportions of debt than companies with high fixed costs.
- | Companies with subscription-based revenue models are better able to support high proportions of debt than companies with pay-per-use revenue models.

The types of assets companies use to generate revenues also affect their ability to issue and service debt. For example, creditors tend to view tangible assets as better collateral than intangible assets, especially when those assets can be sold for cash, if necessary, without losing significant value (i.e., are more liquid) or are more readily substituted for similar assets (i.e., are more fungible). A company that owns its productive assets outright as opposed to using assets owned by others (such as a franchise model) has more collateral, which improves access to debt financing and reduces borrowing costs.

In addition to the types of assets, other issuer-specific conditions include the levels of existing debt and volatility of revenues and earnings. High levels of debt or high volatility of earnings indicates a lower ability to issue additional debt. Leverage (e.g., debt-to-equity or debt-to-operating profit) and coverage ratios (e.g., interest coverage = $\text{EBIT} / \text{interest expense}$) are used for analyzing debt capacity.

We can see the effects of these various factors on the debt-to-equity ratios of companies at different stages of their company's life cycle. Consider the following three stages:

Start-up stage. Sales are just beginning, and operating earnings and cash flows tend to be low or negative. Business risk is relatively high. Company debt is quite risky and, if issued, would require high interest rates. Assets, both accounts receivable and fixed assets, typically are low; therefore, they are not available as collateral for debt. For these reasons, start-up companies are financed almost exclusively with equity. In some cases, high-growth companies with rapidly rising stock prices may find it possible to issue **convertible debt**, which allows the company to use a lower-cost source of capital and avoid immediate dilution of existing shareholders' interests. Leasing assets is another source of debt financing available to companies in this stage of the life cycle.

Growth stage. Revenue and cash flow are rising, and business risk is somewhat reduced. Debt financing cost is somewhat reduced (i.e., usage is conservative), and investors may be willing to lend to the company, often with the loans secured by fixed assets or accounts receivable.

Mature stage. In this stage, revenue growth is slowing and business risk is much lower. Cash flow is significant and relatively stable. Debt financing, including unsecured debt, is widely available at a relatively low cost.

The cost of capital is also influenced by top-down factors. Top-down factors are macroeconomic factors that affect the benchmark interest rate (e.g., US Treasury rate) and credit spreads to benchmark rates. These include inflation, the real GDP growth rate, monetary policy, and exchange rates. In business cycle downturns, debt investors demand greater yield spreads to benchmark bonds from corporate borrowers to compensate them for the increased risk of default. This is especially true for companies in cyclical industries. Some industries may be favorably affected by the economic environment. For example, oil industry profitability is linked to oil prices, and spreads for issuers in the industry tend to narrow when oil prices rise.



MODULE QUIZ 25.1

1. Alpaca's capital structure is 60% debt with an interest rate of 6%. Alpaca's cost of equity is 12%, and the corporate tax rate is 15%. Alpaca's WACC is *closest* to:
 - A. 6.3%.
 - B. 7.9%.
 - C. 9.7%.
2. A company is *most likely* to be financed only by equity during its:
 - A. start-up stage.
 - B. growth stage.
 - C. mature stage.

MODULE 25.2: CAPITAL STRUCTURE THEORIES

LOS 25.c: Explain the Modigliani–Miller propositions regarding capital structure.

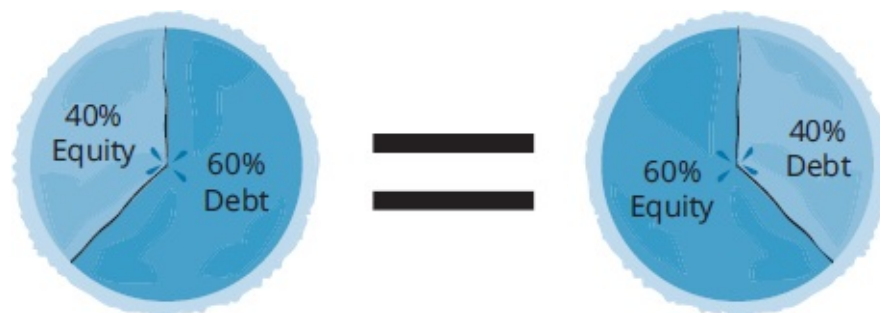
MM Proposition I (No Taxes): Capital Structure Irrelevance

In 1958, Nobel laureates Franco Modigliani and Merton Miller (we will refer to them as MM) published their seminal work on capital structure theory. In it, MM demonstrate that under certain assumptions, the value of a firm is unaffected by its capital structure. This result is referred to as MM I. The assumptions that lead to MM I are as follows:

- | *Capital markets are perfectly competitive.* There are no transactions costs, taxes, or bankruptcy costs.
- | *Investors have homogeneous expectations.* They have the same expectations with respect to cash flows generated by the firm.
- | *There is riskless borrowing and lending.* Investors can borrow and lend at the risk-free rate.
- | *There are no agency costs.* There are no conflicts of interest between managers and shareholders.
- | *Investment decisions are unaffected by financing decisions.* Operating income is independent of how the firm is financed.

Intuitively, we can explain MM I in terms of a pie. MM I essentially concludes that the amount of pie available (value of the firm) does not depend on how it is sliced (the capital structure). The value of a firm does not change depending on how the claims to its earnings are divided. This idea is illustrated in Figure 25.1.

Figure 25.1: MM Capital Structure Irrelevance Proposition



Consider why the pie analogy holds. The operating earnings (EBIT) of a firm are available to providers of capital. In a company with no debt, all of the operating earnings are available to equity holders, and the value of the company is the discounted present value of these earnings, with a discount rate that depends on the risk of (uncertainty about) earnings. If a company is financed partly by debt and partly by equity, operating earnings are divided between debtholders and equity holders. If one entity purchased all the debt and all the equity of the firm,

it would be entitled to all of the EBIT of the firm, so the total value of its holdings must equal the value of an all-equity firm—which also has a claim to all the operating earnings of the firm. Under the assumption that operating earnings are unaffected by financing decisions, the total value of debt and equity will be unaffected by the proportions of debt and equity in a firm’s capital structure.

MM Proposition II: Cost of Equity and Leverage

MM’s second proposition (MM II) is framed in terms of a firm’s cost of capital, rather than firm value. Based on the same assumptions as MM I, MM II states that the cost of equity increases linearly as a company increases its proportion of debt financing. The cash flows promised to bondholders have priority, so that equity holders receive the remaining cash flows after the claims of debtholders have been met. Because the cash flows promised to debtholders are more certain (less risky) than the residual cash flows promised to equity holders, the cost of debt will be less than the cost of equity. The greater the amount of debt in a firm’s capital structure, the more uncertain are the residual cash flows to equity holders. MM II tells us that as companies increase the proportion of debt financing, the risk of the cash flows to equity holders increases, which increases the cost of equity.

The conclusion of MM II is that the decrease in financing costs from using a larger proportion of (lower-cost) debt is just offset by the increase in the cost of equity, resulting in no change in the firm’s WACC.

Given MM II (that the value of the firm, and therefore its WACC, is unaffected by changes in the proportion of debt financing), we can state the relation that must hold between a company’s debt-to-equity ratio and its cost of equity:

$$r_e = r_0 + \frac{D}{E}(r_0 - r_d)$$

where:

r_e = cost of equity

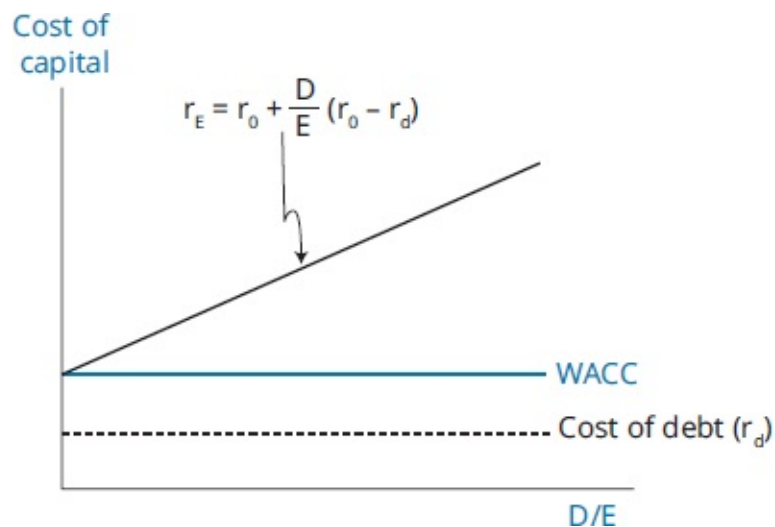
r_0 = cost of equity with no debt (all equity)

r_d = cost of debt

$\frac{D}{E}$ = debt-to-equity ratio

As leverage (the debt-to-equity ratio) increases, the cost of equity increases, but the cost of debt and WACC are unchanged. This relationship between the cost of equity financing and the debt-to-equity ratio is illustrated in Figure 25.2.

Figure 25.2: MM Proposition II (No Taxes)



MM II is consistent with MM I; if the benefits of greater use of lower-cost debt financing are just offset by the increased cost of equity, the proportions of debt versus equity in the firm's capital structure do not affect the firm's overall cost of capital or the value of the firm.

MM With Taxes: Value Is Maximized at 100% Debt

As is often done in theoretical work, MM started with several simplifying assumptions. The next step is to examine the effects of relaxing some of those assumptions. Here, we examine the MM propositions under the assumptions that earnings are taxed and that interest payments to debtholders are tax deductible. Under the tax code of most countries, interest payments are a pretax expense and are therefore tax deductible, while dividends paid to equity holders are not tax deductible.

This differential tax treatment encourages firms to use debt financing because debt provides a **tax shield** that adds value to the company. The tax shield is equal to the tax rate multiplied by the amount of debt in the capital structure. In other words, the value of a levered firm is equal to the value of an unlevered firm plus the value of the tax shield provided by debt financing.

To continue our analogy of a pie, with the introduction of taxes, the government gets a slice of the pie. When debt financing is used, the government's slice of the pie is smaller, so that the amount of pie available to debt and equity holders is greater. The use of debt reduces taxes due to deductibility of interest expense

giving rise to debt tax shield. If a firm were 100% financed with debt, the taxes avoided would be at a maximum, and the after-tax cash flows of the firm (size of the pie) would be maximized. The conclusion of *MM I with taxes* is that the value of the firm is maximized with 100% debt financing.

If we assume a positive tax rate, the formula to solve for return on equity can be used to illustrate *MM II with taxes*:

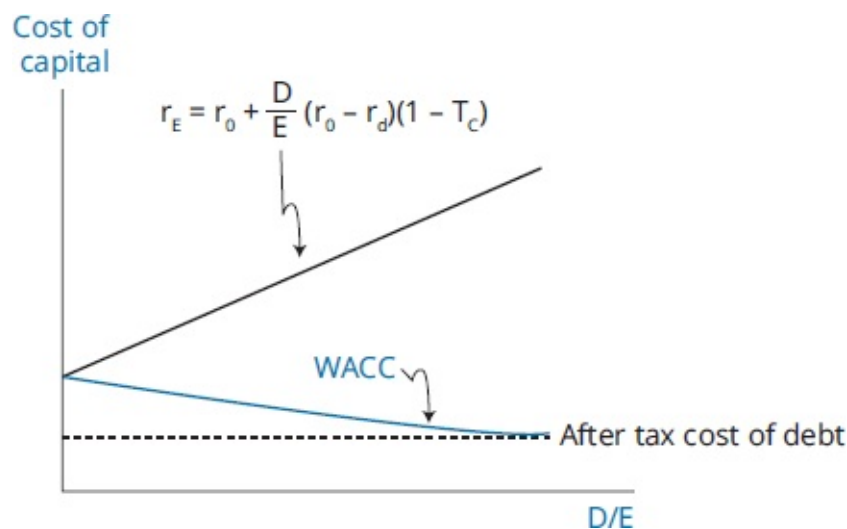
$$r_e = r_0 + \frac{D}{E}(r_0 - r_d)(1 - T_C)$$

where:

T_C = tax rate

Figure 25.3 illustrates that the tax shield provided by debt causes the WACC to decline as leverage increases. The value of the firm is maximized at the point where the WACC is minimized, which is 100% debt.

Figure 25.3: MM Proposition II (With Taxes)



In practice, we do not find that companies finance their assets with 100% debt, or close to it. MM suggested that differential investor tax rates on dividends and interest income could explain differences in capital structures.

Current theory suggests that differences in value-maximizing choices of how much financial leverage to use are the result of additional costs of using debt financing that we have not yet considered—we assumed that there are no bankruptcy costs. However, we can expect these costs, also called the **costs of financial distress**, to increase at higher levels of debt financing. Costs of financial distress are the increased costs a company faces when earnings decline

to the point where the firm has trouble paying its fixed financing costs (interest on debt). The expected costs of financial distress for a firm have two components:

Costs of financial distress and bankruptcy can be direct or indirect. Direct costs of financial distress include the cash expenses associated with the bankruptcy, such as legal fees and administrative fees. Indirect costs include foregone investment opportunities and the costs that result from losing the trust of customers, creditors, suppliers, and employees. Additionally, during periods of financial distress, conflicts of interest between managers (who represent equity owners) and debtholders impose additional costs, referred to as the **agency costs of debt**.

The probability of financial distress is related to the firm's use of operating and financial leverage. In general, higher amounts of financial leverage increase the probability of financial distress (higher probability that cash flows will fall to an amount that is insufficient to make their promised debt payments). Other factors to consider include the quality of a firm's management and the company's corporate governance structure. Lower-quality management and poor corporate governance lead to a higher probability of financial distress.

Higher expected costs of financial distress tend to discourage companies from using large proportions of debt in their capital structures, all else equal. The expected costs of financial distress shrink our pie as the proportion of debt financing is increased.

LOS 25.d: Describe optimal and target capital structures.

Static Tradeoff Theory

The **static tradeoff theory** seeks to balance the costs of financial distress with the tax shield benefits from using debt. There is an amount of debt financing at which the increase in the value of the tax shield from additional borrowing is exceeded by the value reduction of higher expected costs of financial distress. This point represents the **optimal capital structure** for a firm, where the WACC is minimized and the value of the firm is maximized.

Accounting for the costs of financial distress, the expression for the value of a levered firm becomes the following:

$$V_L = V_U + (t \times \text{debt}) - \text{PV (costs of financial distress)}$$

We illustrate this relationship in Figure 25.4.

Figure 25.4: Static Tradeoff Theory: Cost of Capital vs. Capital Structure

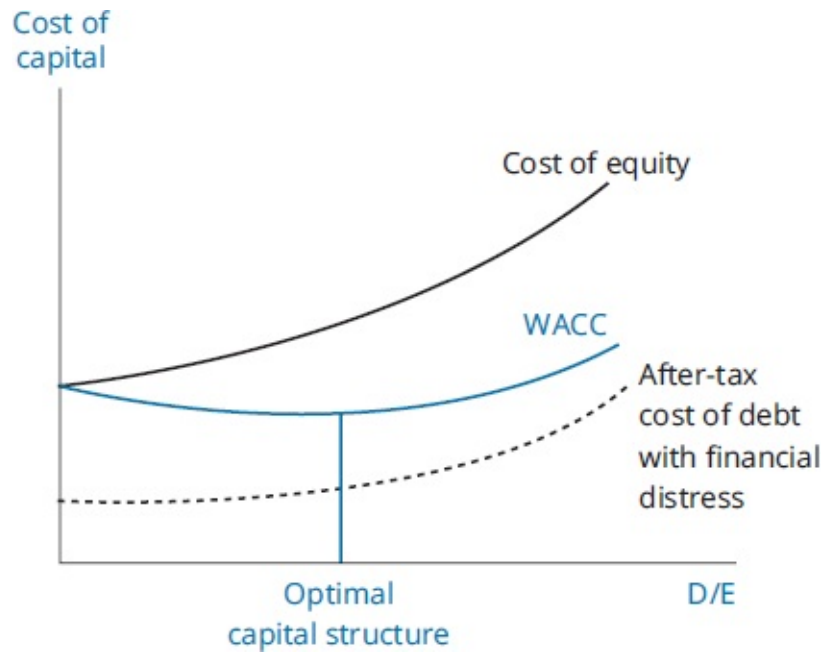
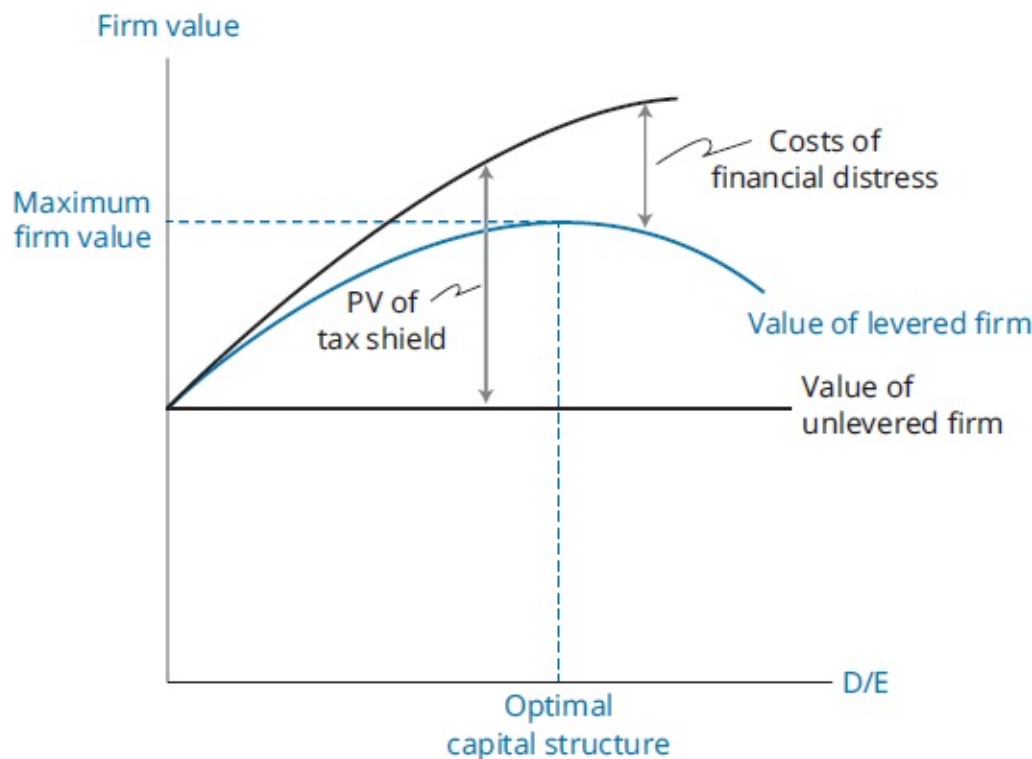


Figure 25.5 illustrates the tradeoff between firm value and capital structure. Note that each firm's optimal capital structure depends on its business risk (operating risk and sales risk), tax rate, corporate governance, industry influences, and other factors.

Figure 25.5: Static Tradeoff Theory: Firm Value vs. Capital Structure



To summarize:

- | *MM's propositions with no taxes or costs of financial distress* are that a company's capital structure is irrelevant because its WACC and its value (the discounted present value of its operating earnings) are unchanged by changes in capital structure.
- | *MM's propositions with taxes but without costs of financial distress* are that a company's WACC is minimized and its value is maximized with 100% debt.
- | *Static tradeoff theory* indicates that firm value initially increases (and WACC decreases) with additional debt financing, but company value decreases at some point when the increase in the expected value of financial distress outweighs the tax benefits of additional debt.

Target capital structure is the capital structure that a firm seeks to achieve on average over time to maximize firm value. That is, it reflects management's beliefs about its optimal capital structure as well as other internal and external factors.

For analysis, the weights to use when estimating a firm's WACC should be based on its target capital structure, the proportions (based on market values) of debt, preferred stock, and equity that the firm expects to achieve over time. An analyst

may use management's stated target proportions of equity and debt; however, most firms do not provide these.

In the absence of stated capital structure weights, an external analyst must estimate a firm's target capital structure. Alternatives for estimating target capital structure include the following:

- | An analyst may simply use the firm's current capital structure (based on market values) as the best indication of its target capital structure.
- | If there has been a noticeable trend in the firm's capital structure, an analyst may incorporate this trend into her estimate of the firm's target.
- | An analyst may use the average capital structure weights for a firm's industry.

In contrast with independent analysts, company managers often focus on book values of debt and equity. Reasons book values may be appropriate for internal analysis include:

- | Short-term fluctuations in the market values of debt and equity do not really affect a company's appropriate level of debt.
- | While market values reflect the perspectives of investors in a company, management's perspective has more to do with how the company needs to deploy its capital to projects.
- | Credit rating agencies use book values in their measures of credit quality. Managers may make capital structure decisions with these measures in mind.

In practice, a firm's actual capital structure tends to fluctuate around the target capital structure. Market fluctuations, especially in the market value of a firm's equity, may cause the firm's capital structure to vary from the target.

Management may choose to exploit opportunities in a specific financing source. For example, a temporary rise in the firm's stock price may create a good opportunity to issue additional equity, which would result in a higher percentage of equity than the target. Also, external capital is raised in minimum-size lots; hence, adherence to precise weights may not be possible.

Costs of asymmetric information arise from the fact that managers typically have more information about a company's prospects and future performance than owners or creditors. Firms with complex products or little transparency in financial statements tend to have higher costs of asymmetric information, which result in higher required returns on both debt and equity capital.

Because shareholders and creditors are aware that asymmetric information problems exist, these investors look for management behavior that signals what knowledge or opinions management may have about the firm's prospects. For example, taking on the commitment to make fixed interest payments through debt financing sends a signal that management is confident in the firm's ability to make these payments in the future. By contrast, issuing equity is typically viewed as a negative signal that managers believe a firm's stock is overvalued. The cost of asymmetric information increases with the proportion of equity in the capital structure.

Agency costs of equity are related to conflicts of interest between managers and owners. Managers who do not have a stake in the company do not bear the costs associated with excessive compensation or taking on too much (or too little) risk. Because shareholders are aware of this conflict, they take steps to reduce these costs. The result is called the **net agency cost of equity**. Net agency costs of equity have three components:

Monitoring costs are associated with supervising management and include the expenses of reporting to shareholders and paying the board of directors. Strong corporate governance systems reduce monitoring costs.

Bonding costs relate to assuring shareholders that the managers are working in the shareholders' best interest. Examples of bonding costs include premiums for insurance to guarantee performance and implicit costs associated with noncompete agreements.

Residual losses may occur even with adequate monitoring and bonding provisions because such provisions do not provide a perfect guarantee.

According to the **free cash flow hypothesis**, the use of debt forces managers to be disciplined with regard to how they spend cash because they have less free cash flow to use for their own benefit. It follows that greater amounts of financial leverage tend to reduce agency costs.

Pecking order theory, based on asymmetric information, is related to the signals that management sends to investors through its financing choices. According to pecking order theory, managers prefer to make financing choices that are least likely to send negative signals to investors. Financing choices under pecking order theory follow a hierarchy based on visibility to investors. Internally generated capital is most preferred, debt is the next-best choice, and external equity is the least preferred financing option. Pecking order theory

implies that the capital structure is a by-product of individual financing decisions.



MODULE QUIZ 25.2

1. A company's optimal capital structure:
 - A. maximizes firm value and minimizes the weighted-average cost of capital.
 - B. minimizes the interest rate on debt and maximizes expected earnings per share.
 - C. maximizes expected earnings per share and maximizes the price per share of common stock.
2. Which of the following statements regarding Modigliani and Miller's Proposition II with no taxes is *most accurate*?
 - A. A firm's cost of debt financing increases as a firm's financial leverage increases.
 - B. A firm's weighted-average cost of capital is not affected by its choice of capital structure.
 - C. A firm's cost of equity financing increases as the proportion equity in a firm's capital structure is increased.
3. To determine their target capital structures in practice, it is *least likely* that firms will:
 - A. use the book value of their debt to make financing decisions.
 - B. match the maturities of their debt issues to specific firm investments.
 - C. determine an optimal capital structure based on the expected costs of financial distress.
4. The pecking order theory of financial structure decisions:
 - A. is based on information asymmetry.
 - B. suggests that debt is the first choice for financing an investment of significant size.
 - C. suggests that debt is the riskiest and least preferred source of financing.

KEY CONCEPTS

LOS 25.a

$$WACC = [\text{weight of debt} \times \text{pretax cost of debt} \times (1 - \text{tax rate})] + (\text{weight of equity} \times \text{cost of equity})$$

LOS 25.b

Capital structures vary among companies. Internal factors that affect capital structures include the characteristics of the business, the company's existing debt level, their corporate tax rate, and the company's life cycle stage. External factors include market and business cycle conditions.

A company's ability to issue debt is greater with predictable cash flows sufficient to make required debt payments, and with liquid tangible assets that the company can pledge as collateral for debt.

New companies with few assets and negative or uncertain cash flows will use little to no debt. Growth companies with positive cash flows and decreasing business risk may use lower levels of debt. Mature companies with predictable cash flows tend to use significantly more debt.

LOS 25.c

MM's propositions with no taxes are that a company's capital structure is irrelevant, because its WACC and firm value (the discounted present value of its operating earnings) are unchanged by changes in capital structure.

MM's propositions with taxes are that a company's WACC is minimized and its value is maximized with 100% debt financing.

LOS 25.d

Static tradeoff theory adds the expected costs of financial distress to the model. It indicates that firm value initially increases (and WACC decreases) with additional debt financing, but that company value decreases at some point with additional debt as the increase in the expected costs of financial distress outweigh the increase in tax benefits from additional debt.

In practice, a company's capital structure will fluctuate around the target due to management's exploitation of market opportunities and market value fluctuations (especially of equity) over time.

Pecking order theory is based on information asymmetry between firm management and investors and suggests that management's choice of financing method signals their beliefs about firm value. The theory concludes that retained earnings are the most preferred source of funds, followed by debt financing, and then issuing new equity.

Under the free cash flow hypothesis, the agency costs of equity, which arise because management and shareholders may have conflicting interests, are reduced by increased debt issuance.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 25.1

. **B** Weight of equity = $1 - 0.60 = 0.40$, or 40%.

$$\text{WACC} = [0.60 \times 0.06 \times (1 - 0.15)] + (0.4 \times 0.12) = 7.86.$$

(LOS 25.a)

- . **A** During the start-up stage, a firm is unlikely to have positive earnings and cash flows or significant assets that can be pledged as debt collateral, so firms in this stage are typically financed by equity only. (LOS 25.b)

Module Quiz 25.2

- . **A** The optimal capital structure minimizes the firm's WACC and maximizes the firm's value (stock price). (LOS 25.c)
- . **B** MM's Proposition II (with no taxes) states that capital structure is irrelevant because the decrease in a firm's WACC from additional debt financing is just offset by the increase in WACC from a decrease in equity financing. The cost of debt is held constant, and the cost of equity financing increases as the proportion of *debt* in the capital structure is increased. (LOS 25.c)
- . **C** While it is a useful theoretical concept, in practice, determining an optimal capital structure based on the cost savings of debt and the expected costs of financial distress is not feasible. Because debt rating companies often use book values of debt, firms often use book values of debt when choosing financing sources. It is common for firms to match debt maturities to the economic lives of specific investments. (LOS 25.d)
- . **A** Pecking order theory is based on information asymmetry and the resulting signals that different financing choices send to investors. It suggests that retained earnings are the first choice for financing an investment, and issuing new equity is the least preferred choice. (LOS 25.d)

READING 26

BUSINESS MODELS

MODULE 26.1: BUSINESS MODEL FEATURES AND TYPES

LOS 26.a: Describe key features of business models.

A successful firm must provide a product or service, find customers, deliver the product or service, and make a profit. A **business model** explains how a firm either does or proposes to do this.

“How we will provide it, sell it, and make a profit” is clearly an oversimplification, but this is the essence of a business model. In practice, the answers to these questions have many facets. Here, we present a framework that incorporates some of the complexities involved. It involves answering the questions of *who*, *what*, *how*, *where*, and *how much*.

A business model should do the following:

- | *Identify the firm’s potential customers (who)*, how they are acquired, the cost of customer acquisition, and how the company will monitor and maintain customer satisfaction. Potential customers can be defined in innumerable ways, ranging from every consumer within a geographic area, to dog owners, to only the home country’s military in the case of a weapon. This characterization of customer groups is called **segmenting**.
- | *Describe the key assets and suppliers of the firm (how)*. Key assets may be, for example, a patent, software, or skilled employees. Key suppliers may be a battery manufacturer for an electric vehicle company, a lithium miner for a battery maker, or a supplier of large excavation machinery for a lithium miner.
- | *Describe the firm’s product or service (what)*, how it meets a need for its potential customers, and what differentiates its products from those of competitors (e.g., low price, premium quality, innovative features).

- | *Explain how the firm will sell its product or service (where),* such as online, physical location, direct mail, trade shows, or through sales representatives; whether the firm will sell directly to the buyers (**direct sales**) or use intermediaries such as wholesalers, retailers, agents, or franchisees; and how the firm will deliver its product or service. The answers to these questions comprise a firm's **channel strategy**. A strategy that includes both digital and physical channels, such as internet sales with delivery at a physical location, is referred to as an **omnichannel strategy**.

Firms that sell to other businesses are said to be **B2B** (business to business) firms, while firms that sell to consumers are said to be **B2C** (business to consumer) firms.

- | *Explain its pricing strategy (how much) and why buyers will pay that price for its product, given the competitive landscape of the market.*

Pricing Strategies

Companies with undifferentiated products, called **commodity producers**, are price takers (e.g., oil and gas, home loans). Industries with few competitors or highly differentiated products enjoy pricing power (e.g., patented drugs). Companies may be able to charge a premium price by including services or features that are valued by customers.

Price discrimination refers to setting different prices for different customers or identifiable groups of customers. Common examples are **tiered pricing** (based on volume of purchases); **dynamic pricing** (depending on the time of day or day of the week), such as peak and off-peak pricing and low-priced airline tickets for very early or very late flights; **value-based pricing** (e.g., a new drug that offers a marginally better protection compared to existing drugs); and **auction pricing** (e.g., eBay).

Pricing models for multiple products include the following:

- | **Bundling.** Where multiple products are complimentary (e.g., a furnished apartment), bundling the products may be a profitable strategy.
- | **Razors-and-blades.** A company may find it profitable to sell a piece of equipment for a relatively low price (low margins) and make profits by selling a consumable used with the equipment. Printers and ink cartridges, and an e-reader and e-books are common examples.

- | **Add-on pricing.** Options or add-ons priced with high margins are added to the product after the purchase decision has been made. An example is the many pricey options that may be offered after a customer has decided to purchase an automobile.

Other pricing models include the following:

- | **Penetration pricing.** A company offers a product at low margins or even at a loss for a period of time to grow market share and achieve greater scale of operations. Netflix followed this strategy to grow its subscriber base rapidly.
- | **Freemium pricing.** Offer a product with basic functionality at no cost, but sell or unlock other functionality for a fee. Video game makers have used this strategy to encourage wide usage and then profit on sales of greater functionality (e.g., upgraded weapons).
- | **Hidden revenue.** Online content may be “free” but generate revenue through ads. For example, an internet search is free to the user while the search engine’s revenue comes from selling user data.

Models that offer alternatives to outright purchases include the following:

- | **Subscription model.** Microsoft’s model for software has changed from selling the software to a subscription (paying monthly for access) to their Office suite of software.
- | **Licensing and franchising.** For a biotech company that has developed a new and effective drug, it may be most profitable to license the production of the drug to an established drug maker with a large sales force and established distribution channels, rather than developing those resources itself for the single drug. A franchisee typically is permitted to sell in a specific area and pays a percentage of sales to the franchisor, which provides some level of product and marketing support.

Value Proposition and Value Chain

A firm’s **value proposition** refers to how customers will value the characteristics of the product or service, given the competing products and their prices. How the firm executes its value proposition is referred to as its **value chain**. A firm’s value chain comprises the assets of the firm and how its organization will add value and exploit the firm’s competitive advantage. The value chain

encompasses the quality of the product, its functionality, service that is included, the sale process, and pricing relative to the competition.

In his 1985 book *Competitive Advantage*, Michael Porter presents five activities (value chain analysis) in which firms should strive to execute well:

Inbound logistics

Operations

Outbound logistics

Marketing

Sales and service

LOS 26.b: Describe various types of business models.

Conventional business models tend to be industry-specific. Examples include those followed by natural resource producers, manufacturers, distributors, retailers, banks, brokers, service providers, and software. Other business models include the following:

- | *Private label manufacturers or contract manufacturers.* Companies produce products for others to market under their own brand name (e.g., Costco's Kirkland-branded products).
- | *Licensing agreements.* A company brand is used by another company on its products for a fee, such as a lunch box branded with a Marvel character.
- | *Value-added resellers.* They offer such things as installation, service, support, or customization for complex equipment.

Innovations in business models often occur due to new entrants sometimes bringing new technology into the industry to challenge the incumbents. Examples include pricing of software as a service (SaaS), ultra-low-cost airlines, and discount brokers.

Network effects refer to the increase in the value of a network as its user base grows. Examples of this include WhatsApp, eBay, and Facebook. Network effects support an initial strategy of penetration pricing. Network effects capitalize on both size and scope. Some networks are two-sided or multi-sided—such as Airbnb, which has a multitude of hosts and guests.

Crowdsourcing models benefit from user contributions—content in the case of Wikipedia, traffic conditions and events in the case of Waze, and product improvements or new applications in the case of open-source software.



MODULE QUIZ 26.1

1. A business model is *least likely* to include details about a company's:
 - A. largest customers.
 - B. workforce characteristics.
 - C. revenue and expense estimates.
2. A pricing strategy of offering a basic product at no cost, and unlocking other functionality for a fee, is *most accurately* called:
 - A. optional products.
 - B. freemium pricing.
 - C. subscription pricing.
3. A start-up business provides consumers with professional chefs at home. For its business model to work, the company needs a sufficient number of chefs available to their customers when they have a need. Similarly, sufficient demand from customers is needed to keep the chefs interested in making themselves available. Which of the following is the company's business model *most likely* reliant on?
 - A. Licensing.
 - B. Bundling.
 - C. Network effects.

KEY CONCEPTS

LOS 26.a

A business model should identify a firm's potential customers, describe its products or services and explain how it will sell them, describe its key assets and suppliers, and explain its pricing strategy.

A value proposition refers to how a firm's customers will value the characteristics of the product or service. A value chain refers to how a firm executes its value proposition.

LOS 26.b

Apart from industry-specific conventional business models, other business models include private label manufacturers, value-added resellers, and licensing agreements.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 26.1

- . **C** Detailed forecasts of revenue in expenses would be in a financial plan, but typically not in a business model. A firm's largest customers and information about its workforce and its value are likely elements of a business model. (LOS 26.a)
- . **B** Freemium pricing offers a product with basic functionality at no cost, but sells or unlocks other functionality for a fee. (LOS 26.a)
- . **C** Network effects refer to the increase in the value of the business as its user base grows. In this instance, the network is two-sided; a sufficient number of customers and service providers are needed for the business to succeed. (LOS 26.b)

Topic Quiz: Corporate Finance

You have now finished the Corporate Finance topic section. Please log into your Schweser online dashboard and take the Topic Quiz on this section. The Topic Quiz provides immediate feedback on how effective your study has been for this material. Questions are more exam-like than typical Module Quiz or QBank questions; a score of less than 70% indicates that your study likely needs improvement. These tests are best taken timed; allow 1.5 minutes per question.

FORMULAS

Quantitative Methods

$$r_{\text{price}} = \frac{P_1 - P_0}{P_0}$$

$$r_{\text{distribution}} = \frac{\text{income}}{P_0}$$

$$r_{\text{total}} = r_{\text{price}} + r_{\text{distribution}}$$

$$\text{arithmetic mean return} = \frac{(R_1 + R_2 + R_3 + \dots + R_n)}{n}$$

$$\text{geometric mean return} = \sqrt[n]{(1 + R_1) \times (1 + R_2) \times (1 + R_3) \times \dots \times (1 + R_n)} - 1$$

$$\text{annualized return} = (1 + \text{return during the period})^{365/\text{days}} - 1$$

continuously compounded return:

$$R_{\text{cc}} = \ln(1 + \text{return during the period}) = \ln\left(\frac{\text{ending value}}{\text{beginning value}}\right)$$

$$(1 + \text{nominal risk-free rate}) = (1 + \text{real risk-free rate})(1 + \text{expected inflation rate})$$

$$\text{real risk-free rate} \approx \text{nominal risk-free rate} - \text{expected inflation rate}$$

$$\text{gross return} = \text{total return} - \text{transaction costs}$$

$$\text{net return} = \text{gross return} - \text{management fees and administrative costs}$$

$$\text{after-tax return} = (\text{price return})(1 - \text{capital gains tax rate}) + (\text{capital distribution return})(1 - \text{capital distribution tax rate})$$

$$\text{leveraged return} = \frac{r_P(V_D + V_E) - r_D V_D}{V_E}$$

time value of money:

$$FV = PV(1 + r)^t$$

$$PV = \frac{FV}{(1 + r)^t} = FV(1 + r)^{-t}$$

time value of money with continuous compounding:

$$FV = PV \times e^{rt}$$

$$PV = FV \times e^{-rt}$$

$$PV \text{ of a perpetuity} = \frac{\text{payment}}{r}$$

constant growth dividend discount model:

$$V_0 = \frac{D_1}{r_e - g_c}$$

sample mean and population mean:

$$\bar{X} = \sum_{i=1}^n \frac{x_i}{n}$$

$$\mu = \sum_{i=1}^n \frac{x_i}{N}$$

range = maximum value – minimum value

$$\text{mean absolute deviation} = \frac{\sum_{i=1}^n |X_i - \bar{X}|}{n}$$

$$\text{median absolute deviation} = \text{Med}(|X_i - \text{Med}(X)|)$$

population standard deviation:

$$\sigma = \sqrt{\sum_{i=1}^n \frac{(X_i - \mu)^2}{n}}$$

sample standard deviation:

$$s = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}}$$

population covariance:

$$\sigma_{XY} = \sum_{i=1}^n \frac{(X_i - \mu_X)(Y_i - \mu_Y)}{n}$$

correlation coefficient:

$$\rho_{XY} = \frac{s_{XY}}{s_X s_Y}$$

semivariance:

$$\sigma_{\text{semi}}^2 = \sum_{i: X_i < \bar{X}} \frac{(X_i - \bar{X})^2}{n}$$

target downside deviation:

$$s_{\text{target}} = \sqrt{\frac{\sum_{\text{all } X_i < B}^n (X_i - B)^2}{n - 1}}$$

where B = the target

coefficient of variation = $\frac{\text{standard deviation of } x}{\text{average value of } x}$

expected value of a random variable:

$$E(X) = \sum_i [x_i \times P(x_i)]$$

unconditional variance:

$$\begin{aligned}\sigma_X^2 &= \text{Var}(X) = E[X - E(X)]^2 \\ &= E(X^2) - [E(X)]^2\end{aligned}$$

unconditional covariance:

$$\begin{aligned}\sigma_{XY} &= \text{Cov}(X, Y) = E\{[X - E(X)] \times [Y - E(Y)]\} \\ &= E(XY) - E(X)E(Y)\end{aligned}$$

conditional mean:

$$E(X|S) = \sum_{i=1}^n X_i \times P(X_i|S)$$

conditional variance:

$$\text{Var}(X|S) = E\{[X - E(X|S)]^2\}$$

Bayes' Formula:

$$P(A|B) = P(B|A) \frac{P(A)}{P(B)}$$

standard error of the sample mean:

$$s_x = \frac{s}{\sqrt{n}}$$

confidence interval for a population mean:

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

Value at Risk using parametric method:

$$\text{VaR} = z_{\alpha} \times \sigma \times P$$

where:

z_{α} = the critical z-value corresponding to a significance of α in a single tail

σ = the standard deviation of the portfolio

P = current portfolio value

$$\text{test statistic for the population mean} = \frac{\text{sample mean} - \text{hypothesized value}}{\text{standard error of sample mean}}$$

variance of returns for a two-asset portfolio:

$$\sigma_P^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \text{Cov}_{A,B}$$

covariance of returns for two assets:

$$\sigma_{ij} = \rho_{ij} \times \sigma_i \times \sigma_j$$

weights of minimum-variance portfolio of two assets:

$$w_A = \frac{\sigma_B^2 - \sigma_{A,B}}{\sigma_A^2 + \sigma_B^2 - 2\sigma_{A,B}}$$

$$w_B = 1 - w_A$$

$$\text{Sharpe ratio} = \frac{E(r_P) - r_f}{\sigma_P}$$

weights of risky and risk-free assets consistent with expected rate of return r_P :

$$w_{\text{risky}} = \frac{r_P - r_f}{r_{\text{risky}} - r_f}, \text{ and } w_{\text{risk-free}} = 1 - w_{\text{risky}}$$

Capital Market Line:

$$E(r_P) = r_f + \frac{E(r_M) - r_f}{\sigma_M} \sigma_P$$

where:

r_M = expected return on the market portfolio

σ_M = standard deviation of the market portfolio

Capital Asset Pricing Model (CAPM):

$$E(R_i) = R_f + \beta_i [E(R_{\text{market}}) - R_f]$$

$$\beta_i = \frac{\sigma_{i,m}}{\sigma_m^2} = \frac{\sigma_i}{\sigma_m} \rho_{i,m}$$

Economics

$$\text{real P/B exchange rate} = \text{nominal P/B exchange rate} \times \left(\frac{\text{CPI}_{\text{base currency}}}{\text{CPI}_{\text{price currency}}} \right)$$

$$\text{arbitrage-free forward exchange rate relationship: } \frac{\text{forward}_{d/f}}{\text{spot}_{d/f}} = \frac{(1 + r_{\text{domestic}})}{(1 + r_{\text{foreign}})}$$

Corporate Finance

$$\text{cash conversion cycle} = \text{days of inventory on hand} + \text{days sales outstanding} \\ - \text{days payables outstanding}$$

$$\text{cost of supplier financing} = \left(1 + \frac{a}{1-a} \right)^{\left(\frac{365}{c-b} \right)} - 1$$

where:

a = percent discount

b = days until discount expires

c = days until full payment is due

$$\text{total working capital} = \text{current assets} - \text{current liabilities}$$

$$\text{net working capital} = \text{current assets (except cash and marketable securities)} \\ - \text{current liabilities (excluding short-term and current debt)}$$

$$\text{current ratio} = \frac{\text{current assets}}{\text{current liabilities}}$$

$$\text{quick ratio} = \frac{(\text{cash and marketable securities} + \text{accounts receivable})}{\text{current liabilities}}$$

$$\text{cash ratio} = \frac{\text{cash and marketable securities}}{\text{current liabilities}}$$

net present value (NPV):

$$\text{NPV} = \text{CF}_0 + \frac{\text{CF}_1}{(1+k)^1} + \frac{\text{CF}_2}{(1+k)^2} + \dots + \frac{\text{CF}_n}{(1+k)^n} = \sum_{t=0}^n \frac{\text{CF}_t}{(1+k)^t}$$

where:

CF_0 = initial investment outlay

CF_t = after-tax cash flow at time t

k = required rate of return

$$\text{return on invested capital} = \frac{\text{net operating profit after tax}}{\text{average book value of total capital}}$$

APPENDICES

APPENDIX A: AREAS UNDER THE NORMAL CURVE

Most of the examples in this book have used one version of the z-table to find the area under the normal curve. This table provides the cumulative probabilities (or the area under the entire curve to left of the z-value).

Probability Example

Assume that the annual earnings per share (EPS) for a large sample of firms is normally distributed with a mean of \$5.00 and a standard deviation of \$1.50. What is the approximate probability of an observed EPS value falling between \$3.00 and \$7.25?

If $EPS = x = \$7.25$, then $z = (x - \mu)/\sigma = (\$7.25 - \$5.00)/\$1.50 = +1.50$

If $EPS = x = \$3.00$, then $z = (x - \mu)/\sigma = (\$3.00 - \$5.00)/\$1.50 = -1.33$

Solving Using the Cumulative Z-Table

For z-value of 1.50: Use the row headed 1.5 and the column headed 0 to find the value 0.9332. This represents the area under the curve to the left of the critical value 1.50.

For z-value of -1.33: Use the row headed 1.3 and the column headed 3 to find the value 0.9082. This represents the area under the curve to the left of the critical value +1.33. The area to the left of -1.33 is $1 - 0.9082 = 0.0918$.

The area between these critical values is $0.9332 - 0.0918 = 0.8414$, or 84.14%.

Hypothesis Testing—One-Tailed Test Example

A sample of a stock's returns on 36 nonconsecutive days results in a mean return of 2.0%. Assume the population standard deviation is 20.0%. Can we say with 95% confidence that the mean return is greater than 0%?

$H_0: \mu \leq 0.0\%$, $H_a: \mu > 0.0\%$. The test statistic = z-statistic

$$= \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}} = (2.0 - 0.0) / (20.0 / 6) = 0.60$$

The significance level = $1.0 - 0.95 = 0.05$, or 5%. Because we are interested in a return greater than 0.0%, this is a one-tailed test.

Using the Cumulative Z-Table

Because this is a one-tailed test with an alpha of 0.05, we need to find the value 0.95 in the cumulative z-table. The closest value is 0.9505, with a corresponding critical z-value of 1.65. Because the test statistic is less than the critical value, we fail to reject H_0 .

Hypothesis Testing—Two-Tailed Test Example

Using the same assumptions as before, suppose that the analyst now wants to determine if he can say with 99% confidence that the stock's return is not equal to 0.0%.

$H_0: \mu = 0.0\%$, $H_a: \mu \neq 0.0\%$. The test statistic (z-value) = $(2.0 - 0.0) / (20.0 / 6) = 0.60$. The significance level = $1.0 - 0.99 = 0.01$, or 1%. Because we are interested in whether or not the stock return is nonzero, this is a two-tailed test.

Using the Cumulative Z-Table

Because this is a two-tailed test with an alpha of 0.01, there is a 0.005 rejection region in both tails. Thus, we need to find the value 0.995 ($1.0 - 0.005$) in the table. The closest value is 0.9951, which corresponds to a critical z-value of 2.58. Because the test statistic is less than the critical value, we fail to reject H_0 and conclude that the stock's return equals 0.0%.

CUMULATIVE Z-TABLE

Standard Normal Distribution

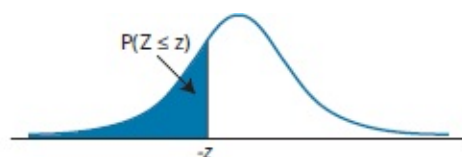
$$P(Z \leq z) = N(z) \text{ for } z \geq 0$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

Standard Normal Distribution

$$P(Z \leq z) = N(z) \text{ for } z \geq 0$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.7	0.2420	0.2389	0.2358	0.2327	0.2297	0.2266	0.2236	0.2207	0.2177	0.2148
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1057	0.1038	0.1020	0.1003	0.0985
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.4	0.0082	0.0080	0.0078	0.0076	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010

APPENDIX B: STUDENT'S T-DISTRIBUTION

Level of Significance for One-Tailed Test						
df	0.100	0.050	0.025	0.01	0.005	0.0005

Level of Significance for Two-Tailed Test						
df	0.20	0.10	0.05	0.02	0.01	0.001
1	3.078	6.314	12.706	31.821	63.657	636.619
2	1.886	2.920	4.303	6.965	9.925	31.599
3	1.638	2.353	3.182	4.541	5.841	12.294
4	1.533	2.132	2.776	3.747	4.604	8.610
5	1.476	2.015	2.571	3.365	4.032	6.869
6	1.440	1.943	2.447	3.143	3.707	5.959
7	1.415	1.895	2.365	2.998	3.499	5.408
8	1.397	1.860	2.306	2.896	3.355	5.041
9	1.383	1.833	2.262	2.821	3.250	4.781
10	1.372	1.812	2.228	2.764	3.169	4.587
11	1.363	1.796	2.201	2.718	3.106	4.437
12	1.356	1.782	2.179	2.681	3.055	4.318
13	1.350	1.771	2.160	2.650	3.012	4.221
14	1.345	1.761	2.145	2.624	2.977	4.140
15	1.341	1.753	2.131	2.602	2.947	4.073
16	1.337	1.746	2.120	2.583	2.921	4.015
17	1.333	1.740	2.110	2.567	2.898	3.965
18	1.330	1.734	2.101	2.552	2.878	3.922
19	1.328	1.729	2.093	2.539	2.861	3.883
20	1.325	1.725	2.086	2.528	2.845	3.850
21	1.323	1.721	2.080	2.518	2.831	3.819
22	1.321	1.717	2.074	2.508	2.819	3.792
23	1.319	1.714	2.069	2.500	2.807	3.768
24	1.318	1.711	2.064	2.492	2.797	3.745
25	1.316	1.708	2.060	2.485	2.787	3.725
26	1.315	1.706	2.056	2.479	2.779	3.707
27	1.314	1.703	2.052	2.473	2.771	3.690
28	1.313	1.701	2.048	2.467	2.763	3.674
29	1.311	1.699	2.045	2.462	2.756	3.659
30	1.310	1.697	2.042	2.457	2.750	3.646
40	1.303	1.684	2.021	2.423	2.704	3.551
60	1.296	1.671	2.000	2.390	2.660	3.460
120	1.289	1.658	1.980	2.358	2.617	3.373
∞	1.282	1.645	1.960	2.326	2.576	3.291

APPENDIX C: F-TABLE AT 5% (UPPER TAIL)

F-Table, Critical Values, 5% in Upper Tail

Degrees of freedom for the numerator along top row

Degrees of freedom for the denominator along side row

	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40
1	161	200	216	225	230	234	237	239	241	242	244	246	248	249	250	251
2	18.5	19.0	19.2	19.2	19.3	19.3	19.4	19.4	19.4	19.4	19.4	19.4	19.4	19.5	19.5	19.5
3	10.1	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	2.01	1.96
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.03	1.98	1.94
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.01	1.96	1.91
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.98	1.94	1.89
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.79	1.74	1.69
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.70	1.65	1.59
120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.50
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.52	1.46	1.39

APPENDIX D: F-TABLE AT 2.5% (UPPER TAIL)

F-Table, Critical Values, 2.5% in Upper Tails

Degrees of freedom for the numerator along top row

Degrees of freedom for the denominator along side row

	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40
1	648	799	864	900	922	937	948	957	963	969	977	985	993	997	1001	1006
2	38.51	39.00	39.17	39.25	39.30	39.33	39.36	39.37	39.39	39.40	39.41	39.43	39.45	39.46	39.46	39.47
3	17.44	16.04	15.44	15.10	14.88	14.73	14.62	14.54	14.47	14.42	14.34	14.25	14.17	14.12	14.08	14.04
4	12.22	10.65	9.98	9.60	9.36	9.20	9.07	8.98	8.90	8.84	8.75	8.66	8.56	8.51	8.46	8.41
5	10.01	8.43	7.76	7.39	7.15	6.98	6.85	6.76	6.68	6.62	6.52	6.43	6.33	6.28	6.23	6.18
6	8.81	7.26	6.60	6.23	5.99	5.82	5.70	5.60	5.52	5.46	5.37	5.27	5.17	5.12	5.07	5.01
7	8.07	6.54	5.89	5.52	5.29	5.12	4.99	4.90	4.82	4.76	4.67	4.57	4.47	4.41	4.36	4.31
8	7.57	6.06	5.42	5.05	4.82	4.65	4.53	4.43	4.36	4.30	4.20	4.10	4.00	3.95	3.89	3.84
9	7.21	5.71	5.08	4.72	4.48	4.32	4.20	4.10	4.03	3.96	3.87	3.77	3.67	3.61	3.56	3.51
10	6.94	5.46	4.83	4.47	4.24	4.07	3.95	3.85	3.78	3.72	3.62	3.52	3.42	3.37	3.31	3.26
11	6.72	5.26	4.63	4.28	4.04	3.88	3.76	3.66	3.59	3.53	3.43	3.33	3.23	3.17	3.12	3.06
12	6.55	5.10	4.47	4.12	3.89	3.73	3.61	3.51	3.44	3.37	3.28	3.18	3.07	3.02	2.96	2.91
13	6.41	4.97	4.35	4.00	3.77	3.60	3.48	3.39	3.31	3.25	3.15	3.05	2.95	2.89	2.84	2.78
14	6.30	4.86	4.24	3.89	3.66	3.50	3.38	3.29	3.21	3.15	3.05	2.95	2.84	2.79	2.73	2.67
15	6.20	4.77	4.15	3.80	3.58	3.41	3.29	3.20	3.12	3.06	2.96	2.86	2.76	2.70	2.64	2.59
16	6.12	4.69	4.08	3.73	3.50	3.34	3.22	3.12	3.05	2.99	2.89	2.79	2.68	2.63	2.57	2.51
17	6.04	4.62	4.01	3.66	3.44	3.28	3.16	3.06	2.98	2.92	2.82	2.72	2.62	2.56	2.50	2.44
18	5.98	4.56	3.95	3.61	3.38	3.22	3.10	3.01	2.93	2.87	2.77	2.67	2.56	2.50	2.44	2.38
19	5.92	4.51	3.90	3.56	3.33	3.17	3.05	2.96	2.88	2.82	2.72	2.62	2.51	2.45	2.39	2.33
20	5.87	4.46	3.86	3.51	3.29	3.13	3.01	2.91	2.84	2.77	2.68	2.57	2.46	2.41	2.35	2.29
21	5.83	4.42	3.82	3.48	3.25	3.09	2.97	2.87	2.80	2.73	2.64	2.53	2.42	2.37	2.31	2.25
22	5.79	4.38	3.78	3.44	3.22	3.05	2.93	2.84	2.76	2.70	2.60	2.50	2.39	2.33	2.27	2.21
23	5.75	4.35	3.75	3.41	3.18	3.02	2.90	2.81	2.73	2.67	2.57	2.47	2.36	2.30	2.24	2.18
24	5.72	4.32	3.72	3.38	3.15	2.99	2.87	2.78	2.70	2.64	2.54	2.44	2.33	2.27	2.21	2.15
25	5.69	4.29	3.69	3.35	3.13	2.97	2.85	2.75	2.68	2.61	2.51	2.41	2.30	2.24	2.18	2.12
30	5.57	4.18	3.59	3.25	3.03	2.87	2.75	2.65	2.57	2.51	2.41	2.31	2.20	2.14	2.07	2.01
40	5.42	4.05	3.46	3.13	2.90	2.74	2.62	2.53	2.45	2.39	2.29	2.18	2.07	2.01	1.94	1.88
60	5.29	3.93	3.34	3.01	2.79	2.63	2.51	2.41	2.33	2.27	2.17	2.06	1.94	1.88	1.82	1.74
120	5.15	3.80	3.23	2.89	2.67	2.52	2.39	2.30	2.22	2.16	2.05	1.94	1.82	1.76	1.69	1.61
∞	5.02	3.69	3.12	2.79	2.57	2.41	2.29	2.19	2.11	2.05	1.94	1.83	1.71	1.64	1.57	1.48

APPENDIX E: CHI-SQUARED TABLE

Values of χ^2 (Degrees of Freedom, Level of Significance)

Probability in Right Tail

Degrees of Freedom	0.99	0.975	0.95	0.9	0.1	0.05	0.025	0.01	0.005
1	0.000157	0.000982	0.003932	0.0158	2.706	3.841	5.024	6.635	7.879
2	0.020100	0.050636	0.102586	0.2107	4.605	5.991	7.378	9.210	10.597
3	0.1148	0.2158	0.3518	0.5844	6.251	7.815	9.348	11.345	12.838
4	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.554	0.831	1.145	1.610	9.236	11.070	12.832	15.086	16.750
6	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.647	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	4.107	5.009	5.892	7.041	19.812	22.362	24.736	27.688	29.819
14	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289	42.796
23	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.558
25	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	12.878	14.573	16.151	18.114	36.741	40.113	43.195	46.963	49.645
28	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.994
29	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.335
30	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
50	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490

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